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Key Data for Ionizing-Radiation Dosimetry: Measurement Standards and Applications

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**KEY DATA FOR IONIZING-RADIATION
DOSIMETRY: MEASUREMENT STANDARDS
AND APPLICATIONS**

**THE INTERNATIONAL COMMISSION ON RADIATION
UNITS AND MEASUREMENTS
(Published October 2016)**

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KEY DATA FOR IONIZING-RADIATION DOSIMETRY: MEASUREMENT STANDARDS AND APPLICATIONS

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[‡]Dr. Paul passed away in December 2015.

International Commission on Radiation Units and Measurements

Introduction

The International Commission on Radiation Units and Measurements (ICRU), since its inception in 1925, has had as its principal objective the development of internationally acceptable recommendations regarding:

- (1) quantities and units of ionizing radiation and radioactivity,
- (2) procedures suitable for the measurement and application of these quantities in clinical radiology and radiobiology, and
- (3) physical data needed in the application of these procedures, the use of which tends to assure uniformity in reporting.

The Commission also considers and makes similar types of recommendations for the radiation protection field. In this connection, its work is performed in cooperation with the International Commission on Radiological Protection (ICRP).

Policy

ICRU endeavors to collect and evaluate the latest data and information pertinent to the problems of radiation measurement and dosimetry and to recommend the most acceptable numerical values for physical reference data and techniques for current use.

The Commission's recommendations are kept under continual review in order to keep abreast of the rapidly expanding uses of radiation.

ICRU feels that it is the responsibility of national organizations to introduce their own detailed technical procedures for the development and maintenance of standards. However, it urges that all countries adhere as closely as possible to the internationally recommended basic concepts of radiation quantities and units.

The Commission maintains and develops a system of quantities and units and concepts (*e.g.*, for radiation therapy) and guidance for measurement procedures and techniques having the widest possible range of applicability. Situations can arise from time

to time for which an expedient solution of a current problem is required.

ICRU invites and welcomes constructive comments and suggestions regarding its recommendations and reports. These may be transmitted to the Chairman.

Current Program

The Commission recognizes its obligation to provide guidance and recommendations in the areas of radiation therapy, radiation protection, and the compilation of data important to these fields, and to scientific research and industrial applications of radiation. Increasingly, the Commission is focusing on the problems of protection of the patient and evaluation of image quality in diagnostic radiology and radiation oncology. These activities do not diminish the ICRU's commitment to the provision of a rigorously defined set of quantities and units useful in a very broad range of scientific endeavors.

The Commission is currently engaged in the formulation of ICRU Reports treating the following subjects:

Bioeffect Modeling and Biologically Equivalent Dose Concepts in Radiation Therapy
Monitoring and Assessment of Radiation Releases to the Environment
Operational Radiation Protection Quantities for External Radiation
Prescribing, Recording, and Reporting Ion-Beam Therapy
Prescribing, Recording, and Reporting Stereotactic Treatments with Small Photo Beams
Retrospective Assessment of Individual Doses for Acture Exposures to Ionizing Radiation

The Commission continually reviews progress in radiation science with the aim of identifying areas in which the development of guidance and recommendations can make an important contribution.

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ICRU's Relationship with Other Organizations

In addition to its close relationship with the ICRP, ICRU has developed relationships with national and international agencies and organizations. In these relationships, ICRU is looked to for primary guidance in matters relating to quantities, units, and measurements for ionizing radiation, and their applications in the radiological sciences. In 1960, through a special liaison agreement, ICRU entered into consultative status with the International Atomic Energy Agency (IAEA). The Commission has a formal relationship with the United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR), whereby ICRU observers are invited to attend annual UNSCEAR meetings. The Commission and the International Organization for Standardization (ISO) informally exchange notifications of meetings, and ICRU is formally designated for liaison with two of the ISO technical committees ICRU is a member of Consultative Committee for Units (CCU) – BIPM and Consultative Committee for Ionizing Radiation (CCRI(I) – BIPM and Observer to CCRI(II) and CCRI (III). ICRU also enjoys a strong relationship with its sister organization, the National Council on Radiation Protection and Measurements (NCRP). In essence, ICRU and NCRP were founded concurrently by the same individuals. Presently, this long-standing relationship is formally acknowledged by a special liaison agreement. ICRU also exchanges reports with the following organizations:

Bureau International des Poids et Mesures
European Commission
International Council for Science
International Electrotechnical Commission
International Labour Office
International Organization for Medical Physics
International Radiation Protection Association
International Union of Pure and Applied Physics
United Nations Educational, Scientific and Cultural Organization

The Commission has found its relationship with all of these organizations fruitful and of substantial benefit to the ICRU program.

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In addition to the direct monetary support provided by these organizations, many organizations provide indirect support for the Commission's program. This support is provided in many forms, including, among others, subsidies for (1) the time of individuals participating in ICRU activities, (2) travel costs involved in ICRU meetings, and (3) meeting facilities and services.

In recognition of the fact that its work is made possible by the generous support provided by all of the organizations supporting its program, the Commission expresses its deep appreciation.

Hans-Georg Menzel
Chairman, ICRU
Heidelberg, Germany

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Preface

This Report is the result of a specific request by the Consultative Committee on Ionizing Radiation (CCRI), established by the International Committee for Weights and Measures (CIPM), the body that supervises the work of the International Bureau of Weights and Measures (BIPM). The histories of the CCRI and ICRU are connected. ICRU evolved from a committee set up at the 1925 meeting of the First International Congress of Radiology convened to address units and measurements in order to describe x rays that were then finding growing use by the medical profession. This International X-Ray Unit Committee met in 1928 and recommended the use of an early version of the röntgen as a dosimetric unit in radiation therapy and suggested that ionization chambers be used for comparisons of standards. Their next meeting in 1931 discussed the comparisons of free-air ionization chambers then in place in England, Germany, France, and the USA, and changed the name of the Committee to the International Committee for Radiological Units. This transformed in several stages to the current name, with “Measurements” being added in 1956 to better describe the Commission’s scope of activities. ICRU recognized that comparisons of national standards would better be carried out within the framework of the Meter Convention, and ICRU suggested that the General Conference on Weights and Measures extend the activities of the BIPM to measurement standards for ionizing radiation. Thus, the CIPM established in 1958 the Consultative Committee for Measurement Standards of Ionizing Radiation (which became the CCRI in 1997) to carry out international comparisons of these standards. During its ongoing work, the CCRI has looked to ICRU as the authoritative source of internationally accepted values of key parameters needed in measurement standards for ionizing radiation.

ICRU Report 10b (1964) and ICRU Report 14 (1969) elaborated the use of the Bragg–Gray ionization chambers for the dosimetry of photons with energies above about 500 keV, which results in the need for electron stopping-power ratios. Because these early Reports were focused on the determination of absorbed dose, the use of W to convert ionization

charge to energy, inherent in these treatments, was also the subject of evaluation. The quantity kerma, adopted in ICRU Report 10a (1962), set the stage for the eventual replacement, through the use of W_{air} , of exposure by air kerma as the quantity of interest in national measurement standards based on free-air chambers for x-ray beams with generating potentials up to about 500 kV. The result has been that electron stopping powers and W_{air} values have become key data in the establishment of national measurement standards for the dosimetry of ionizing radiation.

ICRU continued its work in the development of the physical data needed in radiation dosimetry into more recent years with, in addition to other Reports in which similar data were also included, the publication of ICRU Report 31, *Average Energy Required to Produce an Ion Pair* (1979), and Report 37, *Stopping Powers for Electrons and Positrons* (1984). Augmenting Report 37 to include standard reference data for other charged particles being used in radiation therapy has been Report 49, *Stopping Powers and Ranges for Protons and Alpha Particles* (1993), and Report 73, *Stopping of Ions Heavier Than Helium* (2005). Such data have tended to serve as conventional values used universally in the international harmonization of national standards for air kerma and absorbed dose, and they provide fundamental information used in radiation-transport calculations. However, they do require updating from time to time to take into account the introduction of new pertinent results (such as had been done in 1985 by the CCRI for W_{air}). The accumulation of new results and analyses called into question, in particular, the use of the electron stopping powers for graphite given in ICRU Report 37 for the evaluation of air kerma using graphite-walled ionization chambers. This was the primary motivation for the CCRI request that ICRU develop a recommendation on this and related data.

In the development of this Report, ICRU was most fortunate to have the leadership of Mitio Inokuti, a member of the ICRU since 1985, who was also the Vice Chairman of the Report Committees preparing Reports 37 and 49. He was instrumental in organizing and guiding the efforts for the current report until his death on 4 June 2009. The present volume, in large

measure an extension of his impressive scientific work, is gratefully dedicated to the accomplishments of our colleague and friend Mitio Inokuti.

ICRU extends its condolences to the family and friends of Helmut Paul, formerly with the Johannes Kepler University Linz, who passed away 21 December 2015. Helmut was a Consultant on this ICRU Report, a member of the Report Committee for Report 73, and was largely responsible for important corrections to results given in that Report, later issued by the ICRU in 2009.

Several individuals provided important material and advice that helped in the preparation of this Report: Gerhard Stucki, formerly with the Swiss Federal Institute of Metrology, METAS; Malcolm McEwen of the National Research Council Canada; and Josep Sempau of the Universitat Politècnica de Catalunya. ICRU wishes to thank them for their contributions.

Stephen M. Seltzer
David T. Burns

Abstract

Ionizing radiation dosimetry is used to describe those measurements or calculations that provide information on the energy deposited by the interaction of ionizing radiation with matter. Radiation dosimetry is of importance in several areas, including radiation therapy, radiation protection, and the industrial use of radiation. Several techniques have been developed for measuring energy deposition, with ionization chambers and calorimeters being the most important. These techniques form the basis of primary measurement standards for ionizing radiation, and these primary standards require values and uncertainty estimates for certain key parameters used to relate the result of measurement to the desired quantity.

The calculation of the interaction of radiation with matter using Monte Carlo techniques is now well developed and can be used to study energy deposition for problems where measurements are difficult or impossible. Both measurements and calculations require knowledge of basic quantities related to the interaction of radiation with matter. These include data on photon cross sections, electron stopping powers, and the average energy to create an ion pair, to name a few. These data are often referred to as “key data.”

This Report examines key data for stopping powers for charged particles ranging from electrons to carbon ions. Values and uncertainties are assigned to the mean excitation energies for air, graphite, and liquid water, and tables of stopping powers covering the energy range from 1 keV to 1 GeV, or higher, are provided. Photon cross sections for air, water, and graphite are reviewed, examined, and compared with relevant measurements to estimate their uncertainties. Values are recommended for the average energy to create an ion pair in air, W_{air} . The available data for the chemical yield for Fricke dosimetry and for the heat defects for graphite and liquid water are summarized, as is the humidity correction factor for air-filled ionization chambers and the correction to the measured charge due to the initial ion pairs created by an incident photon. Data for the correction in photon and electron beams at low energies for the deviation of W_{air} from the recommended asymptotic value are also summarized.

The impact of the recommended changes is discussed. Important changes are an increase in the uncertainty for air-kerma measurements with free-air chambers and a decrease of about 0.7 % in ^{60}Co air-kerma measurements. The recommended stopping powers for graphite and liquid water differ by up to 1 % from those previously recommended. For radiation dosimetry based on ionization chambers calibrated against absorbed-dose-to-water calorimeters, changes in the measured absorbed dose to water will not exceed 0.5 %.

1. Introduction

1.1 The Need for Recommendations on Key Data

In metrology, a primary standard is a standard that establishes the unit for a quantity without reference to other standards for the same unit. Generally, primary standards require values and uncertainty estimates for certain key parameters used to relate the result of measurement to the desired quantity. The dosimetry of ionizing radiation involves primary standards for the quantities air kerma and absorbed dose in water that require photon and electron interaction coefficients for graphite, air, and liquid water. Ionometric standards also require a value for W_{air} , the mean energy expended to produce an ion pair in dry air, as well as k_h , which accounts for the effect of humidity (ambient water vapor) on W_{air} and thus the response of an air-filled ionization chamber. Calorimetric standards require knowledge of h , the heat defect, which accounts for the difference between the energy imparted by the ionizing radiation and the energy that appears as heat. Other data that might be considered key arise from the use of other types of measurement standards, notably the radiation chemical yield, G , associated with Fricke dosimetry.

In subsequent steps of the traceability chain, either at the level of secondary standard laboratories or the end user (for example, at a hospital or an industrial facility), the same basic interaction data are needed for reference dosimetry or beam calibration. This is the case in most applications, such as radiation therapy, diagnostic radiology, radiation protection, radiation biology, and industrial radiation processing. Consistency throughout the entire dosimetry chain is ensured with the use of the same key data.

Monte Carlo techniques are increasingly used to model radiation transport for a wide range of applications (Jenkins *et al.*, 1989). Although the calculations can be done to high precision, accurate cross-section data are required if the calculated results are to represent physical reality (Salvat and Fernández-Varea, 2009).

1.2 Reasons for Reevaluating Key Data

Electron stopping powers are currently based on the recommendations of ICRU Report 37 (ICRU,

1984a), which also includes recommended values for the mean excitation energy, I , of elements and compounds. Of particular interest in dosimetry is the use therein of the value $I_g = 78$ eV for graphite, with a standard uncertainty of about 4 eV. The adoption of stopping powers using this value had implications for the W_{air} value used for photon and electron beams because much of the information on W_{air} is derived from the comparison of graphite calorimeters and graphite-walled ionization chambers. This method essentially determines the product of W_{air} and the graphite-to-air stopping-power ratio, $s_{g,\text{air}}$. A re-evaluation of W_{air} by Boutillon and Perroche-Roux (1987) incorporating the stopping powers of ICRU Report 37 resulted in the formal adoption in 1985 of the value $W_{\text{air}} = 33.97$ eV, with standard uncertainty 0.05 eV, by the Consultative Committee on Measurement Standards for Ionizing Radiation (CCEMRI, 1985).

A significant newer measurement of I_g was made by Bichsel and Hiraoka (1992), who published the value $I_g = 86.9$ eV with a relatively small uncertainty of 1.2 eV. Adoption of this value rather than $I_g = 78$ eV would change $s_{g,\text{air}}$ for ^{60}Co γ -ray reference beams by about 1.4 %, although the consequent change to the relevant standards must also take into account the significant correlation between $s_{g,\text{air}}$ and W_{air} . Furthermore, MacPherson (1998) published the results of stopping-power measurements in graphite, indicating that the electronic stopping power should be calculated using the crystalline graphite density rather than its bulk density. Although results for both densities were given in ICRU Report 37 (ICRU, 1984a), it is the latter that was recommended in that Report. The effect of this change is about 0.2 % for ^{60}Co gamma radiation, but approaches 1 % for high-energy electrons. Discussion of these matters at the biennial meetings of the CCEMRI (which became the CCRI, the Consultative Committee on Ionizing Radiation) led in 2001 to a formal CCRI recommendation that the ICRU convene the appropriate expertise to report on this matter. In the meantime, new results became available from a number of sources, and analyses of accumulated data were made by Burns (2009, 2012) and by Thomson and Rogers (2010), with recommendations relevant to the present Report.

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In addition to the need for recommended values and uncertainties for W_{air} , I_g , and the product $W_{\text{air}} s_{g,\text{air}}$, primary standards for air kerma and absorbed dose in water for photon and electron beams also make use of a number of other parameters, as will be discussed in Section 3. As well as measuring I_g , Bichsel and Hiraoka (1992) also determined a value for the mean excitation energy for liquid water of $I_w = 79.7$ eV with an uncertainty of 0.5 eV, higher than the value $I_w = 75$ eV recommended in ICRU Report 37 (ICRU, 1984a). Since then, there have been a number of new determinations of I_w using proton and heavy-ion beams. This renewed interest in I_w has been stimulated by the increasing use of such beams in radiation therapy and the importance of correctly simulating the position of the Bragg peak. The relatively low value of $I_w = 67.2$ eV implicit in the heavy-ion stopping-power data of ICRU Report 73 (ICRU, 2005) was the subject of some controversy, leading to the issuance of an erratum (Sigmund *et al.*, 2009), in which a significantly larger value was used to generate new tables. In making recommendations, consideration is also given to estimates of mean excitation energies derived from dielectric-response functions and *ab initio* calculations (see Section 5.2).

Regarding the dosimetry of proton and heavier ion beams, in addition to the substantial impact of the diversity of estimates for I_w (Andreo, 2009), over the past 10 to 15 years there has been some controversy over the W_{air} value for protons. A comprehensive review of the literature and an analysis of data obtained by comparing calorimetric and ionometric methods was made in ICRU Report 59 (ICRU, 1998a), and the value $W_{\text{air}} = 34.8$ eV was recommended for humid air. However, based on the same basic data, the international code of practice given in the International Atomic Energy Agency's Report TRS-398 (IAEA, 2000a) adopted $W_{\text{air}} = 34.23$ eV for dry air, smaller by about 1.7 %. As discussed by Medin *et al.* (2000), this difference arises not only from the humidity effect, but also from the use of different proton stopping powers and from the use of a statistical analysis based on the weighted median (Müller, 2000a; 2000b). Following further discussion (Andreo *et al.*, 2000; Verhey, 2000), Jones (2006) recommended the value $W_{\text{air}} = 34.2$ eV for dry air, in agreement with that adopted in the IAEA protocol, which was then adopted in ICRU Report 78 (2007). The present Report Committee was also charged with reviewing this issue and providing recommendations.

With recommendations on I values later in this Report, new analyses of the electronic stopping powers for electrons in graphite, air, and liquid water include more recent evaluations of the radiative stopping power for these materials. As will be seen in

Section 3, photon-interaction cross sections for the same key materials are also needed in the development of dosimetry measurement standards, as well as in clinical dosimetry. Although fairly complete and reasonably consistent data on these quantities have been developed over the past few decades, the opportunity is taken to review the calculations of photon-interaction coefficients with an emphasis on graphite, air, and water. Perhaps more importantly, the role of uncertainties in measurement standards has increased significantly in recent years, and the Report Committee was charged with making recommendations on the uncertainties associated with new evaluations of the key parameters. While such estimates can be problematic, particularly for parameters based heavily on theoretical models, the adoption of new parameter values in dosimetry requires a statement of the associated uncertainty, and this is best estimated as part of the process of generating new values. In this respect, it is significant that electron stopping powers and photon-interaction coefficients generally appear as ratios in the determination of dosimetric standards and in practical radiation dosimetry. This has the effect of reducing the overall uncertainty through correlation, and estimates of uncertainty are made for the key ratios discussed in Section 3 taking such correlation into account.

1.3 Relation to Earlier ICRU Reports

As indicated from the preceding paragraphs, the present Report has strong links with previous ICRU Reports on stopping powers of electrons and positrons (ICRU, 1984a), protons (ICRU, 1993), and heavier ions (ICRU, 2005; Sigmund *et al.*, 2009). Although there are no ICRU Reports exclusively devoted to key data for photons, Report 17 (ICRU, 1970) on kilovoltage x-ray dosimetry provides interaction coefficients for photon energies between 1 keV and 150 keV, including those for graphite, air, and water. Comprehensive tabulations of attenuation coefficients for water and air for photon energies above 10 keV have been included in ICRU Reports 44 (ICRU, 1989) and 46 (ICRU, 1992) devoted to tissue substitutes for radiation dosimetry. Ratios of mass stopping powers and mass energy-absorption coefficients for electron- and photon-beam dosimetry have been included explicitly or indirectly in ICRU Reports 35 (ICRU, 1984b) and 64 (ICRU, 2001), respectively; these data have been superseded in current dosimetry protocols, for example, the report of Task Group 51 of the American Association of Physicists in Medicine (Almond *et al.*, 1999; McEwen *et al.*, 2014a) and IAEA TRS-398 (IAEA, 2000a). With regard to the mean energy required to produce an ion pair in air, ICRU Report 31 (ICRU,

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1979) remains one of the most comprehensive sources of data. Finally, definitions in the present Report rest on the most recent edition of the ICRU Report on Fundamental Quantities and Units for Ionizing Radiation, ICRU Report 85a (ICRU, 2011), which supersedes the widely used ICRU Report 60 (ICRU, 1998b).

1.4 Organization of Report

The remainder of this Report is organized as follows: Section 2 gives a brief summary of the definitions of the quantities required in this Report. Although most of the definitions and relationships between the quantities are well established, there continue to be new developments, such as the recognition that the conventional relationship between exposure and air kerma is only approximate. Many of the key data play important roles in the realization of primary standards, so Section 3 reviews the primary standards for air kerma and absorbed dose to water and shows how the key data enter the measurement equations. An important quantity, not

only for measurement standards but also for Monte Carlo calculations, is the charged-particle stopping power. Section 4 briefly reviews the theory of charged-particle stopping power in which the mean excitation energy and the density-effect correction are the non-trivial quantities; the review is not intended to be in-depth, but rather to provide a sufficient understanding of the methods used in this Report that mainly extend those used in ICRU Report 37 (ICRU, 1984a), ICRU Report 49 (ICRU, 1993), and ICRU Report 73 (ICRU, 2005). Section 5 reviews the key data and, when appropriate, gives recommended values and associated uncertainties. Section 6 reviews the theories used to calculate photon cross sections and the resulting attenuation and mass energy-absorption coefficients that play an important role in radiation dosimetry, and includes comparisons with relevant measurements. Section 7 reviews the main impacts that the changes recommended in this Report will have on radiation dosimetry. Tables of pertinent charged-particle stopping powers, incorporating the recommendations made here, are given in the Appendix.

2. Definitions of Basic Quantities and Terms

Although some familiarity with quantities and terms has so far been assumed, the full and accurate development of the discussion of key data requires the definitions of a number of quantities. These, from ICRU Report 85a (ICRU, 2011), are listed below.

2.1 Fluence

The fluence, Φ , is given by $\Phi = dN/da$, where dN is the number of particles incident on a sphere of cross-sectional area da . The energy fluence, Ψ , is given by $\Psi = dR/da$, where dR is the radiant energy incident on a sphere of cross-sectional area da . The radiant energy, R , is the energy (excluding rest energy) of the particles that are emitted, transferred, or received.

The distributions, Φ_E and Ψ_E , of the fluence and energy fluence with respect to energy are given by $\Phi_E = d\Phi/dE$, and $\Psi_E = d\Psi/dE$, where $d\Phi$ is the fluence of particles of energy between E and $E + dE$, and $d\Psi$ is their energy fluence. The relationship between the two distributions is given by $\Psi_E = E\Phi_E$.

2.2 Mass Attenuation Coefficient

The mass attenuation coefficient, μ/ρ , of a material of density ρ , for uncharged particles of a given type and energy, is given by

$$\frac{\mu}{\rho} = \frac{1}{\rho dl} \frac{dN}{N} = \frac{N_A}{M_A} \sigma = \frac{N_A}{M_A} \sum_J \sigma_J, \quad (2.1)$$

where dN/N is the mean fraction of the particles that experience interactions in traversing the distance dl in the material, N_A the Avogadro constant, M_A the molar mass of the target material, and σ the total cross section for an interaction, which is the sum of the component cross sections, σ_J . In this Report, the only uncharged particle considered is the photon, and the component cross sections are for photoelectric absorption, incoherent scattering (mainly Compton), coherent scattering, and pair (and triplet) production.

2.3 Mass Energy-Transfer Coefficient

The mass energy-transfer coefficient, μ_{tr}/ρ of a material of density ρ , for uncharged particles of a given type and energy, is given by

$$\frac{\mu_{tr}}{\rho} = \frac{1}{\rho dl} \frac{dR_{tr}}{R} = \frac{N_A}{M_A} \sum_J f_J \sigma_J, \quad (2.2)$$

where dR_{tr} is the mean energy that is transferred to kinetic energy of charged particles by interactions of the uncharged particles of incident radiant energy R in traversing the distance dl in the material, and f_J is the quotient of the mean energy transferred to kinetic energy of charged particles in an interaction of type J by the kinetic energy of the incident uncharged particle (here, a photon).

2.4 Mass Energy-Absorption Coefficient

A fraction g of the kinetic energy transferred to charged particles is subsequently lost on average in radiative processes (bremsstrahlung, in-flight annihilation, and fluorescence radiations) as the charged particles slow to rest in the material, and this fraction g is specific to the material. The product of μ_{tr}/ρ for a material and $(1-g)$ is called the *mass energy-absorption coefficient*, μ_{en}/ρ , of the material for uncharged particles,

$$\frac{\mu_{en}}{\rho} = \frac{\mu_{tr}}{\rho} (1 - g). \quad (2.3)$$

2.5 Mass Stopping Power

The mass stopping power, S/ρ , of a material of density ρ , for charged particles of a given type and energy, is given by

$$\frac{S}{\rho} = \frac{1}{\rho} \frac{dE}{dl}, \quad (2.4)$$

where dE is the mean energy lost by the charged particles in traversing the distance dl in the material. The quantity $S = dE/dl$ denotes the *linear stopping power*. In general, the stopping power comprises three components

$$\frac{S}{\rho} = \frac{1}{\rho} \left(\frac{dE}{dl} \right)_{\text{el}} + \frac{1}{\rho} \left(\frac{dE}{dl} \right)_{\text{rad}} + \frac{1}{\rho} \left(\frac{dE}{dl} \right)_{\text{nuc}}, \quad (2.5)$$

where $(1/\rho)(dE/dl)_{\text{el}} = (1/\rho)S_{\text{el}}$ is the *mass electronic (or collision¹) stopping power* due to interactions with atomic electrons resulting in ionization or excitation, $(1/\rho)(dE/dl)_{\text{rad}} = (1/\rho)S_{\text{rad}}$ is the *mass radiative stopping power* due to emission of bremsstrahlung in the electric fields of atomic nuclei or atomic electrons, and $(1/\rho)(dE/dl)_{\text{nuc}} = (1/\rho)S_{\text{nuc}}$ is the *mass nuclear stopping power²* due to elastic Coulomb interactions in which recoil energy is imparted to atoms.

A related quantity is the restricted mass electronic stopping power, defined as the mean energy loss by charged particles per pathlength in the material due to ionization or excitation involving energy transfers up to a selected cutoff energy. Expressions for light and heavy charged particles are given in Section 4.1.

2.6 Linear Energy Transfer

The linear energy transfer (LET), L_{Δ} , of a material, for charged particles of a given type and energy, is given by

$$L_{\Delta} = \frac{dE_{\Delta}}{dl}, \quad (2.6)$$

where dE_{Δ} is the mean energy lost by the charged particles due to electronic interactions in traversing the distance dl , minus the mean sum of the kinetic energies in excess of Δ of all electrons released by the charged particles. The LET, L_{Δ} , can also be expressed by $L_{\Delta} = S_{\text{el}} - (dE_{\text{ke},\Delta}/dl)$, where S_{el} is the linear electronic stopping power and $dE_{\text{ke},\Delta}$ is the mean sum of the kinetic energies, greater than Δ , of all the electrons released by the charged particle in traversing a distance dl . If no energy cutoff is imposed, the unrestricted LET, L_{∞} , is equal to S_{el} , and may be denoted as L .

2.7 Cema³

The cema, C , for ionizing charged particles, is given by

$$C = \frac{dE_{\text{el}}}{dm}, \quad (2.7)$$

¹The older term was “collision stopping power.” Because all interactions can be considered “collisions,” the more specific term “electronic” is preferred.

²The established term “mass nuclear stopping power” can be misleading because this quantity does not pertain to nuclear interactions.

where dE_{el} is the mean energy lost in electronic interactions in the mass dm of a material by the charged particles, except secondary electrons, incident on dm . The energy lost by charged particles in electronic interactions includes the energy expended to overcome the binding energy and the initial kinetic energy of the liberated electrons, referred to as secondary electrons. Thus, energy subsequently lost by all secondary electrons is excluded from dE_{el} .

The cema can be expressed in terms of the distribution, Φ_E , of the charged-particle fluence, with respect to energy. According to the definition of cema, the distribution Φ_E does not include the contribution of secondary electrons to the fluence, but the contributions of all other charged particles, such as secondary protons, α particles, tritons, and ions produced in nuclear interactions, are included in the cema. The cema, C , is thus given by

$$C = \int \Phi_E \frac{S_{\text{el}}}{\rho} dE = \int \Phi_E \frac{L_{\infty}}{\rho} dE, \quad (2.8)$$

where S_{el}/ρ is the mass electronic stopping power of a specified material for charged particles of energy E , and L_{∞} the corresponding unrestricted LET. In general, the cema is evaluated as the sum of contributions by all species of charged particles, except liberated secondary electrons.

For charged particles of high energies, it might be undesirable to disregard energy transport by secondary electrons of all energies. A modified concept, *restricted cema*, C_{Δ} , is then defined⁴ as

$$C_{\Delta} = \int_{\Delta} \Phi'_E \frac{L_{\Delta}}{\rho} dE + \Delta \cdot \Phi'_E(\Delta) \frac{L_{\infty}(\Delta)}{\rho}, \quad (2.9)$$

This differs from Eq. (2.8) in that L_{∞} is replaced by L_{Δ} , the distribution Φ'_E now includes secondary electrons liberated in dm with kinetic energies greater than Δ , and the track-end term accounts for local energy deposition by electrons with energies less than Δ (Kellerer *et al.*, 1992). For $\Delta = \infty$, restricted cema is identical to cema.

The quantities cema and restricted cema can be used as approximations to absorbed dose from charged particles. Equality of absorbed dose and cema is approached to the degree that *secondary-charged-particle equilibrium* (CPE) exists and that radiative losses and those due to elastic nuclear interactions are negligible. Secondary CPE is achieved at a point if the fluence of secondary charged particles is

³Converted energy per mass.

⁴This formulation, which includes the so-called track-end term on the far right of Eq. (2.9), is more general than that given in ICRU Report 85a (ICRU, 2011).

constant within distances equal to their maximum range. For restricted cema, only partial secondary CPE, up to kinetic energy Δ , is required.

2.8 Radiation Chemical Yield

The radiation chemical yield, $G(x)$, of an entity, x , is given by

$$G(x) = \frac{n(x)}{\bar{\varepsilon}}, \quad (2.10)$$

where $n(x)$ is the mean amount of substance of that entity produced, destroyed, or changed in a system by the mean energy imparted, $\bar{\varepsilon}$, to the matter of that system.

2.9 Mean Energy Expended in a Gas per Ion Pair Formed

The mean energy expended in a gas per ion pair formed, W , is given by

$$W = \frac{T}{N}, \quad (2.11)$$

where N is the mean total liberated charge of either sign, divided by the elementary charge, when the initial kinetic energy T of a charged particle introduced into the gas is completely dissipated in the gas. It follows from the definition of W that the ions produced by bremsstrahlung or other secondary radiation emitted by the initial and secondary charged particles are included in N . The charge of the initial charged particle is not included in N .

At low incident energies such that T approaches the ionization potential of the gas, a significant fraction of this energy is lost to processes that do not liberate charge. In this domain, W increases rapidly as the incident energy is decreased.

In certain cases, it could be necessary to focus attention on the variation in the mean energy expended per ion pair along the path of the particle; then the concept of a differential W is required, as defined in ICRU Report 31 (ICRU, 1979). The differential value, w , of the mean energy expended in a gas per ion pair formed is

$$w = \frac{dT}{dN}, \quad (2.12)$$

where dT is the mean energy lost by a charged particle of kinetic energy T in traversing a layer of gas of infinitesimal thickness, and dN the mean total liberated charge of either sign divided by the elementary charge when dT is completely dissipated in the gas.

The relationship between W and w is given by

$$W(T) = \frac{T}{\int_{I_1}^T \frac{dT'}{w(T')}}, \quad (2.13)$$

where I_1 is the lowest ionization potential of the gas, and T' the instantaneous kinetic energy of the charged particle as it slows down.

2.10 Kerma⁵

The kerma, K , for ionizing uncharged particles, is given by

$$K = \frac{dE_{tr}}{dm}, \quad (2.14)$$

where dE_{tr} is the mean sum of the initial kinetic energies of all the charged particles liberated in the mass dm of a material by the uncharged particles incident on dm . The special name for the unit of kerma is gray (Gy). The quantity dE_{tr} includes the kinetic energy of the charged particles emitted in the decay of excited atoms/molecules⁶ or in nuclear de-excitation or disintegration.

For a fluence, Φ , of uncharged particles of energy E , the kerma, K , in a specified material is given by

$$K = \Phi E \frac{\mu_{tr}}{\rho} = \Psi \frac{\mu_{tr}}{\rho}, \quad (2.15)$$

where μ_{tr}/ρ is the mass energy-transfer coefficient of the material for these particles.

In dosimetric calculations, the kerma, K , is usually expressed in terms of the distribution, Φ_E , of the uncharged-particle fluence with respect to energy. The kerma, K , is then given by

$$K = \int \Phi_E E \frac{\mu_{tr}}{\rho} dE = \int \Psi_E \frac{\mu_{tr}}{\rho} dE, \quad (2.16)$$

where μ_{tr}/ρ is the mass energy-transfer coefficient of the material for uncharged particles of energy E .

Although kerma is a quantity that concerns the initial transfer of energy to matter, it is sometimes used as an approximation to absorbed dose. The numerical value of the kerma approaches that of the absorbed dose to the degree that charged-particle equilibrium (CPE) exists, that radiative losses are negligible, and that the kinetic energy of the uncharged particles is large compared with the binding energy of the liberated charged particles. Charged-particle equilibrium exists at a point if the

⁵Kinetic energy released per mass.

⁶For example, Auger, Coster-Kronig, shake-off electrons.

charged-particle fluence rate, differential in energy and direction, is constant within distances equal to the maximum charged-particle range.

A quantity related to the kerma, termed the *collision kerma*, has long been in use as an approximation to absorbed dose (Attix, 1979a; 1979b) when radiative losses are not negligible. The collision kerma,⁷ K_{col} , excludes the radiative losses by the liberated charged particles, and—for a fluence, Φ , of uncharged particles of energy E in a specified material—is given by

$$K_{\text{col}} = \Phi E \frac{\mu_{\text{en}}}{\rho} = \Phi E \frac{\mu_{\text{tr}}}{\rho} (1 - g) = K(1 - g), \quad (2.17)$$

where μ_{en}/ρ is the mass energy-absorption coefficient of the material for uncharged particles of energy E , and g the fraction of the kinetic energy of liberated charged particles that would be lost in radiative processes in that material.

In dosimetric calculations, the collision kerma, K_{col} , can be expressed in terms of the distribution, Φ_E , of the uncharged-particle fluence with respect to energy as

$$K_{\text{col}} = \int \Phi_E E \frac{\mu_{\text{en}}}{\rho} dE = \int \Phi_E E \frac{\mu_{\text{tr}}}{\rho} (1 - g) dE = K(1 - \bar{g}), \quad (2.18)$$

where $\bar{g}$ is the mean value of g averaged over the distribution of the kerma with respect to the electron energy. Under conditions of CPE, the numerical value of the collision kerma approaches that of the absorbed dose for higher secondary-electron energies at which radiative losses are not negligible.

2.11 Exposure

The exposure, X , is defined as

$$X = \frac{dq}{dm}, \quad (2.19)$$

where dq is the absolute value of the mean total charge of the ions of one sign produced when all the electrons and positrons liberated or created by photons incident on the mass dm of dry air are completely stopped in dry air. The ionization produced by electrons emitted in atomic/molecular relaxation processes is included in dq ; the ionization due to photons emitted by radiative processes (*i.e.*,

bremsstrahlung and fluorescence photons) is not to be included in dq .

The quantities exposure and air kerma can be related through the use of W_{air}/e , where W_{air} is the mean energy expended in dry air per ion pair formed when the initial kinetic energy of a charged particle is completely dissipated in the air, and e is the elemental charge. Then

$$K_{\text{air}} \cong X \cdot (W_{\text{air}}/e)/(1 - \bar{g}_{\text{air}}). \quad (2.20)$$

In Eq. (2.20), the quantity $\bar{g}_{\text{air}}$ is the fraction, averaged over the distribution of the air kerma with respect to the electron (positron) energy, of the kinetic energy of electrons (and positrons) liberated by the photons that is lost in radiative processes (mainly bremsstrahlung) in dry air.

The approximate equality in Eq. (2.20) is used here to reflect the fact that exposure includes the charge of electrons or ions liberated by the incident photons, whereas W_{air} pertains only to the charge produced during the slowing down of these electrons. Thus, the initial charge created by the interaction of the photons should be discounted when transferring the exposure measurement for the determination of air kerma. This difference, although relatively small, tends to become more significant as the photon energy decreases. The initial-ion correction has been considered by Büermann *et al.* (2006) and by Takata and Begum (2008), and is discussed in more detail in Section 5.5.

Additionally, W_{air} is not constant as perhaps implied in Eq. (2.20) but, as noted in Section 5.4.1, is known to increase at low energies (ICRU, 1979). At energies for which the variation of W_{air} with energy becomes important, one should consider also the effect of this increase.

2.12 Absorbed Dose

The absorbed dose, D , is given by

$$D = \frac{d\bar{\varepsilon}}{dm}, \quad (2.21)$$

where $d\bar{\varepsilon}$ is the mean energy imparted by ionizing radiation to matter of mass dm . The energy imparted, ε , to the matter in a given volume is the sum of all energy deposits in the volume, thus

$$\varepsilon = \sum_i \varepsilon_i. \quad (2.22)$$

The energy deposit, ε_i , is the energy deposited in a single interaction, i , thus

$$\varepsilon_i = \varepsilon_{\text{in}} - \varepsilon_{\text{out}} + Q, \quad (2.23)$$

⁷In the present Report we retain the term “collision kerma,” although noting that it was presumably named in analogy to the “collision stopping power,” which was changed to “electronic stopping power” in ICRU Report 60 (ICRU, 1998).

Definitions of Basic Quantities and Terms

where ε_{in} is the energy of the incident ionizing particle (excluding rest energy), ε_{out} the sum of the energies of all charged and uncharged ionizing particles leaving the interaction (excluding rest

energy), and Q the change in the rest energies of the nucleus and of all elementary particles involved in the interaction ($Q > 0$: decrease in rest energy; $Q < 0$: increase in rest energy).

3. Realization of Quantities by Primary Standards Laboratories

In principle, both kerma and absorbed dose can be determined for any material at any energy. Given the considerations noted above regarding secondary-photon contributions and charged-particle equilibrium (CPE), it is not surprising that the relationship between kerma and absorbed dose changes significantly with energy and material. Because of the very limited beam penetration and the relatively low absorbed-dose rates involved, absorbed dose is very difficult to realize directly for kilovoltage x rays, although a few water calorimeter standards have been developed (see, *e.g.*, de Prez and de Pooter, 2008; Krauss *et al.*, 2012; Rapp *et al.*, 2013). Most primary standards for these radiations are based on kerma, in particular on the determination of the air kerma using a free-air ionization chamber. At ^{60}Co γ -ray energies, the need for CPE would require a prohibitively large free-air chamber, and cavity-ionization chambers are used as primary standards. In more recent years, the direct determination of absorbed dose by graphite and water calorimetry has produced standards with an overall uncertainty that matches, and in some cases reduces, that derived from a determination of air kerma. Nevertheless, air kerma for ^{60}Co γ rays remains a very important reference quantity, particularly for standards laboratories. At the high energies produced by particle accelerators, the determination of air kerma free in air in these beams is no longer used to determine absorbed dose. In more recent years, absorbed-dose standards at high energies have been used, either directly for instrument calibrations or more commonly to determine values for correction factors that convert an ionization-chamber absorbed-dose calibration for ^{60}Co γ rays for use at higher energies (McEwen, 2010; Seuntjens *et al.*, 2000). Several review articles describing the standards used for radiation dosimetry have been published in a special issue of *Metrologia* (Sharpe, 2009).

3.1 Air-Kerma Standards

3.1.1 Free-Air Ionization Chambers

Free-air ionization chambers are discussed in detail in the review paper of Burns and Büermann

(2009) and references therein. The free-air chamber is so-called because from the point of view of radiation interactions, it is wall-less; the contributing primary photons and the liberated electrons interact only in air. The majority of free-air chambers in current use are of the parallel-plate type.

A schematic representation of a parallel-plate free-air chamber is shown in Fig. 3.1. The main elements are an entrance diaphragm with aperture radius r_{ap} and a pair of planar electrodes separated by a distance d . An electric field exists between these electrodes, produced by applying a polarizing potential to one of the electrodes while the other is maintained at ground (or virtual ground) potential. An isolated section of the latter electrode, of length l in the beam direction, is connected to a sensitive current-measuring apparatus and is referred to as the collecting electrode, while the remainder of the plate is the guard electrode. The collecting electrode and the electric field lines define a region in which charge is collected, referred to as the collecting region, and is shown shaded in the figure.

Primary photons enter the chamber through the aperture and interact with the air of the chamber to produce secondary electrons. If the electrode separation is sufficiently large, the secondary electrons slow to rest within the chamber, ionizing the air, as do all electrons liberated in the slowing-down process. If the distance between the diaphragm and the collecting region is greater than the maximum secondary-electron range, CPE will exist in the collecting region, as illustrated by the three identical secondary electron tracks in the figure (strictly speaking, as the photon fluence is attenuated only transient CPE exists). Under these conditions, the charge collected is a measure of the total charge liberated in photon interactions over the length l of the collecting region. The free-air chamber was originally designed to determine exposure, and after combining Eqs. (2.19) and (2.20), the air-kerma at the reference plane⁸ is determined using the

⁸The reference plane, or measurement point, is located at the plane within the aperture that defines the beam diameter. See Fig. 7 in Burns and Büermann (2009).

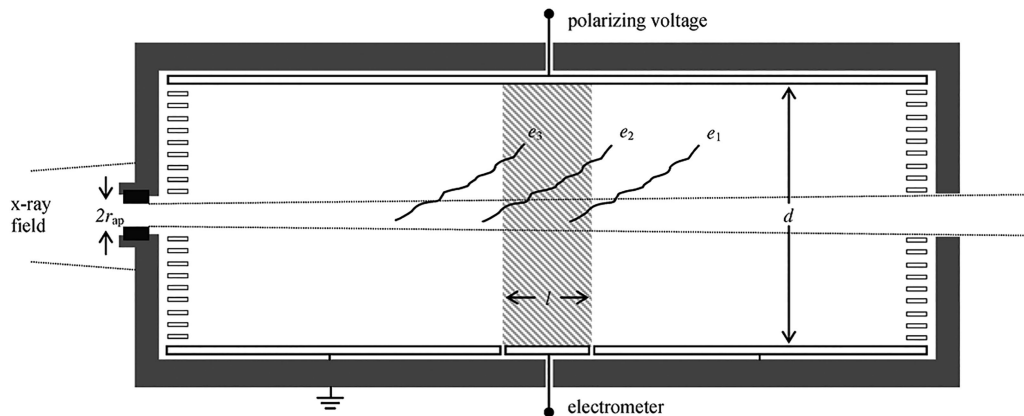


Figure 3.1. Schematic diagram of a free-air chamber. In the course of slowing down, electrons ionize the air and the ionization charge is collected and measured as a current. [Figure from Burns and Büermann (2009).]

measurement equation

$$K_{\text{air}} = (W_{\text{air}}/e) \frac{q_{\text{net}}}{m_{\text{air}}(1 - \bar{g}_{\text{air}})} \prod_i k_i$$

$$= (W_{\text{air}}/e) \frac{q_{\text{net}}}{\rho_{\text{air}} V_{\text{eff}}(1 - \bar{g}_{\text{air}})} \prod_i k_i, \quad (3.1)$$

where q_{net} is the measured net charge (the measured charge corrected for ambient and system-generated background) and the product of factors k_i corrects for the limitations of the free-air chamber. On the right-hand side of Eq. (3.1), the mass m_{air} of dry air is evaluated as the product $\rho_{\text{air}} V_{\text{eff}}$, where the effective collecting volume $V_{\text{eff}} = \pi r_{\text{ap}}^2 l$, and ρ_{air} is the mass density of dry air at the reference temperature and pressure, from which it follows that the measured charge must be corrected for deviations from reference conditions (including a correction for humidity).

The dual requirements for an attenuation length greater than the electron range and an electrode separation such that secondary electrons cannot reach either electrode limit the maximum photon energy at which a given free-air chamber can be operated. In general, small, transportable free-air chambers are used for low-energy x-ray beams up to 50 kV, and a larger instrument is employed for medium-energy x rays, typically up to around 300 kV (even 400 kV).

3.1.2 Cavity-Ionization Chambers

Cavity-ionization chambers are discussed in the review article of Büermann and Burns (2009) and the references therein. An idealized cavity chamber is displayed schematically in Fig. 3.2. The thickness of the wall must not be less than the maximum range in the wall material of the secondary electrons produced in photon interactions in the wall, and the

cavity must be small enough (or the cavity and wall materials sufficiently similar) so as not to perturb the secondary electron fluence. Thus, the energy deposition in the cavity is characteristic of the secondary electrons entering from the wall.

In the cavity gas, the absorbed dose is simply

$$D_{\text{gas}} = \frac{q_{\text{gas}}}{m_{\text{gas}}} (W_{\text{gas}}/e). \quad (3.2)$$

The Bragg–Gray cavity theory (Gray, 1936) relates the absorbed dose in the small gas cavity to the absorbed dose in the wall:

$$D_{\text{wall}} = D_{\text{gas}} \frac{(\overline{S/\rho})_{\text{wall}}}{(\overline{S/\rho})_{\text{gas}}}, \quad (3.3)$$

where $\overline{S/\rho}_{\text{wall}}/\overline{S/\rho}_{\text{gas}}$ is the ratio of the mean electron-fluence-weighted electron mass electronic

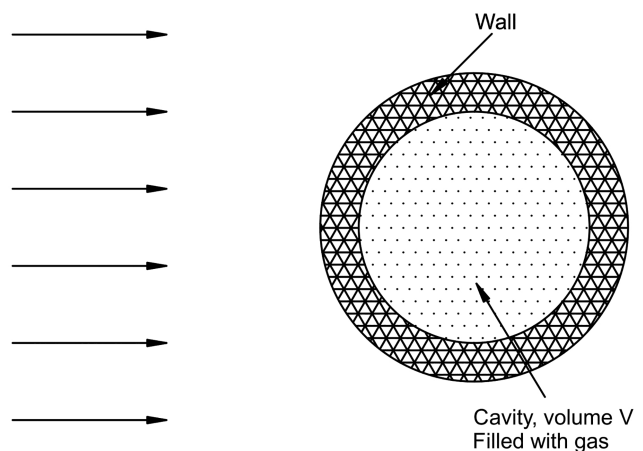


Figure 3.2. Idealized cavity chamber, consisting of the wall and a gas-filled cavity of volume V . The chamber is irradiated by a broad parallel beam of photons. An actual chamber has a stem that connects a conductor to a central electrode, electrically isolated from the wall. [Adapted from Büermann and Burns (2009).]

stopping power of the wall to that of the gas, determined as a Spencer–Attix (Nahum, 1978; Spencer and Attix, 1955) stopping-power ratio [c.f., ICRU Report 35 (ICRU, 1984b)] according to

$$\frac{(\overline{S/\rho})_{\text{wall}}}{(\overline{S/\rho})_{\text{gas}}} = \frac{\int_{\Delta}^{T_{\text{max}}} \Phi_T^{\text{gas}} (S_{\Delta}^{-}(T)/\rho)_{\text{wall}} dT + \Phi_T^{\text{gas}} (\Delta) (S_{\text{el}}^{-}(\Delta)/\rho)_{\text{wall}} \Delta}{\int_{\Delta}^{T_{\text{max}}} \Phi_T^{\text{gas}} (S_{\Delta}^{-}(T)/\rho)_{\text{gas}} dT + \Phi_T^{\text{gas}} (\Delta) (S_{\text{el}}^{-}(\Delta)/\rho)_{\text{gas}} \Delta} \quad (3.4)$$

where Φ_T^{gas} is the total (including secondary) electron fluence differential in energy in the gas, and S_{Δ}^{-}/ρ is the restricted mass electronic stopping power of electrons (see Section 4.1). The integrals on the right-hand side of the fraction can be identified with the restricted cemas in each medium, see Eq. (2.9). The cut-off energy Δ is traditionally chosen so that the csda range in the detector material of an electron with that kinetic energy matches the mean chord length across the volume of the detector.

Under conditions of CPE, the absorbed dose in the wall can be recast in terms of the kerma (see Section 2.10):

$$D_{\text{wall}} = K_{\text{wall}}(1 - \bar{g}_{\text{wall}}). \quad (3.5)$$

The kerma in the wall and free in air (in the absence of the chamber) are each given by Eq. (2.16), so that

$$\begin{aligned} \frac{K_{\text{air}}}{K_{\text{wall}}} &= \frac{\int \Phi_E E \left(\frac{\mu_{\text{tr}}(E)}{\rho} \right)_{\text{air}} dE}{\int \Phi_E E \left(\frac{\mu_{\text{tr}}(E)}{\rho} \right)_{\text{wall}} dE} \\ &= \frac{(\overline{\mu_{\text{en}}/\rho})_{\text{air}}/(1 - \bar{g}_{\text{air}})}{(\overline{\mu_{\text{en}}/\rho})_{\text{wall}}/(1 - \bar{g}_{\text{wall}})}, \end{aligned} \quad (3.6)$$

where Eq. (2.3) has been used. Then one combines Eqs. (3.2) through (3.6) to obtain the generalized result for K_{air} :

$$K_{\text{air}} = \frac{q_{\text{net}}}{\rho_{\text{gas}} V} \frac{(W_{\text{gas}}/e)}{1 - \bar{g}_{\text{air}}} s_{\text{wall,gas}} (\mu_{\text{en}}/\rho)_{\text{air,wall}}, \quad (3.7)$$

where q_{net} is the measured net ionization charge, and where the mass of gas has been replaced by the product of the cavity volume, V , and density of the gas in the cavity, ρ_{gas} . Note that for later convenience, we adopt for the ratio of spectrum-weighted averages the shorthand notation $(x)_{\text{a,b}} \equiv \frac{(\bar{x})_{\text{a}}}{(\bar{x})_{\text{b}}}$. Thus, $s_{\text{wall,gas}}$ is the ratio of the mean mass electronic stopping power (where one here uses $s = S_{\text{el}}/\rho$) of the wall material to that of the cavity gas, and $(\mu_{\text{en}}/\rho)_{\text{air,wall}}$ is

the ratio of the mean mass energy-absorption coefficients for air to that for the wall material. These ratios are of the specified quantities, each averaged over the relevant electron and photon fluence spectra.

Without exception, primary air-kerma cavity-ionization standards are graphite-walled and air-filled, and Eq. (3.7) becomes

$$K_{\text{air}} = \frac{q_{\text{net}}}{\rho_{\text{air}} V} \frac{(W_{\text{air}}/e)}{1 - \bar{g}_{\text{air}}} s_{\text{g,air}} (\mu_{\text{en}}/\rho)_{\text{air,g}} \prod_i k_i. \quad (3.8)$$

Equation (3.8) reflects that idealized conditions can be only approximated in real chambers and radiation fields. For practical graphite-walled cavity chambers filled with humid laboratory air and used in real radiation fields, a number of correction factors, k_i , have been introduced into Eq. (3.8) to account for known deviations from idealized conditions (see, e.g., Büermann and Burns, 2009). In particular, the humidity correction factor, k_h , accounts for the effects of humid *versus* dry air (see Sec. 5.6). Thus, all of the other terms in Eq. (3.8) apply to dry air.

3.2 Absorbed-Dose Standards

Reference dosimetry for radiation therapy requires measurement standards that can be used to establish the absorbed dose to water. Historically, an ionization chamber calibrated against a ^{60}Co air-kerma standard was used, along with key data to convert the measured ionization-chamber response in a water phantom to absorbed dose in water. More recently, absorbed-dose standards have been developed, particularly for photon and electron beams, and these are widely used as the basis for reference dosimetry (Almond *et al.*, 1999; IAEA, 2000a; ICRU, 2001). Three techniques are used to establish primary standards for absorbed dose: an ionization chamber, the Fricke dosimeter, and calorimetry. Seuntjens and Duane (2009) provide an overview of these three techniques and outline under what circumstances each can be considered a primary standard. Additional details are provided in the following Sections.

3.2.1 Cavity-Ionization Chamber

The first method is based on the use of an ionization chamber with a well-established cavity volume. The Bureau International des Poids et Mesures (BIPM) standard for absorbed dose in water, illustrated in Fig. 3.3, is based on this approach (Boutillon and Perroche, 1993) and is further described in Seuntjens and Duane (2009).

The graphite-walled parallel-plate cavity chamber (with a waterproof envelope) is positioned at a

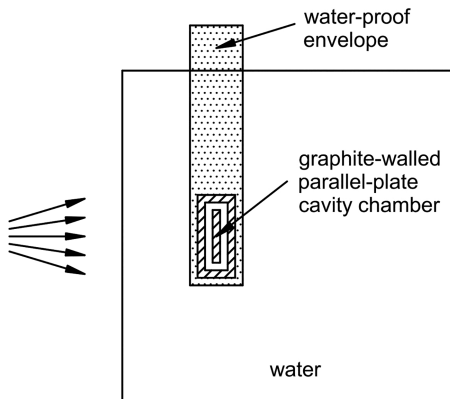


Figure 3.3. Arrangement used at the BIPM for measuring the absorbed dose in a water phantom.

reference depth in a water phantom. Because the cavity volume is known accurately, the energy absorbed in the air in the cavity can be obtained from a measurement of the charge collected, which is used to estimate the absorbed dose in the water.

In this method, the collision kerma introduced in Eq. (2.17) is used. The graphite collision kerma at the reference point, with the chamber air cavity replaced by graphite, is given by

$$K_{\text{col},g} = \frac{D_g}{\beta_g}, \quad (3.9)$$

where D_g is related to the cavity absorbed dose through the Bragg–Gray relationship of Eq. (3.3). Equation (3.9) in effect defines β_g as the absorbed-dose-to-collision-kerma ratio at the reference point in graphite. From Eq. (3.6), the ratio of the water and graphite collision kermas at this reference point in graphite is

$$\begin{aligned} \frac{K_{\text{col},w\text{-in-}g}}{K_{\text{col},g}} &= \frac{\int \Phi_{E,g} E \left(\frac{\mu_{\text{en}}(E)}{\rho} \right)_w dE}{\int \Phi_{E,g} E \left(\frac{\mu_{\text{en}}(E)}{\rho} \right)_g dE} \\ &= (\mu_{\text{en}}/\rho)_{w,g}, \end{aligned} \quad (3.10)$$

where the notation $K_{\text{col},w\text{-in-}g}$ is used to emphasize that the water collision kerma is evaluated using the photon fluence spectrum $\Phi_{E,g}$ present in graphite. Using a similar formulation, the ratio of the water collision kerma at the reference point in water, in the absence of the chamber, to that in graphite is

$$\frac{K_{\text{col},w}}{K_{\text{col},w\text{-in-}g}} = \frac{\int \Phi_{E,w} E \left(\frac{\mu_{\text{en}}(E)}{\rho} \right)_w dE}{\int \Phi_{E,g} E \left(\frac{\mu_{\text{en}}(E)}{\rho} \right)_w dE} = \bar{\Psi}_{w,g}, \quad (3.11)$$

where the water collision kerma in the numerator is now evaluated using the photon spectrum $\Phi_{E,w}$ present at the reference point in water. The factor $\bar{\Psi}_{w,g}$ defined by Eq. (3.11) is essentially a ratio of weighted energy fluences. Using the same formulation as Eq. (3.9), the absorbed dose to water is related to the water collision kerma by

$$D_w = \beta_w K_{\text{col},w}. \quad (3.12)$$

Equations (3.2), (3.3) (for a graphite wall and air cavity), and (3.9) to (3.12) are combined to obtain the measurement equation for the absorbed dose in water as

$$D_w = \frac{q_{\text{net}}}{m_{\text{air}}} (W_{\text{air}}/e) s_{g,\text{air}} (\mu_{\text{en}}/\rho)_{w,g} \bar{\Psi}_{w,g} \beta_{w,g} \prod_i k_i, \quad (3.13)$$

where q_{net} is the measured net charge [as in Eqs. (3.1) and (3.7)], $(\mu_{\text{en}}/\rho)_{w,g}$ and $\bar{\Psi}_{w,g}$ are as defined by Eqs. (3.10) and (3.11), respectively, and $\beta_{w,g}$ is the ratio of absorbed-dose-to-collision-kerma ratios. Among the correction factors, k_i , the most significant is k_{cav} , which accounts for the presence of the air cavity [*i.e.*, the extent to which the ideal conditions for the Bragg–Gray relationship of Eq. (3.3) are not met].

3.2.2 Fricke Dosimetry

The Fricke dosimeter is a dilute aqueous system in which the radiation-induced conversion of ferrous to ferric ions is proportional to the absorbed dose. Its response is stable and reproducible, it is closely water-equivalent, and it is capable of high precision.

The dosimeter solution is composed of 1 mol m^{-3} ferrous sulfate (or ferrous ammonium sulfate) and 1 mol m^{-3} NaCl dissolved in air-saturated 400 mol m^{-3} (0.8 N) sulphuric acid. The best precision is obtained when the solution is irradiated in glass vials. Adequate precision has also been achieved using Lucite holders (Austerlitz *et al.*, 2008; Salata *et al.*, 2014) as well as thin polyethylene bags (Salata *et al.*, 2014; Stucki and Vörös, 2007). An overview of the Fricke dosimetry system in use at the National Research Council Canada is given by Olszanski *et al.* (2002). Irradiation causes oxidation of ferrous (Fe^{2+}) ions to ferric (Fe^{3+}) ions. The ferric-ion concentration is usually determined by direct spectrophotometric analysis of the irradiated solution, based on the well-known absorption spectrum of the ferric and ferrous ions. The change in absorbance due to the irradiation, ΔA , is proportional to the energy absorbed from the radiation field, and thus the absorbed dose in the Fricke solution,

D_F can be obtained as

$$D_F = \frac{\Delta A}{l \rho_F \Delta \epsilon G(\text{Fe}^{3+})}, \quad (3.14)$$

where l is the optical path length in the solution, ρ_F the mass density of the Fricke solution, $\Delta \epsilon$ the difference of molar absorption coefficients⁹ for ferric ions and for ferrous ions, and $G(\text{Fe}^{3+})$ the chemical yield of the ferric ion. The yield, $G(\text{Fe}^{3+})$, plays the same role for Fricke dosimetry as W_{air} does for ionization-chamber dosimetry. In principle, if Fricke dosimetry is to be used as a primary standard, $G(\text{Fe}^{3+})$ must be measured without reference to any other determination of the absorbed dose. This has been done at some standards laboratories using a total-absorption technique in which an electron beam with well-known energy is completely absorbed in Fricke solution (Feist, 1982; Stucki and Vörös, 2007). In practice, a consensus value of $G(\text{Fe}^{3+})$ can also be used to establish an operational standard, just as is done with ionization dosimetry.

For radiation therapy, the Fricke dosimeter is used to establish the absorbed dose in water at a point. In this case, the absorbed dose in water, D_w , is given by

$$D_w = D_F p_F p_{\text{wall}} k_{\text{dd}}, \quad (3.15)$$

where p_F converts the absorbed dose to Fricke solution in a wall-less vessel to the absorbed dose to water in the same location, p_{wall} corrects for the effect of vial wall on the dosimeter response, and k_{dd} accounts for absorbed-dose non-uniformities over the irradiated solution. The factors p_F and p_{wall} can be calculated using Monte Carlo techniques (Ma *et al.*, 1993), while k_{dd} is obtained from measured absorbed-dose distributions.

3.2.3 Calorimetry

The third and most widely used technique for realizing absorbed dose is based on calorimetry. Figure 3.4 shows the main components of a typical water calorimeter.

A calorimetric standard makes use of the fact that, in many materials, most of the energy absorbed from the radiation field appears in the form of heat. If the conversion to heat is complete, the absorbed dose to material m is given by

$$D_m = c_m \Delta T, \quad (3.16)$$

⁹The term “molar extinction coefficient” has often been used in the literature; IUPAC (1997) recommends that “molar absorption coefficient” be used instead.

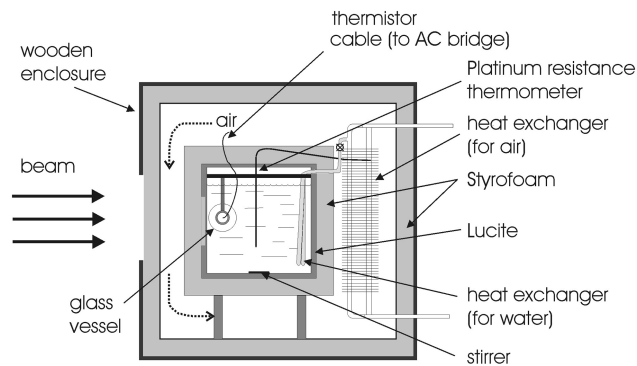


Figure 3.4. Schematic drawing of the water calorimeter in use at the National Research Council of Canada. The glass vessel is filled with high-purity water with a well-known heat defect. The change in temperature is measured by thermistor probes located near the center of the glass vessel. The whole assembly is maintained at 4°C to eliminate convection.

where c_m is the specific heat capacity of the material, and ΔT the measured temperature rise. The specific heat capacity can be measured without reference to a standard of absorbed dose, and ΔT can be made traceable to the SI unit of temperature. The main challenge in calorimetry is to determine the fraction of the energy absorbed from the radiation field that does not appear as heat, referred to as the heat defect, and defined as

$$h = \frac{\varepsilon_i - \varepsilon_h}{\varepsilon_i}, \quad (3.17)$$

where ε_i is the energy imparted by the radiation to a small mass of material, and ε_h is the energy appearing as heat. The heat defect for various materials can be considered key data for which consensus values are appropriate. When h is not zero, Eq. (3.16) can be generalized as

$$D_m = c_m \Delta T / (1 - h). \quad (3.18)$$

Although in principle an absorbed-dose calorimeter can be made from any material, the most widely used are graphite and water. Graphite is chosen because it is well specified (*i.e.*, can be made quite pure and uniform), has a relatively low specific heat capacity that produces temperature changes more easily measured accurately [see Eq. (3.18)], and has radiation-interaction characteristics that are reasonably similar to those of water. Being a solid, graphite permits the construction of well-defined absorbing elements such that the core, in which the temperature is to be determined, can be thermally isolated from surrounding material. Measurement and theory also indicate that any heat defect in graphite should be small. The main disadvantage of graphite calorimetry is that a conversion process is necessary to

obtain the absorbed dose in water. Transfer ionization chambers and Monte Carlo techniques are used for this purpose.

Because the absorbed dose in water is the quantity of interest, particularly for radiotherapy, several laboratories have adopted water calorimetry as their standard for D_w . The principal advantage of water as the calorimetric medium is that it gives the absorbed dose in water directly. The main technical challenges are to construct a temperature probe and water-containment vessel that have a minimal effect on the temperature measurement, to quantify the residual perturbative effects, and to establish the heat defect. Despite their differences, the uncertainties associated with the determination of the absorbed dose in water are similar for both graphite and water calorimetry.

3.3 Summary

The previous subsections show that the determinations of air kerma and absorbed dose require a number of subsidiary quantities. It is not practical for each laboratory to determine them independently, and—in practice—consensus values are adopted.

Two of the most important quantities in the measurement equations are W_{air} and $s_{g,\text{air}}$. While in principle each of them can be determined independently, uncertainties can be reduced if they are evaluated

as a product. Indeed, it is seen from Eqs. (3.8) and (3.13) that for cavity standards for air kerma and absorbed dose, it is the product $W_{\text{air}} s_{g,\text{air}}$ that is required. The consensus value for W_{air} of 33.97 eV (with relative standard uncertainty of 0.15 %) currently recommended is based on the critical analysis by Boutillon and Perroche-Roux (1987) of a number of experiments, of which some determined W_{air} alone, and others, carrying much of the weight, determined the product $W_{\text{air}} s_{g,\text{air}}$ from comparisons of graphite cavity chambers and graphite calorimeters. In the analysis of Boutillon and Perroche-Roux (1987), values for $s_{g,\text{air}}$ were taken from ICRU Report 37 (ICRU, 1984a), using the then recommended I values for air and graphite as well as the bulk density for graphite in the evaluation of the density effect.

An important consequence of this analysis is that the use of the recommended value for W_{air} of 33.97 eV in Eqs. (3.1) and (3.8) is dependent on the use of the electronic stopping powers of ICRU Report 37 (ICRU, 1984a). As will be seen in Section 4.1, the electronic stopping power depends on the value adopted for the mean excitation energy and the density of the medium. In particular, an I value for graphite of 78 eV and its bulk density was used in ICRU Report 37. It follows that any re-evaluation of the graphite stopping power will necessitate a re-evaluation of W_{air} .

4. Charged-Particle Stopping Powers and Related Quantities

Charged-particle stopping powers have been treated in detail in ICRU Report 37 (ICRU, 1984a), Report 49 (ICRU, 1993), and Report 73 (ICRU, 2005; Sigmund *et al.*, 2009). In the present Report, consideration has been confined to electrons (and positrons), protons, and α particles for three key materials: graphite, air, and liquid water. Additionally, information on C ions in these materials has been included in recognition of their increasing use in radiation therapy. The stopping power quantifies the average energy loss per pathlength of charged particles in matter. In principle, the stopping power consists of three contributions, namely the electronic (or collision), the radiative, and the nuclear stopping power. The nuclear stopping power is negligible for electrons; it can be significant for light and heavy ions (ICRU, 1993; 2005), but only at rather low kinetic energies. A recommendation issued jointly by the ICRU and IAEA (Wambersie *et al.*, 2004) calls “light ions” those nuclei with an atomic number equal to, or smaller than, that of neon nuclei ($Z = 10$), leaving the name of “heavy ions” to all heavier nuclei. Results for the nuclear stopping power from ICRU Report 49 (1993) will be included in the tables presented in this Report for completeness. The radiative stopping power for electrons and positrons can be quite significant; for light and heavy ions, it is smaller to first approximation by a factor of $(m_e/M)^2$, where m_e is the rest mass of the electron and M the rest mass of the incident ion. Thus, the radiative stopping power can be ignored for protons and heavier ions at the kinetic energies considered in this Report.

4.1 Electronic Stopping Power

The electronic (collision) stopping power is due to energy transfers from the incident particle to atomic electrons. With $d\sigma/d\omega$ denoting the differential cross section (per atomic electron) for inelastic collisions resulting in an energy transfer of ω , the mass electronic stopping power can be written as

$$\frac{1}{\rho} S_{\text{el}}(T) = NZ \int_0^{\omega_{\text{max}}} \omega \frac{d\sigma}{d\omega} d\omega, \quad (4.1)$$

where T is the initial kinetic energy of the particle, ω_{max} the largest possible energy loss in an inelastic collision with an atomic electron, N the number of atoms per mass of the medium, and Z the atomic number. The quantity $N = N_A/M_A = (uA)^{-1}$, where N_A is the Avogadro constant, M_A the molar mass (*e.g.*, g mol^{-1}), A the relative atomic mass (or atomic weight), and u the atomic mass constant, $1/12$ of the mass of an atom of ^{12}C .

Bethe’s (Bethe, 1930; Fano, 1963) evaluation of Eq. (4.1) essentially divides the energy transfers, ω , into two classes, depending on if they are smaller or larger than a cut-off value, ω_c , where ω_c must be large compared with the binding energies of the atomic electrons. Thus

$$\frac{1}{\rho} S_{\text{el}} = \frac{1}{\rho} S_{\text{el}}(\omega < \omega_c) + \frac{1}{\rho} S_{\text{el}}(\omega > \omega_c). \quad (4.2)$$

The main result of Bethe theory is

$$\begin{aligned} \frac{1}{\rho} S_{\text{el}}(\omega < \omega_c) &= \frac{2\pi r_e^2 m_e c^2}{\beta^2} \frac{Z}{uA} z^2 \left[\ln \left(\frac{2m_e c^2 \beta^2 \omega_c}{(1 - \beta^2) I^2} \right) - \beta^2 \right], \\ & \quad (4.3) \end{aligned}$$

where r_e is the classical electron radius, c is the speed of light *in vacuo* ($m_e c^2$ is the electron rest energy), β is here and in the remainder of the Report the velocity of the incident particle, v , divided by the speed of light, z is the charge of the incident particle in units of the charge of the electron, and I is the mean excitation energy of the medium. Using values for the physical constants from Mohr *et al.* (2012), $2\pi r_e^2 m_e c^2 / u = 0.153537 \text{ MeV cm}^2 \text{ g}^{-1}$. Note that Eq. (4.3) is valid for all charged particles (light and heavy) but only when requirements for the use of the first Born approximation are satisfied: (i) The velocity of the incident particle is large compared with the velocities of the atomic electrons; applying this to K-shell electrons then requires $(\alpha Z/\beta) \ll 1$ (see, *e.g.*, Inokuti, 1971), where α is the fine-structure constant ($\cong 1/137$). (ii) For heavy particles, the amplitude of the wave scattered by the field of the atomic electron

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is small compared with the amplitude of the undisturbed incident wave; thus, $(\alpha z/\beta) \ll 1$ (Bethe and Ashkin, 1953). For the charged particles considered in this Report, condition (i) is the most restrictive for electrons (and positrons), protons, and α particles. For C ions, condition (ii) becomes restrictive and, for the materials of interest in this Report, limits the applicability of the first Born approximation to velocities somewhat similar to those of condition (i). The component due to close collisions is evaluated as if the atomic electrons were free and at rest:

$$\frac{1}{\rho} S_{\text{el}}(\omega > \omega_c) = NZ \int_{\omega_c}^{\omega_{\text{max}}} \omega \frac{d\sigma}{d\omega} d\omega, \quad (4.4)$$

where $d\sigma/d\omega$ is now the differential cross section for energy transfer ω in a collision with a free electron at rest, and

$$\omega_{\text{max}} = \frac{2\tau(\tau+2)m_e c^2}{[1 + 2(\tau+1)(m_e/M) + (m_e/M)^2]}, \quad (4.5)$$

where $\tau = T/(Mc^2)$ is the ratio of the kinetic energy of the incident particle to that of its rest energy, and m_e/M is the ratio of the mass of the electron to that of the incident particle.

4.1.1 Electrons and Positrons

For electrons and positrons, Eq. (4.5) reduces to $\omega_{\text{max}} = T$. For electrons, large energy transfers of Eq. (4.4) are governed by the Møller (1932) cross section,

$$\begin{aligned} \frac{d\sigma}{d\omega} &= \frac{2\pi r_e^2 m_e c^2}{\beta^2} \\ &\times \left\{ \frac{1}{\omega^2} + \frac{1}{(T-\omega)^2} - \frac{2\tau+1}{(\tau+1)^2} \frac{1}{\omega(T-\omega)} + \left(\frac{\tau}{\tau+1} \right)^2 \right\}. \end{aligned} \quad (4.6)$$

For positrons, large energy transfers are governed by the Bhabha (1936) cross section,

$$\begin{aligned} \frac{d\sigma}{d\omega} &= \frac{2\pi r_e^2 m_e c^2}{\beta^2} \frac{1}{T^2} \\ &\times \left\{ \left(\frac{T}{\omega} \right)^2 - \frac{\tau(\tau+2)T}{(\tau+1)^2 \omega} + \frac{1}{2} \left(\frac{\tau}{\tau+1} \right)^2 - \left(\frac{\tau}{\tau+2} \right) \right. \\ &\times \left[\left(\frac{\tau+3}{\tau+1} \right) \frac{T}{\omega} - 2 \frac{\tau(\tau+2)}{(\tau+1)^2} + \left(\frac{\tau}{\tau+1} \right)^2 \frac{\omega}{T} \right] + \left(\frac{\tau}{\tau+2} \right)^2 \\ &\times \left[\frac{1}{2} + \frac{1}{\tau+1} + \frac{3}{2(\tau+1)^2} - \left(\frac{\tau}{\tau+1} \right)^2 \frac{\omega}{T} \left(1 - \frac{\omega}{T} \right) \right] \Bigg\}. \end{aligned} \quad (4.7)$$

For incident electrons, because of the indistinguishability of the incident and ejected target electrons, the stopping power is evaluated for the faster of these electrons, and ω_{max} is by convention taken to be $T/2$. For incident positrons, there are no exchange effects, and $\omega_{\text{max}} = T$. The final expression for the stopping power can be written

$$\begin{aligned} \frac{1}{\rho} S_{\text{el}}^{\pm}(T) &= \frac{2\pi r_e^2 m_e c^2}{\beta^2} \frac{Z}{uA} \\ &\times \left[\ln \left(\frac{T}{I} \right)^2 + \ln \left(1 + \frac{\tau}{2} \right) + F^{\pm}(\tau) - \delta \right], \end{aligned} \quad (4.8)$$

where for electrons

$$F^{-}(\tau) = (1 - \beta^2)[1 + \tau^2/8 - (2\tau + 1) \ln 2], \quad (4.9)$$

and for positrons

$$\begin{aligned} F^{+}(\tau) &= 2 \ln 2 - \left(\frac{\beta^2}{12} \right) \\ &\times \left[23 + \frac{14}{(\tau+2)} + \frac{10}{(\tau+2)^2} + \frac{4}{(\tau+2)^3} \right]. \end{aligned} \quad (4.10)$$

In Eq. (4.8), δ is the density-effect correction, a function of τ , to be discussed in Section 4.3.

The restricted mass electronic stopping power is defined as the mean energy loss by charged particles per pathlength in the material due to ionization or excitation involving energy transfers, ω , up to a chosen cutoff energy Δ . For electrons and positrons, this quantity can be written (ICRU, 1984a)

$$\begin{aligned} \frac{1}{\rho} S_{\Delta}^{\pm}(T) &= \frac{2\pi r_e^2 m_e c^2}{\beta^2} \frac{Z}{uA} \\ &\times \left[\ln \left(\frac{T}{I} \right)^2 + \ln \left(1 + \frac{\tau}{2} \right) + H^{\pm}(\tau, \eta) - \delta \right], \end{aligned} \quad (4.11)$$

where $\eta = \Delta/T$ is the fractional cutoff energy,

$$\begin{aligned} H^{-}(\tau, \eta) &= -1 - \beta^2 + \ln[4(1 - \eta)\eta] + (1 - \eta)^{-1} \\ &+ (1 - \beta^2) \left[\frac{\tau^2 \eta^2}{2} + (2\tau + 1) \ln(1 - \eta) \right], \end{aligned} \quad (4.12)$$

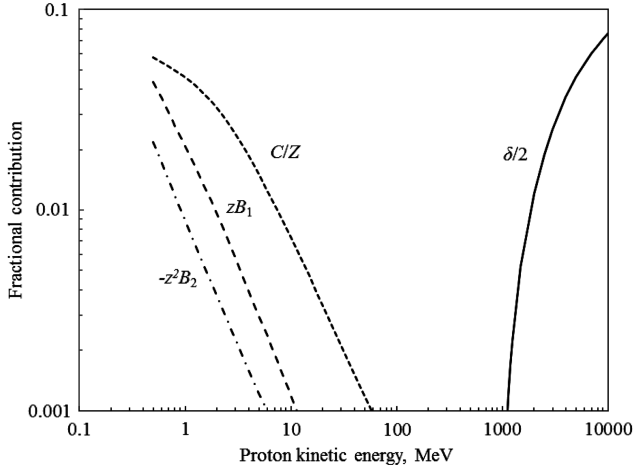


Figure 4.1. Corrections as fractions of the quantity $\ln(2m_e c^2 \beta^2 / (1 - \beta^2) I) - \beta^2$ in Eq. (4.18), shown for protons in liquid water. The corrections include the density-effect correction ($\delta/2$), the shell correction (C/Z), the Barkas correction (zB_1), and the Bloch correction (z^2B_2). Note that the Bloch correction is always a negative quantity.

and

$$\begin{aligned}
 H^+(\tau, \eta) = & \ln(4\eta) \\
 & - \beta^2 \left[1 + (2 - \xi^2)\eta - (3 + \xi^2) \left(\frac{\xi\tau}{2} \right) \eta^2 \right. \\
 & \left. + (1 + \xi\tau) \left(\frac{\xi^2\tau^2}{3} \right) \eta^3 - \left(\frac{\xi^3\tau^3}{4} \right) \eta^4 \right],
 \end{aligned} \quad (4.13)$$

where $\xi = (\tau + 2)^{-1}$.

It should be noted that, except for $\Delta = \omega_{\max}$, there is formally a difference between the linear energy transfer, L_Δ , defined in Section 2.6 and the restricted electronic stopping power, S_Δ , as defined in ICRU Report 37 (1984a) and here: L_Δ excludes secondary electrons with kinetic energies in excess of Δ , whereas S_Δ excludes energy transfers in excess of Δ (Kellerer *et al.*, 1992) and hence L_Δ is greater than S_Δ . The difference, involving the binding energies of the secondary electrons, is usually quite small numerically but can become significant when Δ becomes comparable to the binding energies.

4.1.2 Heavy Charged Particles

For $m_e \ll M$, which is always the case for the incident heavy charged particles considered here, a very good approximation for the maximum energy transfer [from Eq. (4.5)] is

$$\omega_{\max} \cong 2\tau(\tau + 2)m_e c^2 = \frac{2m_e c^2 \beta^2}{1 - \beta^2}. \quad (4.14)$$

The energy-transfer cross section is

$$\frac{d\sigma}{d\omega} = \frac{2\pi r_e^2 m_e c^2}{\beta^2} z^2 \frac{1}{\omega^2} \left(1 - \beta^2 \frac{\omega}{\omega_{\max}} \right), \quad (4.15)$$

with the result [from Eqs. (4.2) through (4.4) and (4.14) and (4.15)]

$$\frac{1}{\rho} S_{\text{el}} = \frac{4\pi r_e^2 m_e c^2}{\beta^2} \frac{Z}{uA} z^2 \left[\ln \left(\frac{2m_e c^2 \beta^2}{(1 - \beta^2) I} \right) - \beta^2 \right]. \quad (4.16)$$

The accuracy of Eq. (4.16) can be improved by the addition of a number of corrections: a shell correction to compensate for the fact that the incident-particle velocity is not sufficiently large compared with the velocity of the electrons in inner shells of target atoms; a density-effect correction that accounts for the reduction in the stopping power due to the polarization of the medium; and corrections that account for departures from the first-Born approximation. Thus, a more general form of Eq. (4.16) is

$$\frac{1}{\rho} S_{\text{el}} = \frac{4\pi r_e^2 m_e c^2}{\beta^2} \frac{Z}{uA} z^2 B(\beta), \quad (4.17)$$

where $B(\beta)$ is the stopping number per electron, and is conventionally expressed as

$$B(\beta) = \ln \left(\frac{2m_e c^2 \beta^2}{(1 - \beta^2) I} \right) - \beta^2 - \frac{\delta}{2} - \frac{C}{Z} + zB_1 + z^2B_2. \quad (4.18)$$

Above, zB_1 is the Barkas correction, z^2B_2 the Bloch correction, and C/Z the shell correction, all of which have been described in ICRU Report 37 (ICRU, 1984a). The density-effect correction, δ , has been included in Eq. (4.18). However, even with the shell, Barkas, and Bloch corrections, the so-called modified Bethe–Bloch formula of Eqs. (4.17) and (4.18) cannot be reliably extended down to very low energies (say, for which β is less than from about 0.04 to 0.1, depending on the particle), and results must be augmented by measured data as indicated in ICRU Report 49 (ICRU, 1993). If one considers a cutoff Δ for energy transfers ($\Delta < \omega_{\max}$), the restricted mass electronic stopping power of heavy charged particles

is given by

$$\frac{1}{\rho}S_{\Delta}(T) = \frac{4\pi r_e^2 m_e c^2}{\beta^2} \frac{Z}{uA} z^2 \times \left[\frac{1}{2} \ln \left(\frac{2m_e c^2 \beta^2 \Delta}{(1-\beta^2)I^2} \right) - \beta^2 - \frac{\delta}{2} - \frac{C}{Z} + zB_1 + z^2 B_2 \right]. \quad (4.19)$$

Methods to evaluate the shell, Barkas, and Bloch corrections have been outlined in ICRU Report 49 (ICRU, 1993), with those for the shell correction being the most elaborate. These methods are incorporated into the computer code BEST (BEthe STopping) developed by M. J. Berger¹⁰ and H. Bichsel for the generation of mass electronic stopping powers in ICRU Report 49, which includes the evaluation of the density-effect correction (see Section 4.3). The present Report makes use of the BEST code, from which the relative contributions of these corrections have been calculated and illustrated in Fig. 4.1 for protons in liquid water.

It is pointed out in ICRU Report 73 (ICRU, 2005) that there are further requirements for the heavy ions (*e.g.*, C ions). These criteria restrict the use of Bethe stopping-power theory to even higher energies, energies that depend not only on the stopping medium but also on the incident particle. In fact, in ICRU Report 73, an alternative approach was taken to evaluate the electron stopping power for these heavy ions, as briefly explained below.

4.1.3 Carbon Ions

Although the mass electronic stopping power of any light ion can be calculated from the modified Bethe–Bloch formula, Eqs. (4.17) and (4.18), the result of the perturbative expansion becomes increasingly inaccurate as the ion charge z increases, especially at low energies. The failure of the formalism can be attributed to the growing importance of non-perturbative effects.

ICRU Report 73 (ICRU, 2005) adopted a different theoretical approach to evaluate the mass electronic stopping power of ions heavier than α particles. Specifically, the binary stopping theory of Sigmund and Schinner (2000) was employed to produce tables of S_{el}/ρ for the ions Li to Ar. The binary stopping theory elaborates the ideas of Bohr (1913); the Barkas–Andersen effect (Barkas correction) is inherent in the treatment, and shell and density-effect corrections are incorporated separately. The

theoretical scheme has been implemented in the computer code PASS (Sigmund and Schinner, 2002), which was used to produce the tables in ICRU Report 73. PASS also includes an additional correction¹¹ for highly relativistic heavy ions taken from Lindhard and Sørensen (1996). The code does not use the I value of the medium directly as input information, but needs instead a set of subshell excitation energies E_j and oscillator strengths f_j for its model. The choice of $\{E_j, f_j\}$ does, however, lead to I values that are, in general, different from those recommended in ICRU Report 49. In the case of liquid water, the value $I_w = 67.2$ eV was used, which has been misunderstood as a recommendation by ICRU, and thus the source of significant controversy. In an Errata and Addenda for ICRU Report 73, Sigmund *et al.* (2009) clarified this issue and recalculated the tables of mass electronic stopping powers of Li to Ar ions in liquid water¹² using a set of excitation energies and oscillator strengths adjusted to yield the value $I_w = 78$ eV.

A different model, the three-body classical-trajectory Monte Carlo method, has been recently employed by Liamsuwan and Nikjoo (2013) to calculate cross sections for target ionization, electron capture, and projectile-electron loss, and the resultant total stopping cross sections, for bare and dressed carbon ions in molecular water and in neon. This method gives detailed doubly differential cross sections, particularly useful for radiobiological Monte Carlo track-structure calculations.

At sufficiently high energies, the mass electronic stopping power and the restricted mass electronic stopping power for carbon ions can be evaluated using Eqs. (4.17) and (4.19), respectively. To be consistent with the approach used in ICRU Report 73 and also used here, the relativistic-Mott-scattering correction indicated in footnote 11 and in Section A.3 should be included.

4.2 Mean Excitation Energy

The mean excitation energy, I , of a substance plays a central role in the electronic stopping power formulas for electrons and light ions. It can be obtained in different ways.

¹¹This correction takes into account the use of the relativistic Dirac equation in the scattering of the target atomic electrons in the potential of the incoming ion, *i.e.*, Mott scattering.

¹²Sigmund *et al.* (2009) also revised their results for water vapor and corrected the earlier tables for fluorine ions in nylon. There is an apparent typographical mistake in Table 1 of Sigmund *et al.* (2009). The last line should indicate a value of $h\omega$ of 32.92 eV in order to produce $I = 78$ eV from their model.

¹⁰Personal communication from M. Berger to H. Paul (1993–1994). The code has been updated with current values of constants from CODATA 2010 (Mohr *et al.*, 2012) for use in this Report.

In principle, I can be extracted from experiments in which the measured electronic stopping power of swift light ions is fitted to Eq. (4.17) leaving I as an empirical parameter (see, e.g., ICRU, 1993; Ziegler, 1999). For such an evaluation, the shell, Barkas, and Bloch corrections have to be provided. This procedure is delicate, and often it is preferable to fit the measured range of the ions in the material of interest. In this case, multiple-angular-scattering effects have to be estimated.

As pointed out in ICRU Report 37 (1984a), there are alternative approaches for estimating I , including the use of the oscillator-strength distribution (OSD) for a gas and the dielectric-response function (DRF) for a condensed medium.

For gases, the mean excitation energy can be obtained from

$$\ln I = \int_0^\infty \frac{df}{dE} \ln E \, dE / \int_0^\infty \frac{df}{dE} dE, \quad (4.20)$$

where df/dE is the density of optical dipole oscillator strength per energy above the ground state, and I and E are expressed in the same units. Information on the OSD can come from *ab initio* calculations or from measurements of photoelectric cross sections and electron inelastic-scattering cross sections. Moments of the OSD (i.e., $\int_0^\infty (df/dE) E^n dE$) can also be used to determine the mean excitation energy.

For condensed media, the mean excitation energy can be obtained from knowledge of the DRF (ICRU, 1984a; Inokuti and Smith, 1982) through the relation

$$\ln I = \frac{\int_0^\infty dE E \operatorname{Im}[-1/\epsilon(E)] \ln E}{\int_0^\infty dE E \operatorname{Im}[-1/\epsilon(E)]}, \quad (4.21)$$

with the optical energy-loss function (OELF) given by

$$\operatorname{Im} \left[\frac{-1}{\epsilon(E)} \right] \equiv \frac{\epsilon_2(E)}{\epsilon_1^2(E) + \epsilon_2^2(E)}, \quad (4.22)$$

where $\epsilon(E) = \epsilon_1(E) + i\epsilon_2(E)$ is the complex DRF in the long-wavelength limit, and E is the energy transfer.

4.3 Density-Effect Correction

When a charged particle moves through a medium, the polarization of atoms in the medium decreases the field acting on the particle so as to reduce the stopping power. This reduction is large in a condensed medium compared with that of a gas with the same composition, and is thus called the

density effect. This decrease in S_{el} is called the density-effect correction and was first calculated by Fermi (1940). It depends on the specific dielectric properties of the traversed material and is also a function of the particle's speed. Fermi's density-effect correction can be evaluated from the OELF (Fano, 1963; ICRU, 1984a; Inokuti and Smith, 1982) by means of

$$\delta(\beta) = \frac{2}{\pi E_p^2} \int_0^\infty \operatorname{Im} \left[\frac{-1}{\epsilon(E)} \right] \ln \left(1 + \frac{l^2}{E^2} \right) E \, dE - \frac{l^2}{E_p^2} (1 - \beta^2), \quad (4.23)$$

where E_p is the plasma energy, and l is a real-valued function of β^2 , defined as the positive root of the equation

$$1 - \beta^2 \epsilon(il) = 0. \quad (4.24)$$

In the limit $\beta \rightarrow 1$,

$$\delta(\beta) \rightarrow 2 \ln \left[\frac{1}{(1 - \beta^2)^{1/2}} \frac{E_p}{I} \right] - 1, \quad (4.25)$$

with the result that the mean excitation energy, I , disappears from the stopping power equations, and—thus in the limit of very high energies—the stopping power depends on the properties of the medium only through the plasma energy. The plasma energy used here is

$$E_p = \left(\frac{4\pi\hbar^2 e^2 n_e}{m_e} \right)^{1/2} = 28.8159 \left(\frac{\rho}{g/cm^3} \frac{Z}{A} \right)^{1/2}, \quad (4.26)$$

where n_e is the total number of electrons per volume.

A difference arises between conducting and non-conducting media in the solution of Eq. (4.24). For conductors, the DRF is such that the equation has a root for any value of β , and the density effect is present even at arbitrarily low energies. For insulators, a root exists only above a minimum value of $\beta_0 = [\epsilon(0)]^{-1/2}$, which is a property of the medium, so there is no density effect for $\beta < \beta_0$.

A simple model of the DRF (see Sternheimer *et al.*, 1982, and references therein), described as a sum of delta functions, has been extensively used to tabulate the density-effect correction δ of elements and compounds in ICRU Report 37 (ICRU, 1984a) for electrons and positrons and in Report 49 (ICRU, 1993) for protons and α particles. This approach constitutes a reasonable compromise when information

on the experimental OELF is lacking, and yields very good results for the density-effect correction (Sternheimer *et al.*, 1982).

Indeed, for most materials, experimental information on the OELF is scarce or fragmentary. In Sternheimer's (1952) simplified general scheme to calculate δ , the OELF is replaced by a number, n , of dispersion oscillators with the result

$$\delta(\beta) = \sum_1^n f_i \ln \left(1 + \frac{l^2}{l_i^2} \right) - l^2(1 - \beta^2), \quad (4.27)$$

where l is now the solution of the equation

$$\frac{1}{\beta^2} - 1 = \sum_1^n \frac{f_i}{(\mu_{\text{St}} E_i / E_p)^2 + l^2}, \quad (4.28)$$

where μ_{St} is an adjustment factor to ensure that the model gives the desired value of the mean excitation energy, *i.e.*,

$$\begin{aligned} \sum_1^{n-1} f_i \ln \left[(\mu_{\text{St}} E_i)^2 + \frac{2}{3} f_i E_p^2 \right]^{1/2} + f_n \ln(f_n^{1/2} E_p) \\ = \ln I. \end{aligned} \quad (4.29)$$

For a conductor, f_n is taken as n_c/Z , where n_c is the effective number of conduction electrons per atom. For a non-conductor, $n_c = 0$, the summation in Eq. (4.28) extends to n , and the last term on the right-hand side disappears. In Eq. (4.29),

$l_i^2 = (\mu_{\text{St}} E_i / E_p)^2 + \left(\frac{2}{3} \right) f_i$ for $E_i > 0$, and $l_n^2 = f_n$ for $E_n = 0$.

In the evaluations done in ICRU Reports 37 and 49, and now here, values of E_i were taken from Carlson's (1975) tables of subshell binding energies for single atoms, with Zf_i the occupation numbers for the subshell; the number of conduction electrons for metals and semiconductors was taken to be equal to the occupation number in the outermost subshell (with $E_n = 0$), and compounds have been treated as insulators.

According to that particular algorithm, the semi-metal graphite was then assumed to have two conduction electrons, whereas—corresponding to its structure—only one of its valence electrons is responsible for conductivity. Thus, the tables given later in this Report assume one conduction electron in the calculation of the density-effect correction for graphite. This change results in relative differences of no more than about 0.1 % (less for most kinetic energies) for values of S_{el}/ρ (the relative differences in δ itself are significant only at low kinetic energies for which their values become too small to have an appreciable effect). Such differences in S_{el}/ρ are

barely distinct from round-off effects associated with the four significant figures given in the tables for charged particles in this Report.

ICRU Report 37 drew attention to another issue in evaluating the density effect for inhomogeneous media, such as graphite. The bulk density of graphite is seldom larger than about 1.8 g cm^{-3} , whereas the density of the grains that form the bulk material is 2.265 g cm^{-3} . ICRU Report 37 gave tables of stopping powers for densities of 1.7 g cm^{-3} and 2.265 g cm^{-3} , suggesting that it was better to use the bulk density; for protons and α particles, ICRU Report 49 provided stopping power data for carbon densities of 1.7 g cm^{-3} and 2.0 g cm^{-3} .

Since then, MacPherson (1998) measured the stopping powers with a relative standard uncertainty of about 0.5 % for several materials using electrons with kinetic energies from 8 MeV to 30 MeV. In general, good agreement was found between the measured stopping powers and the calculated values given in ICRU Report 37. However, agreement for graphite required the use of the crystalline density when evaluating the density effect,¹³ as shown in Figure 4.2 MacPherson also reported measurements for pyrolytic graphite, which has a density close to the crystalline density for graphite. As also shown in Fig. 4.2, the stopping power for this material was also consistent with the ICRU Report 37 stopping power, evaluated using the crystalline density. Burns (2009, 2012) also used results based on the crystalline density in his analyses of measured values of $W_{\text{air}} s_{\text{g,air}}$ to determine consistent results for W_{air} and its uncertainty.

Alanine pellets used for radiation dosimetry consist of a mixture of crystalline grains and a low- Z binder (typically paraffin or polyethylene) and have a bulk density considerably lower than the density of the alanine crystals. Recent work studying the relative response of alanine dosimeters¹⁴ to high-energy photon and electron beams found that agreement between measurement and Monte Carlo

¹³This might be understood by considering that this type of graphite contains trapped porosity. For example, an admixture of only 0.018 % by weight of air will reduce the density from that for the crystallite (2.265 g cm^{-3}) to 1.7 g cm^{-3} , typical of bulk graphite. The polarization of the atoms in bulk graphite, which accounts for the density effect, are thus predominately due to the dielectric properties of the graphite crystal structure. Such a perspective considers bulk graphite as a mixture that can then be handled by Bragg additivity [see Eqs. (5.1)], with the result that the presence of the air in this case has a negligible effect and the crystallite density governs the evaluation of the density-effect correction.

¹⁴The response of these dosimeters is derived from the electron-paramagnetic-resonance (EPR) spectrum of a long-lived radical in the irradiated crystalline alanine; the binder is EPR-silent.

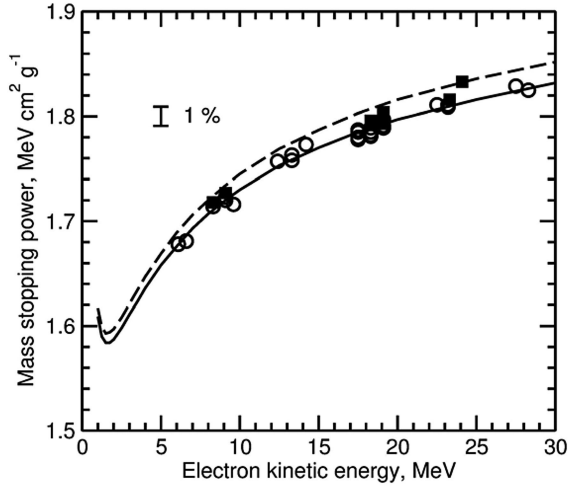


Figure 4.2. Measured electronic stopping power for graphite (open circles) compared with ICRU 37 values. The solid line is the ICRU 37 mass electronic stopping power evaluated using the grain density (2.265 g cm^{-3}) of graphite. The dashed line shows the ICRU 37 stopping power evaluated using the bulk density (1.7 g cm^{-3}). The solid squares show the results of measurements using pyrolytic graphite and are consistent with the crystalline density. [Figure from MacPherson (1998).]

simulations is consistent with the use of the alanine crystalline density and an assumption of no change in its intrinsic energy dependence (Zeng *et al.*, 2005; Anton *et al.*, 2013).

4.4 Mass Radiative Stopping Power of Electrons and Positrons

Because of the acceleration caused by the local electrostatic fields in matter, fast electrons and positrons emit bremsstrahlung (braking radiation), thus reducing their kinetic energy. Conservation of energy and linear momentum implies that this emission can occur only in the field of a massive charged particle, the atomic nucleus or an atomic electron. An excellent review of the theory and available experimental measurements of bremsstrahlung emission can be found in the book of Haug and Nakel (2004). The evaluation of the mass radiative stopping powers for electrons and positrons was rather thoroughly outlined in ICRU Report 37 (1984a), based on information then available, and those methods are followed here except for one, relatively minor, difference.

The radiative stopping power can be written

$$\frac{1}{\rho} S_{\text{rad}}(T) = \alpha r_e^2 \frac{N_A}{uA} (T + m_e c^2) [Z^2 \varphi_{\text{rad,n}} + Z \varphi_{\text{rad,e}}], \quad (4.30)$$

where T is the incident kinetic energy of the electron (or positron), and α is the fine-structure constant. In

Eq. (4.30), the dimensionless, scaled, radiative energy-loss cross section, φ_{rad} , has been separated into two components,

$$\varphi_{\text{rad,n}} = \frac{(\alpha r_e^2 Z^2)^{-1}}{(T + m_e c^2)} \int k \frac{\sigma_n}{k} dk, \quad (4.31)$$

for electron–nuclear bremsstrahlung, where $d\sigma_n/dk$ is the cross section for bremsstrahlung emission in the screened Coulomb field of the atomic nucleus, differential in emitted photon energy k , and

$$\varphi_{\text{rad,e}} = \frac{(\alpha r_e^2)^{-1}}{(T + m_e c^2)} \int k \frac{d\sigma_e}{dk} dk, \quad (4.32)$$

for electron–electron bremsstrahlung, where $d\sigma_e/dk$ is the cross section for bremsstrahlung emission in the screened field of an electron, differential in emitted photon energy k .

4.4.1 Bremsstrahlung in the Field of the Atomic Nucleus

4.4.1.1 Incident Electrons. Seltzer and Berger (1985) describe the synthesis of information then available on $d\sigma_n/dk$. For $T \geq 50 \text{ MeV}$, they use the analytical Born-approximation cross section for the unscreened nucleus of Bethe and Heitler (1934) corrected for screening effects from high-energy approximations with screening (Bethe, 1934) and for Coulomb effects from high-energy approximations with Fury–Sommerfield–Maue Coulomb wave functions (Bethe and Maximon, 1954; Davies *et al.*, 1954; Olsen, 1955; Olsen *et al.*, 1957) augmented by further considerations for the tip region of the spectrum as $k/T \rightarrow 1$ (*i.e.*, the high-frequency limit).

For $T \leq 2 \text{ MeV}$, Seltzer and Berger (1985) used the numerical partial-wave results of Pratt *et al.* (1977) for $d\sigma_n/dk$. The work of Pratt and colleagues, described in Tseng and Pratt (1971), Lee *et al.* (1976), and Kissel and MacCallum (1977), is based on the numerical evaluation of the S matrix with electron wave functions in partial-wave series from the numerical solution of the radial Dirac equation for the screened nucleus. Through interpolation, Pratt *et al.* (1977) prepared a comprehensive set of data for $2 \leq Z \leq 92$, for $0 \leq k/T \leq 1$, and for $1 \text{ keV} \leq T \leq 2 \text{ MeV}$. Seltzer and Berger (1985) used extrapolation to extend the results of Pratt *et al.* to the case $Z = 1$ and $Z = 93$ to 100.

A complete set of cross sections, differential in emitted photon energy, k , was then created (Seltzer and Berger, 1986) for $1 \leq Z \leq 100$ and $1 \text{ keV} \leq T \leq 10 \text{ GeV}$, through the use of least-squares cubic-spline interpolation in T across the gap region, $2 \text{ MeV} < T < 50 \text{ MeV}$, followed by smoothing in k/T again with

least-squares cubic-splines. This procedure resulted in small irregularities in T -dependence, usually less than 1 %, amounting to from 2 % to 3 % in the worst cases. With this information, the radiative energy-loss cross section, $\varphi_{\text{rad},n}$, was also obtained to be used in Eq. (4.30).

For ICRU Report 37 (1984a), the results of Seltzer and Berger (1985) were not yet available and a somewhat different procedure was used. The information on $d\sigma_n/dk$ was used to evaluate $\varphi_{\text{rad},n}$ in the high-energy region, $T \geq 50$ MeV, and in the low-energy region, $T \leq 2$ MeV, and then in the gap region, $2 \text{ MeV} < T < 50 \text{ MeV}$, through the use of least-squares cubic-spline interpolation in T of $\varphi_{\text{rad},n}$. Thus, there are differences among the two versions of $\varphi_{\text{rad},n}$, mostly small and certainly within the uncertainty inherent in the theoretical and numerical treatments. In the present Report, the data of Seltzer and Berger (1986) are adopted for the radiative energy-loss cross section as it is consistent with the full database of differential cross sections now used in many Monte Carlo codes.

4.4.1.2 Incident Positrons. As described in ICRU Report 37 (1984a), the very limited results from partial-wave calculations for $d\sigma_n/dk$ for incident positrons of Feng *et al.* (1981) were used to obtain $\varphi_{\text{rad},n}$, and a universal scaling of the ratio of the radiative energy-loss cross sections for positrons to electrons was deduced to cover all Z and T in the ranges of interest. Although the additional calculations of Kim *et al.* (1986) were then stated to confirm this scaling law, these data were only later incorporated into the procedure for fitting the positron/electron ratio. These somewhat different ratios are given in Table 4.1. Thus, in the present Report, these

Table 4.1. Positron-to-electron ratios, $\varphi_{\text{rad},n}^+/\varphi_{\text{rad},n}^-$, of the radiative energy-loss cross sections for bremsstrahlung in the field of the screened nucleus. Scaling the incident kinetic energy in terms of T/Z^2 was found to produce an approximately universal curve to be applied to all Z .

T/Z^2	$\varphi_{\text{rad},n}^+/\varphi_{\text{rad},n}^-$	T/Z^2	$\varphi_{\text{rad},n}^+/\varphi_{\text{rad},n}^-$
1×10^{-8}	0	1×10^{-4}	0.4463
2×10^{-8}	0.0016	2×10^{-4}	0.5297
5×10^{-8}	0.0098	5×10^{-4}	0.6441
1×10^{-7}	0.0200	1×10^{-3}	0.7274
2×10^{-7}	0.0346	2×10^{-3}	0.8025
5×10^{-7}	0.0613	5×10^{-3}	0.8808
1×10^{-6}	0.0881	1×10^{-2}	0.9227
2×10^{-6}	0.1211	2×10^{-2}	0.9523
5×10^{-6}	0.1754	5×10^{-2}	0.9771
1×10^{-5}	0.2251	1×10^{-1}	0.9882
2×10^{-5}	0.2825	2×10^{-1}	0.9955
5×10^{-5}	0.3704	$\geq 5 \times 10^{-1}$	1.0

newer ratios—coupled to slightly different values of the radiative energy-loss cross section for incident electrons—result in somewhat different values of the radiative stopping powers for incident positrons, but within the stated relative uncertainty of 10 % (ICRU, 1984a).

4.4.2 Bremsstrahlung in the Field of the Atomic Electrons

A theoretical description of bremsstrahlung emission in collisions of electrons and positrons with free electrons is obtained (Haug, 2006) by representing electron states by plane waves. The transition matrix element is difficult to calculate because eight different Feynman diagrams contribute to the transition amplitude. Haug and Nakel (2004) give a covariant formula, depending on five invariants, for the squared transition matrix element for electron–electron bremsstrahlung, averaged over initial spin states, summed over final spin states of the electron and summed over polarization states of the emitted photon. This formula is extremely complicated due to the effects of recoil and exchange.

The emission of bremsstrahlung in the field of electrons bound in atoms can be described by using the impulse approximation (Haug and Keppler, 1984). Alternatively, the effect of binding can be accounted for by using the approximation of Wheeler and Lamb (1939), in which binding effects are introduced by simply multiplying the differential cross section for a target electron free and at rest by the incoherent scattering function, $S(q, Z)$, where q is the momentum transfer to the target electron. It should be noted that this approximation partially disregards the motion of the target electrons. Seltzer and Berger (1985) derived a screening and binding correction from the Wheeler–Lamb approximation, which they added to the Haug (1975) differential cross section for free electrons at rest to get the differential cross section for bremsstrahlung emission by incident electrons in the field of atomic electrons. Their treatment included also the Coulomb correction of Maxon and Corman (1967), although it has only a small effect. More recently, Tessier and Kawrakow (2008) described more elaborate calculations of electron–electron bremsstrahlung based on the Wheeler–Lamb approximation.

A more piecemeal approach to electron–electron bremsstrahlung was taken in ICRU Report 37 (1984a), where the high-energy region and the low-energy region were considered separately and the ratio $\varphi_{\text{rad},e}/\varphi_{\text{rad},n}$ determined for the gap region by least-squares cubic-spline interpolation. Thus, there can be differences in the electron–electron component of the radiative energy-loss cross sections used

Table 4.2. Ratios, $\varphi_{\text{rad,e}}^+/\varphi_{\text{rad,n}}^-$, used to estimate the radiative energy-loss cross sections for bremsstrahlung in the field of the atomic electrons for incident positrons.

T/MeV	$\varphi_{\text{rad,e}}^+/\varphi_{\text{rad,n}}^-$	T/MeV	$\varphi_{\text{rad,e}}^+/\varphi_{\text{rad,n}}^-$
0.001	0.502	0.06	0.594
0.0015	0.505	0.08	0.605
0.002	0.509	0.1	0.613
0.003	0.515	0.15	0.631
0.004	0.520	0.2	0.643
0.005	0.524	0.3	0.663
0.006	0.528	0.4	0.679
0.008	0.534	0.5	0.681
0.01	0.539	0.6	0.701
0.015	0.550	0.8	0.718
0.02	0.558	1.0	0.732
0.03	0.570	1.5	0.762
0.04	0.580	2.0	0.785
0.05	0.587	3.0	0.823

in that Report from those of Seltzer and Berger (1985; 1986) that are used in the present Report.

The numerical estimates of positron–electron bremsstrahlung are even more approximate as there is little in the way of reliable data. A few facts were noted in ICRU Report 37 (1984a), including: (1) at high energies, one expects little difference between the scaled radiative energy-loss cross sections for positrons and electrons, *i.e.*, $\varphi_{\text{rad,e}}^+ \cong \varphi_{\text{rad,e}}^-$, and (2) at very low energies, because of the formal identity of the non-relativistic electron–proton and the positron–electron bremsstrahlung cross sections, $\varphi_{\text{rad,e}}^+ \cong \frac{1}{2} \varphi_{\text{rad,n}}^-$. In that Report, these two limiting cases were connected by assuming that $\varphi_{\text{rad,e}}^+/\varphi_{\text{rad,e}}^-$ is equal to $\varphi_{\text{rad,e}}^+/\varphi_{\text{rad,n}}^-$ for $T \geq 5$ MeV, with values simply taken from a smooth curve to extend down to $\frac{1}{2}$ at extremely low energies. These ratios, given in Table 4.2, are used also in the present Report.

4.4.3 Radiative Stopping Power for Compounds and Mixtures

With the lack of better information, the mass radiative stopping power of compounds and mixtures is evaluated according to Bragg’s additivity approximation, *i.e.*,

$$\frac{1}{\rho} S_{\text{rad}}(T) = \sum_i \zeta_i \left[\frac{1}{\rho} S_{\text{rad}}(T) \right]_i, \quad (4.33)$$

where ζ_i is the fraction by weight and $[(1/\rho)S_{\text{rad}}(T)]_i$ the radiative stopping power of the i th atomic constituent.

4.5 Ranges and Radiation Yields

Quantities associated with the stopping power are included in the results of new evaluations to be given in the Appendix, based on the I values for air, graphite, and liquid water discussed in Section 5. If one assumes that charged particles lose energy continuously along their track, with the mean energy loss per pathlength given by the stopping power, it is straightforward to evaluate quantities that at least roughly characterize the transport of the charged particle. This approximation is called the continuous-slowning-down approximation (csda).

The (mass) *csda range* is defined as

$$\rho r_0(T_0 \rightarrow T_f) = \rho \int_{T_f}^{T_0} \frac{1}{S(T)} dT, \quad (4.34)$$

where T_0 is the initial kinetic energy and T_f is some final kinetic energy of interest (*e.g.*, the lowest ionization potential for the medium, or—in the following—an energy considered to be effectively zero), and $(1/\rho)S(T)$ is the mass stopping power (in principle, the sum of electronic, radiative, and nuclear components). The stopping powers needed for the evaluation of Eq. (4.34) are available only down to some intermediate energy; the tables to be given later in this Report extend down to a kinetic energy of 1 keV. As done in the previous ICRU Reports (ICRU 1984a, 1993), a crude approximation is used in which it is assumed that $1/S(T)$ is zero at $T = 0$ and can be interpolated linearly up to its value at 1 keV. The range is then given by

$$\begin{aligned} \rho r_0(T_0 \rightarrow 0) \\ = \rho \left[\int_{1 \text{ keV}}^{T_0} \frac{1}{S(T)} dT + \frac{1}{2} \frac{1 \text{ keV}}{S(1 \text{ keV})} \right]. \end{aligned} \quad (4.35)$$

This quantity approximates the total (rectified) pathlength traveled by the particle; except at rather low energies, it is close to, but somewhat larger than, the mean pathlength traveled if energy-loss fluctuations were taken into account.

The *csda range* should not be confused with the depth of penetration (along the initial direction), which is affected by multiple angular deflections. For heavy charged particles, a *detour factor* (also evaluated in the *csda*) has been defined (see ICRU, 1993) as the ratio of the mean penetration depth to the *csda range*, and those values given in ICRU Report 49 will be included here in the final tables. For electrons and positrons, which undergo much more significant angular deflections, the *detour factor* is of little utility and—as done previously—will be ignored for those particles.

The *radiation yield* is the fraction of the initial kinetic energy, T_0 , that is converted to bremsstrahlung energy as the charged particle slows down to rest. In the csda, the radiation yield is given by

$$Y(T_0) = \frac{1}{T_0} \int_0^{T_0} \frac{S_{\text{rad}}(T)}{S(T)} dT. \quad (4.36)$$

As indicated earlier, the radiation yield is of interest mainly for electrons and positrons, for which the total stopping power can be written $S(T) = S_{\text{el}}(T) + S_{\text{rad}}(T)$. For the evaluation of Eq. (4.36), the data given down to a kinetic energy of 1 keV were used and the integrand at energies below that again assumed to vary linearly from the value at 1 keV to zero at $T = 0$.

5. Recommended Values for Key Data

This Section presents the analyses that support new recommendations for the numerical values, and the associated uncertainties, of the key parameters considered in this Report. These key data include the mean excitation energies for air, I_{air} , and for graphite, I_g , that are needed in the determination of stopping-power ratios, the average energy to create an ion pair in air, W_{air} , which is needed for the determination of air kerma, and the mean excitation energy for liquid water, I_w , which is central to calculations that support charged-particle therapy and is important for certain factors entering in the determination of absorbed dose in water. Recommendations are also given for the humidity correction factor, k_h , for air-filled ionization chambers, the ferric ion yield, $G(\text{Fe}^{3+})$, required for Fricke dosimetry and the heat defect, h , for graphite and water calorimetry. Data for k_w , the correction in photon and electron beams at low energies for the deviation of W_{air} from the high-energy value, and for k_{ii} , the correction to the measured charge due to the ion pairs created by an incident photon, are also summarized.

5.1 Mean Excitation Energy of Dry Air

Dry air at sea level is a mixture of N_2 , O_2 , Ar, and CO_2 , with fractions by volume assumed to be (as in ICRU, 1984a) 0.7809, 0.2095, 0.0093, and 0.0003, respectively (small additions of trace molecules, being inconsequential, are ignored). Being a true mixture, the I value of dry air can be obtained simply from the I values of the constituent molecules through implementation of Bragg additivity for the electronic stopping power [as done in Eq. (4.33) for the radiative stopping power],

$$\frac{1}{\rho}S_{\text{el}}(T) = \sum_i \zeta_i \left[\frac{1}{\rho}S_{\text{el}}(T) \right]_i, \quad (5.1)$$

which is equivalent to replacing, in the electronic stopping-power formulas Eqs. (4.8) and (4.16)

through (4.18), the quantities Z/A , $\ln I$, and δ by

$$\langle Z/A \rangle = \sum_i \zeta_i (Z_i/A_i), \quad (5.2)$$

$$\ln I = \left[\sum_i \zeta_i (Z_i/A_i) \ln I_i \right] / \langle Z/A \rangle, \quad (5.3)$$

$$\delta = \left[\sum_i \zeta_i (Z_i/A_i) \delta_i \right] / \langle Z/A \rangle, \quad (5.4)$$

where ζ_i is the fraction by weight and Z_i , A_i , I_i , and δ_i pertain to the i th constituent.

The I value for dry air given in ICRU Report 37 (1984a) and used in ICRU Report 49 (ICRU, 1993) was 85.7 eV, based on the then available results from analyses of stopping-power and range measurements for protons and α particles and from analyses of dipole-oscillator-strength distributions (OSD). Since then some new values for constituent molecules and atoms have been reported, notably those from Shiomi-Tsuda *et al.* (1999) and from Kamakura *et al.* (2006). Table 5.1 lists reported I values for the constituents of dry air.

Table 5.1 illustrates some of the issues involved in arriving at a recommended I value and its standard uncertainty (listed in the last two columns). This Report strives to use and recommend a standard uncertainty (*i.e.*, a coverage factor of unity) consistent with the Guide to the Reporting of Uncertainty in Measurement (JCGM, 2008), popularly known as the GUM.¹⁵ Many references do not indicate the nature of quoted uncertainties. In this case, the stated uncertainties were assumed to have the nature of standard uncertainties. For those older experimental values taken from ICRU Report 37 (ICRU, 1984a), the uncertainties stated in that Report have been reduced in the present Report by the factor of 1.6.¹⁶ If

¹⁵The GUM recommends the reporting of a combined standard uncertainty, u_c (comprising Type A and Type B), which predicts an interval having a confidence of approximately 67 % if the underlying probability distribution is approximately normal and the effective number of degrees of freedom is of significant size.

¹⁶Footnote 10 in ICRU Report 37 (1984a) suggests that the stated uncertainties be interpreted as corresponding to a 90 % confidence interval and, reflecting even the uncertainty in that

Table 5.1. Reported mean excitation energies, I , for constituents of dry air. Listed results include those from the analyses of measurements of stopping powers (S) and ranges (R) for protons (p) and α particles (α). Standard uncertainties, u_c , for the I values are listed, as well as the relative contributions by corrections [see Eq. (4.18) and Fig. 4.1], calculated with the BEST code, that affect the analyses of measured stopping powers.

Source	Method	Relative contribution of corrections, %	I/eV	u_c/eV	Recommended	
					I/eV	u_c/eV
N₂						
From experiment						
Brolley and Ribe (1955)	S: 4.43 MeV p	1.8	88 ^a	9	82.3	1.2
Martin and Northcliffe (1962)	S: (4–40) MeV α	0.8–4.9	79 ^a	7		
Swint <i>et al.</i> (1970)	S: (0.4–3.4) MeV p	2.3–4.5	76 ^a	9		
Besenbacher <i>et al.</i> (1979)	S: (0.04–1) MeV p; (0.1–1.2) MeV α	>4.8; >7.5	86.7 ^a	(15) ^b		
Baumgart <i>et al.</i> (1983)	S: (0.066–0.8) MeV p	>5	83.5	(15) ^b		
Shiomi-Tsuda <i>et al.</i> (1999)	S: (4–13) MeV p	0.6–2	85.6	3.5		
From OSD						
Dalgarno <i>et al.</i> (1967)	OSD		82.1	2 ^a	95.2	1.0
Zeiss <i>et al.</i> (1977)	OSD		81.8	2 ^a		
Sauer <i>et al.</i> (1995)	OSD		83.7	(4) ^b		
Kamukura <i>et al.</i> (2006)	OSD		83.0	(4) ^b		
O₂						
From experiment						
Brolley and Ribe (1955)	S: 4.43 MeV p	1.8	95 ^a	8	187	3
Swint <i>et al.</i> (1970)	S: (0.4–3.4) MeV p	2.3–4.5	90 ^a	10		
Besenbacher <i>et al.</i> (1979)	S: (0.04–1) MeV p; (0.1–1.2) MeV α	>4.8; >7.5	102.1 ^a	(15) ^b		
Baumgart <i>et al.</i> (1983)	S: (0.066–0.8) MeV p	>5	96.0	(15) ^b		
Shiomi-Tsuda <i>et al.</i> (1999)	S: (4–13) MeV p	0.6–2	96.3	3.7		
From OSD						
Zeiss <i>et al.</i> (1977)	OSD		95.0	2 ^a	187	3
Kumar <i>et al.</i> (1996)	OSD		95.0	1		
Kamukura <i>et al.</i> (2006)	OSD		96.5	(2) ^c		
Ar						
From experiment						
Brolley and Ribe (1955)	S: 4.43 MeV p	1.8	190 ^a	15	87	2
Martin and Northcliffe (1962)	S: (4–40) MeV α	0.8–4.9	190 ^a	7		
Hanke and Bichsel (1970)	S: 1–9 MeV p	1.2–9.0	188 ^a	10		
Besenbacher <i>et al.</i> (1979)	S: (0.04–1) MeV p; (0.1–1.2) MeV α	>4.8; >7.5	194 ^a	(15) ^b		
Baumgart <i>et al.</i> (1983)	S: (0.066–0.8) MeV p	>5	190.0	(15) ^b		
From OSD						
Langhoff and Yates (1972)	Moment theory		111	57	87	2
Bell <i>et al.</i> (1972)	From HF wave functions		174	3.5		
Eggarter (1975)	OSD		179	(4) ^c		
Dehmer <i>et al.</i> (1975)	From HS wave functions		177.4	(4) ^c		
Kumar and Meath (1985)	OSD		186.2	(4) ^c		
Kamukura <i>et al.</i> (2006)	OSD		191	(2) ^c		
CO₂						
From experiment						
Thompson (1952)	ΔR : (340–200) MeV p	0.02–0.03	85.0 ^a	2	87	2
Porter (1994)	S: p; α data of Reiter <i>et al.</i> (1990)	>10.4; >7.9	90	(12) ^b		
From OSD						
Jhanwar <i>et al.</i> (1986)	OSD		87.1	(2) ^c		
Kamukura <i>et al.</i> (2006)	OSD		89.6	(2) ^c		

^aAs reported in ICRU Report 37 (ICRU, 1984a).

^bApproximate estimation done here based on consideration of the contribution of corrections.

^cEstimation done here based on those quoted for similar results.

no uncertainties were given by the authors, estimates of standard uncertainties have been made that were guided, and judged to be consistent with, other analyses. Thus, such recommendations as given in this Report are a result of both objective and subjective considerations.

Note the following for the results in Table 5.1. The recommended values for I and u of the constituents are from the weighted means and the corresponding variances, respectively, except for Ar. In the case of Ar, the results from OSD analyses are inconsistent with one another as judged by a Birge ratio¹⁷ of 2.3 and are also largely inconsistent with those from experiment, except for the OSD results of Kamukura *et al.* (2006); the moment-theory result of Langhoff and Yates (1972) is clearly an outlier, but its large uncertainty makes it inconsequential in the analysis. An analysis of the OSD results using the Mandel–Paule algorithm (see footnote 17) results in the value 182 eV with standard uncertainty 4 eV. Combining this with the experimental values in a weighted mean gives the final recommended result in Table 5.1. For CO₂, the existence of only one reliable experimental value gives too much weight to the OSD results; this was reduced by combining the two OSD results into the single mean value 88.3 eV (standard uncertainty 2 eV). In any event, the value used for CO₂ has no impact on the final value for air.

The results for dry air are summarized in Table 5.2. Using Eqs. (5.2) and (5.3), the value obtained for the mean excitation energy of dry air should be 85.9 eV. This is very close to the value of 85.7 eV given in ICRU Report 37 (ICRU, 1984a). It should be noted that other choices in the analysis of the data in Table 5.1 would change the outcome only slightly, by about 0.5 eV. In view of this small effect, remaining consistent with the recommendation in ICRU Report 37 would seem to combine the judgment of the two report committees, and it is this recommended value that is given in Table 5.2. Nevertheless, it is recommended here to replace the stated uncertainty of 1.7 eV in the earlier Report by

estimate, that the standard uncertainty can be obtained by multiplying “by about one half.” For the more conservative reduction in uncertainty used in the current Report, the factor ~ 1.6 obtained from the relation between the 90 % and 68 % confidence intervals for a normal distribution is used.

¹⁷In a number of analyses in the present Report, the self-consistency of data from different sources is judged using the Birge ratio (Birge, 1932), which compares the variance of a data set with that expected from the stated uncertainties and is essentially the square root of the reduced χ^2 . When the Birge ratio is significantly greater than unity, the Mandel–Paule algorithm (Mandel and Paule, 1970; Paule and Mandel, 1982) is invoked. This introduces an additional “inter-source” uncertainty for each datum such that the re-evaluated Birge ratio is unity.

Table 5.2. Data used in the evaluation of the mean excitation energy of dry air.

Constituent	Fraction by weight	$\langle Z/A \rangle$	Recommended	
			I/eV	u/eV
N ₂	0.755267	0.499761	82.3	1.2
O ₂	0.231450	0.500019	95.2	1.0
Ar	0.012827	0.450586	187	3
CO ₂	0.000456	0.499889	87	2
Dry air	1	0.499190	85.7	1.2

the standard uncertainty of 1.2 eV resulting from the present analysis.

5.2 Mean Excitation Energy of Graphite

There has been evidence for some time that the best estimate of the product $W_{\text{air}} s_{\text{g,air}}$ for ⁶⁰Co γ radiation is lower than the value currently used for all primary-standard cavity ionization chambers. Many of the data are summarized in papers by Seltzer and Bergstrom (2003), by Burns (2009), and by Thomson and Rogers (2010), and conclusions are presented in relation to the value for W_{air} , the value of the graphite mean excitation energy, I_{g} , and the choice of either the bulk or the grain mass density, ρ_{g} , when evaluating the density-effect correction in the electronic stopping power for graphite. In this Section, we consider the data that can be used to determine a value for I_{g} .

ICRU Report 37 (ICRU, 1984a) used four measured values for I_{g} from which a value of 78 eV was obtained. The stated uncertainty of 7 eV is to be interpreted as representing a confidence level of 90 %, so the standard uncertainty is taken to be 4 eV. Since the publication of ICRU Report 37, three new determinations of I_{g} have been reported. These new values are all larger than 78 eV and have relatively small uncertainties.

The first four entries in Table 5.3 correspond to the data set used for the ICRU Report 37 evaluation. Taking a weighted mean using the standard uncertainties gives 78.7 eV, close but not identical to the ICRU Report 37 result. The Birge ratio for the data set has a value of 2.4, indicating that the estimated uncertainties for some of the data points are too small. Using the Mandel–Paule algorithm leads to a revised mean of 80.6 eV with a standard uncertainty of 4.2 eV. This uncertainty is consistent with the value given in ICRU Report 37, although the revised mean is 2.6 eV larger. The authors of ICRU Report 37 used a more subjective approach to determine consensus values for I_{g} and its uncertainty.

For the present evaluation, it is assumed that the mean excitation energy of aluminum is among the

Table 5.3. Compilation of data used to establish a recommended value for the mean excitation energy of graphite, I_g .

Source	Method	I_g/eV	Stated uncertainty (eV)	Assumed u_c/eV
Data used in ICRU Report 37				
Bakker and Segré (1951)	340 MeV protons	91.7 ^a	8.6 ^b	5.4
Bakker and Segré (1951)	340 MeV protons	83.3 ^c	7.6 ^b	4.8
Bichsel and Tschalär (1965)		80.0	2.4 ^b	1.5
Nordin and Henkelman (1979)	60 MeV positive and negative pions	70.8 ^b	4.0 ^b	2.5
Data used for this Report				
Bakker and Segré (1951)	340 MeV protons	83.3 ^c	7.6 ^b	4.8
Bichsel and Tschalär (1965)		80.0	2.4 ^b	1.5
Nordin and Henkelman (1979)	60 MeV positive and negative pions	73.5 ^c	—	3.1
Bichsel and Hiraoka (1992)	70 MeV protons, based on $I_{\text{Al}} = 166 \text{ eV}$	86.9	1.7	1.7
Sakamoto <i>et al.</i> (1996)	Low-energy protons	79.9	1.5	1.5
Büermann and Hilgers (2007)	Comparison of cavity and free-air chambers	82.8	2.2	2.2

^aAssuming $I_{\text{Cu}} = 322 \text{ eV}$.^bAs stated in ICRU Report 37 (1984a). Standard uncertainty obtained by dividing by 1.6.^cAssuming $I_{\text{Al}} = 166 \text{ eV}$.

best known. Thus, we do not consider the value from Bakker and Segré (1951) relative to copper because it is the same measured data that yields the value relative to aluminum. ICRU Report 37 used the value from Nordin and Henkelman (1979) relative to liquid water. However, these authors also report data for aluminum, and we have taken the value for I_g based on the measured graphite-to-aluminum ratio in conjunction with the value $I_{\text{Al}} = 166 \text{ eV}$ with standard uncertainty 1.3 eV.

We have used the values recommended by Burns (2012) for the work of Sakamoto *et al.* (1996) and for Büermann and Hilgers (2007).

Taking a weighted mean of the data in the bottom section of Table 5.3 leads to 81.5 eV, and a standard uncertainty of 0.8 eV. The Birge ratio for the data set is 2.0, which leads using the Mandel–Paule method to a slightly revised mean of 81.2 eV and a standard uncertainty of 1.8 eV. This result is close to the value $I_g = 81.1 \text{ eV}$ with standard uncertainty 2.0 eV obtained by Burns (2012) using a slightly different set of input values for I_g . The analysis by Burns included data for W_{air} and for the product $W_{\text{air}} s_{g,\text{air}}$, and used a minimization process to obtain values for I_g and W_{air} consistent with the entire data set. A subsequent work (Burns *et al.*, 2014) analyzed the results of a single experiment in accelerator photon beams (again in terms of I_g and W_{air}) to obtain the independent estimate $I_g = 81.1 \text{ eV}$ with standard uncertainty 1.8 eV.

In view of the above, the value $I_g = 81.0 \text{ eV}$ with a standard uncertainty of 1.8 eV is now recommended. The data set, mean value, and associated uncertainty are shown graphically in Fig. 5.1.

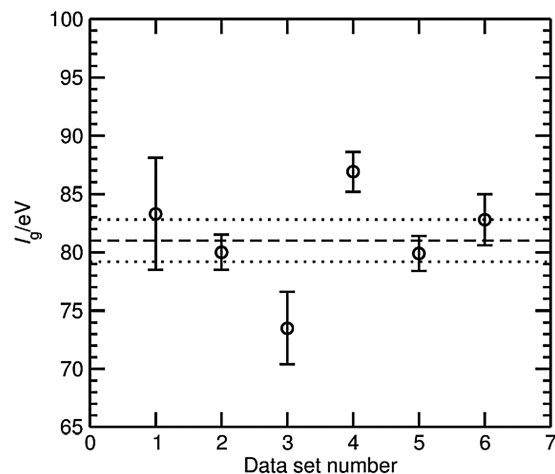


Figure 5.1. Data, from Table 5.3, used to establish a recommended value for the mean excitation energy of graphite. The uncertainty bars represent estimated standard uncertainties. The weighted mean is shown by the dashed line, and the dotted lines indicate the standard uncertainty of the mean.

5.3 Mean Excitation Energy of Liquid Water

The uncertainties achievable in radiation dosimetry at the time of the development of ICRU Report 37 (ICRU, 1984a) were such that the uncertainty for I_w was not considered to play a significant role. However, present-day requirements for stopping-power ratios with uncertainties of a few tenths of a percent have increased the importance of a robust estimate for I_w and its uncertainty, as changes in I_w affect the values for water-to-air stopping-power ratios, a key quantity for dosimetry based on ionization chambers for all particle types. Furthermore, for protons and heavier charged particles, I_w is critical for the position of the Bragg peak (Andreo,

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2009), and significant uncertainties in the I values for liquid water and for human tissue affect the spatial accuracy of clinical dose delivery (Andreo, 2009; Besemer *et al.*, 2013).

Since the recommendations of ICRU Report 37 (ICRU, 1984a), there have been a significant number of new estimates for I_w , based either on stopping power and range measurements for protons and heavier charged particles at high energies, where the conditions for Bethe's theory are valid and the influence of the corrections in the stopping-power expression small, or on dielectric-response functions (DRFs) from experimental data for photons at low energies. In the latter case, the results are influenced considerably by the models and data used, choices that can produce a range of values for I_w . Moreover, many such determinations lack uncertainty estimates, notably those based on DRFs. While the dosimetric consequences of this diversity of values have been investigated, there has been no proposal for a harmonized value taking all the available data into account.

The recommendation in ICRU Report 37 is $I_w = 75$ eV with a standard uncertainty estimated to be about 2 eV. As described by Berger and Paul (1995), this value was obtained from an analysis of four sets of values, two from DRFs (Ashley, 1982; Ritchie *et al.*, 1978) and two from stopping-power and range measurements (Thompson, 1952, using protons, and Nordin and Henkelman, 1979, using pions). Subsequently, ICRU Report 49 (ICRU, 1993) included a footnote referring to a new experimental determination by Bichsel and Hiraoka (1992), using 70 MeV protons, who had determined the value $I_w = 79.7$ eV with the very much smaller uncertainty of 0.5 eV (however, ICRU Report 49 did not make a recommendation for this new value and retained $I_w = 75$ eV from ICRU Report 37). The Bichsel and Hiraoka value was later supported by Bichsel *et al.* (2000), using carbon ions, with the result $I_w = 80$ eV and uncertainty 1.3 eV, although significant approximations were made. Each of these estimations is toward the upper limit of the recommendation in ICRU Reports 37 and 49.

An implicit recommendation through the use of $I_w = 67.2$ eV was included in ICRU Report 73 (ICRU, 2005), which was subsequently questioned by Paul *et al.* (2007) who suggested the value $I_w = 80.8$ eV with an uncertainty of 2 eV. This estimate was based on the DRF-based values of 81.8 eV by Dingfelder *et al.* (1998) and the Bichsel and Hiraoka (1992) value of 79.7 eV. Later Paul (2007) gave support to the work of Sihver *et al.* (1998), concluding that the value $I_w = 75$ eV (from ICRU Reports 37 and 49) "agrees best with Sihver's experimental value." The indirect determination by Sihver *et al.* (1998) was obtained by comparing an analytical

model with depth-ionization curves for 195 MeV and 270 MeV carbon ions. The same data were used by Krämer and Scholz (2000) at the GSI Helmholtz Center for Heavy Ion Research in Darmstadt, Germany, to benchmark their model for treatment planning, which was found to require the value $I_w = 77$ eV to reproduce the depths of the measured Bragg peaks. The low value of 67.2 eV implicit in ICRU Report 73 (ICRU, 2005) was subsequently addressed by ICRU (Sigmund *et al.*, 2009), and the tentative value $I_w = 78$ eV was recommended, based on measurements for protons and heavier ions by Scharadt *et al.* (2008) described in a GSI report.

A recent work by Emfietzoglou *et al.* (2009), based on a DRF deduced from the inelastic-x-ray-scattering-spectroscopy (IXSS) data for liquid water of Hayashi *et al.* (2000), suggested the value $I_w = 77.8$ eV. Unusually for DRF-based analyses, these authors also estimated an uncertainty of 1 eV. This significant work included a compilation of published values, mostly derived using analogous numerically based approaches and based on one of two sets of experimental reflectance and spectroscopy data (the so-called REF and IXSS data sets). These values for I_w ranged from the 72.5 eV of Tan *et al.* (2004) to the 82.4 eV of an earlier work by Emfietzoglou *et al.* (2005); this range in itself raises the question of how the uncertainty of 1 eV should be interpreted.

A compilation of the experimental and DRF-based values for I_w described above, done for the present Report, was published separately by Andreo *et al.* (2013) and is given here in Table 5.4; such a compilation was initiated by Vatnitskiy *et al.* (2011). The compilation excludes values for which important corrections were not taken into account, for example, multiple-scattering corrections in low-energy electron measurements or corrections to the stopping powers of heavy charged particles at low energies (see, *e.g.*, Fig. 4.1), or for which the experimental uncertainties of stopping-power measurements were substantial, leading to a much higher uncertainty in the value obtained for I_w .

An important aspect of the analysis by Andreo *et al.* (2013) was an uncertainty estimate for the DRF-based values. This estimate was based on a study of the results of a set of publications by Emfietzoglou and collaborators (included in Table 5.4), in which different physical models and optical data were used that were considered representative of the entire set of DRF-based values. The use of different models was assessed to introduce a standard uncertainty of 2.5 eV, and the different optical data gave 3.3 eV. The combined result for the DRF-based data was the value $I_w = 78.4$ eV with a standard uncertainty of 4.2 eV (which is in fact close to the standard deviation of the DRF-based data set).

Table 5.4. Compilation of experimental (measured energy-range distributions) and dielectric-response-function (DRF)-based published values of the mean excitation energy of liquid water, I_w , used in the present analysis (adapted from Andreo *et al.*, 2013).

Source	Method	I_w /eV	Stated uncertainty (eV)	Assumed u_c /eV
From experiment				
Thompson (1952)	340 MeV, 200 MeV protons	75.4 ^a	1.9 ^b	1.0
Nordin and Henkelman (1979)	60 MeV positive and negative pions	74.7 ^b	2.7 ^b	1.4
Bichsel and Hiraoka (1992)	70 MeV protons, based on $I_{Al} = 166$ eV	79.7	0.5	0.3
Krämer <i>et al.</i> (2000); Sihver <i>et al.</i> (1998)	195 MeV and 270 MeV carbon ions	77.0	—	2.0 ^c
Bichsel <i>et al.</i> (2000)	290 MeV carbon ions	80.0	1.3	0.7
Kumazaki <i>et al.</i> (2007)	150 MeV, 190 MeV, 230 MeV protons	78.4	1.0	0.8
Schardt <i>et al.</i> (2008)	¹ H, ³ He, ⁷ Li, ¹² C, ¹⁶ O ions up to 400 MeV	78.0	1.0	1.0
From DRF				
Ritchie <i>et al.</i> (1978)	REF optical data ^d	75.0	—	4.2
Ashley (1982)	Semi-empirical model	75.4	—	4.2
LaVerne and Mozumder (1986)	Semi-empirical model	74.9	—	4.2
Dingfelder <i>et al.</i> (1998)	Semi-empirical model	81.8	—	4.2
Akkerman and Akkerman (1999)	Semi-empirical model	74.0	—	4.2
Tan <i>et al.</i> (2004)	Semi-empirical model	72.5	—	4.2
Emfietzoglou (2003)	REF optical data ^d	80.9	—	4.2
Emfietzoglou and Nikjoo (2005)	REF optical data ^d	85.4	—	4.2
Emfietzoglou and Nikjoo (2005)	IXSS optical data ^e	80.7	—	4.2
Emfietzoglou <i>et al.</i> (2005)	IXSS optical data ^e	82.4	—	4.2
Emfietzoglou <i>et al.</i> (2009)	IXSS optical data ^e	77.8	1.0	4.2
Garcia-Molina <i>et al.</i> (2009)	IXSS optical data ^e	79.4	—	4.2

^aBased on $I_{Cu} = 322$ eV.

^bAs stated in ICRU Report 37 (ICRU 1984a) and in Berger and Paul (1995).

^cThis value is taken as the standard deviation of the experimental data set and is consistent with the authors' statement of a Bragg-peak shift of 0.5 mm for a 2 eV change in I_w .

^dReflectance measurements by Heller *et al.* (1974).

^eInelastic x-ray-scattering spectroscopy by Hayashi *et al.* (2000).

For the experimental values, Andreo *et al.* (2013) used a statistical analysis based on the Mandel–Paule method (see footnote 17) to obtain the value $I_w = 77.8$ eV; the standard uncertainty was estimated to be 2.0 eV. The experimental result was combined with that of the DRF-based values noted above to yield, as a weighted mean, the final rounded value $I_w = 78$ eV with its standard uncertainty of 2 eV. Figure 5.2 illustrates the results of the analysis.

As shown in Table 5.4, the uncertainties for the experimental values used in the analysis by Andreo *et al.* (2013) were modified in the spirit of footnote 10 in ICRU Report 37 (ICRU, 1984a), where the suggestion is made that the estimates of uncertainty given by the authors of that document for their results given in two important tables by subjectively combining published information on I values correspond to a confidence level of about 90 %, and that such values could be converted to standard uncertainties by multiplying them by a factor of about one-half. For the present Report, the robustness of this analysis for I_w is tested by applying the Mandel–Paule algorithm to a modified set of experimental results as given in Table 5.5. These differ from the results of Table 5.4 in two respects.

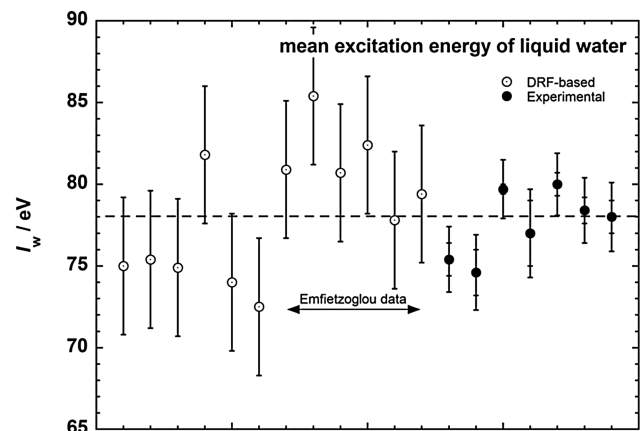


Figure 5.2. Values of the mean excitation energy of liquid water, I_w , derived from experimental and DRF-based published values (from Table 5.4). The assumed experimental uncertainties have been augmented as plotted in the figure to yield a Birge ratio of unity. Identical uncertainties are assumed for all the DRF-based values, obtained from an analysis of the data of Emfietzoglou and co-workers using different physical models and optical data. The dashed line indicates the adopted value. [From Andreo *et al.* (2013).]

First, the reduction by a factor of two has been removed, and the uncertainties stated by the

Table 5.5. Alternative version of values from experiment given in Table 5.4.

Source	Method	I_w/eV	Stated uncertainty(eV)
From experiment			
Thompson (1952)	340 MeV, 200 MeV protons	75.4	1.9
Nordin and Henkelman (1979)	60 MeV positive and negative pions	72.6	2.2
Bichsel and Hiraoka (1992)	70 MeV protons, based on $I_{\text{Al}} = 166 \text{ eV}$	79.7	0.5
Krämer <i>et al.</i> (2000); Sihver <i>et al.</i> (1998)	195 MeV and 270 MeV carbon ions	77.0	2.0
Bichsel <i>et al.</i> (2000)	290 MeV carbon ions	80.0	1.3
Kumazaki <i>et al.</i> (2007)	150 MeV, 190 MeV, 230 MeV protons	78.4	1.0
Schardt <i>et al.</i> (2008)	^1H , ^3He , ^7Li , ^{12}C , ^{16}O ions up to 400 MeV	78.0	1.0

authors assumed to be standard (*i.e.*, coverage factor of unity). Secondly, the results of Nordin and Henkleman (1979) have been treated as described in Section 5.2, that is, using only the results for liquid water relative to aluminum, which yields the result $I_w = 72.6 \text{ eV}$ with a standard uncertainty of 2.2 eV. Using the Mandel–Paule algorithm for these revised data, the weighted mean of the experimental values in Table 5.5 is $I_w = 77.8 \text{ eV}$.

These analyses indicate that the Andreo *et al.* result of $I_w = 78 \text{ eV}$ with standard uncertainty 2 eV is rather robust and as such is the recommended value in the present Report.

Interestingly, this coincides with the value used in the Errata and Addenda to ICRU Report 73 (Sigmund *et al.*, 2009), but with a robust uncertainty estimate and, as in ICRU Report 37 (ICRU, 1984a), includes both experimental and DRF-based values. The impact of this 4 % relative increase in I_w is to decrease the water-to-air stopping-power ratios for all charged-particle beams (see Section 7).

5.4 Average Energy to Create an Ion Pair in Dry Air

As noted in Section 2.9, W and w (differential in energy) relate the charge produced to the energy dissipated as charged particles slow down in a medium. Ionization and excitation of atoms or molecules are the dominant non-elastic processes for charged particles slowing down in matter; bremsstrahlung production can be important for high-energy electrons, but the photons interacting in the medium produce secondary electrons that ionize and excite as well. Many of the secondary electrons ejected in the primary ionizing event carry sufficient energy to cause further ionizations and, according to the definition, all of the ion pairs produced by the primary charged particle are to be included in the determination of W . To calculate W from first principles requires accurate knowledge of cross sections, both for all channels of ionization and those that govern the slowing-down processes. Because our current knowledge of cross sections does not meet the

required level of accuracy, values of W for metrological applications still depend mainly on measurements.

Although it is possible to satisfy the conditions of the definition for low-energy charged particles, it becomes much more difficult for high-energy electrons as bremsstrahlung losses are significant. In fact, the only measurements that fully satisfy the definition are for electrons with kinetic energies below about 10 keV. Techniques at higher energies are based on equating the energy deposited, determined by some technique such as calorimetry, with a measurement of the charge deposited in a gas cavity. Although charged-particle equilibrium can be used to suggest that W_{air} measured in this way is equivalent to what a direct measurement would yield, this has not been rigorously proven or tested. Direct measurements with low-energy electrons combined with indirect measurement at higher energies suggest that W_{air} might become independent of electron energy for energies above several tens of keV, but there is no firm theoretical basis for this observation. The review article by Burns and Büermann (2009) has a discussion of some of the challenges associated with measuring W_{air} .

5.4.1 W_{air} for Electrons

This Section examines the available data on W_{air} for electrons. For energies below about 10 keV, W_{air} changes rapidly with energy¹⁸ and has been

¹⁸The magnitude of W is always greater than I_1 , the first ionization potential of a molecule, and it approaches a constant at high energies. As the kinetic energy decreases below the energy above which the asymptotic value applies, the W value increases because the ratio of cross sections for ionization to that for non-ionizing processes decreases as a larger fraction of the initial particle energy is dissipated in non-ionizing events/processes such as excitation and elastic scattering. General theoretical results for electrons with T significantly larger than I_1 suggest the simple approximation $W(T) = W_{\infty}T/(T - U)$, where W_{∞} is the asymptotic W value for large T , and U is of the order of the average kinetic energy of a sub-ionization electron (Inokuti, 1975). The fitting function, Eq. (5.5), more accurately follows the available measurements and is reasonably compatible with this general theoretical prediction (*i.e.*, A_3 is small, and A_2 is of the order of -1).

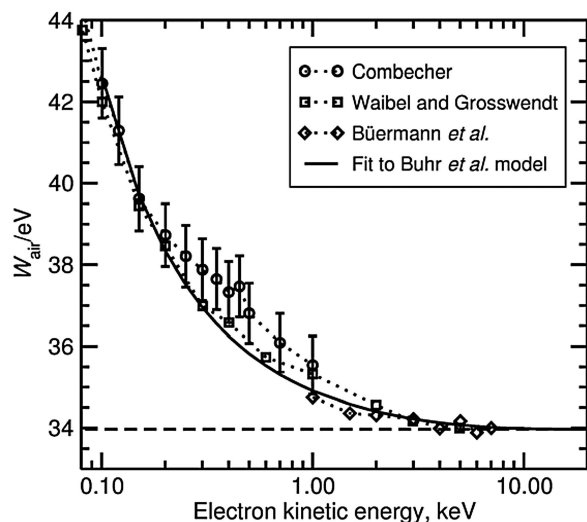


Figure 5.3. Low energy data for W_{air} from Combecher (1980), Waibel and Grosswendt (1978), and Büermann *et al.* (2006). The uncertainties for the data from Waibel and Grosswendt and from Büermann *et al.* are generally smaller than the symbols and are not shown. The function describing the fit is given in the text.

measured directly from about 15 eV to 7 keV. The available data in this energy range are shown in Fig. 5.3. For some applications, it is useful to have an empirical fit that adequately represents the measured data. Buhr *et al.* (2012) proposed the use of

$$W(T) = W_{\infty} \frac{1}{1 - (T/A_1)^{A_2 + A_3 T}}, \quad (5.5)$$

where W_{∞} is the asymptotic value of W_{air} , taken to be 33.97 eV, and A_1 , A_2 , and A_3 are free parameters to be obtained by fitting the measured data. A weighted, least-squares fit to the data shown in Fig. 5.3 gives $A_1 = 14.393(18)$ eV, $A_2 = -0.8179(87)$, and $A_3 = -3.4(1.5) \times 10^{-5}$ eV $^{-1}$, where the numbers in parentheses are the standard uncertainties in common notation. Because Buhr *et al.* (2012) used a subset of the data shown in Fig. 5.3, the coefficients they report are slightly different, but the fits agree to better than 0.2 %.

For electron energies well above 10 keV, the presently recommended value is 33.97 eV with a standard uncertainty of 0.05 eV (CCEMRI, 1985) and, in the absence of any data to the contrary, W_{air} is taken to be independent of energy. Since the time of that recommendation, historical measurements have been re-analyzed using new data and Monte Carlo techniques. As noted at the beginning of Section 5.2, this has led to significant changes in previously reported values of $W_{\text{air}} s_{\text{g,air}}$. In addition, new measurements have been reported that can be analyzed to obtain W_{air} . The data are summarized in Table 5.6 and in Fig. 5.4 and are discussed below.

Rogers (1993) re-analyzed the data sets used by Boutillon and Perroche-Roux (1987) using newer values for some parameters and updating the uncertainty estimates. In particular, he examined results for W_{air} obtained using ^{35}S β rays with a mean energy of 48.8 keV. References to the original papers are given in Table 5.6. The weighted mean of the ^{35}S results is 33.90 eV with a standard uncertainty of 0.16 eV.

Next in increasing energy is a single measurement using ^{137}Cs γ rays. Thomson and Rogers (2010) re-analyzed many of the experiments that measured the product $W_{\text{air}} s_{\text{g,air}}$, including the ^{137}Cs measurement of Engelke and Hohlfeld (1971). They calculated the graphite-to-air stopping-power ratios appropriate for ^{137}Cs and ^{60}Co γ rays and reported a value for $W_{\text{air}} s_{\text{g,air}}$ that would apply had the measurements been carried out using ^{60}Co γ rays. Using the values of I_{g} and I_{air} recommended in the present Report to calculate $s_{\text{g,air}}$, we obtain $W_{\text{air}} = 34.03$ eV with a standard uncertainty of 0.15 eV (0.44 %). Note that the estimated relative standard uncertainty of $s_{\text{g,air}}$ is 0.32 % with contributions of 0.28 % and 0.16 % from I_{g} and I_{air} , respectively.

There have been a large number of measurements of $W_{\text{air}} s_{\text{g,air}}$ carried out using ^{60}Co γ rays. Thomson and Rogers (2010) have re-analyzed the historical papers and identified important corrections that had been missed or calculated improperly. Using the values of I_{g} and I_{air} and of the graphite density recommended in the present Report, we have extracted values of W_{air} as shown in Table 5.6.

Burns (2012) describes three recent measurements of $W_{\text{air}} s_{\text{g,air}}$. One is work by Büermann *et al.* (2007) that compared a water calorimeter measurement with an ionometric determination of absorbed dose to water for ^{60}Co radiation. While the authors had used the graphite bulk density, Burns (2012) made a re-analysis using the graphite grain density and obtained $W_{\text{air}} s_{\text{g,air}} = 33.55$ eV. Taking correlation into account, the relative standard uncertainty was evaluated to be 0.41 %. Using the values for I_{g} and I_{air} and of the graphite density recommended in the present Report for calculating $s_{\text{g,air}}$ leads to $W_{\text{air}} = 33.79$ eV with a standard uncertainty of 0.18 eV.

Burns (2012) made a similar comparison of the BIPM ionometric absorbed-dose standard for ^{60}Co radiation with the mean of the national primary standards for absorbed dose in water (graphite and water calorimeters) that have taken part in the BIPM.RI(I)-K4 comparison series, the results of which appear in the BIPM online key comparison database (KCDB, 2013). The result is $W_{\text{air}} s_{\text{g,air}} = 33.77$ eV with a relative standard uncertainty of 0.22 % (0.07 eV), resulting in $W_{\text{air}} = 34.02$ eV with standard uncertainty 0.13 eV.

Table 5.6. Data for W_{air} for electrons.

Source	Comment	W_{air}/eV	u_c/eV
Bay <i>et al.</i> (1957)	^{35}S β rays, as reported by Rogers (1993)	33.60	0.21
Gross <i>et al.</i> (1957)	"	34.63	0.33
Jesse (1958)	"	33.90	0.34
Engelke and Hohlfield (1971)	^{137}Cs γ rays, <i>via</i> Thomson and Rogers (2010) ^a	34.03	0.15
Myers <i>et al.</i> (1961)	^{60}Co γ rays, as reported by Rogers (1993)	33.85	0.34
Bewley (1963)	^{60}Co γ rays, <i>via</i> Thomson and Rogers (2010) ^a	34.11	0.23
Petree and Lamperti (1967)	"	33.83	0.15
Guiho and Simoen (1975)	"	33.81	0.20
Niatel <i>et al.</i> (1985) #1	"	33.83	0.14
Niatel <i>et al.</i> (1985) #2	"	33.84	0.21
Büermann <i>et al.</i> (2007)	^{60}Co γ rays, <i>via</i> Burns (2012) ^a	33.79	0.18
Burns (2012)	^{60}Co γ rays, data set 9 in Burns (2012) ^a	34.02	0.13
"	^{60}Co γ rays, data set 10 in Burns (2012) ^{a,b}	34.05	0.13
Burns <i>et al.</i> (2014) ^a	6 MV x rays ^b	33.99	0.10
"	10 MV x rays ^b	34.01	0.09
"	25 MV x rays ^b	34.06	0.08
Cojocaru and Ross (2012)	High-energy electrons	33.84	0.14

^aThomson and Rogers (2010), Burns (2012), and Burns *et al.* (2014) report values for $W_{\text{air}} s_{\text{g,air}}$. Values for W_{air} have been obtained using the values for I_{g} and I_{air} (and their uncertainties) recommended in the present Report.

^bAlthough the calorimetric results of Burns (2012) (^{60}Co) and Burns *et al.* (2014) (high-energy x-ray beams) are represented here by four entries, for the purpose of investigating an energy dependence of W_{air} , the four are strongly correlated and only the single value 34.03(09) eV is included in the weighted mean.

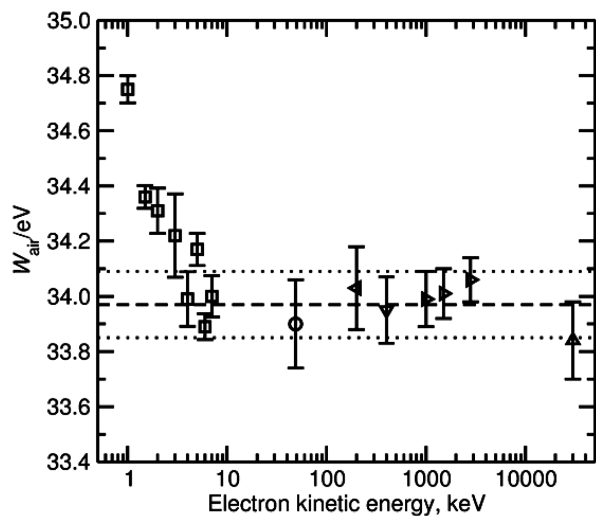


Figure 5.4. Values of W_{air} extracted from measured data. Symbols are as follows: square—low-energy electrons; circle— ^{35}S β rays; left triangle— ^{137}Cs γ rays; down triangle— ^{60}Co γ rays; right triangle—MV x-ray beams; up triangle—high-energy electrons. Values are plotted as a function of the approximate mean electron energy. The dashed line is the recommended value (33.97 eV) for W_{air} for electron kinetic energies above 10 keV, and the dotted lines indicate the standard uncertainty of the mean (0.12 eV or 0.35 %).

The BIPM graphite calorimeter (Picard *et al.*, 2009) has been used to establish the graphite absorbed-dose rate in the BIPM reference ^{60}Co beam. These measurements have been made in

combination with a reference ionization chamber that is designed to directly replace the calorimeter core and whose absolute volume is known. As described in Burns (2012), the results of these measurements have been combined with detailed Monte Carlo simulations of both arrangements to derive the value $W_{\text{air}} s_{\text{g,air}} = 33.80$ eV with standard uncertainty 0.06 eV, resulting in $W_{\text{air}} = 34.05$ eV with standard uncertainty 0.13 eV.

The weighted mean of all the ^{60}Co data, taking correlation in the uncertainties of I_{g} and of I_{air} into account, is 33.95 eV with a standard uncertainty of 0.12 eV, and this value is plotted in Fig. 5.4.

The BIPM calorimeter and reference ionization chamber have also been used in clinical photon beams from 6 MV to 25 MV (Burns *et al.*, 2014) in five different accelerator facilities. Combined with corresponding Monte Carlo calculations, values for $W_{\text{air}} s_{\text{g,air}}$ were obtained for 14 radiation beams over this energy range. For the purpose of the present Report, the results for W_{air} (assuming the values for I_{g} and I_{air} recommended here) have been divided into three nominal ranges, 6 MV, 10 MV, and 25 MV as given in Table 5.6.

Cojocaru and Ross (2012) re-analyzed the results for W_{air} obtained by Domen and Lamperti (1976) using a graphite calorimeter and an ionization chamber with known air volume in electron beams with kinetic energies from 15 MeV to 50 MeV. The original results had been re-analyzed by Svensson and Brahme (1986), and Cojocaru and Ross (2012)

addressed remaining problems related to stopping powers, correction factors, and material densities to obtain the value $W_{\text{air}} = 33.84$ eV with a standard uncertainty of 0.14 eV. In this case, the quantity measured is more closely identified with w_{air} rather than W_{air} , but it does not strictly correspond to either definition.

In constructing Fig. 5.4, an estimate must be made of the mean electron energy associated with the photon beams. For ^{137}Cs γ rays, the value of 200 keV was taken from Davis *et al.* (2003). For ^{60}Co γ rays and the three MV x-ray beams, values of 400 keV, 1000 keV, 1500 keV, and 2800 keV were taken from Anton *et al.* (2013).

Figure 5.4 indicates that W_{air} shows no significant energy dependence for electron energies well above 10 keV. The spread of the data in this energy range is consistent with the stated uncertainties, and the weighted mean value (33.94 eV) is consistent with the presently recommended value of 33.97 eV. However, the uncertainty is increased from 0.05 eV to 0.12 eV. The main reason for this increase is that Boutillon and Perroche-Roux (1987) underestimated the uncertainty of the graphite-to-air stopping-power ratio arising from the uncertainties of the I values for graphite and air.

5.4.2 Effective W_{air} for Use with Monoenergetic Photon Beams

As indicated in Eqs. (3.1) and (3.8), the determination of air kerma requires knowledge of W_{air} . If the electrons set in motion by the photons have energies above about 10 keV, using the constant high-energy value of W_{air} should be adequate. For low-energy photon beams, however, one needs a correction factor, k_W , that takes into account the low-energy W_{air} data of Fig. 5.3 averaged over the initial electron spectrum. The PENELOPE Monte Carlo code (Salvat, 2015a; 2015b) has been used to calculate k_W . This code uses photon-interaction cross sections discussed in Section 6 of this Report and includes a treatment of atomic relaxation subsequent to photoionization (see also Section 5.5). These cross sections, however, are essentially applicable to atoms rather than to the major molecular constituents, N_2 and O_2 , of air. Because free air chambers and graphite cavity chambers require charged-particle equilibrium, only the first generation of electrons set in motion by the photons is required. Thus to estimate the initial electron spectrum, electron transport was disabled in the calculation, as was the creation of any secondary photons. One million photon histories were simulated, each of which generated one or more electrons. For example, Fig. 5.5 shows the electron spectrum generated by 7 keV photons in air.

The number of ion pairs, N_{ip} , for each photon beam, can be calculated assuming either the high-energy constant value for W_{air} (denoted as N_{ip}^c) or a value given by Eq. (5.5) (denoted as N_{ip}^v). Thus, k_W is given by

$$k_W = N_{\text{ip}}^c / N_{\text{ip}}^v, \quad (5.6)$$

and is shown graphically in Fig. 5.6. Note that k_W shows some structure near 3.2 keV because of the change in electron emission associated with the argon K edge. In most cases when k_W is important,

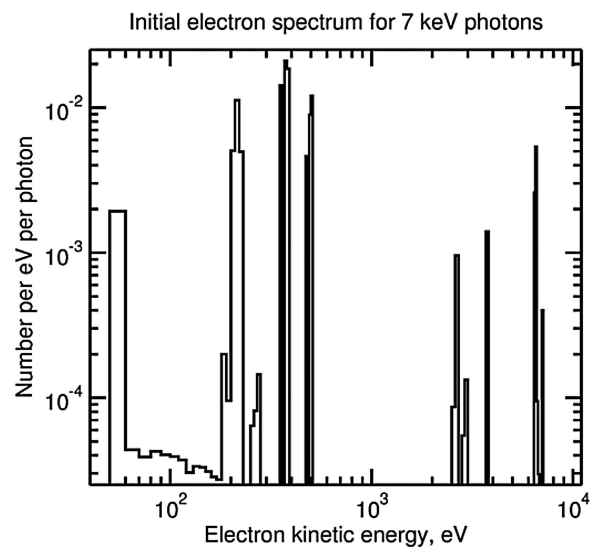


Figure 5.5. Spectrum of first-generation electrons set in motion by 7 keV photons in air. The data were generated by the PENELOPE Monte Carlo code and shows that the spectrum consists mainly of a series of lines corresponding to the emission of photo- and Auger electrons.

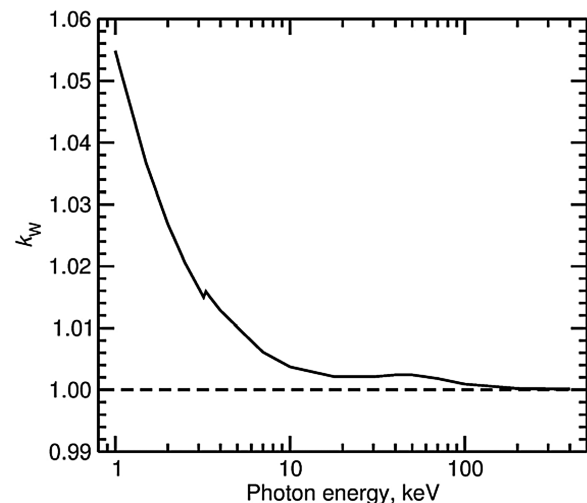


Figure 5.6. Correction factor, k_W , for photon air-kerma determinations due to the energy dependence of W_{air} . Results are shown as a function of the monoenergetic photon energy.

the initial-ionization correction (see Section 5.5) is likely to be important as well.

5.4.3 $W_{\text{air}} s_{\text{g,air}}$ for ^{60}Co γ rays

The air kerma produced by ^{60}Co γ rays is generally established using graphite cavity chambers and requires knowledge of W_{air} and $s_{\text{g,air}}$. It turns out that the product of these quantities has a smaller uncertainty for ^{60}Co γ rays than is obtained from combining the individual values of W_{air} and $s_{\text{g,air}}$. The reason for this is that there have been several high-precision measurements of the absorbed dose to graphite based on graphite calorimetry and the absorbed dose to water based on water calorimetry, from which $W_{\text{air}} s_{\text{g,air}}$ can be obtained by carrying out ionization chamber measurements using a graphite chamber. Burns (2012) has tabulated four such sets of data and reports the best estimate as 33.72(3) eV, consistent with the recommendations of this Report ($W_{\text{air}} = 33.97$ eV, $I_{\text{g}} = 81$ eV, and the use of the graphite grain density when evaluating the stopping power). Note that the relative standard uncertainty of the product is 0.08 %, whereas the combined uncertainty from W_{air} and $s_{\text{g,air}}$ independently is 0.41 %. Thomson and Rogers (2010) have shown that this result is only weakly dependent on the spectrum of the ^{60}Co source and the geometry of the graphite cavity chamber used for the measurements. Thus, this product can be expected to apply to almost any graphite chamber used as a primary standard for air kerma in ^{60}Co γ rays. However, Thomson and Rogers (2010) also show how to adjust the product for slight spectral differences or for cavity chambers with geometries very different from that of the reference chamber used in the measurement, the BIPM cavity standard.

5.4.4 W_{air} for Protons and Heavy Ions

In proton-therapy dosimetry, the appropriate conversion factor for dose calculations is the differential value w_{air} because protons lose only a fraction of their energy in traversing the gas. However, for high-energy protons (>10 MeV), the value of w_{air} approaches the value of W_{air} .

The heavy charged particles considered for use in radiotherapy include He, C, Ne, Si, and Ar ions. The appropriate kinetic energies to obtain a sufficient beam penetration are typically 250 MeV/u for He ions, 200 MeV/u to 400 MeV/u for C ions, 620 MeV/u for Ne ions, 800 MeV/u for Si ions, and 860 MeV/u for Ar ions. For these ions, W_{air} is approximately constant if the energy is above about 20 MeV/u. Several compilations and reviews, such as ICRU Report 31 (ICRU, 1979) and IAEA TECDOC 799

(IAEA, 1995), provide data for values of W_{air} for different charged particles in various gases.

The current W_{air} value for protons originates from the re-evaluation by Medin *et al.* (2000) of a compilation in ICRU Report 59 (ICRU, 1998a). This set of data was updated and statistically analyzed in IAEA TRS-398 (IAEA, 2000a), leading to the recommended current value (34.23 eV). A subsequent analysis by Jones (2006) suggested using only calorimetric data, obtaining for this specific data set, a W_{air} value consistent with that in TRS-398, which was endorsed in ICRU Report 78 (2007). The data set for the determination of W_{air} is based on the expressions commonly used in radiotherapy for the determination of absorbed dose using ionization chambers, which is compared with a calorimetric-determined absorbed dose. As these expressions include a dependence on calibration coefficients, water-to-air stopping-power ratios, and chamber-specific perturbation correction factors, the impact of their changes on W_{air} following the adoption of new key data will be analyzed in Section 7. The new value of W_{air} for protons becomes 34.44 eV with a standard uncertainty of 0.14 eV or 0.4 %.

In the case of carbon ions, the determination of values of W_{air} at energies used in radiotherapy are scarce. Based on the sets of data available, IAEA TRS-398 (IAEA, 2000a) recommended a value of 34.50 eV. Using an analysis identical to that for proton beams, adopting the new key data yields a new value of 34.71 eV with a standard uncertainty of 0.52 eV or 1.5 %.

5.5 Initial-Ionization Correction for Air-Filled Ionization Chambers

It was pointed out in Section 2.11 that the relationship between the charge collected in an air-filled ionization chamber and the air kerma is not as simple as sometimes assumed. The complication can be understood by considering the example of a 3 keV photon interacting in a free-air chamber. The photoelectric effect is the dominant interaction process, and thus the 3 keV photon will set in motion an electron with a kinetic energy of approximately 3 keV. This electron, as it slows down, will create on average about 88 ion pairs. The charge due to these 88 ion pairs will be collected by the ion chamber, but so will the charge due to the first ion pair created by the photoelectric interaction. W_{air} , the average energy to create an ion pair, is defined and measured by considering only the ion pairs created by an electron losing all its energy by slowing down in air. Thus, initial ion pairs should not be included in the determination of

air kerma, which for the example of the 3 keV photon leads to a correction of about 1 %.

An approximate estimation of the correction factor for the initial ion pair, k_{ii} , was included by Büermann *et al.* (2006) in their measurements of the mass energy-absorption coefficient of air for photons with energies from 3 keV to 10 keV. A more detailed analysis, including the effect of Auger-electron emission, was carried out by Takata and Begum (2008),¹⁹ who recognized that there is a fortuitous cancellation of the rather large effects of k_{ii} and k_W and calculated the combined effect of the two corrections. In a later publication extending the work of Büermann *et al.* (2006), Buhr *et al.* (2012) calculated an effective value of the combined effects of k_{ii} and k_W . All of these calculations relied on cross sections and transition probabilities applicable to single atoms rather than to the molecular constituents of air.

A calculation of k_{ii} has been done for this Report using the PENELOPE Monte Carlo model indicated in Section 5.4.2. For such a calculation, the fluorescence yields and atomic relaxation following an inner-shell ionization is pertinent; the PENELOPE code uses the data of Perkins *et al.* (1991). Thus, for the dominant photoeffect process, this calculation also is based on cross-section information for single atoms of C, N, O, and Ar rather than for the molecular constituents of air. The use of this approximation is perhaps somewhat mitigated by the fact that the dominant K-shell photoeffect involved is expected to be largely the same for the atoms and molecules; however, the atomic data assumed for the subsequent Auger cascade are more questionable.

If N_e represents the total number of electrons set in motion, then k_{ii} is given by

$$k_{ii} = N_{ip}^v / (N_e + N_{ip}^v), \quad (5.7)$$

where N_{ip}^v , the total number of ion pairs created, is defined in Section 5.4.2. Although N_{ip}^v depends on knowledge of the energy dependence of W_{air} , for photon energies above 2 keV, the value obtained for k_{ii} would differ by less than 0.1 % if N_{ip}^c were used instead. The variation of k_{ii} with photon energy is shown in Fig. 5.7.

Figure 5.8 and Table 5.7 compare values for the combined correction as calculated by Takata and Begum, by Buhr *et al.*, and by the methods used in this Report. The sudden change in the values calculated by Takata and Begum (2008) near 7 keV is because their fit of W_{air} is forced to the value of 33.97 eV at this energy. Above 7 keV, all the models agree to about 0.1 %.

¹⁹They also suggest that the exposure needs to be corrected for the effect of the initial ion production; however, no such correction is needed in that case.

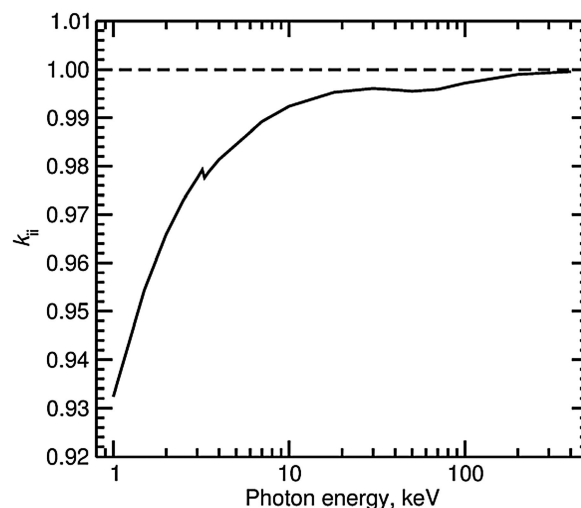


Figure 5.7. The initial-ion correction factor, k_{ii} , as a function of monoenergetic photon energy. The structure near 3.2 keV is due to the change in electron emission associated with the argon K edge.

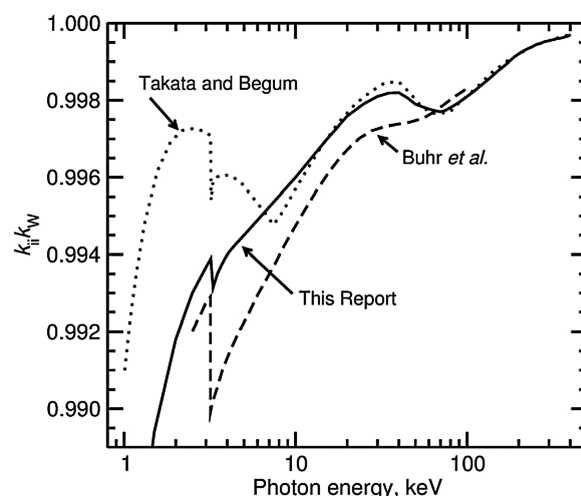


Figure 5.8. Combined values of the correction factors k_{ii} and k_W for low-energy monoenergetic photon beams, comparing the results of Takata and Begum (2008), Buhr *et al.* (2012), and the work of this Report.

In the case of a graphite-walled cavity chamber used to realize air kerma, charged-particle equilibrium is established in graphite; thus, k_{ii} and k_W should be calculated for the initial electron spectrum produced in graphite. Although the values of k_{ii} and k_W in graphite each differ by up to 0.2 % compared to the corresponding values for air, the product differs by less than 0.1 %. As a result, the data in Fig. 5.8 and Table 5.7 could be used both for free-air chambers and approximately for graphite-walled cavity chambers. However, graphite-walled cavity chambers are used primarily in γ -ray beams comprised of

Table 5.7. Values of the product $k_{ii}k_W$ as a function of photon energy, E .

E/keV	Takata and Begum (2008) ^a	Buhr <i>et al.</i> (2012)	This Report
1.0	0.9909		0.9835
1.5	0.9959		0.9894
2.0	0.9971		0.9918
2.5	0.9973	0.9920	0.9930
3.0	0.9972	0.9928	0.9937
3.2		0.9899	0.9939
Ar K edge		0.9930	0.9939
Ar K edge		0.9897	0.9930
3.3	0.9959	0.9902	0.9931
3.5	0.9960	0.9906	0.9935
4	0.9961	0.9913	0.9940
5	0.9958	0.9923	0.9945
6	0.9953	0.9930	0.9950
8	0.9949	0.9940	0.9956
10	0.9957	0.9948	0.9960
15	0.9970	0.9960	0.9970
20	0.9977	0.9967	0.9976
30	0.9984	0.9973	0.9981
40	0.9985	0.9974	0.9982
50	0.9980	0.9975	0.9979
60	0.9977	0.9977	0.9978
80	0.9977	0.9980	0.9978
100	0.9981	0.9983	0.9981
150	0.9988		0.9987
200	0.9992		0.9992
300	0.9995		0.9995
400	0.9997		0.9997
Assumed Ar K edge energy, keV	3.2059	3.1776	3.2063

^aValues read from their figure; the precision is too poor to give in the vicinity of the Ar K edge.

photons with energies mainly above 300 keV for which the correction $k_{ii}k_W$ is essentially unity.

5.6 Humidity Correction for Air-Filled Ionization Chambers

Dosimetry measurements for photons and charged particles often rely on ionization chambers filled with atmospheric air. As a result, the ionization chamber response will depend on the ambient air temperature, pressure, and humidity, and needs to be corrected to reference conditions. These conditions are generally specified as a temperature, T_{ref} , of either 22 °C or 20 °C, a pressure, P_{ref} , of 101.33 kPa, and air free of any water vapor. The correction factor for temperature and pressure is obtained assuming that the ideal gas law applies and is given by

$$k_{PT} = \frac{T/^{\circ}\text{C} + 273.15}{T_{\text{ref}}/^{\circ}\text{C} + 273.15} \frac{P_{\text{ref}}}{P}, \quad (5.8)$$

where T and P denote the ambient conditions of temperature and pressure, respectively. The simple expression given by Eq. (5.8), which essentially corrects the air density at measurement to that for reference conditions, can be insufficient for walled chambers used in air-kerma measurements of low-energy x-ray beams when the density of the ambient air is rather low (La Russa *et al.*, 2007). This deficiency is usually not of concern for primary standards based on the use of free-air chambers for such beams, but should be kept in mind particularly when transferring the measurements to high-altitude regions.

The effect of humidity on ionization chamber response is more difficult to obtain. For a given temperature and pressure, the density of humid air is different from that of dry air. Furthermore, the interaction cross sections, as well as W_{gas} , the average energy to create an ion pair, depend on the water-vapor content. The variation of density with humidity can be calculated using the ideal gas law, while the Bragg additivity rule can be used to compute the photon mass energy-absorption coefficients or the charged-particle stopping powers for humid air. However, the variation of W_{air} with humidity cannot be reliably calculated and must be measured. Using a free-air chamber and 50 kV x rays, Niatel (1969) measured the change in W_{air} as a function of the relative humidity, and the result is shown in Fig. 5.9. The curve labeled regular mixture was based on the assumption that humid air is a simple mixture of dry air and water vapor, *i.e.*, in the absence of the so-called Jesse effect, which describes the complex interactions of metastable, resonance, and excited states in gas mixtures that lead to changes in ionization yield.

Assuming the same variation applies to all types of beams, it is possible to derive how the response of a free-air chamber or a cavity chamber depends on the relative humidity. For a free-air chamber, the humidity correction factor, k_h , is given by

$$k_h = \frac{W_{\text{humid-air}}}{W_{\text{dry-air}}} \frac{(\bar{\mu}_{\text{en}})_{\text{dry-air}}}{(\bar{\mu}_{\text{en}})_{\text{humid-air}}}, \quad (5.9)$$

where the linear energy-absorption coefficient, $\bar{\mu}_{\text{en}}$, is averaged over the photon spectrum. For a cavity chamber, the result is

$$k_h = \frac{W_{\text{humid-air}}}{W_{\text{dry-air}}} \frac{(\bar{S})_{\text{dry-air}}}{(\bar{S})_{\text{humid-air}}}, \quad (5.10)$$

where $\bar{S}$ is the spectrum-averaged linear stopping power. Niatel (1975) showed that Eq. (5.10) gave results that were in good agreement with measurements obtained using cavity chambers irradiated by

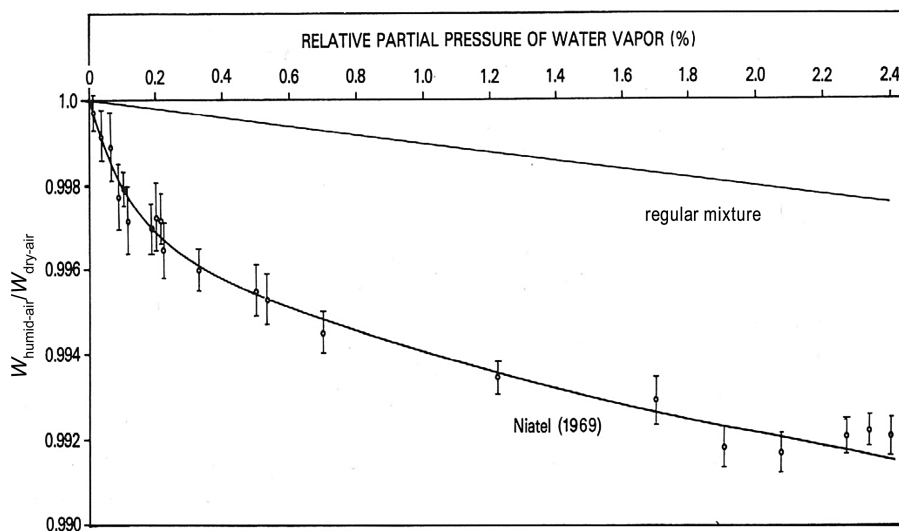


Figure 5.9. Data obtained by Niatel (1969) using a free-air chamber and 50 kV x rays for the variation of the mean energy to create an ion pair in air as a function of water-vapor content. [Adapted from Niatel (1969).]

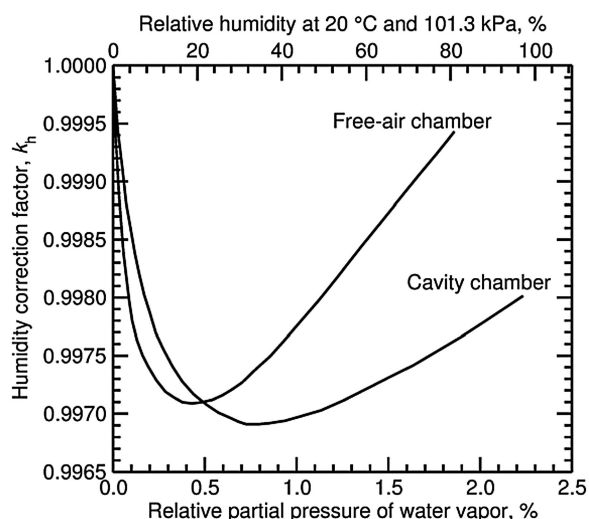


Figure 5.10. The humidity-correction factor, k_h , as a function of the water-vapor content in air for free-air chambers and cavity chambers. The top scale can be used if the air temperature is close to 20 °C, while the bottom scale should be used otherwise. [Adapted from ICRU Report 31 (ICRU, 1979).]

^{60}Co and ^{137}Cs γ rays. Values for k_h are shown graphically in Fig. 5.10 for a free-air chamber irradiated with 50 kV x rays and a cavity chamber irradiated with ^{60}Co γ rays. These data have been extracted from a similar figure published in ICRU Report 31 (ICRU, 1979).

Because k_h is never more than about 0.3 % different from unity, the relative humidity need not be measured with great accuracy. It should be noted that the top scale in Figure 5.10 can be used only if the air temperature is about 20 °C. For the same value of the relative humidity, the amount of water vapor in the air varies significantly with the

temperature. For example, for a relative humidity of 50 % at a temperature of 20 °C, the relative partial pressure of the water vapor is about 1.2 %, while at 30 °C, it is 2.2 %. This would lead to a change of about 0.2 % in k_h for a free-air chamber.

Rogers and Ross (1988) showed that the ratio of linear stopping powers in Eq. (5.10) is very insensitive to the electron energy. Thus, the cavity-chamber curve in Fig. 5.10 can be used for any photon- or electron-beam qualities encountered in radiation therapy. In order to clarify the situation for heavy charged particles, it would be helpful to have data similar to that shown in Fig. 5.10., but obtained with a cavity chamber irradiated in a proton or heavy-ion beam.

5.7 Heat Defect

The heat defect, h , is defined by Eq. (3.17) and is the most challenging correction to evaluate for absorbed-dose calorimetry. A heat defect will arise from any process that leads to a difference between the energy imparted by the radiation field and the energy appearing as heat. When less energy appears as heat than was imparted by the ionizing radiation, endothermic processes are present and the heat defect is positive. Conversely, if more heat is generated than was imparted, exothermic processes are present and the heat defect is negative.

In liquids, such as water, the most likely source of a heat defect will be radiation-induced chemical reactions (Ross and Klassen, 1996). In plastics, cross-linking, *i.e.*, the formation of double bonds and hydrogen evolution, can contribute (Säbel *et al.*, 1974). In crystalline solids, ionizing radiation can

lead to lattice displacements that contribute to a heat defect (Panta *et al.*, 1997). More generally, radiation-induced optical emission could contribute in a transparent medium as could the generation of acoustic energy. The heat defect can be several percent of the deposited energy. Ideally, one would choose a calorimetric material for which the heat defect is close to zero and has a well-established value and uncertainty.

Although some effort has been devoted to constructing tissue-equivalent calorimeters based on plastic materials (see, *e.g.*, McDonald and Domen, 1986), at the present time all calorimetric primary standards for absorbed dose in water are constructed of either graphite or water.

5.7.1 Heat Defect of Graphite

Most graphite calorimeters follow the design principles laid out by Domen and Lamperti (1974). The radiation-induced temperature change is measured in an absorbing element that is surrounded by a series of vacuum gaps. Because air has been removed from the vicinity of the absorbing element, it is unlikely that chemical reactions could contribute to a heat defect in the graphite. In a study using high-energy and high-dose-rate electron beams, Domen and Lamperti (1976) discuss the possibility of a graphite heat defect. Their calorimeter had been opened to the atmosphere twice during an extensive series of measurements. They report that the mean difference between the measurements before and after exposure to air was $0.2\% \pm 0.2\%$ and conclude that there was no evidence of a heat defect due to radiation-induced reactions of graphite with oxygen.

McEwen *et al.* (1998) built a graphite calorimeter for electron beams in which the gaps between the absorbing element and the body of the calorimeter were not evacuated but left filled with ambient air. They do not estimate the effect of radiation-induced reactions between the oxygen and graphite. Because any reaction can only occur near the surface of the absorber, it is unlikely that any thermal effect would be large, but it might not be negligible. The oxidation of graphite during irradiation has been studied in the context of nuclear reactors (Davidge *et al.*, 1959).

Ionizing radiation can produce lattice displacements in crystalline materials. The energy required to produce a displacement is typically about 25 eV, so from a kinematic point of view electrons with energies as low as 150 keV can produce lattice defects. However, the cross section for defect production is much greater for high-LET radiation, such as neutrons or protons, than for electrons (see, *e.g.*, Akkerman *et al.*, 2001). Neutron-induced lattice displacements in graphite used in nuclear reactors can

lead to stored energy, referred to as the Wigner effect. Using available data on the stored energy as a function of the neutron fluence, Ramanathan (2008) estimated a heat defect of 0.14 % for a graphite calorimeter used to measure neutron absorbed dose. Panta *et al.* (1997) calculated the cross section required to produce a lattice displacement in graphite and estimated the heat defect for a 10 MeV electron beam to be about 0.1 %. We note that because the non-ionizing energy-loss stopping power is more than an order of magnitude greater for neutrons than for electrons (Akkerman *et al.*, 2001), these two results seem to be inconsistent.

There have been several efforts to measure the graphite heat defect, and the results have been summarized by Panta *et al.* (1997). Except for the results of Schulz *et al.* (1990), the results are consistent with a heat defect of zero, although the uncertainties range from 0.6 % to 2 %. Schulz *et al.* compared the thermal response of graphite and aluminum using a low-energy proton beam and, assuming the heat defect of aluminum is zero, estimated a heat defect for graphite of $(0.43 \pm 0.34)\%$. If considered to be statistically significant, this result might reflect the fact that the production of lattice displacements is much greater for protons than for electrons. It is unclear if one can assume that no energy is trapped in lattice displacements in aluminum.

It is now quite common to assume that the heat defect of a graphite calorimeter used for photon or electron measurements is zero (see, *e.g.*, Seuntjens and Duane, 2009). Most experimental data are consistent with this assumption, although the estimated uncertainty might be as large as 0.5 %. If the calorimeter is operated in vacuum, then the only likely heat defect mechanism arises from lattice displacements. For low-LET radiation, estimates indicate that the energy stored in lattice defects is less than about 0.1 % of the absorbed energy. However, it might be significantly larger for high-LET radiation, and more study is needed.

5.7.2 Heat Defect of Water

Ross *et al.* (1989) considered three mechanisms that might contribute to a heat defect in water: radiation-induced optical emission; acoustic energy generated by energetic charged particles; and radiation-induced chemical reactions. They concluded that chemical reactions were likely to be the only significant source of a heat defect in liquid water.

Because of its importance for nuclear reactors, the radiation chemistry of water has been extensively studied, and detailed computer models are available to predict the reaction products as a function of time. The first application of these models to

water calorimetry was by Fletcher (1982) who calculated the heat defect for several aqueous systems in response to the pioneering water-calorimeter measurements by Domen (1980). Unfortunately, applying Fletcher's calculated values of the heat defect to Domen's measurements led to discrepancies of several percent compared with conventional determinations of the absorbed dose in water (Ross and Klassen, 1996).

Because Domen's calorimeter was a large, open design, it was difficult to control the presence of impurities or dissolved gases. To better test the model predictions, Ross *et al.* (1984) constructed a small, sealed calorimeter that permitted the water to be saturated with different gas mixtures. In a series of papers, Klassen and Ross (1991; 1997; 2002) reported detailed model calculations for several aqueous systems and compared their relative response using the sealed calorimeter. In general, they found that model predictions and measurements agreed to better than 0.5 %. They also noted that trace impurities can have a major impact on the measured response and is likely to explain the discrepancies observed by Domen. Klassen and Ross identified three systems that were well suited for water calorimetry. These are: pure water, with no dissolved gases, or saturated with an inert gas; water saturated with hydrogen gas; and water saturated with a gas mixture composed of 43 % hydrogen molecules and 57 % oxygen molecules.

Because of the recognition of the role that impurities might play in generating an unknown heat defect, all water calorimeters are now built with a sealed vessel that is carefully cleaned and contains a well-defined aqueous system. If the aqueous system consists of either of the first two identified by Ross and Klassen, the reaction products reach a steady state after a small accumulated dose, and thus the heat defect is zero. On the other hand, the third system has a finite heat defect of about -2.3 %, which is largely independent of the accumulated dose. Empirically, the H_2/O_2 system was found to be least sensitive to residual impurities. Furthermore, carrying out measurements with at least two systems gives added confidence in the results if they are consistent with the predicted heat defect.

The Physikalisch-Technische Bundesanstalt (PTB) (Krauss, 2006; Krauss and Kramer, 2003) studied the predicted and measured behavior of an aqueous system in which the water had been saturated with a gas mixture consisting of mainly hydrogen, but with a small admixture of oxygen. In this case, the model predicts a sharp exothermic spike at a certain value of the accumulated dose after which the heat defect is predicted to be zero. Krauss suggested that if the calorimeter showed this response, one could

have added confidence that the water was free of impurities and had zero heat defect. Surprisingly, this system showed a response that differed by about 0.35 % from that expected from the three systems defined by Klassen and Ross. The reason for this difference remains unresolved.

Ross *et al.* (2007) have constructed and tested vessels that are permanently sealed and for which glass is the only material in contact with the aqueous system. Vessels containing water with no dissolved gases and water saturated with hydrogen gas have been constructed. The response of these vessels has been observed to be stable for more than a decade and consistent with results obtained with conventional vessels for which the solution can be changed.

Klassen and Ross (2002) concluded that pure water, water saturated with an inert gas, or water saturated with hydrogen gas are the best choices for water calorimetry. These systems all reach a steady state after a modest accumulated dose, and this prediction is independent of the details of the model for the radiation chemistry. For photon and electron beams typical of those used for radiation therapy, the uncertainty assigned to the heat defect of these systems is 0.15 % (Picard *et al.*, 2010). The H_2/O_2 system is valuable as a check, but its heat defect is model-dependent, making it more difficult to estimate the uncertainty.

The heat defect of an aqueous system will depend on the LET of the radiation field because the initial yield, G , of the reactive species that escape from the vicinity of the particle track depends on the LET, as shown in Fig. 5.11. These data can be used with the

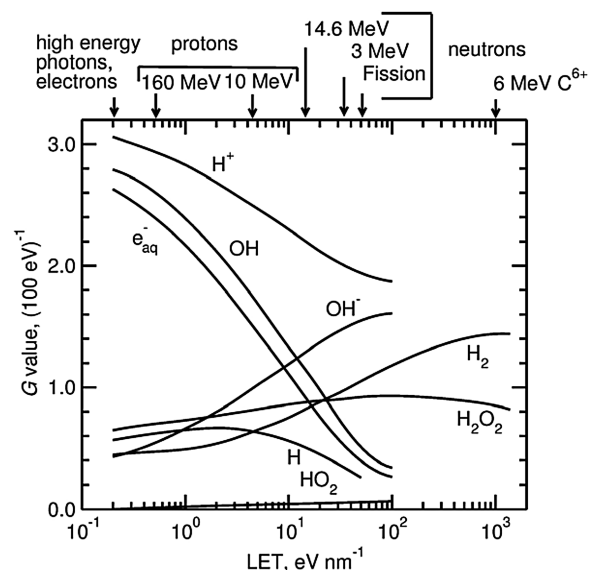


Figure 5.11. Values of G for the species emerging from the vicinity of the interaction sites produced by energetic particles passing through liquid water. The labels at the top of the graph identify particles that give rise to the LET values indicated on the abscissa. [Figure from Ross and Klassen (1996).]

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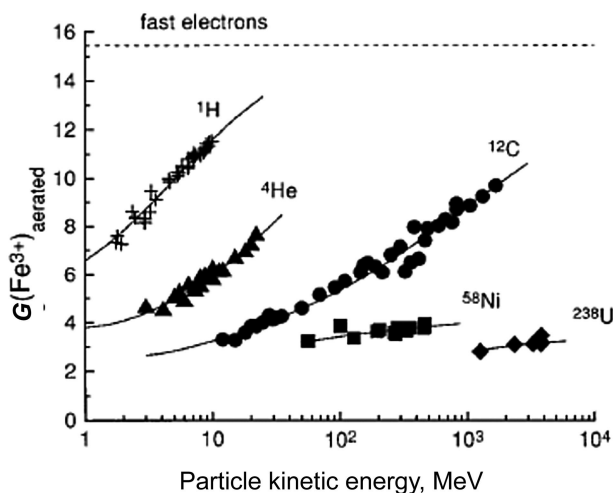


Figure 5.12. Data showing how $G(\text{Fe}^{3+})$ (expressed in ions/100 eV) depends on the type of charged particle and its energy. $G(\text{Fe}^{3+})$ decreases as the LET of the charged particle increases. [Figure from LaVerne and Schuler (1996).]

established reaction set to calculate the heat defect for an arbitrary value of LET. Klassen and Ross (1988) calculated the heat defect for two aqueous systems and for low, medium, and high values of LET. These results can serve as a guide if water calorimetry were to be applied to radiation beams with high LET.

5.8 Radiation Chemical Yield for Fricke Dosimetry

The value of $G(\text{Fe}^{3+})$ depends on the LET of the radiation beam, as shown in Fig. 5.12, which is taken from the work of LaVerne and Schuler (1996). Because of the large variations of $G(\text{Fe}^{3+})$ with particle type and energy, Fricke dosimetry is not often used for the dosimetry of heavy charged particles. On the other hand, for the photons and electrons most often used for radiation therapy, $G(\text{Fe}^{3+})$ is approximately constant, and Fricke dosimetry has found a number of applications.

Although there have been efforts to calculate $G(\text{Fe}^{3+})$ from first principles, the uncertainties are too large to be useful for radiation dosimetry. Instead, $G(\text{Fe}^{3+})$ must be established by measuring the energy deposited in the Fricke solution by the ionizing radiation. One approach is to use calorimetry. Ross *et al.* (1989) used a small, stirred-water calorimeter that was alternatively filled with water or Fricke solution to establish a value for $G(\text{Fe}^{3+})$ for 20 MV x-ray beams. Alternatively, a conventional graphite or water calorimeter can be used to establish the absorbed dose in water at a reference point, and the Fricke solution can be calibrated against this value (Cottens *et al.*, 1980; Klassen *et al.*, 1999).

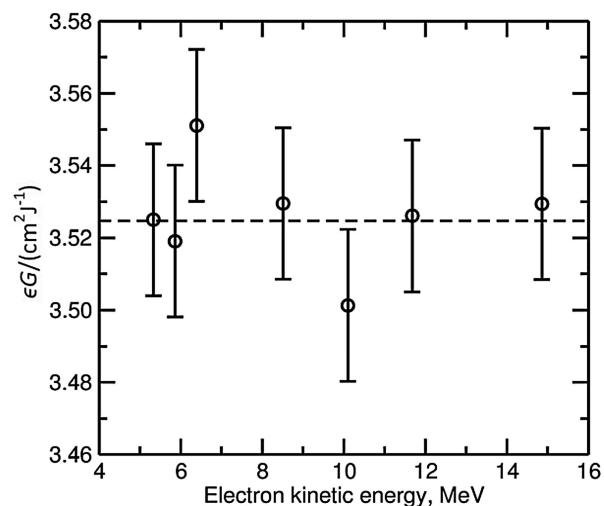


Figure 5.13. Values of $\epsilon G(\text{Fe}^{3+})$, as measured using total absorption of electron beams at the Swiss standards laboratory, METAS. Relative results were reported by Stucki and Vörös (2007); Stucki completed the data analysis to obtain absolute values.

For electron beams, a different approach has been used in which the beam is allowed to completely dissipate its energy in a well-defined volume of Fricke solution (Feist, 1982; Stucki and Vörös, 2007). For low- and medium-energy x-ray beams, the absorbed dose in the Fricke solution is established using ionization chambers (Fregene, 1967).

Shalek and Smith (1969) summarized the available data for $G(\text{Fe}^{3+})$ for both photon and electron beams prior to 1969. As the photon or electron energy decreases, $G(\text{Fe}^{3+})$ becomes smaller because of the increasing LET of the electrons set in motion in the Fricke solution. However, for photon or electron beams typically used for external-beam radiation therapy, the data would suggest that $G(\text{Fe}^{3+})$ does not depend strongly on beam energy. Figure 5.13 shows data obtained for electron beams using the total absorption method by the Swiss standards laboratory, METAS. The experimental details and relative values of $G(\text{Fe}^{3+})$ as a function of electron energy were published by Stucki and Vörös (2007). Stucki²⁰ completed the data analysis to obtain absolute values of $\epsilon G(\text{Fe}^{3+})$. Within the uncertainties, $G(\text{Fe}^{3+})$ showed no dependence on the incident energy of the electron beam. Stucki²⁰ has reported that the weighted mean of the data in Fig. 5.13 yields a value for $\epsilon G(\text{Fe}^{3+})$ of $3.525 \text{ cm}^2 \text{ J}^{-1}$ with a standard uncertainty of $0.021 \text{ cm}^2 \text{ J}^{-1}$. This result is consistent with ICRU Report 35 (ICRU, 1984b), which recommended a value for $\epsilon G(\text{Fe}^{3+})$ of $3.52 \text{ cm}^2 \text{ J}^{-1}$ for all electron energies from 1 MeV to 30 MeV. No uncertainty estimate was provided.

²⁰Personal communication from Dr. Gerhard Stucki, January 2015.

Table 5.8. Values of $G(\text{Fe}^{3+})$ for a range of photon and electron beam qualities. Most entries are from the report by McEwen *et al.* (2014b). The results for beam qualities indicated by an asterisk (*) are from Klassen *et al.* (1999). The value for ^{192}Ir was obtained by interpolation. The value for high-energy electrons is expected to apply to electron beams of 5 MeV and greater. The last entry is from Stucki²⁰ and is the weighted mean for electron energies from 5.33 MeV to 14.86 MeV.

Beam quality	Effective photon energy (MeV)	Effective electron energy (keV)	$G(\text{Fe}^{3+})$ ($\mu\text{mol J}^{-1}$)	u_c/G (%)
250 kV x-ray beams (HVL = 2.03 mm Cu)	0.109	25	1.586	0.8
250 kV x-ray beams (HVL = 2.68 mm Cu)	0.127	33	1.570	0.6
^{192}Ir γ rays	0.38	120	1.589	0.6
^{60}Co γ rays*	1.25	420	1.613	0.3
^{60}Co γ rays	1.25	420	1.610	0.5
20 MV x-ray beams*	6.7	2300	1.625	0.4
30 MV x-ray beams*	7.5	3200	1.621	0.4
18 MeV electrons ($R_{50} = 7.0$ cm)	—	6800	1.622	0.5
22 MeV electrons ($R_{50} = 8.8$ cm)	—	8700	1.618	0.5
Electrons, 5.33 MeV to 14.86 MeV	—	—	1.621	0.6

As pointed out in Section 3.2.2, $\Delta\epsilon$ is the difference of the molar absorption coefficients for the ferric and ferrous ions. At 303 nm, the wavelength most often used for Fricke dosimetry, $\epsilon(\text{Fe}^{2+})$ is less than 0.05 % of $\epsilon(\text{Fe}^{3+})$ (see, *e.g.*, Fricke and Hart, 1966) and is often ignored; thus the term $\Delta\epsilon$ becomes simply $\epsilon(\text{Fe}^{3+})$. Because it is then the product $\epsilon(\text{Fe}^{3+}) G(\text{Fe}^{3+})$ that appears in Eq. (3.14), it is common to give the value of $\epsilon G(\text{Fe}^{3+})$ rather than $G(\text{Fe}^{3+})$ itself. For a calibrated and well-behaved spectrophotometer, a value of $\epsilon(\text{Fe}^{3+})$ of $2174(2) \text{ M}^{-1}\text{cm}^{-1}$ (Cottens *et al.*, 1980) is widely used. Note that these values for $\epsilon(\text{Fe}^{3+})$ and $G(\text{Fe}^{3+})$ assume that both the irradiation and read-out temperatures are 25 °C, and corrections must be applied if either temperature is different (Shortt, 1989).

From a series of measurements using a liquid ionization chamber, Mattsson *et al.* (1990) suggested that the same value for $G(\text{Fe}^{3+})$ [or, equivalently, $\epsilon G(\text{Fe}^{3+})$] could be used for all photon beams with energies above that of ^{60}Co γ rays, as well as for all electron beams used for radiation therapy. From a series of measurements based on water calorimetry, Klassen *et al.* (1999) showed that $G(\text{Fe}^{3+})$ increases by about 0.7 % as the photon beam energy increases from ^{60}Co to 20 MV x rays. They showed that, within the uncertainties, this result was consistent with earlier work and also with theoretical expectations.

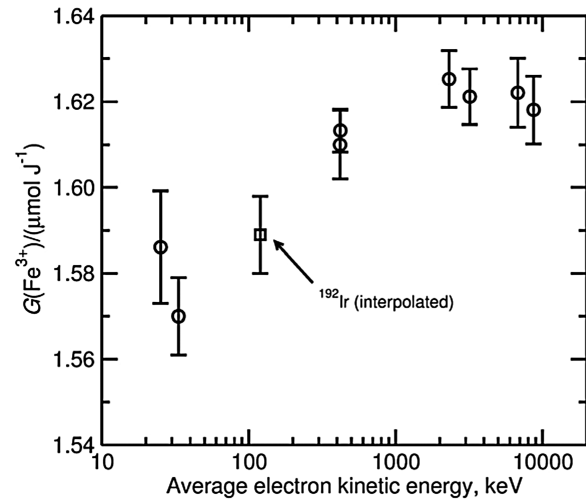


Figure 5.14. Values of $G(\text{Fe}^{3+})$ from Table 5.8, plotted as a function of the average electron energy in the Fricke solution. The value for ^{192}Ir was obtained by interpolating between the values for 250 kV x-rays (HVL = 2.68 mm Cu) and for ^{60}Co γ rays.

Cojocaru *et al.* (2010) used water calorimetry to measure $G(\text{Fe}^{3+})$ for an 18 MeV electron beam and obtained a result consistent with the work of Klassen *et al.* for 20 MV and 30 MV x-ray beams.

Shalek and Smith (1969) summarized measured data for $G(\text{Fe}^{3+})$ for photons as low as 10 keV and electrons as low as 5 keV. They show that $G(\text{Fe}^{3+})$ decreases by more than 20 % from its limiting value at high energies. It is not practical to re-analyze these important historical data using updated estimates for the absorbed dose in the Fricke solution. In any case, Fricke dosimetry is not routinely used for very low-energy beams. However, there is interest in using Fricke dosimetry for ^{192}Ir brachytherapy dosimetry (Austerlitz *et al.*, 2007; 2008), and this requires knowledge of $G(\text{Fe}^{3+})$ for a photon energy of about 380 keV. El Gamal *et al.* (2013) measured $G(\text{Fe}^{3+})$ for 250 kV x-ray beams (effective energy of 126 keV) and interpolated between this and ^{60}Co to obtain $G(\text{Fe}^{3+})$ for ^{192}Ir .

A report by McEwen *et al.* (2014b) reviewed results obtained at the National Research Council Canada and compared them with historical data. These values for $G(\text{Fe}^{3+})$ cover the range of photon and electron beams of relevance to radiation therapy and are listed in Table 5.8.

The relationship between $G(\text{Fe}^{3+})$ and the mean electron energy in the Fricke solution is shown in Fig. 5.14. $G(\text{Fe}^{3+})$ can be considered constant if the electron energy is greater than about 2 MeV. It decreases slowly with decreasing energy, and is about 3 % below its asymptotic value for 30 keV electrons. A smooth curve through these data is likely to yield a value of $G(\text{Fe}^{3+})$ accurate to about 0.5 % for any beam quality from 250 kV x-ray beams to 30 MeV electrons.

6. Photon Interaction Coefficients

As indicated in Section 2.2, the mass attenuation coefficient is given in terms of the sum of cross sections for photon interactions in the material, *i.e.*, $(\mu/\rho) = (N_A/M_A) \sum_J \sigma_J$. However, for this report, the quantity of key importance for the dosimetry of ionizing radiation, particularly for primary standards, is the mass energy-transfer coefficient, $(\mu_{tr}/\rho) = (N_A/M_A) \sum_J f_J \sigma_J$, where f_J is the fraction of energy transferred to charged particles in the interaction of type J , and the closely related mass energy-absorption coefficient, $(\mu_{en}/\rho) = (\mu_{tr}/\rho)(1 - g)$, where g is the fraction of energy transferred to charged particles that is subsequently lost on average in radiative processes. Traditionally, the component cross sections, σ_J , considered have been those for the interaction of the photon with atomic electrons and the Coulomb fields of the nucleus and atomic electrons, namely photoelectric absorption, coherent (Rayleigh) scattering²¹, incoherent (mostly Compton) scattering, and pair production in the fields of the screened nucleus (pair) and atomic electrons (triplet). Interactions with nucleons (photonuclear²² cross sections) are usually not included in critically evaluated compilations. It should be noted that f_J is considered to be zero for coherent scattering, which removes consideration of coherent scattering in the calculation of μ_{tr} and μ_{en} .

A recent review by Pratt (2014) highlights our understanding of atomic photoeffect, Compton-scattering, and Rayleigh-scattering cross sections. He points out that, although much progress has been made in the theory and measurement of these photon-interaction cross sections, both theory and measurement still lack the accuracy to resolve discrepancies at levels of from 1 % to about 10 %. Compilations of such cross-section data are now indispensable in applications of radiation science,

and figure importantly in the realization of measurement standards for ionizing radiation (see Section 3). A useful review of the history of the information used in such compilations can be found in the work of Hubbell (1969; 1999; 2006). Over more than six decades of such development, the results of theory and numerical computation have shown reasonable agreement with measured data and have largely replaced reliance on measurement as the basis for the standard reference data on photon-interaction cross sections. The work of Hubbell *et al.* (1975; 1979), culminating in the online database XCOM (Berger and Hubbell, 1987; Berger *et al.*, 2010), has become a standard source of atomic cross sections for photon interactions.

This Report is concerned with photon energies above about 1 keV and only with air, graphite, and liquid water; however, virtually all materials are involved in application areas of photon transport and dosimetry. Because our knowledge of the cross sections is still not complete, it appears difficult to draw unambiguous conclusions on which data to recommend. Rather, some relevant comparisons are made, and the focus will be on the uncertainties of the quantities pertinent to this Report.

6.1 Photoelectric Absorption

In the photoelectric effect, or photoeffect, a photon of energy E is absorbed by the electrons of the target atom, which is assumed to be in its ground state. The elementary theory of the atomic photoeffect has been described by Pratt *et al.* (1973) and by Scofield (1973). It is based on the independent-particle approximation for the atomic wave functions of the isolated atom, which are calculated from the Dirac–Hartree–Fock–Slater (DHFS) potential. The interaction between the atom and the electromagnetic field is treated as a perturbation to first order. This theory describes processes in which a single photon is absorbed by one of the atomic electrons, and leads to results equivalent to those obtained from the one-active-electron approximation. That is, in calculations of the photoeffect, one can consider that the active electron, initially in a

²¹The related process of Delbrück scattering is ignored, as best estimates indicate that it is a negligible contribution (Hubbell and Bergstrom, 2004).

²²As pointed out by Hubbell (1982), photonuclear cross sections can contribute as much as from 5 % to 10 % to the total cross section at energies between about 10 MeV and 30 MeV, depending on the target nuclide. See IAEA (2000b) and Plujko *et al.* (2011) for some recent compilations of photonuclear-absorption cross sections.

bound orbital of the i th shell is raised to an excited orbital, which can be either bound (excitation) or free (ionization). Because a sum over all orbitals of a closed shell renders the calculation of angular integrals trivial, calculations usually provide the total cross section $\sigma_{\text{pe},i}(E)$ for absorption by the electrons in each shell, i , that is, for transitions that generate a vacancy in that shell.

Dirac–Hartree–Fock–Slater orbitals are expected to be less accurate than those resulting from a multi-configuration Dirac–Fock (MCDF) calculation, and one might expect that a theory based on MCDF atomic wave functions would yield improved cross sections for the photoeffect. Unfortunately, such a calculation does not seem to be available yet. A method for estimating how the use of more accurate orbitals would affect the cross section for the photoeffect is provided by Pratt’s renormalization screening approximation (Pratt, 1960a), which is discussed by Schmickley and Pratt (1967), Pratt and Tseng (1972), and Pratt *et al.* (1973). The key idea underlying this approximation is that, for photon energies a few keV above the ionization threshold, the dominant contributions to the transition-matrix elements arise from radial distances, r , that are substantially larger than the radius of the nucleus but much smaller than the average radial distance of electrons in bound orbitals. That is, at radii important for the photoeffect, the only effect of screening is a change in the normalization of bound states. Consequently, it is expected that the photoelectric cross section $\sigma_{\text{pe},i}(E)$ will be improved by multiplying it by a factor equal to the ratio of electron densities in MCDF and DHFS orbitals of shell i near $r = 0$ (in the high-energy limit). On the other hand, although the negative DHFS energy eigenvalues are quite close to the experimental shell ionization energies, the differences induce appreciable shifts of the absorption edges. To correct the calculated cross sections for these discrepancies, the energy scale is shifted (or, in the case of XCOM, interpolated or extrapolated) to match experimental edge energies.

The first systematic calculation of cross sections for the photoeffect with the DHFS potential was performed by Scofield (1973) using the Hartree–Slater model. His tables were for the energy range from 1 keV to 1.5 MeV, and included the contributions of individual shells. Scofield also provided renormalization factors obtained from restricted relativistic Hartree–Fock calculations for the elements $1 \leq Z \leq 54$.

For higher energies, the total cross section was extrapolated using the energy dependence given by Hubbell *et al.* (1980), which in turn was obtained by combining Scofield data in the energy interval from 1 MeV to 1.5 MeV with an empirical high-energy

expression fitted to a combination of theoretical calculations and experimental data. The shell cross sections for energies from 1 MeV to 100 GeV were obtained from Scofield’s results at 1 MeV by assuming that in this energy range the relative contribution of each shell remains constant with energy. The XCOM computer code (Berger *et al.*, 2010; Berger and Hubbell, 1987) contains these cross sections for photoelectric absorption, interpolated or extrapolated to edge energies given by Beardon and Burr (1967) for photon energies above 1 keV. An extended database of photoelectric cross sections, mostly calculated with Scofield’s computer code, is included in the Evaluated Photon Data Library (EPDL) (Cullen *et al.*, 1997). The Evaluated Photon Data Library covers a wide photon energy range, which extends from the edge of the outer shell up to 100 GeV. In early compilations, Hubbell *et al.* (1980) used Scofield shell cross sections renormalized according to the normalization screening approximation. However, from comparison with experimental data, Saloman and Hubbell (1986) concluded that the Scofield values without renormalization were in better agreement with the experimental data up to a photon energy of 1 keV than were the renormalized cross sections. Although they could draw no such conclusions at energies above 1 keV, they relied on their low-energy finding, and—consequently—subsequent compilations, including those in Saloman *et al.* (1988), the EPDL database, and the XCOM code, adopted the unrenormalized cross-section values calculated by Scofield. This issue has re-emerged because recent measurements of mass-energy absorption coefficients in air by Buermann *et al.* (2006) and corrected by Buhr *et al.* (2012) clearly favor the renormalized cross sections. However, results of a measurement (Kato *et al.*, 2010) of the mass attenuation coefficient in air at energies for which the photoeffect is dominant do not support this conclusion, as will be discussed later.

Sabbatucci and Salvat (2016) have generated a database of photoelectric cross sections with the DHFS potential for the shells of the ground state configurations of the elements $1 \leq Z \leq 99$. Although the theory employed was the same as in Scofield calculations, the computer code implements more robust numerical algorithms and an adaptive energy grid that allows a more accurate description of the variation of the cross section near absorption edges. The very close agreement of the basic numerical results from the Sabbatucci–Salvat and the Scofield calculations provide remarkable confirmation of independent calculations performed some 40 years apart. Salvat shifted his DHFS cross-section results to subshell ionization energies from Carlson (1975) and applied his MCDF/DHFS renormalization

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factors²³ to obtain renormalized DHFS photoelectric-absorption cross-section tables for $1 \leq Z \leq 99$. A complete database, covering the energy range up to 1 GeV, has been generated by extending the shell-corrected DHFS cross sections to high photon energies by means of the extrapolation expression given by Pratt (1960b). This database is included in the distribution package of the PENELOPE Monte Carlo system (Salvat, 2015b).

The earlier work of Storm and Israel (1970) was done with the Brysk–Zerby computer code (Brysk and Zerby, 1968) and bound-state wave functions of Liberman *et al.* (1965) with a modified exchange term and the substitution of experimental for calculated binding energies. They extended their results for incident photon energies above 200 keV using the results from Rakavy and Ron (1967) and from Schmickley and Pratt (1967) who included electron screening through renormalization of the cross section. Their tables, for $1 \text{ keV} \leq E \leq 100 \text{ MeV}$ and $1 \leq Z \leq 100$, include also the cross sections for incoherent [using the Waller–Hartree (1929) approach, see Section 6.3] and coherent [using atomic-form-factor evaluations, see Section 6.2] scattering, and for pair and triplet production [using data from Hubbell and Berger (1966)]. They give “approximate accuracies” of 3 % for incoherent scattering, 3 % for coherent scattering, 5 % each for pair and triplet production, and for the photoeffect cross section: 10 % for $1 \text{ keV} \leq E < 6 \text{ keV}$, 3 % for $6 \text{ keV} \leq E \leq 200 \text{ keV}$, and 10 % for $200 \text{ keV} < E \leq 100 \text{ MeV}$.

Chantler (1993; 1995; 2000) adopted the work of Cromer, Mann, and Liberman (Cromer, 1965; Cromer and Liberman, 1970; 1981; Cromer and Mann, 1968) to generate data on form factors and the resultant photoelectric-absorption cross section; because his interest was mainly in applications of x-ray diffraction, interferometry, crystallography, *etc.*, the work, for $1 \leq Z \leq 92$, covered photon energies from 1 eV up to 433 keV. Chantler used relativistic, self-consistent-field wavefunctions from the Kohn–Sham potential (Kohn and Sham, 1965) with the Brysk–Zerby code. Chantler (1993; 1995) states that the experimental binding energies of Bearden and Burr (1967) were incorporated into the basic calculation of the shell-by-shell cross sections. His database FFAST (Chantler *et al.*, 2005) is available online (<http://www.nist.gov/pml/data/ffast/index.cfm>). The relative agreement between Chantler’s and Scofield’s total photoeffect cross sections is mostly within 5 %, but much larger relative differences do occur for $Z < 30$ near the upper limit of Chantler’s energy coverage

(up to 50 %), and near some absorption edges (> 50 %). Some of these differences can be attributed to differences in functional shapes and in energy resolution of the two databases, but are typically larger than the uncertainties estimated by Chantler (1995) for his results. Chantler’s FFAST database includes the sum of the cross sections for coherent scattering (using his atomic-form-factor evaluations) and for incoherent scattering [using the Waller–Hartree (1929) approach in conjunction with his incoherent-scattering functions]. Chantler (2000) states that the sum of his scattering cross-section values is accurate “to the order of 5 %.”

The approaches of Scofield, of Storm and Israel, and of Chantler are based on an independent-particle approximation for free atoms, so they inherently ignore molecular and solid-state effects leading to well-known extended x-ray absorption fine structure²⁴ (EXAFS); see, *e.g.*, Lee *et al.* (1981). Although there are similarities in the bases for these calculations, the results vary according to details of the methods used; this is illustrated in Table 6.1 for the low- Z materials of primary interest in this Report.

A few remarks can be made regarding Table 6.1. The renormalization factors for Scofield’s (and Salvat’s) results are close to from 0.96 to 0.97 at all the energies shown. Up to 200 keV, the maximum/minimum ratios vary between 1.03 and 1.09, and the relative standard deviation among the results [excluding renormalized Scofield, to which Salvat (2015b) is nearly identical] vary between 1.4 % and 3.9 %. Compared with the others, Chantler’s FFAST results for the photoeffect show a relative reduction above about 200 keV that grows to the end of his coverage for these materials.

6.2 Coherent (Rayleigh) Scattering

Rayleigh scattering is the elastic scattering of photons, without electronic excitation of the target atom. An incident photon with energy E and momentum $\mathbf{k}$ ($k = E/c$) is absorbed by the target atom and a scattered photon is emitted with the same energy, $E' = E$, in a direction defined by the angle θ with respect to the initial direction. Because the process involves two photons, it is described using second-order perturbation theory. The calculation of second-order transition-matrix elements is complicated because they involve a summation over all possible, positive- and negative-energy intermediate

²³These agree with those given by Scofield quite well except for a few cases, mostly toward the outer shells, for which relative discrepancies can be as much as about 6 %.

²⁴EXAFS refers to oscillations of the x-ray absorption coefficient on the high-energy side of an absorption edge. Such oscillations can extend up to 1000 eV above the edge and can have a magnitude of 10 % or more. The theory of this effect has been reviewed by Rehr and Albers (2000).

KEY DATA FOR IONIZING-RADIATION DOSIMETRY: MEASUREMENT STANDARDS AND APPLICATIONS

Table 6.1a. Photoeffect interaction coefficients for air, as a function of photon energy E . Values are given of $(N_A/M_A)\sigma_{pe}(E) = \mu_{pe}(E)/\rho$, in units of $\text{cm}^2 \text{g}^{-1}$, from a number of calculations. Air

E/keV	Storm and Israel (1970)	Scofield (1973)	Scofield renormalized ^a	Chantler <i>et al.</i> (2005)	Salvat (2015b)
1.0	3.58E+03	3.605E+03	3.497E+03	3.457E+03	3.488E+03
1.5	1.19E+03	1.190E+03	1.154E+03	1.127E+03	1.152E+03
2.0	5.23E+02	5.267E+02	5.112E+02	4.973E+02	5.105E+02
3.0	1.61E+02	1.616E+02	1.568E+02	1.539E+02	1.567E+02
Ar K edge ^b	1.33E+02	1.331E+02	1.292E+02	1.264E+02	1.286E+02
Ar K edge	1.48E+02	1.476E+02	1.436E+02	1.418E+02	1.406E+02
4.0	7.69E+01	7.714E+01	7.505E+01	7.301E+01	7.515E+01
5.0	3.96E+01	3.966E+01	3.859E+01	3.744E+01	3.865E+01
6.0	2.29E+01	2.288E+01	2.227E+01	2.170E+01	2.231E+01
8.0	9.18E+00	9.505E+00	9.255E+00	9.123E+00	9.267E+00
10.0	4.59E+00	4.766E+00	4.641E+00	4.572E+00	4.647E+00
15.0	1.29E+00	1.335E+00	1.301E+00	1.273E+00	1.302E+00
20.0	5.17E-01	5.347E-01	5.211E-01	5.121E-01	5.216E-01
30.0	1.41E-01	1.451E-01	1.414E-01	1.407E-01	1.416E-01
40.0	5.56E-02	5.704E-02	5.561E-02	5.530E-02	5.566E-02
50.0	2.69E-02	2.755E-02	2.686E-02	2.682E-02	2.688E-02
60.0	1.49E-02	1.517E-02	1.480E-02	1.485E-02	1.481E-02
80.0	5.80E-03	5.912E-03	5.766E-03	5.828E-03	5.772E-03
100.0	2.78E-03	2.847E-03	2.777E-03	2.812E-03	2.780E-03
150.0	7.34E-04	7.602E-04	7.417E-04	7.479E-04	7.430E-04
200.0	2.95E-04	3.026E-04	2.952E-04	2.925E-04	2.956E-04
300.0	8.78E-05	8.604E-05	8.396E-05	7.787E-05	8.403E-05
400.0		3.698E-05	3.610E-05	3.047E-05	3.613E-05
500.0		1.998E-05	1.950E-05		1.952E-05

^aRenormalized using factors provided by Scofield (1973).

^bAssumed K-edge energies for Ar are 3.203 keV for Storm and Israel (1970), 3.2029 keV for Chantler *et al.* (2005), and 3.2063 keV for Salvat (2015b); the Scofield (1973) results are shown for the XCOM assumption of 3.203 keV, rather than that inherent in the original Scofield calculation (3.1776 keV).

Table 6.1b. Photoeffect interaction coefficients for carbon, as a function of photon energy E . Values are given of $(N_A/M_A)\sigma_{pe}(E) = \mu_{pe}(E)/\rho$, in units of $\text{cm}^2 \text{g}^{-1}$, from a number of calculations. Carbon

E/keV	Storm and Israel (1970)	Scofield (1973)	Scofield renormalized ^a	Chantler <i>et al.</i> (2005)	Salvat (2015b)
1.0	2.18E+03	2.2098E+03	2.1301E+03	2.0723E+03	2.1131E+03
1.5	6.92E+02	6.9926E+02	6.7401E+02	6.4263E+02	6.7031E+02
2.0	3.00E+02	3.0172E+02	2.9081E+02	2.7693E+02	2.8962E+02
3.0	8.93E+01	8.9658E+01	8.6410E+01	8.4428E+01	8.6169E+01
4.0	3.71E+01	3.7239E+01	3.5889E+01	3.4910E+01	3.5815E+01
5.0	1.86E+01	1.8659E+01	1.7982E+01	1.7578E+01	1.7953E+01
6.0	1.04E+01	1.0545E+01	1.0162E+01	9.9345E+00	1.0149E+01
8.0	4.01E+00	4.2413E+00	4.0872E+00	4.0051E+00	4.0835E+00
10.0	1.97E+00	2.0760E+00	2.0005E+00	1.9584E+00	1.9992E+00
15.0	5.32E-01	5.5825E-01	5.3793E-01	5.2447E-01	5.3777E-01
20.0	2.08E-01	2.1764E-01	2.0972E-01	2.0592E-01	2.0969E-01
30.0	5.52E-02	5.7052E-02	5.4974E-02	5.4754E-02	5.4983E-02
40.0	2.12E-02	2.1931E-02	2.1132E-02	2.1123E-02	2.1141E-02
50.0	1.01E-02	1.0423E-02	1.0043E-02	1.0091E-02	1.0050E-02
60.0	5.57E-03	5.6710E-03	5.4642E-03	5.5185E-03	5.4689E-03
80.0	2.13E-03	2.1699E-03	2.0908E-03	2.1294E-03	2.0938E-03
100.0	9.98E-04	1.0318E-03	9.9413E-04	1.0202E-03	9.9594E-04
150.0	2.59E-04	2.706E-04	2.607E-04	2.680E-04	2.609E-04
200.0	1.04E-04	1.063E-04	1.025E-04	1.038E-04	1.025E-04
300.0		2.980E-05	2.872E-05	2.727E-05	2.875E-05
400.0		1.271E-05	1.225E-05	1.056E-05	1.226E-05
500.0		6.837E-06	6.588E-06		6.597E-06

^aRenormalized using factors provided by Scofield (1973).

Photon-Interaction Coefficients

Table 6.1c. Photoeffect interaction coefficients for water, as a function of photon energy E . Values are given of $(N_A/M_A)\sigma_{pe}(E) = \mu_{pe}(E)/\rho$, in units of $\text{cm}^2 \text{g}^{-1}$, from a number of calculations. Water

E/keV	Storm and Israel (1970)	Scofield (1973)	Scofield renormalized ^a	Chantler <i>et al.</i> (2005)	Salvat (2015b)
1.0	4.05E+03	4.076E+03	3.967E+03	3.925E+03	3.975E+03
1.5	1.36E+03	1.375E+03	1.338E+03	1.296E+03	1.340E+03
2.0	6.12E+02	6.161E+02	6.000E+02	5.864E+02	6.006E+02
3.0	1.91E+02	1.919E+02	1.869E+02	1.833E+02	1.870E+02
4.0	8.19E+01	8.198E+01	7.984E+01	7.690E+01	7.990E+01
5.0	4.18E+01	4.191E+01	4.082E+01	3.915E+01	4.085E+01
6.0	2.40E+01	2.407E+01	2.344E+01	2.256E+01	2.345E+01
8.0	9.69E+00	9.919E+00	9.661E+00	9.447E+00	9.666E+00
10.0	4.75E+00	4.943E+00	4.815E+00	4.750E+00	4.817E+00
15.0	1.32E+00	1.369E+00	1.334E+00	1.303E+00	1.334E+00
20.0	5.25E-01	5.437E-01	5.296E-01	5.201E-01	5.299E-01
30.0	1.41E-01	1.457E-01	1.420E-01	1.412E-01	1.420E-01
40.0	5.52E-02	5.680E-02	5.532E-02	5.502E-02	5.535E-02
50.0	2.65E-02	2.725E-02	2.654E-02	2.650E-02	2.656E-02
60.0	1.46E-02	1.493E-02	1.454E-02	1.459E-02	1.455E-02
80.0	5.65E-03	5.770E-03	5.621E-03	5.685E-03	5.625E-03
100.0	2.69E-03	2.762E-03	2.691E-03	2.729E-03	2.693E-03
150.0	7.05E-04	7.307E-04	7.117E-04	7.192E-04	7.129E-04
200.0	2.82E-04	2.887E-04	2.812E-04	2.792E-04	2.819E-04
300.0	8.39E-05	8.162E-05	7.950E-05	7.359E-05	7.956E-05
400.0	3.51E-05	3.495E-05	3.404E-05	2.857E-05	3.407E-05
500.0		1.884E-05	1.835E-05		1.836E-05

^aRenormalized using factors provided by Scofield (1973).

one-electron states. The calculation scheme adopted to bypass the infinite summation is known as the second-order S -matrix method. Numerical S -matrix calculations were done by Kane *et al.* (1986) using the code of Kissel for the total (Rayleigh, nuclear Thomson, and Delbrück) elastic-scattering cross sections for seven photon energies spanning the range from 59.5 keV to 1.33 MeV, for the atoms of Al, Ca, Zn, Mo, Sn, Nd, Ta, Pb, U, and Lr, and for scattering angles from 0° to 180° . Chatterjee and Roy (1998) expanded these best available results by interpolation and extrapolation to include 14 photon energies from 50 keV to 1.5 MeV and the elements $13 \leq Z \leq 104$.

An approximation to the differential cross section for Rayleigh scattering is given by Cromer and Liberman (1970). The differential cross section per solid angle in the direction of the scattered photon can be expressed as the product of the Thomson (1906) cross section and the square of the atomic form factor (plus corrections),

$$\frac{d\sigma_{\text{coh}}}{d\Omega} = r_e^2 \frac{1 + \cos^2\theta}{2} |F(q, Z) + f' + i f''|^2, \quad (6.1)$$

where the quantity q is the magnitude of the momentum transfer $\mathbf{q} = \mathbf{k} - \mathbf{k}'$, where $\mathbf{k}'$ is the momentum of the scattered photon. Because $k' = k$,

$$q = 2k \cdot \sin(\theta/2). \quad (6.2)$$

The function $F(q, Z)$ in Eq. (6.1) is the atomic form factor, *i.e.*, the Fourier transform of the electron density $\rho(r)$,

$$F(q, Z) = \frac{4\pi}{q} \int_0^\infty \rho(r) \sin(qr) r dr. \quad (6.3)$$

Hubbell *et al.* (1975) have published tables of atomic form factors, obtained mainly from non-relativistic Hartree–Fock electron densities. For $2 \leq Z \leq 6$, the electron densities were derived from configuration-interaction wave functions, so include electron-correlation effects. The EPDL97 library (Cullen *et al.*, 1997) contains a more detailed tabulation of Hubbell’s form factors.

Hubbell and Øverbø (1979) published atomic form factors from relativistic Hartree–Fock electron densities. They show that these factors result in total coherent cross sections that differ from their non-relativistic values by no more than about 1 % for low- Z elements; the relative differences can be as much as 19 % for high- Z elements at photon energies above about 100 keV, but for which the coherent-scattering cross section makes a negligible contribution to the total photon-interaction cross section. XCOM uses these relativistic atomic form factors and disregards the anomalous scattering factor.

The complex quantity $f' + i f''$ is known as the anomalous-scattering factor or as the dispersion correction to the form factor. The adjective “anomalous”

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Table 6.2. Coherent-scattering coefficients for air, carbon, and water, as a function of photon energy E . Values are given of $(N_A/M_A)\sigma_{\text{coh}}(E) = \mu_{\text{coh}}(E)/\rho$, in units of $\text{cm}^2 \text{g}^{-1}$, from XCOM (Hubbell and Øverbø, 1979) and from those used in PENELOPE (Cullen *et al.*, 1997; Salvat, 2015b). Note that there are some small irregularities in the PENELOPE results near the Ar K edge that are not shown.

E/keV	Air		Carbon/graphite		Water	
	XCOM	PENELOPE	XCOM	PENELOPE	XCOM	PENELOPE
1.0	1.362E+00	1.546E+00	1.079E+00	1.231E+00	1.372E+00	1.536E+00
1.5	1.247E+00	1.385E+00	9.586E-01	1.055E+00	1.269E+00	1.418E+00
2.0	1.116E+00	1.208E+00	8.318E-01	8.939E-01	1.150E+00	1.260E+00
3.0	8.634E-01	9.035E-01	6.127E-01	6.435E-01	9.087E-01	9.683E-01
4.0	6.648E-01	6.917E-01	4.602E-01	4.773E-01	7.082E-01	7.424E-01
5.0	5.218E-01	5.396E-01	3.595E-01	3.702E-01	5.579E-01	5.789E-01
6.0	4.207E-01	4.326E-01	2.918E-01	2.993E-01	4.489E-01	4.626E-01
8.0	2.946E-01	3.008E-01	2.096E-01	2.137E-01	3.102E-01	3.167E-01
10.0	2.223E-01	2.260E-01	1.620E-01	1.643E-01	2.305E-01	2.342E-01
15.0	1.314E-01	1.327E-01	9.787E-02	9.873E-02	1.333E-01	1.345E-01
20.0	8.752E-02	8.808E-02	6.478E-02	6.515E-02	8.856E-02	8.912E-02
30.0	4.619E-02	4.622E-02	3.365E-02	3.376E-02	4.694E-02	4.694E-02
40.0	2.828E-02	2.826E-02	2.045E-02	2.049E-02	2.874E-02	2.875E-02
50.0	1.905E-02	1.903E-02	1.371E-02	1.373E-02	1.936E-02	1.935E-02
60.0	1.370E-02	1.366E-02	9.807E-03	9.821E-03	1.392E-02	1.390E-02
80.0	8.028E-03	8.000E-03	5.711E-03	5.710E-03	8.165E-03	8.143E-03
100.0	5.254E-03	5.234E-03	3.719E-03	3.719E-03	5.349E-03	5.333E-03
150.0	2.396E-03	2.385E-03	1.685E-03	1.684E-03	2.442E-03	2.434E-03
200.0	1.361E-03	1.355E-03	9.541E-04	9.539E-04	1.388E-03	1.383E-03
300.0	6.096E-04	6.065E-04	4.264E-04	4.264E-04	6.215E-04	6.194E-04
400.0	3.438E-04	3.421E-04	2.403E-04	2.401E-04	3.506E-04	3.492E-04
500.0	2.203E-04	2.192E-04	1.539E-04	1.538E-04	2.247E-04	2.238E-04
600.0	1.531E-04	1.523E-04	1.069E-04	1.069E-04	1.561E-04	1.555E-04
800.0	8.618E-05	8.576E-05	6.017E-05	6.019E-05	8.790E-05	8.761E-05
1000.0	5.517E-05	5.492E-05	3.852E-05	3.853E-05	5.627E-05	5.608E-05
1500.0	2.453E-05	2.442E-05	1.712E-05	1.713E-05	2.501E-05	2.493E-05
2000.0	1.380E-05	1.374E-05	9.632E-06	9.637E-06	1.407E-05	1.403E-05
3000.0	6.133E-06	6.106E-06	4.281E-06	4.283E-06	6.255E-06	6.235E-06
4000.0	3.451E-06	3.435E-06	2.408E-06	2.409E-06	3.519E-06	3.507E-06
5000.0	2.208E-06	2.198E-06	1.541E-06	1.542E-06	2.252E-06	2.245E-06
6000.0	1.534E-06	1.527E-06	1.070E-06	1.071E-06	1.564E-06	1.559E-06
8000.0	8.628E-07	8.588E-07	6.022E-07	6.023E-07	8.796E-07	8.769E-07
10 000.0	5.522E-07	5.496E-07	3.853E-07	3.855E-07	5.630E-07	5.612E-07

refers to the fast variation of this quantity for photon energies around the absorption edge. Numerical S -matrix calculations indicate that, for energies below the absorption edge of the K shell, the anomalous scattering factors are practically independent of the scattering angle θ [see, *e.g.*, Kissel *et al.* (1995) and references therein]. The tables of Henke *et al.* (1993) include anomalous scattering factors for the elements $1 \leq Z \leq 92$ and photon energies between 50 eV and 30 keV. An extensive tabulation of anomalous scattering factors, covering all the elements up to $Z = 100$ and energies between 1 eV and 10 MeV has been prepared by Cullen (1989). The PENELOPE database (Salvat, 2015b) uses non-relativistic form factors and anomalous scattering factors from the EPDL database (Cullen *et al.*, 1997). Values from Hubbell and Øverbø (1979) *via* XCOM from the web site of the National Institute of Standards and Technology (NIST) and from EPDL

via PENELOPE (Salvat, 2015b) are compared in Table 6.2. The relative differences become significant ($\geq 1\%$) only for energies below about 15 keV at which coherent scattering becomes relatively unimportant (see Fig. 6.1b).

6.3 Incoherent (Compton) Scattering

The theoretical approaches to the calculation of Compton scattering from bound electrons have been reviewed by Bergstrom and Pratt (1997) and by Pratt *et al.* (2010). In Compton scattering, an incident photon with energy E and linear momentum $\mathbf{k}$ is absorbed, and a scattered photon is emitted with lower energy, E' , and momentum $\mathbf{k}'$ ($k' < k$). The energy difference between the incident and scattered photons is transferred to an atomic electron in a bound orbital of the i th shell with energy $-E_i$,

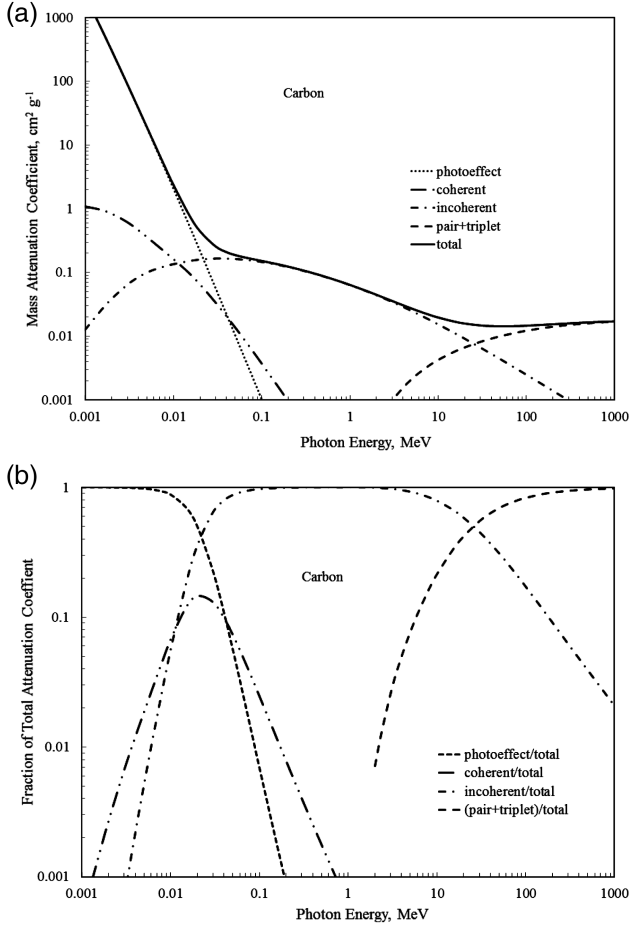


Figure 6.1. Mass attenuation coefficient for carbon as a function of photon energy. Results taken from XCOM for illustration. (a) Partial and total attenuation coefficients. (b) Fractional contribution of the partials to the total attenuation coefficient.

which makes a transition to a final state with energy $E - E' - E_i$.

A limiting situation that can be described analytically is that of Compton scattering by free electrons at rest. In this case, the initial and final states of the target electron can be represented as discrete plane waves, and the differential cross section can be calculated analytically. The resulting differential cross section per solid angle in the direction of the scattered photon is given by the Klein–Nishina (1929) formula,

$$\frac{d\sigma_{\text{inc}}^{\text{KN}}}{d\Omega} = \frac{r_e^2}{2} \left(\frac{E'}{E} \right)^2 \left(\frac{E}{E'} + \frac{E'}{E} - \sin^2 \theta \right), \quad (6.4)$$

where θ is the polar scattering angle, *i.e.*, the angle between the directions of the incident and scattered photons, and the energy of the scattered photon is

given by the usual Compton (1923) relation

$$E' = \frac{E}{1 + (E/m_e c^2)(1 - \cos \theta)}. \quad (6.5)$$

The linear momentum $\mathbf{p}_e$ of the target electron after the collision is equal to the momentum transfer, $\mathbf{q} = \mathbf{k} - \mathbf{k}'$, and its magnitude is given by

$$q = \frac{1}{c} \sqrt{E^2 + (E')^2 - 2EE' \cos \theta}. \quad (6.6)$$

The derivation of the Klein–Nishina formula disregards the fact that transitions of atomic electrons involving energy transfers $E - E'$ below the lowest excitation energy of the active shell are forbidden (binding effect). Moreover, atomic electrons are not at rest but move with a certain momentum distribution, which causes the so-called Doppler broadening of the Compton line (see, *e.g.*, Brusa *et al.*, 1996).

The most elaborate calculations of Compton scattering by bound atomic electrons were performed by Surić *et al.* (1991) and Bergstrom *et al.* (1993) using the second-order *S*-matrix method with DHFS atomic orbitals. These calculations are quite complicated, and the results cannot be reduced to a compact form suited for practical dosimetry calculations or Monte Carlo simulations. Consequently, existing interaction databases and simulation codes utilize less-sophisticated interaction models, usually based on the Klein–Nishina formula with corrections to account for electron-binding effects.

XCOM (Berger and Hubbell, 1987; Berger *et al.*, 2010), EPDL (Cullen *et al.*, 1997), and several general-purpose Monte Carlo systems describe Compton scattering using the theory of Waller and Hartree (1929), which accounts for the binding effect but disregards Doppler broadening, *i.e.*, photons scattered through an angle θ are assumed to have the energy E' . This theory leads to the differential cross section

$$\frac{d\sigma_{\text{inc}}}{d\Omega} = \frac{d\sigma_{\text{inc}}^{\text{KN}}}{d\Omega} S(q), \quad (6.7)$$

where q is the momentum transfer to the Compton electron, given by Eq. (6.6), and $S(q)$ is here the incoherent-scattering function, which can be calculated in terms of the atomic ground-state wave function, summed over all final states. Thus, this result inherently includes both ionization (Compton scattering) and excitation (Raman scattering); hence, it is more accurately termed incoherent or inelastic scattering. The incoherent-scattering function is a monotonically increasing function of q that takes values from $S(0) = 0$ to $S(\infty) = Z$. Tables of the incoherent-scattering function for all elements from $Z = 1$ to 100 and the resultant values of σ_{inc} have been

compiled by Hubbell *et al.* (1975) and incorporated in XCOM. Hubbell *et al.* (1975) included electron-correlation effects for Z from 2 to 6, which produce substantial corrections, particularly at low energies, and also included the relatively small radiative and double-Compton corrections of Mork (1971). The EPDL database (Cullen *et al.*, 1997) integrates Eq. (6.7) using the incoherent-scattering functions of Hubbell *et al.* (1975).

Differential cross sections that take into account the motion of the bound electron are provided by an approximate expression of the relativistic-impulse approximation (RIA) formulated by Ribberfors (1975), which accounts for Doppler broadening and binding effects in a natural, and relatively simple, way. These differential cross sections are obtained by considering that electrons in the i th shell move with a momentum distribution $\rho_i(\mathbf{p})$. For an electron in an orbital $\psi_i(\mathbf{r})$, $\chi_i(\mathbf{p}) \equiv |\psi_i(\mathbf{p})|^2$, where $\psi_i(\mathbf{p})$ is the wavefunction in the momentum representation. The differential cross section for Compton scattering by an electron with momentum $\mathbf{p}$ is derived from the Klein–Nishina formula by applying a Lorentz transformation with velocity equal to that of the moving target electron. The impulse approximation to the Compton differential cross section (per electron) of the shell is obtained by averaging over the momentum distribution $\chi_i(\mathbf{p})$ of the electrons in the shell. The differential cross section for Compton scattering by an electron in the i th shell can, to a good approximation, be expressed²⁵ as

$$\frac{d^2\sigma_{\text{inc},i}^{\text{RIA}}}{dE'd\Omega} = \frac{r_e^2 E' m_e}{2 E q} \left[1 + \left(\frac{p_z}{m_e c} \right)^2 \right]^{-1/2} X J_i(p_z), \quad (6.8)$$

where q is given by Eq. (6.6), the quantity p_z is the projection of the initial momentum $\mathbf{p}$ of the active electron on the direction of the vector $-\mathbf{q}$, given by

$$p_z = \frac{EE'(1 - \cos \theta) - m_e(E - E')}{q}, \quad (6.9)$$

X is an analytical but somewhat involved function of E , E' , $\cos \theta$, and p_z , and $J_i(p_z)$ is the Compton profile of the active electron shell,

$$J_i(p_z) \equiv \iint \rho_i(\mathbf{p}) d p_x d p_y = 2\pi \int_{|p_z|}^{\infty} \rho_i(p) p dp, \quad (6.10)$$

where the last equality follows from the spherical symmetry of the electron density $\rho_i(p)$ of the shell. Extensive tables of Hartree–Fock Compton profiles

²⁵Although the impulse-approximation cross sections are, more strictly, only for Compton scattering, the subscript indicating incoherent scattering is retained.

for the elements have been published by Biggs *et al.* (1975). Brusa *et al.* (1996) introduced a simplification of the Ribberfors formula, with an analytical approximation to the Compton profiles of Biggs *et al.* that greatly facilitates the use of the RIA in Monte Carlo simulations, with a loss in accuracy that is practically negligible for most applications. It is this approach that is used in the Monte Carlo code PENELOPE (Salvat, 2015b) to approximate results for real materials, *i.e.*, beyond those for isolated atoms in some cases. For example, although alternatives can be chosen by the user, the default selection in PENELOPE for air (their material number 104) uses 8 orbitals as opposed to some 16 in an atomic representation; for graphite (their material number 175) uses 3 orbitals as in atomic carbon, but assumes $n_c = 2$ conduction electrons with a binding energy corresponding to a free-electron gas given by $(n_c Z)^{1/2} E_p$ (see Eq. 4.26); and for water (their material number 278) uses 2 orbitals as opposed to 4 orbitals in an atomic representation. As indicated in Section 4.3, this Report adopts the assumption of one conduction electron for graphite.

At energies much greater than the binding energies of the atomic electrons, Compton scattering from all the orbital electrons is nearly the same [*i.e.*, $S(q \rightarrow \infty) = Z$ in the integration of Eq. (6.7) to obtain the total incoherent-scattering cross section]. As the photon energy falls below the orbital binding energies, the sub-shell contribution and the total cross section become increasingly sensitive to the assumed binding energies of the outer shells. Thus, Compton-scattering cross sections at low energies become significantly dependent on the model assumed in the calculation. This is illustrated in Table 6.3 for the materials air, graphite (carbon), and water, which compare the values from Hubbell *et al.* (1975) via XCOM, from atomic *ab initio* relativistic-impulse-approximation (RIA) calculations as described by Bergstrom *et al.* (1993), from atomic (and molecular for water²⁶) non-relativistic-impulse-approximation calculations (IA) of Rao *et al.* (2004), and from PENELOPE RIA evaluations with various assumptions of the number of conduction electrons in graphite.

Some observations can be made regarding the results shown in Table 6.3. The PENELOPE results

²⁶These calculations use the methods of Rao *et al.* (2004) based on the Compton profiles of Biggs *et al.* (1975), but assuming the molecular orbitals whose parentage coefficients are derived from LCAO calculations (Siegbahn, 1969), *i.e.*, two electrons each in molecular orbital: 1b₁ (12.62 eV), 1.00 O_{2p}; 3a₁ (14.75 eV), 0.175 H_{1s}, 0.72 O_{2p}, 0.105 O_{2s}; 1b₂ (18.51 eV), 0.41 H_{1s}, 0.59 O_{2p}; 2a₁ (32.4 eV), 0.25 H_{1s}, 0.75 O_{2s}; 1a₁ (539.7 eV), 1.0 O_{1s}, where the binding energies are given in parentheses followed by the parentage coefficients for the indicated atomic orbital.

Photon-Interaction Coefficients

Table 6.3a. Incoherent (Compton)-scattering coefficients for air, as a function of photon energy E . Values are given of $(N_A/M_A)\sigma_{\text{inc}}(E) = \mu_{\text{inc}}(E)/\rho$, in units of $\text{cm}^2 \text{g}^{-1}$, from the NIST XCOM (originally from Hubbell *et al.*, 1975), from *ab initio* RIA calculations as described in Bergstrom *et al.* (1993), and from PENELOPE (Salvat, 2015b). Air

E/MeV	XCOM	<i>Ab initio</i> RIA	PENELOPE	E/MeV	XCOM	<i>Ab initio</i> RIA	PENELOPE
0.0010	1.038E-02	1.048E-02	6.977E-03	1.50	5.162E-02	5.125E-02	5.158E-02
0.0015	2.116E-02	2.158E-02	2.098E-02	2.00	4.407E-02	4.372E-02	4.399E-02
0.0020	3.340E-02	3.320E-02	3.576E-02	3.00	3.467E-02	3.438E-02	3.460E-02
0.0030	5.748E-02	5.642E-02	6.115E-02	4.00	2.892E-02	2.867E-02	2.885E-02
0.0040	7.770E-02	7.601E-02	8.039E-02	5.00	2.497E-02	2.475E-02	2.491E-02
0.0050	9.331E-02	9.081E-02	9.498E-02	6.00	2.207E-02	2.187E-02	2.201E-02
0.0060	1.051E-01	1.019E-01	1.063E-01	8.00	1.806E-02	1.789E-02	1.800E-02
0.0080	1.213E-01	1.175E-01	1.224E-01	10.00	1.538E-02	1.523E-02	1.533E-02
0.0100	1.316E-01	1.284E-01	1.333E-01	15.00	1.138E-02	1.126E-02	1.134E-02
0.0150	1.471E-01	1.456E-01	1.497E-01	20.00	9.135E-03	9.036E-03	9.093E-03
0.0200	1.556E-01	1.548E-01	1.584E-01	30.00	6.652E-03		6.613E-03
0.0300	1.625E-01	1.618E-01	1.648E-01	40.00	5.286E-03		5.250E-03
0.0400	1.632E-01	1.623E-01	1.648E-01	50.00	4.411E-03		4.379E-03
0.0500	1.614E-01	1.605E-01	1.626E-01	60.00	3.801E-03		3.770E-03
0.0600	1.586E-01	1.577E-01	1.595E-01	80.00	2.998E-03		2.971E-03
0.0800	1.523E-01	1.514E-01	1.529E-01	100.00	2.488E-03		2.465E-03
0.1000	1.460E-01	1.451E-01	1.464E-01	150.00	1.772E-03		1.749E-03
0.1500	1.324E-01	1.316E-01	1.326E-01	200.00	1.389E-03		1.368E-03
0.2000	1.217E-01	1.209E-01	1.217E-01	300.00	9.847E-04		9.647E-04
0.3000	1.061E-01	1.053E-01	1.061E-01	400.00	7.712E-04		7.514E-04
0.4000	9.511E-02	9.447E-02	9.509E-02	500.00	6.382E-04		6.184E-04
0.5000	8.688E-02	8.630E-02	8.686E-02	600.00	5.467E-04		5.271E-04
0.6000	8.039E-02	7.985E-02	8.037E-02	800.00	4.269E-04		4.092E-04
0.8000	7.064E-02	7.016E-02	7.060E-02	1000.00	3.512E-04		3.359E-04
1.0000	6.352E-02	6.307E-02	6.347E-02				

for zero conduction electrons in graphite (Table 6.3b) tend to track those done in other approximations for atomic carbon. For all of the materials considered here, the cross sections for incoherent (Compton) scattering agree to within better than about 1 % for energies greater than 50 keV, with agreement falling to within from 4 % to 5 % at 10 keV. At 1 keV, the differences that result from details of the models used are quite dramatic, but the relative contribution of incoherent (Compton) scattering is then very small (see Fig. 6.1b).

6.4 Electron-Positron Pair Production

Thorough reviews of available theories and extensive tables of pair-production total cross sections, evaluated by combining different theoretical approximations, have been published by Hubbell *et al.* (1980). These tables give the separate contributions of pair production in the field of the nucleus and in that of the atomic electrons for $1 \leq Z \leq 100$ and for photon energies from threshold up to 105 MeV. This remains the most authoritative source of pair- and triplet-production data available. Hubbell's data are incorporated in the XCOM database (Berger and Hubbell, 1987; Berger *et al.*, 2010)

and in the EPDL97 database (Cullen *et al.*, 1997). Representative values for air, carbon, and water are given in Table 6.4.

6.5 Mass Attenuation Coefficients

Because available databases are mainly limited to interactions with atoms (except as indicated in Section 6.3), the mass attenuation coefficients for compounds and mixtures are usually obtained from the additivity approximation in which cross sections are approximated by the sum of cross sections, or mass attenuation coefficients, of the atoms in the material, *i.e.*,

$$\begin{aligned} \frac{\mu}{\rho} &= \frac{N_A}{M_A} \sum_J \sigma_J = \sum_J \left(\frac{\mu}{\rho} \right)_J \\ &= N_A \sum_J \sum_i \frac{\zeta_i}{A_i} \sigma_J(Z_i), \end{aligned} \quad (6.11)$$

where A_i is the atomic weight and ζ_i the fraction by mass of the i th constituent atom. Thus, the distinction between “atomic” and “molecular” air, between graphite and carbon, and between liquid water and water vapor is ignored in the remaining discussion of photon-interaction coefficients.

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Table 6.3b. Incoherent (Compton)-scattering coefficients for carbon/graphite, as a function of photon energy E . Values are given of $(N_A/M_A)\sigma_{\text{inc}}(E) = \mu_{\text{inc}}(E)/\rho$, in units of $\text{cm}^2 \text{g}^{-1}$, from the NIST XCOM (originally from Hubbell *et al.*, 1975), from *ab initio* RIA calculations as described in Bergstrom *et al.* (1993), from the IA calculations of Rao *et al.* (2004), and from PENELOPE (Salvat, 2015b).

E/MeV	Atomic carbon			PENELOPE graphite		
	XCOM	<i>Ab initio</i> RIA	Rao <i>et al.</i> IA	Number of conduction electrons		
				0	1	2
0.0010	1.263E-02	1.022E-02	2.344E-02	9.311E-03	2.749E-02	4.239E-02
0.0015	2.507E-02	2.158E-02	3.801E-02	2.235E-02	3.875E-02	5.094E-02
0.0020	3.862E-02	3.491E-02	5.249E-02	3.717E-02	5.196E-02	6.212E-02
0.0030	6.408E-02	6.138E-02	7.631E-02	6.383E-02	7.579E-02	8.340E-02
0.0040	8.448E-02	8.138E-02	9.282E-02	8.357E-02	9.308E-02	9.912E-02
0.0050	9.947E-02	9.545E-02	1.046E-01	9.790E-02	1.054E-01	1.103E-01
0.0060	1.104E-01	1.059E-01	1.136E-01	1.088E-01	1.148E-01	1.187E-01
0.0080	1.252E-01	1.211E-01	1.272E-01	1.248E-01	1.287E-01	1.313E-01
0.0100	1.352E-01	1.323E-01	1.373E-01	1.361E-01	1.388E-01	1.406E-01
0.0150	1.510E-01	1.499E-01	1.535E-01	1.534E-01	1.547E-01	1.556E-01
0.0200	1.595E-01	1.588E-01	1.615E-01	1.621E-01	1.629E-01	1.634E-01
0.0300	1.655E-01	1.646E-01	1.666E-01	1.674E-01	1.678E-01	1.680E-01
0.0400	1.653E-01	1.643E-01	1.660E-01	1.666E-01	1.668E-01	1.670E-01
0.0500	1.630E-01	1.620E-01	1.634E-01	1.639E-01	1.640E-01	1.641E-01
0.0600	1.598E-01	1.589E-01	1.602E-01	1.606E-01	1.607E-01	1.607E-01
0.0800	1.531E-01	1.521E-01	1.533E-01	1.535E-01	1.536E-01	1.536E-01
0.1000	1.466E-01	1.457E-01	1.468E-01	1.469E-01	1.469E-01	1.502E-01
0.1500	1.327E-01	1.319E-01	1.328E-01	1.328E-01	1.328E-01	1.329E-01
0.2000	1.219E-01	1.211E-01	1.219E-01	1.219E-01	1.219E-01	1.219E-01
0.3000	1.062E-01	1.055E-01	1.062E-01	1.062E-01	1.062E-01	1.062E-01
0.4000	9.521E-02	9.456E-02	9.519E-02	9.518E-02	9.518E-02	9.518E-02
0.5000	8.699E-02	8.637E-02	8.695E-02	8.694E-02	8.694E-02	8.694E-02
0.6000	8.047E-02	7.991E-02	8.045E-02	8.043E-02	8.043E-02	8.043E-02
0.8000	7.070E-02	7.021E-02	7.068E-02	7.066E-02	7.066E-02	7.066E-02
1.0000	6.358E-02	6.312E-02	6.354E-02	6.352E-02	6.352E-02	6.352E-02
1.5000	5.169E-02	5.128E-02	5.162E-02	5.162E-02	5.162E-02	5.162E-02
2.0000	4.410E-02	4.374E-02	4.403E-02	4.402E-02	4.402E-02	4.402E-02
3.0000	3.470E-02	3.440E-02	3.463E-02	3.462E-02	3.462E-02	3.462E-02
4.0000	2.894E-02	2.869E-02	2.887E-02	2.887E-02	2.887E-02	2.887E-02
5.0000	2.499E-02	2.477E-02	2.493E-02	2.493E-02	2.493E-02	2.493E-02
6.0000	2.209E-02	2.189E-02	2.203E-02	2.203E-02	2.203E-02	2.203E-02
8.0000	1.807E-02	1.790E-02	1.801E-02	1.801E-02	1.801E-02	1.801E-02
10.0000	1.539E-02	1.524E-02	1.534E-02	1.534E-02	1.534E-02	1.534E-02
15.0000	1.139E-02	1.127E-02		1.134E-02	1.134E-02	1.134E-02
20.0000	9.140E-03	9.042E-03		9.100E-03	9.100E-03	9.100E-03
30.0000	6.653E-03			6.617E-03	6.617E-03	6.617E-03
40.0000	5.290E-03			5.254E-03	5.254E-03	5.254E-03
50.0000	4.415E-03			4.382E-03	4.382E-03	4.382E-03
60.0000	3.804E-03			3.773E-03	3.773E-03	3.773E-03
80.0000	3.000E-03			2.973E-03	2.973E-03	2.973E-03
100.0000	2.490E-03			2.466E-03	2.466E-03	2.466E-03
150.0000	1.773E-03			1.750E-03	1.750E-03	1.750E-03
200.0000	1.390E-03			1.369E-03	1.369E-03	1.369E-03
300.0000	9.852E-04			9.653E-04	9.653E-04	9.653E-04
400.0000	7.716E-04			7.519E-04	7.519E-04	7.519E-04
500.0000	6.388E-04			6.188E-04	6.188E-04	6.188E-04
600.0000	5.470E-04			5.274E-04	5.274E-04	5.274E-04
800.0000	4.272E-04			4.095E-04	4.095E-04	4.095E-04
1000.0000	3.514E-04			3.362E-04	3.362E-04	3.362E-04

Tables of total and partial mass attenuation coefficients of photons in arbitrary materials can be obtained from XCOM, from EPDL, or from the PENELOPE Monte Carlo system (Salvat, 2015b).

These databases use essentially identical cross sections for pair and triplet production, but differ in details for the other interaction mechanisms. Summarizing from the discussions above, for the

Photon-Interaction Coefficients

Table 6.3c. Incoherent (Compton)-scattering coefficients for water, as a function of photon energy E . Values are given of $(N_A/M_A)\sigma_{\text{inc}}(E) = \mu_{\text{inc}}(E)/\rho$, in units of $\text{cm}^2 \text{g}^{-1}$, from the NIST XCOM (originally from Hubbell *et al.*, 1975), from *ab initio* RIA calculations as described in Bergstrom *et al.* (1993), from the IA calculations of Rao *et al.* (2004), and from PENELOPE (Salvat, 2015b).

E/MeV	XCOM	<i>Ab initio</i> RIA	Rao <i>et al.</i>		PENELOPE
			Atomic IA	Molecular IA	
0.0010	1.319E-02	1.088E-02	2.736E-02	1.416E-02	1.094E-02
0.0015	2.673E-02	2.506E-02	4.270E-02	2.796E-02	2.759E-02
0.0020	4.184E-02	4.080E-02	5.862E-02	4.235E-02	4.501E-02
0.0030	7.075E-02	7.046E-02	8.694E-02	7.067E-02	7.623E-02
0.0040	9.430E-02	9.397E-02	1.077E-01	9.413E-02	1.001E-01
0.0050	1.123E-01	1.113E-01	1.225E-01	1.115E-01	1.180E-01
0.0060	1.259E-01	1.240E-01	1.332E-01	1.243E-01	1.311E-01
0.0080	1.440E-01	1.410E-01	1.478E-01	1.416E-01	1.482E-01
0.0100	1.550E-01	1.520E-01	1.574E-01	1.529E-01	1.587E-01
0.0150	1.699E-01	1.681E-01	1.718E-01	1.694E-01	1.733E-01
0.0200	1.774E-01	1.763E-01	1.791E-01	1.776E-01	1.806E-01
0.0300	1.829E-01	1.821E-01	1.841E-01	1.834E-01	1.854E-01
0.0400	1.827E-01	1.818E-01	1.835E-01	1.831E-01	1.846E-01
0.0500	1.803E-01	1.793E-01	1.809E-01	1.805E-01	1.817E-01
0.0600	1.770E-01	1.760E-01	1.774E-01	1.772E-01	1.780E-01
0.0800	1.697E-01	1.687E-01	1.699E-01	1.698E-01	1.703E-01
0.1000	1.626E-01	1.616E-01	1.628E-01	1.626E-01	1.630E-01
0.1500	1.473E-01	1.464E-01	1.474E-01	1.473E-01	1.475E-01
0.2000	1.353E-01	1.345E-01	1.354E-01	1.353E-01	1.354E-01
0.3000	1.179E-01	1.172E-01	1.179E-01	1.179E-01	1.180E-01
0.4000	1.058E-01	1.051E-01	1.057E-01	1.057E-01	1.058E-01
0.5000	9.663E-02	9.596E-02	9.657E-02	9.654E-02	9.660E-02
0.6000	8.939E-02	8.879E-02	8.935E-02	8.933E-02	8.937E-02
0.8000	7.856E-02	7.801E-02	7.850E-02	7.848E-02	7.851E-02
1.0000	7.066E-02	7.013E-02	7.056E-02	7.056E-02	7.058E-02
1.5000	5.742E-02	5.698E-02	5.733E-02	5.732E-02	5.736E-02
2.0000	4.901E-02	4.861E-02	4.891E-02	4.890E-02	4.892E-02
3.0000	3.855E-02	3.822E-02	3.846E-02	3.846E-02	3.847E-02
4.0000	3.216E-02	3.188E-02	3.207E-02	3.207E-02	3.208E-02
5.0000	2.777E-02	2.752E-02	2.769E-02	2.769E-02	2.770E-02
6.0000	2.454E-02	2.432E-02	2.447E-02	2.447E-02	2.448E-02
8.0000	2.008E-02	1.989E-02	2.001E-02	2.001E-02	2.002E-02
10.0000	1.710E-02	1.693E-02	1.704E-02	1.704E-02	1.704E-02
15.0000	1.266E-02	1.253E-02			1.261E-02
20.0000	1.016E-02	1.005E-02			1.011E-02
30.0000	7.396E-03				7.353E-03
40.0000	5.875E-03				5.838E-03
50.0000	4.906E-03				4.869E-03
60.0000	4.225E-03				4.193E-03
80.0000	3.333E-03				3.303E-03
100.0000	2.767E-03				2.741E-03
150.0000	1.970E-03				1.945E-03
200.0000	1.545E-03				1.521E-03
300.0000	1.095E-03				1.073E-03
400.0000	8.574E-04				8.355E-04
500.0000	7.099E-04				6.876E-04
600.0000	6.079E-04				5.861E-04
800.0000	4.747E-04				4.550E-04
1000.0000	3.905E-04				3.736E-04

photoeffect, XCOM and EPDL utilize cross sections calculated by Scofield, omitting the renormalization factor. However, the renormalized Scofield cross sections have been incorporated for a number of comparisons used in this Report. The photoeffect cross sections in the PENELOPE database were

calculated independently, with the normalization correction and energy shifts, up to an energy of about 2 MeV and uses slightly different high-energy extrapolations for the photoeffect cross section. XCOM and EPDL use incoherent-scattering cross sections obtained from the incoherent-scattering-

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Table 6.4. Pair- and triplet-production coefficients for air, carbon, and water, as a function of photon energy E . Values are given of $(N_A/M_A)\sigma_{\text{pair}}(E) = \mu_{\text{pair}}(E)/\rho$ and of $(N_A/M_A)\sigma_{\text{trip}}(E) = \mu_{\text{trip}}(E)/\rho$, both in units of $\text{cm}^2 \text{g}^{-1}$, from the NIST XCOM (originally from Hubbell *et al.*, 1980). Note that the threshold energy for pair production is $2m_e c^2$, and that for triplet production is $4m_e c^2$.

E/MeV	Air		Carbon		Water	
	Pair	Triplet	Pair	Triplet	Pair	Triplet
1.022E+00	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0.000E+00	0.000E+00
1.500E+00	9.848E-05	0.000E+00	7.992E-05	0.000E+00	9.820E-05	0.000E+00
2.000E+00	3.918E-04	0.000E+00	3.187E-04	0.000E+00	3.908E-04	0.000E+00
2.044E+00	4.223E-04	0.000E+00	3.435E-04	0.000E+00	4.212E-04	0.000E+00
3.000E+00	1.120E-03	1.213E-05	9.125E-04	1.214E-05	1.117E-03	1.349E-05
4.000E+00	1.817E-03	4.952E-05	1.482E-03	4.956E-05	1.812E-03	5.507E-05
5.000E+00	2.437E-03	9.869E-05	1.988E-03	9.877E-05	2.431E-03	1.097E-04
6.000E+00	2.996E-03	1.516E-04	2.445E-03	1.517E-04	2.987E-03	1.685E-04
8.000E+00	3.940E-03	2.556E-04	3.218E-03	2.558E-04	3.927E-03	2.843E-04
1.000E+01	4.715E-03	3.516E-04	3.854E-03	3.519E-04	4.699E-03	3.910E-04
1.500E+01	6.165E-03	5.518E-04	5.039E-03	5.525E-04	6.141E-03	6.135E-04
2.000E+01	7.212E-03	7.085E-04	5.901E-03	7.095E-04	7.186E-03	7.878E-04
3.000E+01	8.686E-03	9.423E-04	7.120E-03	9.441E-04	8.661E-03	1.048E-03
4.000E+01	9.707E-03	1.112E-03	7.962E-03	1.115E-03	9.674E-03	1.237E-03
5.000E+01	1.048E-02	1.245E-03	8.594E-03	1.248E-03	1.045E-02	1.385E-03
6.000E+01	1.110E-02	1.352E-03	9.100E-03	1.357E-03	1.106E-02	1.505E-03
8.000E+01	1.202E-02	1.519E-03	9.862E-03	1.525E-03	1.199E-02	1.693E-03
1.000E+02	1.270E-02	1.644E-03	1.042E-02	1.652E-03	1.267E-02	1.834E-03
1.500E+02	1.383E-02	1.861E-03	1.137E-02	1.872E-03	1.380E-02	2.081E-03
2.000E+02	1.454E-02	2.004E-03	1.195E-02	2.018E-03	1.450E-02	2.243E-03
3.000E+02	1.538E-02	2.186E-03	1.268E-02	2.205E-03	1.534E-02	2.453E-03
4.000E+02	1.588E-02	2.299E-03	1.310E-02	2.321E-03	1.583E-02	2.584E-03
5.000E+02	1.621E-02	2.378E-03	1.339E-02	2.403E-03	1.615E-02	2.675E-03
6.000E+02	1.645E-02	2.437E-03	1.360E-02	2.464E-03	1.639E-02	2.742E-03
8.000E+02	1.677E-02	2.518E-03	1.388E-02	2.549E-03	1.670E-02	2.838E-03
1.000E+03	1.698E-02	2.573E-03	1.406E-02	2.606E-03	1.691E-02	2.902E-03
1.500E+03	1.729E-02	2.655E-03	1.433E-02	2.693E-03	1.721E-02	2.999E-03
2.000E+03	1.746E-02	2.702E-03	1.449E-02	2.743E-03	1.738E-02	3.054E-03
3.000E+03	1.764E-02	2.754E-03	1.465E-02	2.798E-03	1.755E-02	3.114E-03
4.000E+03	1.774E-02	2.784E-03	1.474E-02	2.830E-03	1.765E-02	3.150E-03
5.000E+03	1.780E-02	2.803E-03	1.480E-02	2.850E-03	1.771E-02	3.172E-03
6.000E+03	1.785E-02	2.816E-03	1.483E-02	2.864E-03	1.776E-02	3.188E-03
8.000E+03	1.790E-02	2.833E-03	1.489E-02	2.882E-03	1.781E-02	3.208E-03
1.000E+04	1.794E-02	2.844E-03	1.492E-02	2.894E-03	1.784E-02	3.222E-03
1.500E+04	1.799E-02	2.859E-03	1.496E-02	2.911E-03	1.789E-02	3.239E-03
2.000E+04	1.802E-02	2.868E-03	1.499E-02	2.920E-03	1.792E-02	3.250E-03
3.000E+04	1.805E-02	2.877E-03	1.501E-02	2.929E-03	1.795E-02	3.260E-03
4.000E+04	1.806E-02	2.882E-03	1.503E-02	2.935E-03	1.796E-02	3.267E-03
5.000E+04	1.807E-02	2.885E-03	1.503E-02	2.938E-03	1.797E-02	3.270E-03
6.000E+04	1.808E-02	2.887E-03	1.504E-02	2.941E-03	1.798E-02	3.273E-03
8.000E+04	1.809E-02	2.890E-03	1.505E-02	2.943E-03	1.798E-02	3.275E-03
1.000E+05	1.809E-02	2.891E-03	1.505E-02	2.945E-03	1.799E-02	3.278E-03

function approximation, Eq. (6.7), and the PENELOPE Compton-scattering cross sections were calculated from the RIA as described by Brusa *et al.* (1996). Finally, for Rayleigh scattering, XCOM uses relativistic atomic form factors and disregards the anomalous scattering factor, while EPDL and PENELOPE use non-relativistic form factors and anomalous scattering factors from the EPDL database (Cullen *et al.*, 1997).

The partial attenuation coefficients, $(\mu/\rho)_j$, are illustrated in Fig. 6.1a for carbon, using results from XCOM. Further discussion of the interaction

coefficients will be aided by noting the fractional contribution of the partials shown in Fig. 6.1b. Thus, one sees that the photoeffect is dominant for energies from 1 keV to about 10 keV and is significant at energies up to about 40 keV (using a 10 % relative contribution as a rough guide to significance); coherent scattering is perhaps significant for energies from about 15 keV to 40 keV; incoherent scattering is significant from about 15 keV to 150 MeV, dominating (>90 % relative contribution) from about 50 keV to 6 MeV; pair (plus triplet) production is significant above about 6 MeV,

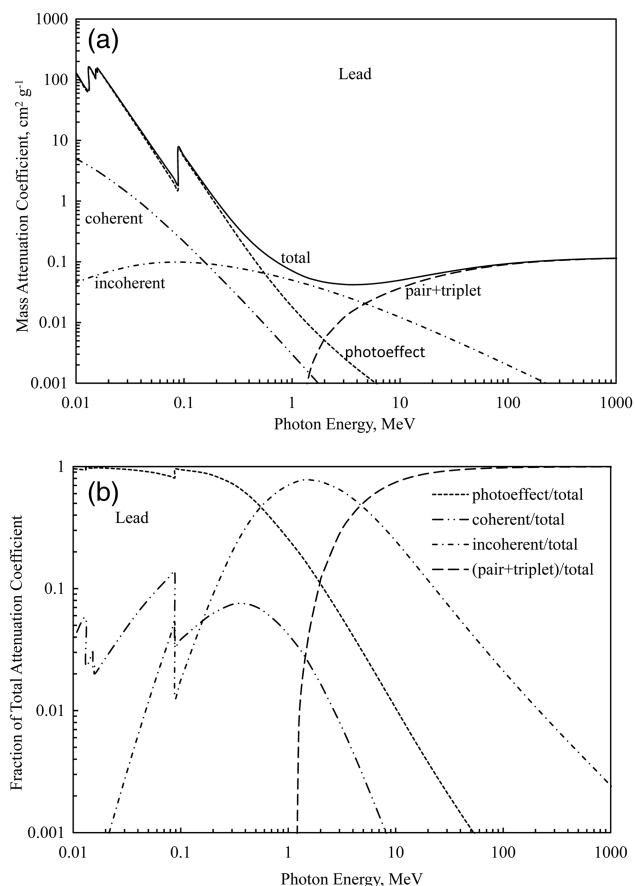


Figure 6.2. Mass attenuation coefficient for lead as a function of photon energy. Results taken from XCOM for illustration. (a) Partial and total attenuation coefficients. (b) Fractional contribution of the partials to the total attenuation coefficient.

dominating above about 150 MeV. The picture is similar for air and water, but different for higher- Z materials (see Fig. 6.2 for lead). In any case, above 1 keV, low energies are dominated by the photoeffect, coherent scattering is relatively small for all energies, incoherent scattering is relatively small below 10 keV, and pair (plus triplet) production dominates above some tens of MeV. Thus, the sometimes significant differences of partial cross sections derived from various approximations, *e.g.*, coherent scattering at all energies and incoherent scattering below 5 keV to 10 keV, have a negligible effect on the results of most dosimetric calculations.

Attenuation coefficients can be measured for the material in question. Measured attenuation coefficients²⁷ are compared with those from the various databases in Fig. 6.3 for air, in Fig. 6.4 for carbon

(graphite), and in Fig. 6.5 for water. These comparisons are quite busy (even though measured results for energies above a few MeV have been ignored) and largely inconclusive. A few points, however, can be made about the results from the various databases. The results from XCOM with renormalized Scofield photoeffect cross sections are quite close to those used in PENELOPE; there are relative differences in these materials of less than 0.8 % for energies from 1 keV to 2 keV due to differences in the photoeffect cross sections, presumably due to residual effects from the imposition of chosen edge energies (interpolation and extrapolation for XCOM; shifting for PENELOPE); there are relative differences between XCOM and PENELOPE in the incoherent-scattering cross section of as large as about 6 % for energies below about 50 keV. Corresponding relative differences of FFAST (Chantler *et al.*, 2005) are mostly within about from 3 % to 5 %. A calculation of the mean absolute relative difference of all the measured results from those from the databases suggest perhaps a slight preference for XCOM with renormalized Scofield photoeffect cross sections (and, hence, also PENELOPE), but the discrepancies shown in Figs. 6.3–6.5 indicate significant inconsistencies in the measured attenuation coefficients themselves, as much as from 10 % to 40 %.

There are two recent measurements (Kato *et al.*, 2010; Peele *et al.*, 2002) of the attenuation coefficient for low-energy (from 1 keV to 4 keV) monochromatized synchrotron radiation in air, shown in Fig. 6.3d with the older measured results in that energy region. The results of Peele *et al.* (2002) are not in agreement with any calculated data (although stated by them to favor FFAST), but the results of Kato *et al.* (2010), along with the older results shown, favor XCOM without the renormalization of the Scofield photoeffect cross sections.

Ali *et al.* (2015) report an analysis of the uncertainty of the XCOM (unrenormalized) attenuation coefficients at energies above about 100 keV for graphite and for lead based on recent transmission measurements with relative standard uncertainties of ~ 0.5 % for 10 MV to 30 MV bremsstrahlung beams from the linac at the National Research Council Canada, and based on comparison with measured data reported between 1950 and 1995 for energies from 100 keV to 25 MeV that are within 4 % of the XCOM values to which IAEA photonuclear cross sections (IAEA, 2000b; Varlamov *et al.*, 1999) have been added. For the case of graphite, of particular interest for this Report, their analysis essentially tests the accuracy of the XCOM incoherent-scattering plus photonuclear-absorption data

²⁷The collection of measured data used in this Report has been greatly facilitated by access to files, both written and electronic, of J. Hubbell; however, it is not complete, as some data have probably been missed and results for energies above a few MeV, which are dominated by pair (and triplet) production and which

can be affected by photonuclear absorption, have not been included.

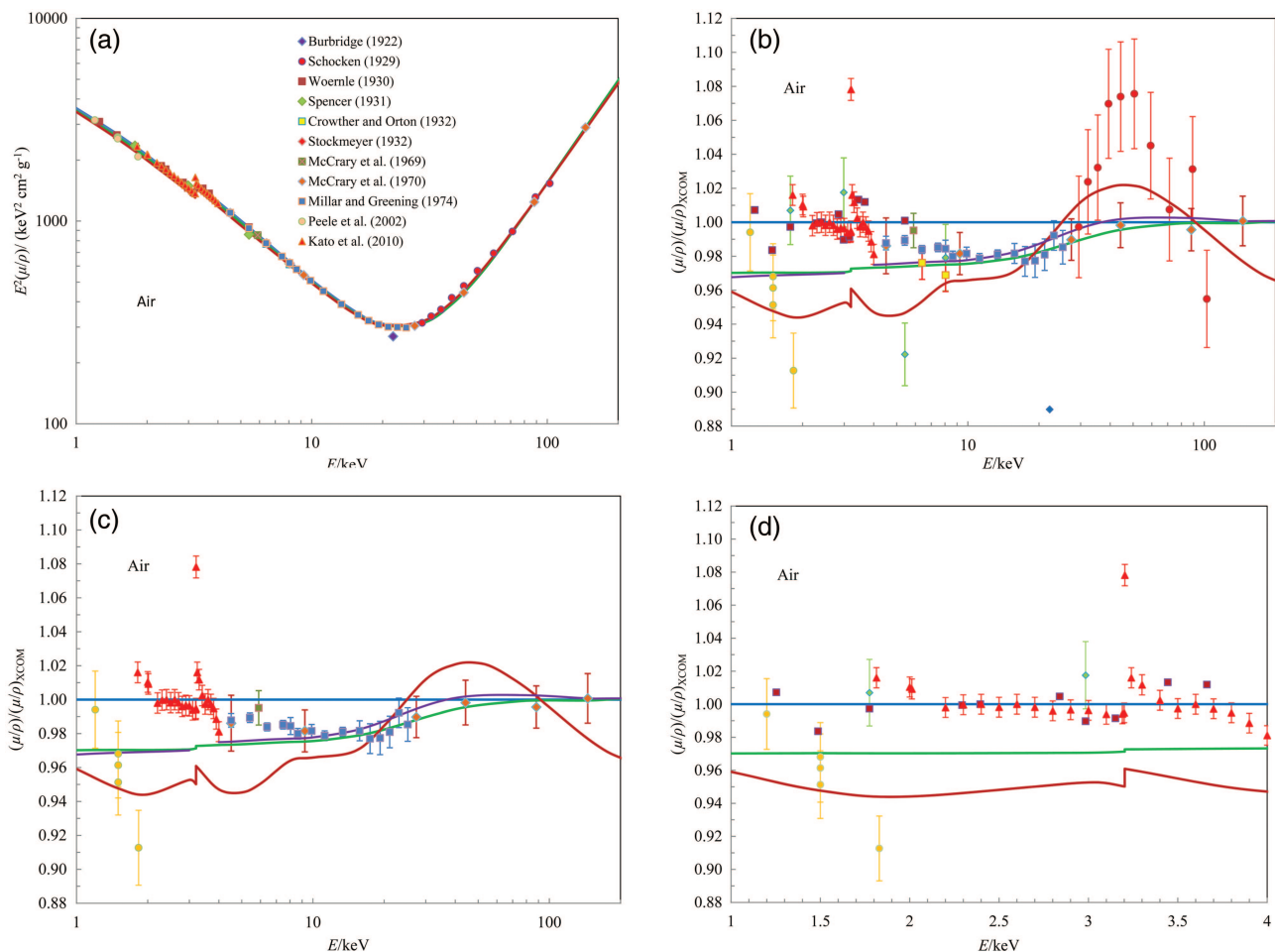


Figure 6.3. Comparison of total attenuation coefficients for air. (a) The measured data since 1922 are compared with XCOM results with unrenormalized (blue curve) and renormalized (green curve) Scofield photoeffect cross sections and with FFAST results of Chantler (red curve). Note that the ordinate is the square of the photon energy, E , times the attenuation coefficient. (b) The measured data since 1922 are shown in the ratio to XCOM results with unrenormalized Scofield photoeffect cross sections (blue line); ratios are given also for XCOM with renormalized Scofield photoeffect cross sections (green curve), for FFAST results from Chantler (red curve), and from PENELOPE (purple curve; ratios not shown near 3.2 keV due to mismatch of assumed K-edge energies). (c) The measured data since 1969 are shown in the ratio to XCOM results with unrenormalized Scofield photoeffect cross sections (blue line); ratios are given also for XCOM with renormalized Scofield photoeffect cross sections (green curve), for FFAST results from Chantler (red curve), and from PENELOPE (purple curve; ratios not shown near 3.2 keV due to mismatch of assumed K-edge energies). (d) The measured data from 1 keV to 4 keV are shown in the ratio to XCOM results with unrenormalized Scofield photoeffect cross sections (blue line); ratios are given also for XCOM with renormalized Scofield photoeffect cross sections (green curve) and for FFAST results from Chantler (red curve). The “outlier” point for Kato *et al.* (2010) at 3.204 keV, just below the Ar K-shell binding energy, is due to resonance excitation of Ar K-shell electrons (1s to 4p), a process not included in the calculations of the photoeffect.

(see Fig. 6.1b), for which they find a relative standard uncertainty of about 0.4 %.

It should be mentioned also that there are measurements of total attenuation coefficients by Chantler’s University of Melbourne group with the x-ray-extended-range technique in the energy region from about 5 keV to about 50 keV for Si (Tran *et al.*, 2003), Cu (Glover *et al.*, 2008), Zn (Rae *et al.*, 2010), Mo (De Jong *et al.*, 2005), Ag (Tran *et al.*, 2005), and Au (Islam *et al.*, 2010); these are stated to be of very high accuracy and have been made fairly recently, well after the

rather comprehensive work of Saloman and Hubbell (1986). The comparisons, not shown in this Report, indicate remarkably good agreement with the results from FFAST (note that the attenuation coefficients in these energy regions are dominated by the photoeffect).

Total attenuation coefficients for air, graphite (carbon), and water from the various databases are given in Table 6.5. Results from FFAST are not included in Table 6.5 as they extend only to a little above 400 keV and so are not suitable for all applications in the dosimetry of ionizing radiation.

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Photon-Interaction Coefficients

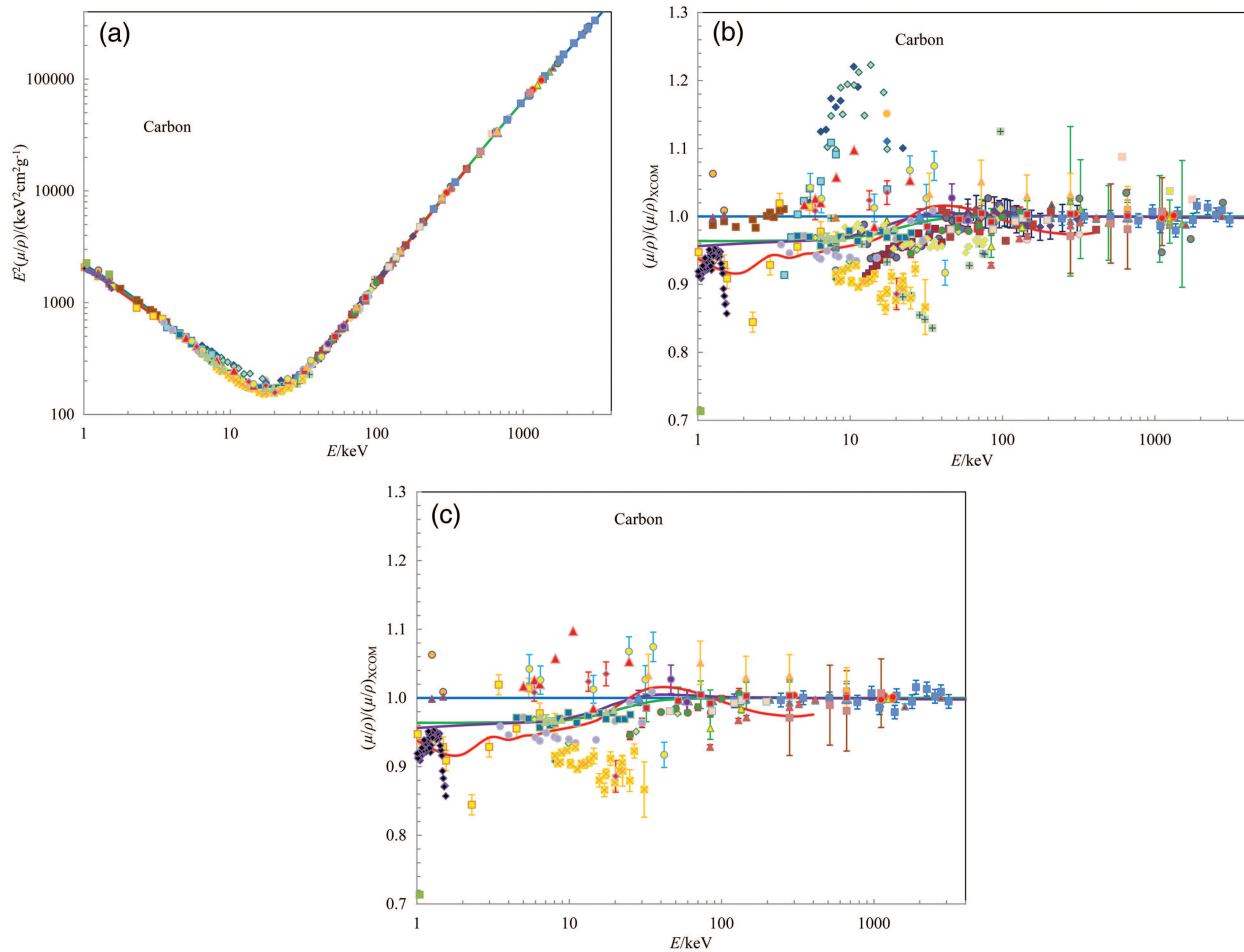


Figure 6.4. Comparison of total attenuation coefficients for carbon. (a) The measured data since 1909 are compared with XCOM results with unrenormalized (blue curve) and renormalized (green curve) Scofield photoeffect cross sections and with FFAST results from Chantler (red curve). Note that the ordinate is the square of the photon energy, E , times the attenuation coefficient. The points are from the measurements of:

- | | | |
|--------------------------------|--|---------------------------------------|
| ◆ Barkla and Sandler (1909) | ◆ Shimizu <i>et al.</i> (1952) | ● Goswami and Chodhuri (1971;1973) |
| ■ Kohlrausch (1917) | ● Chipman (1955) | ▲ Gopal and Sajeevaiah (1973a; 1973b) |
| ● Hewlett (1921) | ■ Batterman (1958) | ■ Millar and Greening (1974) |
| ▲ Taylor (1922) | ▲ Ergun and Tiensuu (1958) | ▲ Joyet <i>et al.</i> (1974) |
| ● Wingardh (1922) | ◆ Roof (1959) | ◆ Calvert <i>et al.</i> (1975) |
| ■ Olson <i>et al.</i> (1923) | ■ Lakshminarayana and Jnanannanda (1961) | ■ Murty <i>et al.</i> (1977) |
| ◆ Allen (1926) | ■ Wiedenbeck (1962) | ■ Berry and Lawrence (1979) |
| ■ Woernle (1930) | ■ Murty (1964) | ■ Puttaswamy <i>et al.</i> (1979) |
| ■ Grosskurth (1934) | ● Ogier <i>et al.</i> (1964) | ■ Rao <i>et al.</i> (1981) |
| ■ Hahn (1934) | ◆ Weissmantel and Wunschmann (1965) | ■ Umesh <i>et al.</i> (1981) |
| ▲ Mayneord and Roberts (1935) | ■ Bearden (1966) | ▲ Gerward (1983) |
| ■ Cuykendall (1936) | ▲ Henke <i>et al.</i> (1967) | ● Nathuram <i>et al.</i> (1984) |
| ■ Andrews (1938) | ● McCrary <i>et al.</i> (1967) | ■ Rao <i>et al.</i> (1984) |
| ◆ Wrede (1939) | ■ Weisweiler (1968) | ▲ Bradley <i>et al.</i> (1986) |
| ▲ Mayneord and Cipriani (1947) | ◆ Senemaud (1969) | ◆ Nathuram <i>et al.</i> (1988) |
| ● Cowan (1948) | ■ Delgrande <i>et al.</i> (1969) | |
| ◆ Colgate (1952) | ▲ Conner <i>et al.</i> (1970) | |
| ▲ Wyard (1952) | ■ Henry and Kennett (1971) | |

(b) The measured data since 1909 are shown in the ratio to XCOM results with unrenormalized Scofield photoeffect cross sections (blue line); ratios are given also for XCOM with renormalized Scofield photoeffect cross sections (green curve), for FFAST results from Chantler (red curve), and for PENELOPE results (purple curve). (c) The measured data since 1964 are shown in the ratio to XCOM results with unrenormalized Scofield photoeffect cross sections (blue line); ratios are given also for XCOM with renormalized Scofield photoeffect cross sections (green curve), for FFAST results from Chantler (red curve), and for PENELOPE results (purple curve).

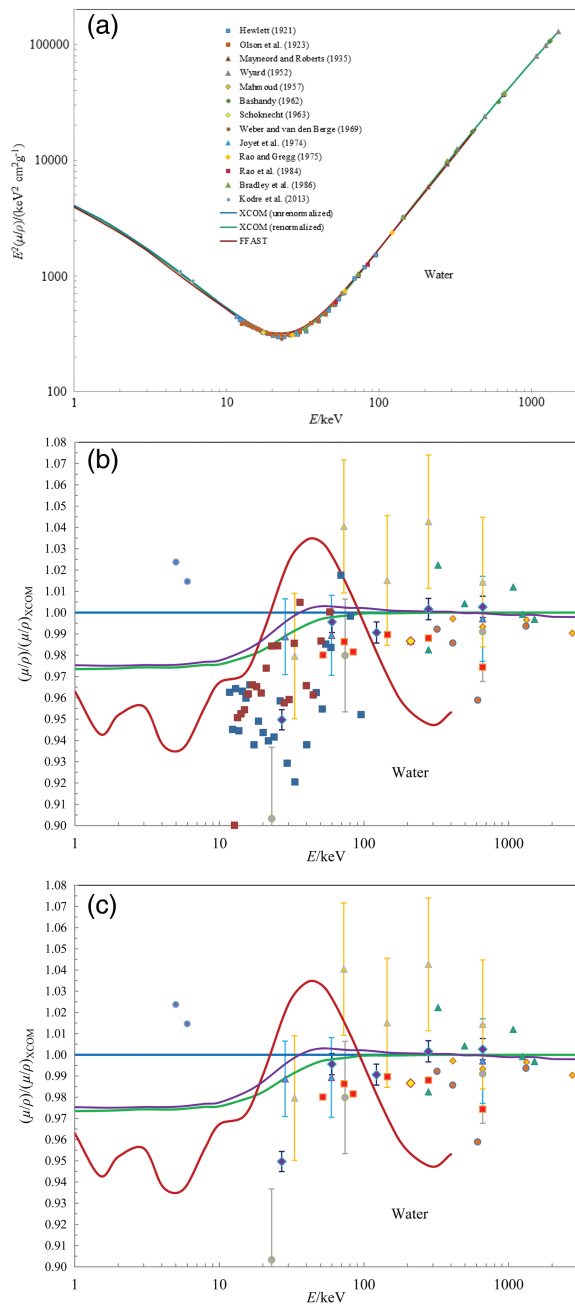


Figure 6.5. Comparison of total attenuation coefficients for water. (a) The measured data since 1921 are compared with XCOM results with unrenormalized (blue curve) and renormalized (green curve) Scofield photoeffect cross sections and with FFAST results from Chantler (red curve). Note that the ordinate is the square of the photon energy, E , times the attenuation coefficient. (b) The measured data since 1921 are shown in the ratio to XCOM results with unrenormalized Scofield photoeffect cross sections (blue line); ratios are given also for XCOM with renormalized Scofield photoeffect cross sections (green curve); for FFAST results from Chantler (red curve), and for PENELOPE results (purple curve). (c) The measured data since 1952 are shown in the ratio to XCOM results with unrenormalized Scofield photoeffect cross sections (blue line); ratios are given also for XCOM with renormalized Scofield photoeffect cross sections (green curve), for FFAST results from Chantler (red curve), and for PENELOPE results (purple curve).

Table 6.5a. Mass attenuation coefficients, (μ/ρ) , in units of $\text{cm}^2 \text{g}^{-1}$ for air, as a function of photon energy E . Values given are from XCOM, the XCOM cross-section database but with renormalized Scofield photoeffect cross sections, and from PENELOPE with their renormalized Scofield photoeffect cross sections and RIA Compton cross sections. Corresponding assumed values of the Ar K edge energy are given in Table 6.1a.

E/MeV	XCOM Unrenormalized	XCOM Renormalized	PENELOPE Renormalized
1.0E-03	3.606E+03	3.499E+03	3.489E+03
1.5E-03	1.191E+03	1.156E+03	1.154E+03
2.0E-03	5.280E+02	5.123E+02	5.117E+02
3.0E-03	1.625E+02	1.577E+02	1.576E+02
Ar K edge	1.339E+02	1.301E+02	1.295E+02
Ar K edge	1.485E+02	1.444E+02	1.415E+02
4.0E-03	7.787E+01	7.579E+01	7.592E+01
5.0E-03	4.027E+01	3.921E+01	3.928E+01
6.0E-03	2.340E+01	2.280E+01	2.284E+01
8.0E-03	9.918E+00	9.671E+00	9.690E+00
1.0E-02	5.120E+00	4.995E+00	5.006E+00
1.5E-02	1.614E+00	1.579E+00	1.585E+00
2.0E-02	7.779E-01	7.642E-01	7.680E-01
3.0E-02	3.538E-01	3.501E-01	3.526E-01
4.0E-02	2.485E-01	2.471E-01	2.488E-01
5.0E-02	2.080E-01	2.073E-01	2.085E-01
6.0E-02	1.875E-01	1.871E-01	1.880E-01
8.0E-02	1.662E-01	1.661E-01	1.666E-01
1.0E-01	1.541E-01	1.540E-01	1.544E-01
1.5E-01	1.356E-01	1.355E-01	1.357E-01
2.0E-01	1.233E-01	1.234E-01	1.234E-01
3.0E-01	1.067E-01	1.068E-01	1.067E-01
4.0E-01	9.549E-02	9.549E-02	9.547E-02
5.0E-01	8.712E-02	8.712E-02	8.710E-02
6.0E-01	8.055E-02	8.056E-02	8.053E-02
8.0E-01	7.074E-02	7.073E-02	7.069E-02
1.0E+00	6.358E-02	6.358E-02	6.353E-02
1.5E+00	5.175E-02	5.175E-02	5.170E-02
2.0E+00	4.447E-02	4.447E-02	4.440E-02
3.0E+00	3.581E-02	3.581E-02	3.573E-02
4.0E+00	3.079E-02	3.079E-02	3.072E-02
5.0E+00	2.751E-02	2.751E-02	2.744E-02
6.0E+00	2.522E-02	2.522E-02	2.516E-02
8.0E+00	2.225E-02	2.225E-02	2.220E-02
1.0E+01	2.045E-02	2.045E-02	2.039E-02
1.5E+01	1.810E-02	1.810E-02	1.805E-02
2.0E+01	1.706E-02	1.706E-02	1.701E-02
3.0E+01	1.628E-02	1.628E-02	1.624E-02
4.0E+01	1.611E-02	1.611E-02	1.607E-02
5.0E+01	1.614E-02	1.614E-02	1.610E-02
6.0E+01	1.625E-02	1.625E-02	1.622E-02
8.0E+01	1.654E-02	1.654E-02	1.651E-02
1.0E+02	1.683E-02	1.683E-02	1.681E-02
1.5E+02	1.746E-02	1.746E-02	1.744E-02
2.0E+02	1.793E-02	1.793E-02	1.791E-02
3.0E+02	1.855E-02	1.855E-02	1.853E-02
4.0E+02	1.895E-02	1.895E-02	1.893E-02
5.0E+02	1.923E-02	1.923E-02	1.921E-02
6.0E+02	1.943E-02	1.943E-02	1.941E-02
8.0E+02	1.971E-02	1.971E-02	1.970E-02
1.0E+03	1.991E-02	1.991E-02	1.989E-02

Photon-Interaction Coefficients

Table 6.5b. Mass attenuation coefficients, (μ/ρ), in units of $\text{cm}^2 \text{g}^{-1}$ for carbon (graphite) and water, as a function of photon energy E . Values given are from XCOM, the XCOM cross-section database but with renormalized Scofield photoeffect cross sections, and from PENELOPE with their renormalized Scofield photoeffect cross sections and RIA Compton cross sections. Corresponding assumed values of the Ar K edge energy are given in Table 6.1a.

E/MeV	Carbon		Graphite		Water	
	XCOM Unrenormalized	XCOM Renormalized	PENELOPE Renormalized	XCOM Unrenormalized	XCOM Renormalized	PENELOPE Renormalized
1.0E-03	2.211E+03	2.131E+03	2.114E+03	4.077E+03	3.969E+03	3.976E+03
1.5E-03	7.004E+02	6.750E+02	6.714E+02	1.376E+03	1.340E+03	1.342E+03
2.0E-03	3.026E+02	2.917E+02	2.906E+02	6.173E+02	6.012E+02	6.019E+02
3.0E-03	9.032E+01	8.709E+01	8.689E+01	1.928E+02	1.879E+02	1.881E+02
4.0E-03	3.778E+01	3.643E+01	3.639E+01	8.278E+01	8.065E+01	8.074E+01
5.0E-03	1.912E+01	1.844E+01	1.843E+01	4.259E+01	4.150E+01	4.155E+01
6.0E-03	1.095E+01	1.056E+01	1.056E+01	2.464E+01	2.402E+01	2.405E+01
8.0E-03	4.576E+00	4.422E+00	4.426E+00	1.037E+01	1.012E+01	1.013E+01
1.0E-02	2.373E+00	2.298E+00	2.302E+00	5.330E+00	5.201E+00	5.210E+00
1.5E-02	8.074E-01	7.868E-01	7.912E-01	1.672E+00	1.637E+00	1.642E+00
2.0E-02	4.420E-01	4.340E-01	4.377E-01	8.098E-01	7.956E-01	7.995E-01
3.0E-02	2.562E-01	2.541E-01	2.565E-01	3.756E-01	3.718E-01	3.744E-01
4.0E-02	2.076E-01	2.069E-01	2.085E-01	2.683E-01	2.668E-01	2.687E-01
5.0E-02	1.871E-01	1.868E-01	1.878E-01	2.269E-01	2.262E-01	2.276E-01
6.0E-02	1.753E-01	1.751E-01	1.759E-01	2.059E-01	2.055E-01	2.065E-01
8.0E-02	1.610E-01	1.609E-01	1.614E-01	1.837E-01	1.835E-01	1.841E-01
1.0E-01	1.514E-01	1.513E-01	1.516E-01	1.707E-01	1.707E-01	1.711E-01
1.5E-01	1.347E-01	1.346E-01	1.348E-01	1.505E-01	1.505E-01	1.507E-01
2.0E-01	1.229E-01	1.230E-01	1.230E-01	1.370E-01	1.370E-01	1.371E-01
3.0E-01	1.066E-01	1.067E-01	1.066E-01	1.186E-01	1.186E-01	1.187E-01
4.0E-01	9.547E-02	9.546E-02	9.543E-02	1.061E-01	1.061E-01	1.061E-01
5.0E-01	8.715E-02	8.715E-02	8.710E-02	9.687E-02	9.687E-02	9.685E-02
6.0E-01	8.058E-02	8.058E-02	8.054E-02	8.956E-02	8.956E-02	8.954E-02
8.0E-01	7.076E-02	7.076E-02	7.072E-02	7.866E-02	7.865E-02	7.861E-02
1.0E+00	6.362E-02	6.362E-02	6.356E-02	7.072E-02	7.072E-02	7.064E-02
1.5E+00	5.179E-02	5.179E-02	5.171E-02	5.754E-02	5.754E-02	5.748E-02
2.0E+00	4.443E-02	4.443E-02	4.435E-02	4.942E-02	4.942E-02	4.932E-02
3.0E+00	3.563E-02	3.563E-02	3.555E-02	3.969E-02	3.969E-02	3.961E-02
4.0E+00	3.047E-02	3.047E-02	3.040E-02	3.403E-02	3.403E-02	3.395E-02
5.0E+00	2.708E-02	2.708E-02	2.701E-02	3.031E-02	3.031E-02	3.024E-02
6.0E+00	2.469E-02	2.469E-02	2.462E-02	2.770E-02	2.770E-02	2.763E-02
8.0E+00	2.154E-02	2.154E-02	2.149E-02	2.429E-02	2.429E-02	2.423E-02
1.0E+01	1.959E-02	1.959E-02	1.954E-02	2.219E-02	2.219E-02	2.213E-02
1.5E+01	1.698E-02	1.698E-02	1.694E-02	1.941E-02	1.941E-02	1.936E-02
2.0E+01	1.575E-02	1.575E-02	1.571E-02	1.813E-02	1.813E-02	1.809E-02
3.0E+01	1.472E-02	1.472E-02	1.468E-02	1.710E-02	1.710E-02	1.706E-02
4.0E+01	1.437E-02	1.437E-02	1.433E-02	1.679E-02	1.679E-02	1.675E-02
5.0E+01	1.426E-02	1.426E-02	1.422E-02	1.674E-02	1.674E-02	1.670E-02
6.0E+01	1.426E-02	1.426E-02	1.423E-02	1.679E-02	1.679E-02	1.676E-02
8.0E+01	1.439E-02	1.439E-02	1.436E-02	1.702E-02	1.701E-02	1.698E-02
1.0E+02	1.457E-02	1.457E-02	1.454E-02	1.727E-02	1.727E-02	1.724E-02
1.5E+02	1.501E-02	1.501E-02	1.499E-02	1.785E-02	1.785E-02	1.783E-02
2.0E+02	1.536E-02	1.536E-02	1.534E-02	1.829E-02	1.829E-02	1.826E-02
3.0E+02	1.586E-02	1.586E-02	1.584E-02	1.888E-02	1.888E-02	1.886E-02
4.0E+02	1.619E-02	1.619E-02	1.617E-02	1.927E-02	1.927E-02	1.925E-02
5.0E+02	1.643E-02	1.643E-02	1.641E-02	1.954E-02	1.954E-02	1.951E-02
6.0E+02	1.661E-02	1.661E-02	1.659E-02	1.974E-02	1.974E-02	1.972E-02
8.0E+02	1.685E-02	1.685E-02	1.684E-02	2.002E-02	2.001E-02	2.000E-02
1.0E+03	1.702E-02	1.702E-02	1.701E-02	2.020E-02	2.020E-02	2.019E-02

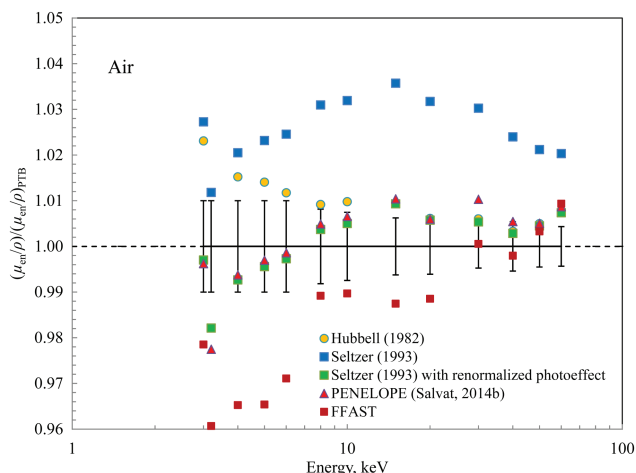


Figure 6.6. Ratios of calculated values of the mass energy-absorption coefficient, μ_{en}/ρ , of air to those measured by the group at the PTB, $(\mu_{\text{en}}/\rho)_{\text{PTB}}$. The relative standard uncertainties of the PTB data are shown for the line at unity, extending from 3 keV to 60 keV. The results of Hubbell (1982) were guided by (see text), and those of the “renormalized” Seltzer (1993) were calculated with, renormalized values of Scofield’s photoeffect cross sections; those from Salvat (2015b) used renormalized photoeffect cross sections from his new, similar calculations of photoeffect cross sections. The results from the various databases are plotted only at energies on the so-called standard grid.

6.6 Mass Energy-Transfer and Energy-Absorption Coefficients

For the photoelectric-absorption, incoherent-scattering, and pair- and triplet-production cross sections, one needs to evaluate the fractions f_J of energy transferred to charged particles and the fraction g of charged-particle kinetic energies subsequently lost in radiative processes in order to calculate the mass energy-transfer and mass energy-absorption coefficients. Coherent scattering is neglected as there is, by definition, no energy loss. These fractions have been considered by many authors over the years at various levels of refinement that have taken into account the available information. More discussion of these evaluations can be found in the work of Seltzer (1993), which used the XCOM cross-section database to produce online tables (<http://www.nist.gov/pml/data/xraycoef/index.cfm>) for the elements from hydrogen to uranium ($Z = 1-92$) and for a list of compounds and mixtures, for photon energies between 1 keV and 20 MeV.²⁸ In the PENELOPE system (Salvat, 2015b), the values of f_J and g values are obtained from Monte Carlo simulations up to energies of 1 GeV. These simulations to generate tables of values can require substantial computation times;

²⁸The code used for these tables can handle any material and energies from 1 keV to 100 MeV (and higher, with some modification).

they include consistently all interactions and are applicable to any material. Slight systematic differences with the NIST values are attributable to the different cross-section models employed in the two calculations.

An important set of results became available from the measurements of Büermann *et al.* (2006), which were expanded and corrected by Buhr *et al.* (2012). This group, from the Physikalisch-Technische Bundesanstalt (PTB), measured the mass energy-absorption coefficient of air with relative standard uncertainties less than 1 % for x-ray energies from 3 keV to 60 keV using monochromatized synchrotron radiation. A comparison of these results with those from pertinent databases is shown in Fig. 6.6. Whereas using the unrenormalized Scofield photoeffect cross sections as done by Seltzer (1993) gives values about from 2 % to 3 % too large, the use of renormalized Scofield photoeffect cross sections clearly provides better agreement with the measured data. The earlier results of Hubbell (1982), which nominally used the renormalized photoeffect cross sections of Scofield, veer off somewhat below 8 keV because of empirical adjustments he made at that time in the low-energy photoeffect cross sections for C, N, O, and Ar (see also, Hubbell, 1977). The results of “renormalized” Seltzer (1993) and of PENELOPE (Salvat, 2015b) are both based on renormalized photoeffect cross sections and agree with the measured data largely²⁹ to within the stated measurement uncertainties; the slight systematic differences between these two calculations are attributable to the different cross-section models employed.

It was possible to include in Fig. 6.6 comparable results from the use of the Chantler *et al.* (2005) FFAST database, but only within some limitations. Because he does not address the mass energy-absorption coefficient and his Compton (incoherent)-scattering cross sections are not given separately from his Rayleigh (coherent)-scattering cross sections, the following was done. As suggested by Chantler (2000), the incoherent-scattering cross sections from the similar approach of XCOM (Hubbell *et al.*, 1975) were used instead and combined with the photoeffect cross sections from FFAST; the energy-transfer fractions, f , and the radiative-loss fractions, g , of Eqs. (2.2) and (2.3) were taken from the work of Seltzer (1993) and applied to the component cross sections. Note that the

²⁹Ratios near the K-shell absorption-edge energy are particularly sensitive to the assumed values of that energy. It should be noted that the photoeffect cross sections in all these databases neglect the excitation of argon atoms to bound states, which might contribute to the noticeable dip in the ratios shown in Fig. 6.6 at and above the K-edge energy of about 3.2 keV. Such an effect can be expected for all gases, because these photoeffect databases are limited to ionization.

Table 6.6a. Mass energy-absorption coefficients, (μ_{en}/ρ), in units of $\text{cm}^2 \text{g}^{-1}$ for air, as a function of photon energy E . Values given are from Seltzer (1993) using the XCOM cross-section database, from the same calculation but with renormalized Scofield photoeffect cross sections, and from PENELOPE with its renormalized Scofield photoeffect cross sections and RIA Compton cross sections. Corresponding assumed values of the Ar K edge energy are given in Table 6.1a. Some of the small differences in the values in the last two columns are within the statistical fluctuations in the PENELOPE results.

E/MeV	XCOM Unrenormalized	XCOM Renormalized	PENELOPE Renormalized
0.0010	3.599E+03	3.491E+03	3.482E+03
0.0015	1.188E+03	1.153E+03	1.151E+03
0.0020	5.262E+02	5.107E+02	5.101E+02
0.0030	1.614E+02	1.567E+02	1.566E+02
Ar K edge	1.330E+02	1.291E+02	1.285E+02
Ar K edge	1.460E+02	1.420E+02	1.393E+02
0.0040	7.636E+01	7.430E+01	7.436E+01
0.0050	3.931E+01	3.826E+01	3.830E+01
0.0060	2.270E+01	2.210E+01	2.213E+01
0.0080	9.446E+00	9.197E+00	9.207E+00
0.010	4.742E+00	4.618E+00	4.625E+00
0.015	1.334E+00	1.300E+00	1.301E+00
0.020	5.389E-01	5.253E-01	5.254E-01
0.030	1.537E-01	1.501E-01	1.507E-01
0.040	6.833E-02	6.690E-02	6.708E-02
0.050	4.098E-02	4.029E-02	4.031E-02
0.060	3.041E-02	3.004E-02	3.007E-02
0.080	2.407E-02	2.392E-02	2.391E-02
0.10	2.325E-02	2.317E-02	2.320E-02
0.15	2.496E-02	2.493E-02	2.491E-02
0.20	2.672E-02	2.672E-02	2.668E-02
0.30	2.872E-02	2.874E-02	2.865E-02
0.40	2.949E-02	2.949E-02	2.948E-02
0.50	2.966E-02	2.966E-02	2.959E-02
0.60	2.953E-02	2.953E-02	2.950E-02
0.80	2.882E-02	2.882E-02	2.883E-02
1.0	2.789E-02	2.789E-02	2.783E-02
1.5	2.547E-02	2.547E-02	2.547E-02
2.0	2.345E-02	2.345E-02	2.344E-02
3.0	2.057E-02	2.057E-02	2.057E-02
4.0	1.870E-02	1.870E-02	1.862E-02
5.0	1.740E-02	1.740E-02	1.735E-02
6.0	1.647E-02	1.647E-02	1.643E-02
8.0	1.525E-02	1.525E-02	1.525E-02
10	1.450E-02	1.450E-02	1.449E-02
15	1.353E-02	1.353E-02	1.351E-02
20	1.311E-02	1.311E-02	1.310E-02
30	1.277E-02	1.277E-02	1.275E-02
40	1.262E-02	1.262E-02	1.264E-02
50	1.252E-02	1.252E-02	1.249E-02
60	1.242E-02	1.242E-02	1.239E-02
80	1.220E-02	1.220E-02	1.220E-02
100	1.195E-02	1.195E-02	1.199E-02

photoeffect strongly dominates the calculation of μ_{en}/ρ in this energy region, contributing from about 97 % to 99 % of the total attenuation coefficient; the values of f are within a fraction of 1 % of unity, and the value of $1-g$ is virtually unity at these energies. Thus, very little bias should be introduced, certainly less than

1 %. Based on these assumptions, the data from FFAST appear too low at energies below about 8 keV.

The comparison of Fig. 6.6 clearly favors the use of renormalized Scofield photoeffect cross sections. However, taking into account the measured results shown in Fig. 6.3d for the same medium confuses a possible conclusion. Thus, while theoretical considerations might also favor the use of renormalized Scofield photoeffect cross sections, it does not seem prudent to make a formal recommendation in this equivocal situation. Rather, values from the available databases are given in Table 6.6.

6.7 Uncertainties of Photon-Interaction Coefficients

Both the theory and the measurement of photon cross sections have advanced, and have reached a level of agreement of from about 1 % to perhaps about 10 %. However, below that level, both are still too inaccurate and inconsistent to make an unambiguous choice among the various available data sets. In view of the situation indicated in Figs. 6.3–6.6, and taking into account the large amount of work done in the past, it is still difficult to provide some reasonable estimates of the uncertainties of the component cross sections that comprise the attenuation coefficients of arbitrary materials or even of the three materials of primary concern in this Report. Guided by the differences in results from calculations done in somewhat different approximations, and concentrating on the energy intervals (above 1 keV) in which the component cross sections are dominant and/or significant, the following estimates are offered for air, graphite, and water, corresponding to a relative standard deviation. For the photoeffect, well away from edge energies, a variation of the estimates suggested by Cullen *et al.* (1989) seems appropriate: 1 keV to 100 keV, ~ 2 % to 3 %; higher energies, ~ 2 % to 5 %. For coherent scattering, the estimate is ~ 0.5 % to 1 % for energies above about 10 keV, growing to about 10 % as the energy decreases to 1 keV. For incoherent scattering, the estimate is about 0.5 % for energies greater than about 50 keV, increasing to perhaps 2 % at 10 keV.³⁰ For pair-plus-triplet production, we have done no new analysis, so the estimate of Gimm and Hubbell (1978) would seem appropriate: about 0.2 % for light elements and about 0.4 % for heavy elements (Pb) for energies above about 10 MeV.

³⁰At lower energies, differences among results shown in Table 6.3b from credible models for carbon/graphite reach a factor of more than 2.5 at 1 keV, but the total interaction cross section at that energy is dominated (by more than three orders of magnitude) by the photoeffect.

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Table 6.6b. Mass energy-absorption coefficients, (μ_{en}/ρ) , in units of $\text{cm}^2 \text{g}^{-1}$ for carbon (graphite) and for water, as a function of photon energy E . Values given are from Seltzer (1993) using the XCOM cross-section database, from the same calculation XCOM but with renormalized Scofield photoeffect cross sections, and from PENELOPE with their renormalized Scofield photoeffect cross sections and RIA Compton cross sections. Corresponding assumed values of the Ar K edge energy are given in Table 6.1a. Some of the small differences in the values in the last two columns are within the statistical fluctuations in the PENELOPE results.

E/MeV	Carbon		Graphite	Water		
	XCOM Unrenormalized	XCOM Renormalized	PENELOPE Renormalized	XCOM Unrenormalized	XCOM Renormalized	PENELOPE Renormalized
0.0010	2.209E+03	2.129E+03	2.112E+03	4.065E+03	3.956E+03	3.963E+03
0.0015	6.990E+02	6.737E+02	6.701E+02	1.372E+03	1.336E+03	1.338E+03
0.0020	3.016E+02	2.907E+02	2.896E+02	6.152E+02	5.991E+02	5.998E+02
0.0030	8.963E+01	8.638E+01	8.615E+01	1.917E+02	1.867E+02	1.868E+02
0.0040	3.723E+01	3.588E+01	3.581E+01	8.191E+01	7.978E+01	7.983E+01
0.0050	1.866E+01	1.798E+01	1.795E+01	4.188E+01	4.079E+01	4.083E+01
0.0060	1.054E+01	1.016E+01	1.015E+01	2.405E+01	2.343E+01	2.345E+01
0.0080	4.243E+00	4.088E+00	4.086E+00	9.915E+00	9.658E+00	9.669E+00
0.010	2.078E+00	2.003E+00	2.002E+00	4.944E+00	4.816E+00	4.819E+00
0.015	5.627E-01	5.424E-01	5.428E-01	1.374E+00	1.338E+00	1.341E+00
0.020	2.238E-01	2.159E-01	2.162E-01	5.503E-01	5.363E-01	5.372E-01
0.030	6.614E-02	6.408E-02	6.426E-02	1.557E-01	1.519E-01	1.522E-01
0.040	3.343E-02	3.264E-02	3.271E-02	6.947E-02	6.798E-02	6.824E-02
0.050	2.397E-02	2.359E-02	2.361E-02	4.223E-02	4.152E-02	4.164E-02
0.060	2.098E-02	2.077E-02	2.077E-02	3.190E-02	3.150E-02	3.147E-02
0.080	2.037E-02	2.029E-02	2.026E-02	2.597E-02	2.582E-02	2.581E-02
0.10	2.147E-02	2.144E-02	2.141E-02	2.546E-02	2.538E-02	2.535E-02
0.15	2.449E-02	2.448E-02	2.446E-02	2.764E-02	2.761E-02	2.763E-02
0.20	2.655E-02	2.655E-02	2.653E-02	2.967E-02	2.965E-02	2.965E-02
0.30	2.870E-02	2.869E-02	2.868E-02	3.192E-02	3.191E-02	3.188E-02
0.40	2.950E-02	2.950E-02	2.948E-02	3.279E-02	3.280E-02	3.273E-02
0.50	2.969E-02	2.969E-02	2.967E-02	3.299E-02	3.299E-02	3.301E-02
0.60	2.956E-02	2.955E-02	2.954E-02	3.284E-02	3.284E-02	3.277E-02
0.80	2.885E-02	2.885E-02	2.883E-02	3.206E-02	3.206E-02	3.203E-02
1.0	2.792E-02	2.792E-02	2.791E-02	3.103E-02	3.103E-02	3.101E-02
1.5	2.551E-02	2.551E-02	2.548E-02	2.833E-02	2.833E-02	2.830E-02
2.0	2.345E-02	2.345E-02	2.342E-02	2.608E-02	2.608E-02	2.607E-02
3.0	2.048E-02	2.048E-02	2.044E-02	2.281E-02	2.281E-02	2.277E-02
4.0	1.849E-02	1.849E-02	1.846E-02	2.066E-02	2.066E-02	2.060E-02
5.0	1.709E-02	1.709E-02	1.705E-02	1.915E-02	1.915E-02	1.911E-02
6.0	1.607E-02	1.607E-02	1.603E-02	1.806E-02	1.806E-02	1.804E-02
8.0	1.468E-02	1.468E-02	1.465E-02	1.658E-02	1.658E-02	1.659E-02
10	1.379E-02	1.379E-02	1.374E-02	1.566E-02	1.566E-02	1.568E-02
15	1.257E-02	1.257E-02	1.255E-02	1.441E-02	1.441E-02	1.442E-02
20	1.197E-02	1.197E-02	1.194E-02	1.382E-02	1.382E-02	1.381E-02
30	1.141E-02	1.141E-02	1.140E-02	1.327E-02	1.327E-02	1.327E-02
40	1.112E-02	1.112E-02	1.113E-02	1.298E-02	1.298E-02	1.299E-02
50	1.091E-02	1.091E-02	1.089E-02	1.279E-02	1.279E-02	1.272E-02
60	1.075E-02	1.075E-02	1.074E-02	1.261E-02	1.261E-02	1.260E-02
80	1.044E-02	1.044E-02	1.045E-02	1.228E-02	1.228E-02	1.232E-02
100	1.014E-02	1.014E-02	1.012E-02	1.194E-02	1.194E-02	1.194E-02

However, it should be pointed out that the key photon-interaction data of concern in this Report is [see Eq. (3.7)] the quantity $(\mu_{\text{en}}/\rho)_{\text{air,g}}/(1 - \bar{g}_{\text{air}})$. The evaluation of this quantity is invariably done with cross sections for both air and graphite derived consistently in the same approximation, which are then rather highly correlated. Seltzer and Bergstrom³¹ analyzed the ratio of the mean values

of (μ_{en}/ρ) for air and graphite for the photon spectra of ^{60}Co , ^{137}Cs , and ^{192}Ir , all of which are dominated by incoherent scattering. They evaluated the incoherent-scattering cross sections consistently in

³¹Presentation at the Workshop on Uncertainties, 17 May 2005, Consultative Committee on Ionizing Radiation Section I [CCRI(I)], Bureau de Poids et Mesures (BIPM), Paris, ICRU 2016 -- All rights reserved.

different approximations of increasing realism [bare Klein–Nishina, Eq. (6.4); incoherent-scattering function times Klein–Nishina, Eq. (6.7); and the RIA (numerical integrations of the cross sections differential in angle and energy)] and assigned the relative differences to a relative standard uncertainty of 0.04 % for the spectrum of a ^{60}Co source, 0.05 % for a ^{137}Cs source, and 0.1 % for an ^{192}Ir source. Assuming a 3 % relative uncertainty in the bremsstrahlung yield, the estimated relative standard uncertainty of $1/(1 - \bar{g}_{\text{air}})$ is about 0.015 %, 0.01 %, and 0.006 % for ^{60}Co , ^{137}Cs , and ^{192}Ir sources, respectively. Independently, and using a

somewhat different approach, Andreo *et al.* (2012) considered (μ_{en}/ρ) values and their ratios for graphite, air, and water, for spectra typical of those used in mammography, in kilovoltage radiography, and those of ^{192}Ir and ^{60}Co sources. Their conclusion was a relative standard uncertainty for $(\mu_{\text{en}}/\rho)_{\text{air,g}}$ of 0.1 % for the γ -ray sources, increasing to 1.5 % for the mammography spectra. Their result of 0.1 % for ^{192}Ir and ^{60}Co sources compared with 0.1 % and 0.04 %, respectively, of Seltzer and Bergstrom (2005) suggests a conservative estimate of 0.1 % for the relative standard uncertainty of $(\mu_{\text{en}}/\rho)_{\text{air,g}}/(1 - \bar{g}_{\text{air}})$ for the γ -ray sources.

7. Impact of Recommendations

This Report has reviewed relevant data and has recommended values of quantities that play an important role in radiation dosimetry, especially those needed in measurement standards. This Section considers some of the implications of the recommended changes for dosimetric measurements and on calculations made in the fields of radiation medicine, industry, and other applications, including radiation research.

Recommended values and uncertainties are given for W_{air} , the average energy required to produce an ion pair, the heat defect of liquid water, h_{w} , and the radiation chemical yield for the Fricke dosimeter, $G(\text{Fe}^{3+})$. A new value is also recommended for the product, $W_{\text{air}} s_{\text{g,air}}$, for ^{60}Co γ rays. The humidity correction, k_{h} , for air-filled ionization chambers is reviewed, but no changes are recommended. However, it is noted that, for high precision, the variation of k_{h} with relative humidity or, more properly, with the partial pressure of water vapor, should be considered. Data for the heat defect of graphite are reviewed, but no definitive conclusions could be reached and more study is recommended.

The value of W_{air} for electrons is left unchanged at 33.97 eV, but its standard uncertainty has been increased from 0.05 eV (0.15 %) to 0.12 eV (0.35 %). This will have an impact on the uncertainty of air-kerma standards based on free-air chambers and will for many standards become the dominant component. The available data for W_{air} indicate that it can be considered constant at high energies. However, for electron energies below about 10 keV, W_{air} cannot be considered constant. Furthermore, as was pointed out in Section 5.5, when the air kerma is obtained from a charge measurement, a correction should be applied for the charge of the initial electrons set in motion by the photons. The combined correction for these last two effects (see Table 5.7) can be significant for low-energy photons (up to 0.7 %) and could give rise to changes in primary standards.

Recommendations have been made for the mean excitation energies for air, graphite, and liquid water as well as for the graphite density to use when evaluating the density effect (2.265 g cm^{-3}). From these

recommendations, tables of the stopping powers for electrons, protons, alpha particles, and carbon ions have been generated (see the Appendix). For air, no change in the value of the mean excitation energy is recommended, *i.e.*, $I_{\text{air}} = 85.7 \text{ eV}$ but now with an uncertainty of 1.2 eV (1.4 %); stopping power values for all particles thus remain unaltered, except for carbon ions, for which an I_{air} value of 82.8 eV was implicitly used in ICRU Report 73 (2005). For electrons in graphite, the change in the electronic stopping powers relative to those given in ICRU Report 37 (1987a) is shown in Fig. 7.1. The value of I_{g} has increased from 78 eV to 81 eV, and the standard uncertainty has decreased from 4 eV to 1.8 eV. The increase in the mean excitation energy and the change in the density used to evaluate the density-effect correction both result in a decrease in the electronic stopping power. For the secondary electrons produced by ^{60}Co γ rays, the electronic stopping power in graphite decreases by about 0.7 %, while for high-energy electrons, the decrease is more than 1 %. For liquid water, there is a 4 % relative increase in I_{w} , from 75 eV, as used in ICRU Report 37, to 78 eV, with a relative standard uncertainty of 2.6 %, which also results in a decrease in the electronic-stopping-power values.

For protons and carbon ions, the change in electronic stopping powers relative to the values given in previous ICRU Reports are shown in Figs. 7.2 and 7.3, respectively, which, in addition to the changes in I values and densities mentioned above, are based on improved calculations using the Bethe–Bloch expression for S_{el}/ρ , see Eqs. (4.17) and (4.18), and are complemented with experimental data at low energies as described in the Appendix.

For photons, following the analysis of photoeffect cross sections with regard to the use of renormalized values and of the two options for determining Compton cross sections (impulse approximation *versus* Waller–Hartree theory), tables of mass energy-absorption coefficients for air, graphite, and water have been given. No recommendations on the choice of these options are given in this Report, but some discussion is included on the effects of considering them. Ratios of the renormalized-to-unrenormalized mass energy-

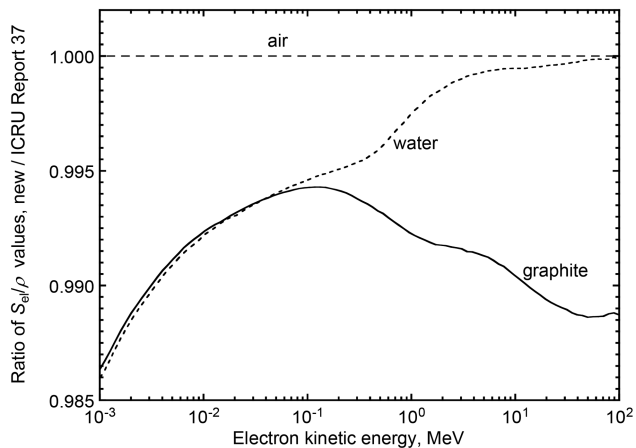


Figure 7.1. Ratios of the mass electronic stopping power for electrons recommended in this Report to those given in ICRU Report 37 (1984a), as a function of the electron energy, for graphite (solid line), liquid water (short dashes), and air (long dashes).

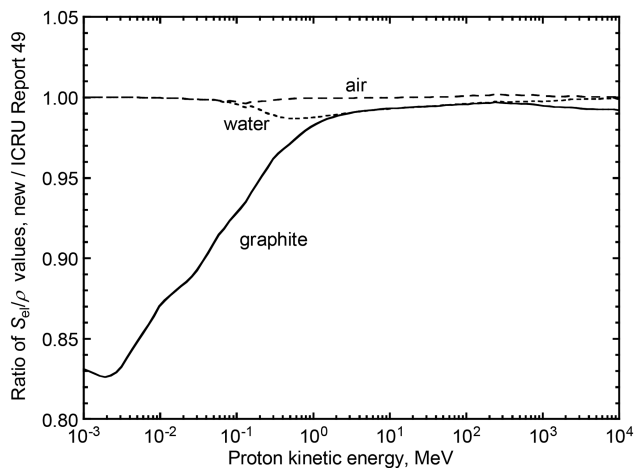


Figure 7.2. Ratios of the mass electronic stopping power for protons recommended in this Report to those given in ICRU Report 49 (1993), as a function of the proton kinetic energy, for graphite (solid line), liquid water (short dashes), and air (long dashes).

absorption coefficients for graphite, water, and air, as well as for air-to-graphite and water-to-air for the widely used NIST values are shown in Fig. 7.4, where for the latter group, relative differences of up to about 1 % can be seen for energies below about 100 keV. With these changes, the fraction g of the photon energy transferred to charged particles and subsequently lost on average in radiative processes remains unaltered.

Table 7.1 summarizes the recommended new key data and the resulting changes compared with the previously recommended data. Changes in stopping powers and mass energy-absorption coefficients are given in the figures above.

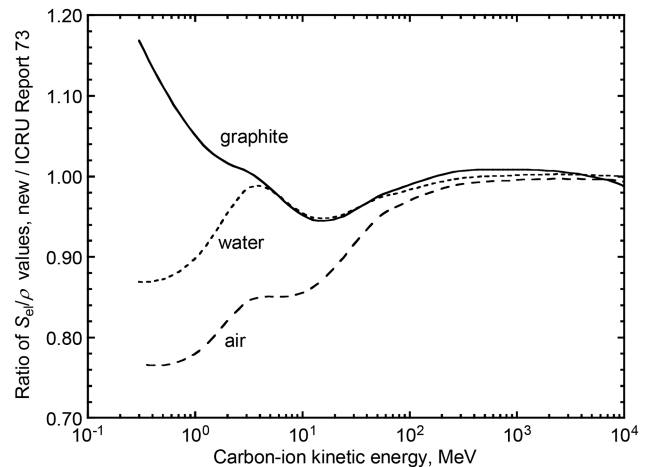


Figure 7.3. Ratios of the mass electronic stopping power for carbon ions recommended in this Report to those given in ICRU Report 73 (2005) for graphite and air, and given in its Errata and Addenda (Sigmund *et al.*, 2009) for water, as a function of the carbon-ion kinetic energy, for graphite (solid line), liquid water (short dashes), and air (long dashes).

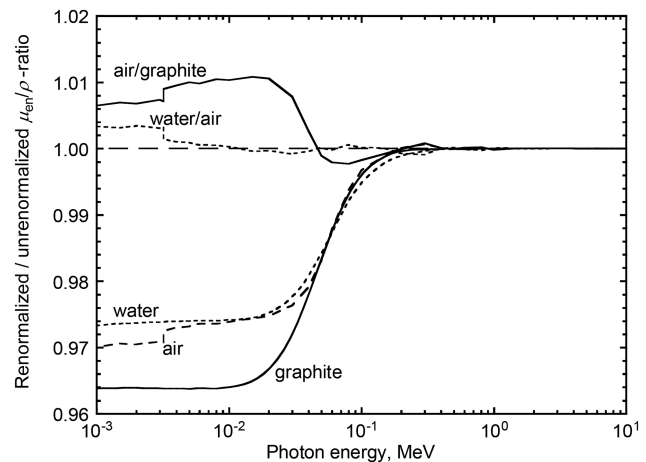


Figure 7.4. Ratios of mass energy-absorption coefficients calculated with renormalized and unrenormalized photoeffect cross sections for graphite (lower solid line), water (lower short dashes), and air (lower long dashes), and for air-to-graphite (upper solid line) and water-to-air (upper short dashes) for the widely used NIST XCOM values. No recommendations regarding the choice of these options are given in this Report.

7.1 Impact on Measurement Standards

The measurement equations at standards laboratories have been given in Section 3 for air-kerma standards and ionometric absorbed-dose-to-water standards, which will be used for analyzing the impact of the values recommended in this Report.

Air-kerma standards based on free-air chambers depend on W_{air} and $\bar{g}_{air}$, see Eq. (3.1). The values of these quantities are not changed, but the relative standard uncertainty of W_{air} has increased from 0.15 % to 0.35 %. This will lead to an increase in the

Table 7.1. Comparison of the values of the key data recommended in this Report with previously recommended values. The data listed include W_{air} , the mean energy expended in air per ion pair formed, h_w , the heat defect in water, $G(\text{Fe}^{3+})$, the radiation chemical yield for Fricke dosimetry, and I , the mean excitation energy given here for air, graphite, and liquid water.

Quantity	Previous	This Report	Relative standard uncertainty (%)	Relative change (%)	Comments
W_{air} for electrons ^a	33.97 eV	33.97 eV	0.35	0	
W_{air} for protons ^a	34.23 eV	34.44 eV	0.4	0.6	See Section 5.4.4
W_{air} for carbon ions ^a	34.50 eV	34.71 eV	1.5	0.6	See Section 5.4.4
h_w (4°C)	0	0	0.15	0	Low-LET radiations
$G(\text{Fe}^{3+})$					See Table 5.8
I_{air}	85.7 eV	85.7 eV	1.40	0	
I_g	78 eV	81 eV	2.2	3.8	See footnote b, below
I_w	75 eV	78 eV	2.6	4.0	

^aHigh-energy (assumed asymptotic) value.

^bThis Report recommends the use of $\rho = 2.265 \text{ g cm}^{-3}$ in the evaluation of the density-effect correction to the electronic stopping power of charged particles in graphite.

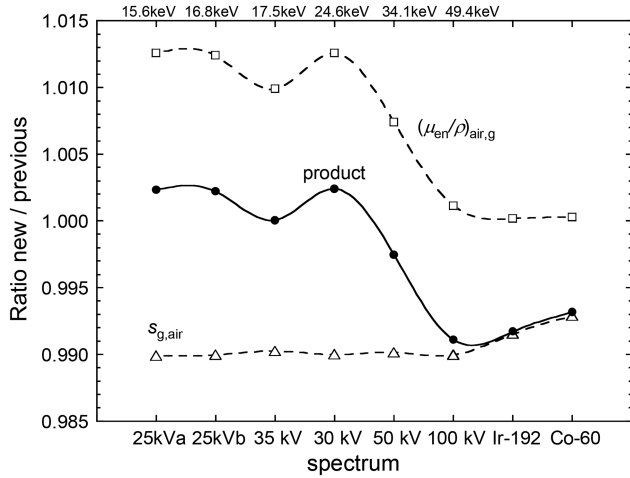


Figure 7.5. Ratios of the air-to-graphite mass energy-absorption coefficients (upper dashed line, squares) using the renormalized (termed new) versus unrenormalized (termed previous) Scofield photoeffect cross sections in NIST XCOM data of Table 6.5a–6.5c, of the graphite-to-air stopping powers using the data recommended in this Report and previous data (lower dashed line, triangles), and of their product (solid line, dots), for kilovoltage x rays, ^{192}Ir , and ^{60}Co spectra. The upper abscissa shows the fluence-weighted mean energy of the kV spectra. (See Andreo *et al.*, 2012, for details on the spectra.)

uncertainty of the measured air kerma for low- and medium-energy x-ray beams. New correction factors, k_{ii} and k_w , should now be included in Eq. (3.1), and for practical dosimetry, they lead to relative changes of up to 0.5 % in the measured air kerma (for synchrotron work using free-air chambers, the effect can be up to 1 %, as indicated in Table 5.7). As illustrated in Fig. 5.8, the product $k_{\text{ii}}k_w$ is largest for low-energy photon beams, but causes a relative change of about 0.2 % even at 100 keV.

Air-kerma standards based on graphite-walled cavity-ionization chambers used for ^{60}Co γ -ray

beams depend on W_{air} , $\bar{g}_{\text{air}}$, $s_{g,\text{air}}$ and $(\mu_{\text{en}}/\rho)_{\text{air,g}}$, see Eq. (3.7). As above, no changes are recommended for W_{air} and $\bar{g}_{\text{air}}$ (although the uncertainty of W_{air} increases). The estimated relative change in $s_{g,\text{air}}$ for a ^{60}Co spectrum is a decrease of about 0.7 %, while the relative change in $(\mu_{\text{en}}/\rho)_{\text{air,g}}$ due to the use of renormalized photoeffect cross sections is practically negligible (see Fig. 7.5). The analysis of Burns (2012) results in the best estimate of W_{air} $s_{g,\text{air}} = 33.72 \text{ eV}$ for ^{60}Co radiation, determined with a relative standard uncertainty of 0.08 %. Adoption of this result would reduce the air-kerma determination for ^{60}Co graphite-cavity standards by about 0.7 %.

Although for energies below that of ^{192}Ir , the standard is normally a free-air chamber, which does not depend on $s_{g,\text{air}}$ and $(\mu_{\text{en}}/\rho)_{\text{air,g}}$ ratios [c.f. Eq. (3.1)], it might be of interest to analyze the spectral dependence of the changes in these quantities at lower energies. Examples are shown in Fig. 7.5 for 25 kV to 100 kV x-ray beams and for the ^{192}Ir spectrum of a brachytherapy source; for completeness, data for a ^{60}Co spectrum are also included in the figure. It can be observed that, at the lowest energies, the changes in $s_{g,\text{air}}$ (new versus previous) and possible differences in $(\mu_{\text{en}}/\rho)_{\text{air,g}}$ if the renormalized Scofield photoeffect cross sections were used practically compensate each other, leading to a net change of less than 0.3 %. However, above about 50 kV, and particularly in the region from 100 kV to ^{192}Ir , the relative change in K_{air} if a cavity-ionization standard was used in this region would be up to 1 %.

Ionometric absorbed-dose-to-water standards used for ^{60}Co γ -ray beams depend on W_{air} , $s_{g,\text{air}}$ and $(\mu_{\text{en}}/\rho)_{\text{air,g}}$, with relevant changes only in $s_{g,\text{air}}$. As above, the stopping-power ratio has an estimated relative decrease of about 0.7 %, which would in itself result in a decrease in the absorbed-dose-to-water for ^{60}Co

graphite-cavity standards by the same amount. It is emphasized, however, that the implementation of the new key data in the BIPM ionometric standard for absorbed dose to water is done in parallel with other changes in associated correction factors (see, e.g., Burns, 2009), leading to a practically negligible net change.

Graphite- and water-calorimetry standards depend on the heat defects, h_g and h_w , respectively, for which no recommended changes are given. Graphite calorimeters require the transfer of absorbed-dose-to-graphite to absorbed-dose-to-water, and often the dose transfer is based on the use of an ionization chamber and expressions that include ratios of stopping powers and energy-absorption coefficients. For the dose transfer in ^{60}Co beams using thick-walled chambers, the impact of the new key data is estimated to be negligible. For high-energy (megavoltage) photon beams, however, a given chamber wall thickness might not be considered thick, in which case, the impact of the new stopping powers will depend on the transfer-chamber constructional details and on the transfer method used, and needs to be investigated for each specific case. Whatever method is used, the water-to-graphite μ_{en} ratio will always appear in the transfer equation; however, as the changes in this ratio occur mostly below 100 keV (see Fig. 7.4), their influence is very small.

7.2 Impact on Practical Dosimetry

The changes indicated above will result in changes to the relevant calibration coefficients N_K and (to a lesser extent) $N_{D,w}$ for secondary dosimetry standards laboratories (accredited laboratories) and thus to users in various applications.

N_K -based dosimetry is used in numerous applications in radiation medicine, for example, radiotherapy with orthovoltage x-ray beams, brachytherapy, diagnostic radiology, and radiation protection. For the determination of air kerma in kilovoltage x-ray beams in these applications, the most relevant change would be that of N_K , which will decrease by up to 0.5 %, depending on the energy spectrum of the beam. For in-phantom measurements, the use of renormalized *versus* unrenormalized photoeffect cross sections in the XCOM-based mass energy-absorption coefficients for water would result in maximum relative changes in the required backscatter factors of about from 0.7 % to 0.8 % (30 kV and 50 kV) and from 0.5 % to 0.6 % (90 kV and 150 kV), whereas μ_{en}/ρ ratios would not change by more than a few tenths of a percent (see Fig. 7.6).

The changes in measurement standards will be relevant for the ionometric dosimetry of high-energy

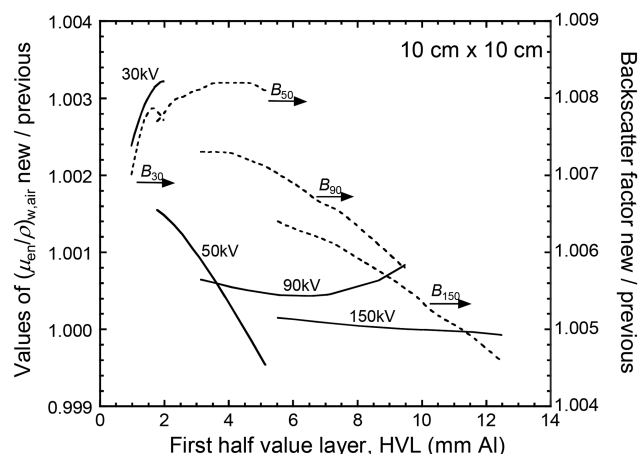


Figure 7.6. Ratios of the water-to-air (solid lines) mass energy-absorption coefficients, left ordinate, and of the backscatter factors in water (dashed lines), right ordinate, for clinical kilovoltage x-ray spectra using the NIST XCOM renormalized (labeled here as new) *versus* unrenormalized (labeled here as previous) data of Table 6.5a–6.5c, as a function of the photon beam quality in terms of the first HVL in aluminum (Benmakhlouf, private communication).

(megavoltage) photon and electron beams, as well as of protons and heavier charged particles. In what follows, the beam quality, specified in terms of the beam quality indices, c.f. IAEA TRS-398 (IAEA, 2000) and Almond *et al.* (1999), will be denoted by Q .

For N_K -based dosimetry, usually with air-filled ionization chambers calibrated in the reference quality Q_0 of ^{60}Co γ rays and measurements made at a reference depth in water, $M_{w,Q}(z_{\text{ref}})$, the absorbed dose can generally be determined as

$$D_{w,Q}(z_{\text{ref}}) = M_{w,Q}(z_{\text{ref}}) N_{D,\text{air},^{60}\text{Co}} \frac{(W_{\text{air}})_Q}{(W_{\text{air}})^{60\text{Co}}} (s_{w,\text{air}})_Q p_{\text{ch},Q}, \quad (7.1)$$

where $N_{D,\text{air},^{60}\text{Co}} = N_K(1 - \bar{g}_{\text{air}}) \prod_i k_i$ is the dose-to-air chamber coefficient (denoted also by N_{gas}); W_{air} has its common meaning, here referred to the beam to be calibrated, of quality Q , and to the reference quality, ^{60}Co ; $(s_{w,\text{air}})_Q$ is the water-to-air stopping-power ratio for the beam quality Q ; $p_{\text{ch},Q}$ is the chamber-specific overall perturbation correction for Q in water; the k_i are specific chamber correction factors for ^{60}Co γ rays in air. The constancy of W_{air} in megavoltage photon and electron beams makes the ratio of these quantities equal to one, but the ratio becomes larger than one for protons and heavier charged particles as will be shown below.

It should be noted that for high-energy radiotherapy beams, N_K -based dosimetry has been superseded in most instances by $N_{D,w}$ dosimetry described below, but Eq. (7.1) has been used frequently for the

determination of $(W_{\text{air}})_Q$ in proton beams as mentioned in Section 5.4.3 and discussed in detail below.

For $N_{D,w}$ -based dosimetry, the expression corresponding to Eq. (7.1) is

$$D_{w,Q}(z_{\text{ref}}) = M_{w,Q}(z_{\text{ref}})N_{D,w,Q_0}k_{Q,Q_0}, \quad (7.2)$$

where k_{Q,Q_0} is the so-called beam-quality factor that converts the calibration coefficient N_{D,w,Q_0} into $N_{D,w,Q}$ for the beam quality Q . Ideally, k_{Q,Q_0} should be determined experimentally at the standards laboratory as a ratio $N_{D,w,Q}/N_{D,w,Q_0}$ using a selected reference quality Q_0 (McEwen, 2010), but in most cases Q_0 is taken to be ^{60}Co and the beam-quality factor is determined by calculation as

$$k_{Q,Q_0} = \frac{(s_{w,\text{air}})_Q}{(s_{w,\text{air}})_{Q_0}} \frac{p_{\text{ch},Q}}{p_{\text{ch},Q_0}} \frac{(W_{\text{air}})_Q}{(W_{\text{air}})_{Q_0}}, \quad (7.3)$$

where the different quantities have the same meanings as in the N_K case. However, current dosimetry protocols for radiotherapy beams (*e.g.*, IAEA TRS-398 and AAPM TG-51) have provided tables of calculated values of k_{Q,Q_0} .

As can be seen from Eqs. (7.1) to (7.3), in addition to the changes in calibration coefficients resulting from the changes in measurement standards, the relevant data for high-energy beams are water-to-air stopping-power ratios and perturbation correction factors (or their product, $s_{w,\text{air}}p_{\text{ch}}$) for all beam types; additionally, $(W_{\text{air}})_Q$ needs to be considered for protons and heavier charged particles.

With regard to $s_{w,\text{air}}$, the most significant impact of the 4 % relative increase in I_w is a relative decrease in the water-to-air stopping-power ratio, which for ^{60}Co radiation is about 0.6 % and for all types of megavoltage radiotherapy photon and electron beams is estimated to be between approximately 0.3 % and 0.5 %. Figure 7.7 shows the ratio of $s_{w,\text{air}}$ at the reference depth resulting from the new I_w value and from that used in ICRU Report 37 (1984a) for photon and electron beams. It is observed that the relative decrease is practically constant for electron beams of different qualities, being about 0.3 %, while it varies from 0.6 % for ^{60}Co γ rays to 0.3 % for the highest photon-beam quality ($\text{TPR}_{20,10} = 0.8$).

The impact of the new key data on perturbation factors is difficult to estimate due to their chamber-specific dependence, governed by the chamber geometry and materials, and also depends to some extent on the method used for their evaluation. The relevance of the new key data is mainly for ^{60}Co γ rays and megavoltage photon beams, and for this Report, the focus is on graphite-walled ionization chambers. Currently, the product $(s_{w,\text{air}}p_{\text{ch}})_Q$ in Eq. (7.3) can be either estimated indirectly from the measured ^{60}Co

calibration coefficients, $N_{D,w}/N_K$ or obtained from Monte Carlo calculations. Estimates by Andreo *et al.* (2013) for a graphite-walled chamber yield a net relative increase in $(s_{w,\text{air}}p_{\text{ch}})_{^{60}\text{Co}}$ of about 0.7 % (*i.e.*, $p_{\text{ch},^{60}\text{Co}}$ increases by about 1.3 %), while for megavoltage photons, it increases by between 0.5 % (low Q) and 0.2 % (high Q). For electron beams, the net relative increase in $(s_{w,\text{air}}p_{\text{ch}})_Q$ is estimated to be of the order of 0.3 %.

Considering the quantities entering in Eqs. (7.2) and (7.3), the estimated change in the absorbed dose to water for photon and electron beams can be obtained as

$$\Delta D_{w,Q} \approx \Delta N_{D,w,Q_0} + \Delta(s_{w,\text{air}}p_{\text{ch}})_Q - \Delta(s_{w,\text{air}}p_{\text{ch}})_{Q_0}, \quad (7.4)$$

which, for a scenario in which N_{D,w,Q_0} is unchanged,

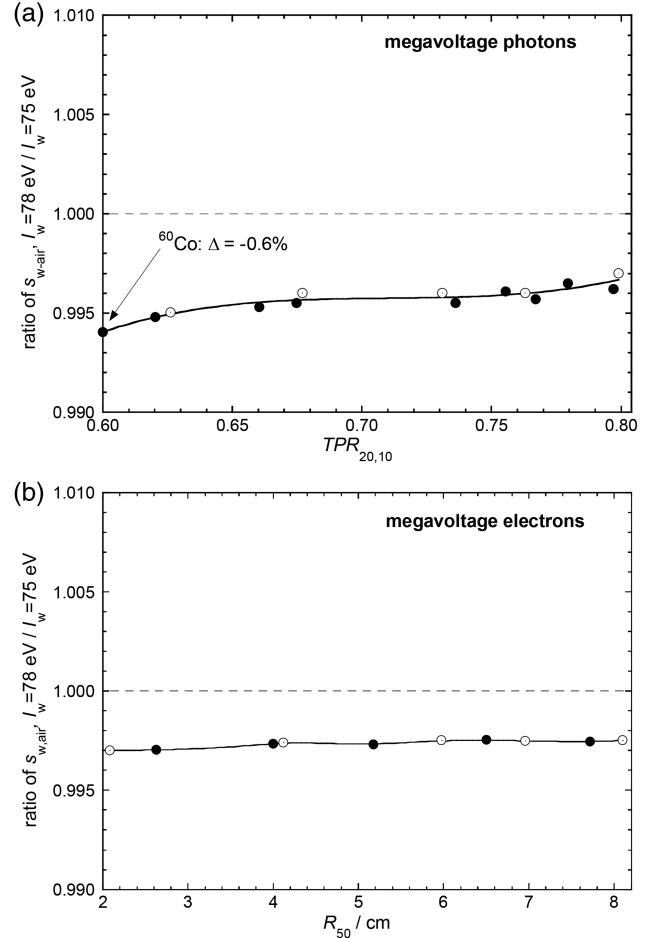


Figure 7.7. Influence of the increase in the I_w value on the water-to-air stopping-power ratio at reference depths for megavoltage photons (a) and electrons (b). The relative differences, from using $I_w = 75 \text{ eV}$ (ICRU Report 37 and ICRU Report 49) and $I_w = 78 \text{ eV}$, vary between -0.6% for ^{60}Co and from -0.3% to -0.5% for high-energy photons and electrons. Details on the input spectra and calculations are given in Andreo *et al.* (2013), from where the figure is adopted.

yields an estimated relative decrease of between 0.2 % and 0.5 % for these beams.

For protons and carbon ions, in addition to a nearly 0.5 % relative decrease in their calculated ranges at energies used in radiotherapy, Fig. 7.8 illustrates the depth variation of $s_{w,air}$ for the new and previous I_w values for two mono-energetic beams, showing in both cases a practically constant relative difference of about from 0.4 % to 0.5 % at the plateau and at the distal edge of the Bragg peak, where an abrupt increase in $s_{w,air}$ occurs (Medin and Andreo, 1997). The influence of the increased I_w on calculated ion-chamber perturbation factors is expected to be well below the differences noted above, becoming negligible for practical purposes.

As mentioned in Section 5.4.4, the value of $(W_{air})_Q$ for protons has most often been determined by comparing ionization-chamber dosimetry, using the $N_K-N_{D,air}$ formalism described by Eq. (7.1), with calorimetry measurements, *i.e.*, $D_{w,Q}^{calor} = D_{w,Q}^{ion}$, from which

$$(W_{air})_Q = \frac{D_{w,Q}^{calor}}{M_{w,Q} N_{D,air,^{60}Co}} \frac{(W_{air})_{^{60}Co}}{(s_{w,air})_Q p_{ch,Q}}, \quad (7.5)$$

where $D_{w,Q}^{calor}$ is the calorimetry-determined absorbed dose to water at the proton beam quality Q and the remaining quantities have been defined for Eq. (7.1). (The subscript, Q , on $(W_{air})_Q$ acknowledges the possibility that W_{air} might depend on the proton energy.) The effect of the changes recommended in this Report on $(W_{air})_Q$ can be determined by using

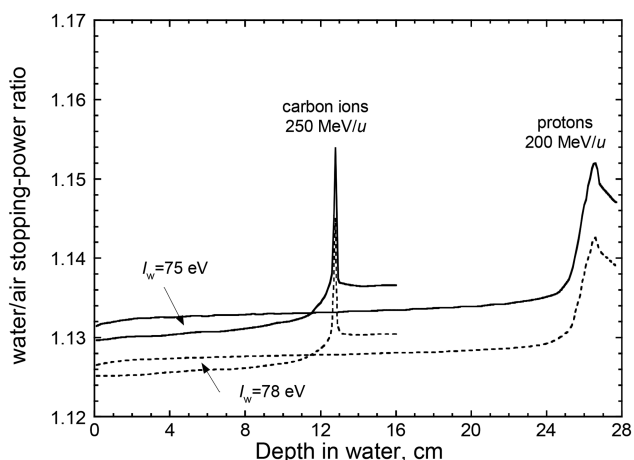


Figure 7.8. Influence of an increase in the I_w value on the water-to-air stopping power ratios for protons (200 MeV/u) and carbon ions (250 MeV/u). The relative difference between using $I_w = 75$ eV (ICRU Report 37 and ICRU Report 49, solid lines), and $I_w = 78$ eV (dashed lines) is approximately 0.4 %. Data computed using the SHIELD-HIT Monte Carlo code, v10. [Figure from Andreo *et al.* (2013).]

Eq. (7.5) to write

$$\Delta(W_{air})_Q \approx \Delta(W_{air})_{^{60}Co} - \Delta(N_{D,air,^{60}Co}) - \Delta(s_{w,air} p_{ch})_Q, \quad (7.6)$$

For graphite-walled chambers, a change in $(s_{w,air})_Q p_{ch,Q}$ would mainly be based on that of $s_{w,air}$ for protons (−0.4 %), given that various studies have indicated that the perturbation correction factor for these chambers in proton beams is close to unity (Palmans *et al.*, 2001; 2011; Verhaegen and Palmans, 2001). Thus, the value of $(W_{air})_Q$ would be affected only by the changes in $N_{D,air,^{60}Co}$ (−0.7 %) and $(s_{w,air})_Q$, as $(W_{air})_{^{60}Co}$ remains constant. Note, however, that all determinations of the $(W_{air})_Q$ value for protons according to Eq. (7.5) had been made prior to the change (+0.5 %) of the BIPM air-kerma standard in 2007 (Burns *et al.*, 2007) and prior to comparable changes to other primary air-kerma standards around the same time; hence, the net relative change in $N_{D,air,^{60}Co}$ from those determinations amounts to −0.2 % instead of −0.7 %. The estimated changes in the different components would then yield an overall relative increase in $(W_{air})_Q$ of about 0.6 %, *i.e.*, from the current 34.23 eV to 34.44 eV with an estimated relative uncertainty of 0.4 %. A small number of publications on the determination of $(W_{air})_Q$ during recent years were based on a comparison of ionization-chamber dosimetry using $N_{D,w,^{60}Co}$ calibration coefficients and water calorimetry. As shown by Andreo *et al.* (2013), the change in $(W_{air})_Q$ determined *via* this route would be entirely consistent with the 0.6 % relative increase, thus confirming the estimate of $(W_{air})_Q = 34.44$ eV. It is emphasized that both routes are based on the assumption of $p_{ch,Q} \approx 1$.

The arguments and methods of estimating given above would also be applicable to calorimetrically determined $(W_{air})_Q$ values for charged particles heavier than protons, unless specific new values of $(W_{air})_Q$ for these particles, different from those in IAEA TRS-398 (IAEA, 2000), would become available. The value of $(W_{air})_Q$ for carbon ions thus would change from 34.50 eV to 34.71 eV, with a relative standard uncertainty of 1.5 %. The relatively small change in $(W_{air})_Q$, due to the cancellation of the various factors, confirms the need for maintaining consistency.

It should be noted that the various earlier data sets and analysis deriving a $(W_{air})_Q$ value have not addressed properly the consistency of the data along the dosimetry chain. For example, in the analysis by Jones (2006), the data had mostly been corrected for the difference in water-to-air stopping-power ratios [those by Medin and Andreo (1997) *versus* Janni (1982) or ICRU Report 49 (ICRU, 1993)]. However, differences in other basic data included in the

Table 7.2. Estimated relative changes in the reference dosimetry of high-energy radiotherapy beams and related quantities following the implementation of the new key data. The estimates are applicable to graphite-walled ionization chambers calibrated in terms of absorbed dose to water ($N_{D,w,^{60}\text{Co}}$).

Quantity	Relative change, %	Comments
D_w for photons		See Eqs. (7.2) and (7.3)
$N_{D,w,^{60}\text{Co}}$	0	
$(s_{w,\text{air}}p_{\text{ch}})_Q$	+0.5	For lower-energy beam qualities
	+0.2	For higher-energy beam qualities
$(s_{w,\text{air}}p_{\text{ch}})_{^{60}\text{Co}}$	+0.7	
Total change in D_w for photons	−0.2	For lower-energy beam qualities
	−0.5	For higher-energy beam qualities
D_w for electrons		See Eqs. (7.2) and (7.3)
$N_{D,w,^{60}\text{Co}}$	0	
$(s_{w,\text{air}}p_{\text{ch}})_Q$	+0.3	
$(s_{w,\text{air}}p_{\text{ch}})_{^{60}\text{Co}}$	+0.7	
Total change in D_w for electrons	−0.4	
W_{air} for protons and carbon ions		See Eq. (7.6).
$N_{D,\text{air}} - N_K$	−0.2	See text following Eq. (7.6)
$(s_{w,\text{air}}p_{\text{ch}})_Q$	−0.4	Protons and carbon ions
Total change in $(W_{\text{air}})_Q$	+0.6	Based on the assumption $p_{\text{ch},Q} \approx 1$
D_w for protons and carbon ions		See Eqs. (7.2) and (7.3).
$N_{D,w,^{60}\text{Co}}$	0	
$(s_{w,\text{air}}p_{\text{ch}})_Q$	−0.4	
$(s_{w,\text{air}}p_{\text{ch}})_{^{60}\text{Co}}$	+0.7	
$(W_{\text{air}})_Q$	+0.6	
Total change in D_w for protons and carbon ions	−0.5	

dosimetry protocols used, especially AAPM Report 16 (AAPM, 1986), the first European Charged Heavy Particle Dosimetry Group (ECHED) protocol (Vynckier *et al.*, 1991), or more recent dosimetry recommendations, had been ignored. Also, the values of $(s_{w,\text{air}})_Q p_{\text{ch},Q}$ used in the derivation of $(W_{\text{air}})_Q$ from a comparison of ionization-chamber dosimetry using the $N_{D,w}$ formalism and calorimetry, might not be consistent with experimental values across all chamber types (at least 10 different chamber types have been involved in the evaluations leading to the data compiled by Jones). Hence, a lack of data consistency in the various steps of the dosimetry chain used to derive $(W_{\text{air}})_Q$, together with the absence of consolidated perturbation correction factors in proton beams for the chamber types used, might indicate the need for a future revision of $(W_{\text{air}})_Q$ when such data become available.

Incorporating the changes discussed so far in the quantities entering in Eqs. (7.2) and (7.3), the

change in the absorbed dose to water for proton beams can be estimated by

$$\Delta D_{w,Q} \approx \Delta N_{D,w,^{60}\text{Co}} + \Delta(s_{w,\text{air}}p_{\text{ch}})_Q - \Delta(s_{w,\text{air}}p_{\text{ch}})_{^{60}\text{Co}} + \Delta(W_{\text{air}})_Q, \quad (7.7)$$

which yields a relative decrease of 0.5 % as $N_{D,w,^{60}\text{Co}}$ remains unchanged (note that $(W_{\text{air}})_{^{60}\text{Co}}$ remains constant). A similar estimate occurs for carbon ions. Note that these estimates ignore the possible changes in chamber-perturbation correction factors in protons and heavier-charged-particle beams.

A summary of the estimated relative changes in the reference dosimetry of high-energy radiotherapy beams following the implementation of the new key data is given in Table 7.2. These correspond to absorbed dose to water and related quantities, and are applicable to $N_{D,w}$ -calibrated graphite-walled ionization chambers.

Appendix. Stopping Power and Range Tables for Charged Particles

New tables of stopping powers and ranges have been developed, using the methods described earlier, for the three materials of interest in this Report: air,³² graphite, and liquid water. The main differences in results from those in the earlier ICRU tabulations are due to the change in I values. However, the choice of the grain density for graphite, as indicated earlier, also affects the mass electronic stopping power through the density-effect correction.

Estimates of the uncertainties associated with the tabulated quantities are given below, generally based on the expected accuracy of either the underlying theoretical treatment or the measured data used to extend coverage of the heavy charged particles to low energies. Uncertainties in the stopping powers and ranges due to uncertainties associated with the I values can be determined from the coefficients of relative change given in the tables. This was done in ICRU Report 37 (ICRU, 1984a) and facilitates also the determination of the quantities for a somewhat different choice of I value. The coefficients are given in terms of $\partial(\log X)/\partial(\log I)$, where X can be the mass electronic stopping power, S_{el}/ρ , the csda range, ρr_0 , and—in the case of electrons and positrons—the radiation yield, Y . The main results are given to four significant figures, more than are warranted by the expected accuracy of the evaluations, in order to facilitate smooth interpolation. Based on practices used in the development of these tables, log-log interpolation is recommended, using natural cubic splines fitted to $\ln X$ as a function of $\ln T$, where X is the variable of interest.

A.1 Electrons and Positrons

Tables A.1–A.6 give results for kinetic energies from 1 keV up to 1 GeV. The relative standard uncertainties for the electronic stopping powers estimated in ICRU Report 37 (1984a) are taken over for this

Report: for $T > 100$ keV, 0.5 % to 1 %; $100 \text{ keV} \geq T > 10$ keV, 1 % to 1.5 %; $10 \text{ keV} > T \geq 1$ keV, 1.5 % to 5 %. These estimates include the uncertainty that results from the propagation of the standard uncertainties for the mean excitation energies through the use of the tabulated coefficients $\partial(\log(S_{\text{el}}/\rho))/\partial(\log I)$.

Radiative stopping powers for electrons and positrons are somewhat different from those tabulated in ICRU Report 37 (ICRU, 1984a), particularly at low energies, for reasons indicated in Section 4.4. These differences are, however, within the estimated uncertainties. Pratt *et al.* (1977) estimate the relative uncertainty of their differential cross sections for bremsstrahlung production in the field of the nucleus, for $T \leq 2$ MeV, to be no greater than 10 %. The numerical treatment used by Seltzer and Berger (1985) to connect these results to those from high-energy theory contributes about 1 % to 2 %, and thus represent a negligible additional contribution. The radiative stopping powers are derived from these cross sections as integrals over the differential cross sections, and should have a smaller relative uncertainty, perhaps about 3 % (ICRU, 1984a). The high-energy-theory results, for $T \geq 50$ MeV, have a relative uncertainty of about 1 %, and the results interpolated across the energy region $2 \text{ MeV} < T < 50 \text{ MeV}$ should have a relative uncertainty intermediate to these estimates, perhaps 1 % to 3 % (ICRU, 1984a). The basis for bremsstrahlung production by positrons is considerably less firm at low energies; for $T \geq 50$ MeV, the cross sections for bremsstrahlung production in the field of the nucleus for positrons are the same as for electrons in the high-energy theory used, except near the high-frequency limit. Thus, the relative uncertainties are estimated to be about 1 % for $T \geq 50$ MeV, 1 % to 5 % for $2 \text{ MeV} < T < 50 \text{ MeV}$, and about 5 % for $T \leq 2$.

A.2 Protons and Alpha Particles

Tables A.7–A.12 give results for kinetic energies from 1 keV up to 10 GeV. The computer code BEST (see Section 4.1.2) was used to evaluate the electronic stopping powers at sufficiently high energies. The

³²Air is assumed to be dry air at sea level; the density assumed for the tables was chosen to correspond to a pressure of 101.325 kPa (760 mmHg or 1 atm) and a temperature of 22 °C, but other combinations are possible.

low-energy electronic stopping powers are the adopted measurement-based results from ICRU Report 49 (see Section 3 in that Report) and have been connected to the high-energy results by interpolating with cubic splines using the same methods and cut-off energies used in ICRU Report 49. One exception to this (see also Berger, 1993) was the use in the present Report of the empirical results of Nečas *et al.* (1993) for the electronic stopping power of protons in graphite. In that case, the modified Bethe results have been used for $T \geq 0.5$ MeV and the empirical data used for $T \leq 0.25$ MeV. Note that in the following tables, values of $\partial(\log X)/\partial(\log I)$ can be given only for sufficiently high energies at which the dependence on the mean excitation energy, I , has been determined through the use of the modified Bethe formula. Thus, the values at low energies have been set to zero.

The relative standard uncertainties for the electronic stopping powers estimated in ICRU Report 49 (ICRU, 1993) are taken over for this Report: 1 % to 3 % for $T \geq 1$ MeV; 3 % to 6 % for $1 \text{ MeV} > T \geq 100 \text{ keV}$; 6 % to 10 % for $100 \text{ keV} > T \geq 10 \text{ keV}$; 10 % to 15 % for $10 \text{ keV} > T \geq 1 \text{ keV}$. These estimates include the uncertainty that results from the propagation of the standard uncertainties for the mean excitation energies through the use of the tabulated coefficients $\partial(\log(S_{el}/\rho))/\partial(\log I)$.

Values for the nuclear stopping powers and the detour factors, taken over from ICRU Report 49, are included for completeness. The relative standard uncertainties for the nuclear stopping powers estimated in ICRU Report 49 (1993) are here interpreted to be: 3 % for $T \geq 100 \text{ keV}$; 3 % to 6 % for $100 \text{ keV} > T \geq 10 \text{ keV}$; 6 % to 12 % for $10 \text{ keV} > T \geq 1 \text{ keV}$.

A.3 Carbon Ions

ICRU Report 73 (2005) includes, down to a kinetic energy of 0.3 MeV, electronic-stopping-power and csda range results calculated for C ions in air, amorphous carbon, and water, with input data that implicitly indicate the use of I values of 82.8 eV, 86.0 eV, and 67.2 eV, respectively. Sigmund *et al.* (2009), in the Errata and Addenda for ICRU Report 73, corrects the results for liquid water with new calculations with the PASS code used in ICRU Report 73, with $I = 78$ eV. This I value agrees with that recommended for liquid water in the current Report, but the I values used in ICRU Report 73 for graphite and air do not.

Efforts to maintain consistency with the format of ICRU Report 49 (1993) prompted investigations into alternative estimates of the electronic stopping

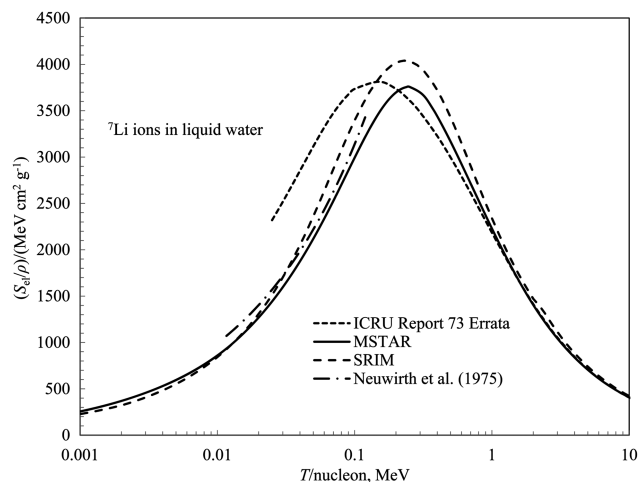


Figure A.1. Electronic stopping powers for Li ions in liquid water, comparing the measured results (with a stated accuracy of the order of 1 %) of Neuwirth *et al.* (1975) with results from the PASS code as used in ICRU Report 73 Errata (Sigmund *et al.*, 2009), from MSTAR, and from SRIM.

power for C ions at high kinetic energies using the BEST code and using semi-empirical methods to extend to kinetic energies below 300 keV, even down to 1 keV, in order to provide data for evaluation of the csda range. An additional correction factor was added to the BEST code to account for relativistic Mott scattering, an effect expected for high-energy heavy ions. For this correction, the formula of Ahlen (1980) was used, which brought the BEST results into relative agreement with those in Sigmund *et al.* (2009) for liquid water with $I = 78$ eV to within about 0.07 % at relativistic energies, very close to round-off differences associated with the use of four significant figures in the respective results, and thus providing confirmation of the two rather different calculations at high kinetic energies.

At low energies, there appear to be no measured results reported for C ions in liquid water; however, there are measured data for Li ions. The PASS results for Li ions in liquid water are compared in Fig. A.1 with the measurements of Neuwirth *et al.* (1975), with the semi-empirical results of the MSTAR code (v. 3.12) of Paul and Schinner (2001; 2002), and results from the SRIM code v.2013.00 (Ziegler *et al.*, 2010). MSTAR is based on the α -particle results from ICRU Report 49 and the use of scaling through an empirically derived function determined for all projectiles from Li to Ar in solid and gaseous targets. SRIM, under continuing development for nearly 30 years, is based on a scaling approach combined with corrections to Bragg's rule, and for phase effects in light target compounds, that have been extracted from measured H, He, and Li stopping powers. The MSTAR and the SRIM results are clearly in better agreement with the measured

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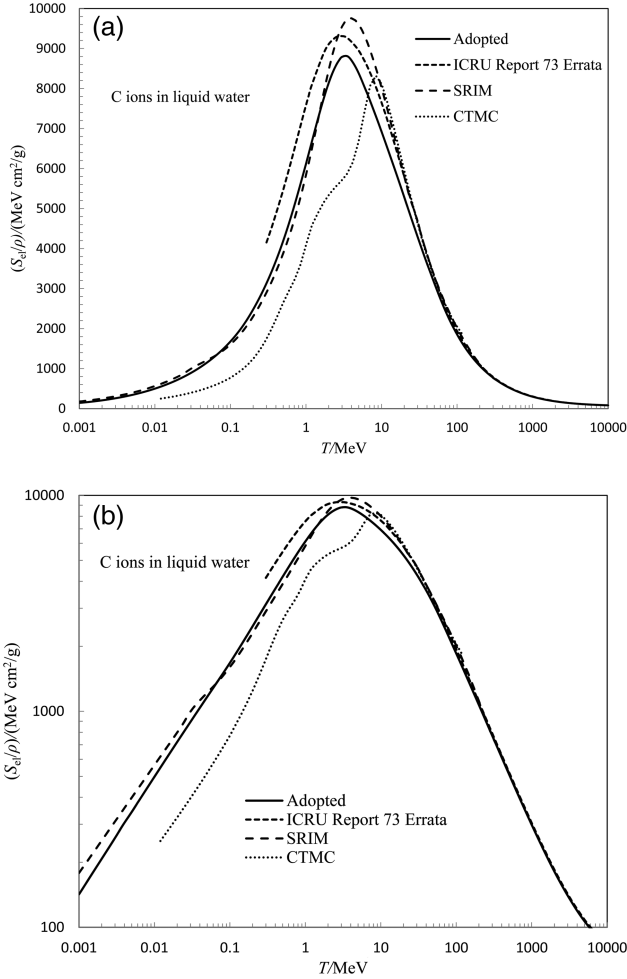


Figure A.2. Electronic stopping power for C ions in liquid water, comparing the results adopted here with those from the ICRU Report 73 Errata (Sigmund *et al.*, 2009), from SRIM, and from the classical-trajectory Monte Carlo (CTMC) calculations of Liamsuwan and Nikjoo (2013). (a) Semi-log plot. (b) Log-log plot.

results, advocating the use of such scaled results for low-energy C ions in liquid water; MSTAR was chosen, as it is more easily understood and accessible for small changes.³³

The low-energy electronic stopping powers from MSTAR have been connected to the high-energy results of the corrected BEST code ($I = 78$ eV) by interpolating the quantity $\beta^2 S_{el}(T)/\rho$ with cubic splines, similar to that done for protons and α particles in ICRU Report 49 (1993), but here for C ions using the scaled empirical data for $T \leq 30$ MeV and

the modified Bethe results for $T \geq 60$ MeV. The adopted results³⁴ are shown in Fig. A.2 and compared with those from Sigmund *et al.* (2009), those from SRIM, and those from the classical-trajectory Monte Carlo (CTMC) calculations of Liamsuwan and Nikjoo (2013).

The same procedure, combining the modified Bethe results from the BEST code ($I = 85.7$ eV) incorporating the relativistic Mott correction with the results from MSTAR, has been used for air. The adopted results are shown in Fig. A.3 and compared with those from Sigmund *et al.* (2009), from SRIM, and the measured results from Fastrup *et al.* (1968) and from Hvelplund (1971).

There appear to be numerous results of measurement for C ions in amorphous carbon, but only those of Baek *et al.* (1987) for C ions in graphite. Results from MSTAR and from SRIM for amorphous carbon agree better than those from ICRU Report 73 with the measured results from some 17 experiments for C ions with kinetic energies up to about 48 MeV (see the website <http://www.exphys.jku.at/stopping/> of H. Paul). The modified Bethe results from the BEST code incorporating the relativistic Mott correction, combined with the results from MSTAR at kinetic energies below 30 MeV, has been used for graphite (with $I = 81.0$ eV). The adopted results are shown in Fig. A.4 and compared with those from SRIM and with the Baek *et al.* (1987) results. The inverted Doppler-shift attenuation method used by Baek *et al.* requires knowledge of the mean lifetime of the $^{12}\text{C}^*$ projectiles produced in the $^{12}\text{C}(\alpha, \alpha')^{12}\text{C}^*$ reaction (*i.e.*, ^{12}C from the 4.438 MeV excited state); the stopping cross sections from their analysis are directly proportional to this parameter. Rather than use the reported value of the mean lifetime, 61 fs, quoted in their paper, Baek *et al.* normalized their results to the stopping cross sections of Ziegler (1977; 1978), from which they deduced an implied mean lifetime of 53.3 fs. However, the mean lifetime of the 4.438 MeV excited state is still accepted as 60.9 fs (see, *e.g.*, the level-width data from F. Ajzenberg-Selov, 1990; her annual updates have since reported no newer values). Thus, the Baek *et al.* data are shown in Fig. A.4 also normalized to the mean lifetime of 60.9 fs.

Nuclear stopping powers for C ions in air, carbon, and water have been evaluated according to the method outlined in ICRU Report 49 (1993), using the universal, scaled³⁵ nuclear stopping power of

³³For consistency with this Report, the MSTAR results were adjusted to reflect the electronic stopping powers for α particles in liquid water and graphite from Tables A.11 and A.12, produced with the new I values. These adjustments cause relative changes of less than about 1 %, which is within the estimated accuracy of MSTAR (Paul and Schinner, 2002), and ensure that the MSTAR results more consistently point to the new high-energy modified Bethe results to which they will be connected.

³⁴The MSTAR results were extended from 3 keV down to 1 keV assuming a power-law dependence as was done for very low kinetic energies in ICRU Report 49 (1993).

³⁵There are misprints in ICRU Report 49 (1993): a section of Table 4.1 has been transposed, and the function $a(z, Z)$ should be taken as $z^{0.23} + Z^{0.23}$ for α particles (*i.e.*, particles heavier than protons, and used here for C ions).

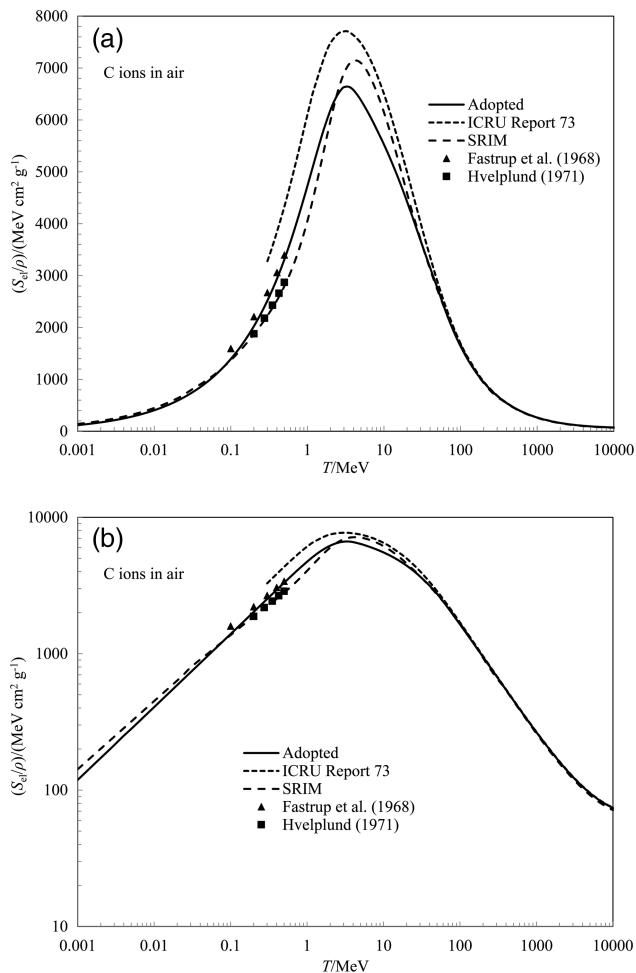


Figure A.3. Electronic stopping power for C ions in dry air, comparing the results adopted here with those from ICRU Report 73 (2005), from SRIM, and from the experiments of Fastrup *et al.* (1968) and of Hvelplund (1971). (a) Semi-log plot. (b) Log-log plot.

Ziegler *et al.* (1985); see Table 4.1 and Eqs. (4.14) and (4.15) in ICRU Report 49. The adopted values of the nuclear stopping powers agree with those from SRIM to within about 1 % up to kinetic energies of 20 MeV and to within about 5 % up to 1 GeV. The ranges are affected by inclusion of the nuclear stopping power, mostly of course at low energies where S_{nuc}/ρ is relatively large. The relative differences in the csda ranges, calculated with and without including S_{nuc}/ρ , are less than 1 % above a kinetic energy of about 40 MeV. Based on the energy scaling involved, the relative standard uncertainties of the nuclear stopping powers should be similar to those given for alpha particles in Section A.2, but for corresponding kinetic energies larger by about a factor of 5.

The final results are tabulated in Tables A.13–A.15. The coefficients of the relative change of S_{el}/ρ

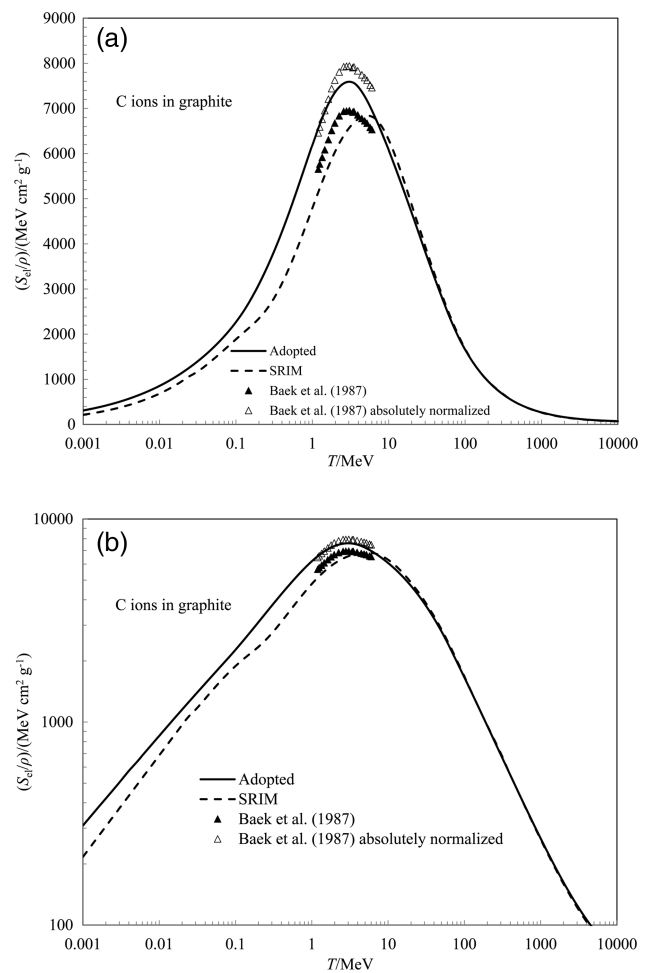


Figure A.4. Electronic stopping power for C ions in graphite, comparing the results adopted here with those from SRIM and with those of Baek *et al.* (1987) (the solid triangles are the Baek *et al.* data as reported, with normalization to then current stopping-power tabulations; the open triangles are their data absolutely normalized to the mean lifetime, 60.9 fs, of the 4.438 MeV excited state of ^{12}C). (a) Semi-log plot. (b) Log-log plot.

with I were evaluated from the BEST calculations down to a kinetic energy of 60 MeV. At kinetic energies below 30 MeV, the MSTAR results were considered to be measured and thus not a function of I . The relative standard uncertainties of the electronic stopping powers are estimated to about 1 % to 2 % for $T > 100$ MeV; 2 % to 5 % for $100 \text{ MeV} \geq T > 10$ MeV; 5 % to 15 % for $10 \text{ MeV} \geq T > 1$ MeV; 15 % to 30 % for $1 \text{ MeV} \geq T > 100$ keV; 30 % to 100 % for $100 \text{ keV} \geq T \geq 1$ keV. These estimates include the uncertainty that results from the propagation of the standard uncertainties for the mean excitation energies through the use of the tabulated coefficients $\partial(\log(S_{\text{el}}/\rho))/\partial(\log I)$.

Appendix

Table A.1. Electrons in air, $I = 85.7$ eV, $\rho = 0.0012$ g cm⁻³.

T	S_{el}/ρ	S_{rad}/ρ	S_{tot}/ρ	ρr_0	Y	δ	$\partial(\log)/\partial(\log I)$		
MeV	MeV cm ² g ⁻¹			g cm ⁻²			S_{el}/ρ	ρr_0	Y
0.0010	1.025E+02	2.883E-03	1.025E+02	4.879E-06	1.407E-05	0.000E+00	-0.383	0.383	0.383
0.0015	7.903E+01	3.171E-03	7.904E+01	1.049E-05	2.077E-05	0.000E+00	-0.332	0.367	0.367
0.0020	6.501E+01	3.351E-03	6.502E+01	1.750E-05	2.705E-05	0.000E+00	-0.303	0.347	0.345
0.0030	4.879E+01	3.562E-03	4.879E+01	3.547E-05	3.883E-05	0.000E+00	-0.270	0.315	0.312
0.0040	3.954E+01	3.679E-03	3.954E+01	5.839E-05	4.991E-05	0.000E+00	-0.251	0.293	0.290
0.0050	3.348E+01	3.752E-03	3.349E+01	8.598E-05	6.045E-05	0.000E+00	-0.237	0.277	0.274
0.0060	2.918E+01	3.801E-03	2.919E+01	1.181E-04	7.058E-05	0.000E+00	-0.228	0.265	0.262
0.0080	2.344E+01	3.858E-03	2.345E+01	1.950E-04	8.983E-05	0.000E+00	-0.214	0.247	0.245
0.0100	1.975E+01	3.889E-03	1.976E+01	2.884E-04	1.080E-04	0.000E+00	-0.204	0.235	0.233
0.0150	1.444E+01	3.919E-03	1.445E+01	5.886E-04	1.502E-04	0.000E+00	-0.189	0.215	0.213
0.0200	1.157E+01	3.933E-03	1.158E+01	9.781E-04	1.891E-04	0.000E+00	-0.179	0.202	0.201
0.0300	8.491E+00	3.958E-03	8.495E+00	2.002E-03	2.607E-04	0.000E+00	-0.167	0.187	0.186
0.0400	6.848E+00	3.981E-03	6.852E+00	3.322E-03	3.266E-04	0.000E+00	-0.160	0.178	0.177
0.0500	5.818E+00	4.006E-03	5.822E+00	4.912E-03	3.883E-04	0.000E+00	-0.155	0.171	0.171
0.0600	5.110E+00	4.039E-03	5.114E+00	6.750E-03	4.468E-04	0.000E+00	-0.151	0.166	0.166
0.0800	4.197E+00	4.116E-03	4.201E+00	1.110E-02	5.566E-04	0.000E+00	-0.145	0.159	0.158
0.1000	3.633E+00	4.205E-03	3.637E+00	1.623E-02	6.591E-04	0.000E+00	-0.140	0.153	0.153
0.1500	2.861E+00	4.463E-03	2.865E+00	3.193E-02	8.930E-04	0.000E+00	-0.133	0.145	0.144
0.2000	2.469E+00	4.759E-03	2.474E+00	5.082E-02	1.105E-03	0.000E+00	-0.128	0.140	0.139
0.3000	2.084E+00	5.463E-03	2.089E+00	9.528E-02	1.494E-03	0.000E+00	-0.122	0.133	0.132
0.4000	1.902E+00	6.283E-03	1.908E+00	1.456E-01	1.859E-03	0.000E+00	-0.118	0.128	0.127
0.5000	1.802E+00	7.194E-03	1.809E+00	1.995E-01	2.214E-03	0.000E+00	-0.114	0.125	0.123
0.6000	1.743E+00	8.181E-03	1.751E+00	2.558E-01	2.566E-03	0.000E+00	-0.112	0.122	0.120
0.8000	1.683E+00	1.033E-02	1.694E+00	3.723E-01	3.270E-03	0.000E+00	-0.107	0.118	0.115
1.0000	1.661E+00	1.267E-02	1.674E+00	4.912E-01	3.982E-03	0.000E+00	-0.104	0.115	0.112
1.5000	1.661E+00	1.919E-02	1.680E+00	7.901E-01	5.815E-03	0.000E+00	-0.099	0.109	0.105
2.0000	1.684E+00	2.644E-02	1.711E+00	1.085E+00	7.718E-03	0.000E+00	-0.095	0.105	0.101
3.0000	1.740E+00	4.247E-02	1.782E+00	1.658E+00	1.168E-02	0.000E+00	-0.090	0.100	0.095
4.0000	1.790E+00	5.994E-02	1.850E+00	2.208E+00	1.579E-02	0.000E+00	-0.087	0.097	0.091
5.0000	1.833E+00	7.841E-02	1.911E+00	2.740E+00	1.997E-02	0.000E+00	-0.084	0.094	0.088
6.0000	1.870E+00	9.762E-02	1.968E+00	3.255E+00	2.420E-02	0.000E+00	-0.082	0.092	0.085
8.0000	1.931E+00	1.378E-01	2.068E+00	4.246E+00	3.268E-02	0.000E+00	-0.080	0.088	0.081
10.0000	1.979E+00	1.796E-01	2.159E+00	5.192E+00	4.114E-02	0.000E+00	-0.078	0.085	0.078
15.0000	2.069E+00	2.891E-01	2.359E+00	7.405E+00	6.181E-02	0.000E+00	-0.074	0.080	0.072
20.0000	2.134E+00	4.032E-01	2.537E+00	9.447E+00	8.161E-02	0.000E+00	-0.072	0.076	0.068
30.0000	2.226E+00	6.394E-01	2.865E+00	1.315E+01	1.184E-01	7.556E-03	-0.062	0.071	0.062
40.0000	2.282E+00	8.825E-01	3.164E+00	1.647E+01	1.517E-01	1.375E-01	-0.044	0.064	0.052
50.0000	2.319E+00	1.130E+00	3.449E+00	1.949E+01	1.821E-01	3.188E-01	-0.035	0.059	0.044
60.0000	2.347E+00	1.381E+00	3.728E+00	2.228E+01	2.100E-01	5.024E-01	-0.030	0.054	0.037
80.0000	2.387E+00	1.890E+00	4.277E+00	2.729E+01	2.594E-01	8.360E-01	-0.024	0.047	0.029
100.0000	2.417E+00	2.405E+00	4.822E+00	3.169E+01	3.018E-01	1.120E+00	-0.022	0.042	0.024
150.0000	2.468E+00	3.713E+00	6.181E+00	4.082E+01	3.857E-01	1.669E+00	-0.019	0.035	0.017
200.0000	2.502E+00	5.035E+00	7.538E+00	4.814E+01	4.484E-01	2.078E+00	-0.017	0.030	0.013
300.0000	2.550E+00	7.698E+00	1.025E+01	5.947E+01	5.368E-01	2.674E+00	-0.015	0.026	0.009
400.0000	2.582E+00	1.037E+01	1.296E+01	6.813E+01	5.971E-01	3.116E+00	-0.013	0.023	0.007
500.0000	2.606E+00	1.306E+01	1.566E+01	7.514E+01	6.413E-01	3.471E+00	-0.011	0.021	0.006
600.0000	2.625E+00	1.575E+01	1.837E+01	8.103E+01	6.754E-01	3.772E+00	-0.009	0.019	0.005
800.0000	2.653E+00	2.114E+01	2.380E+01	9.056E+01	7.251E-01	4.266E+00	-0.007	0.017	0.004
1000.0000	2.674E+00	2.655E+01	2.922E+01	9.813E+01	7.599E-01	4.667E+00	-0.005	0.016	0.003

KEY DATA FOR IONIZING-RADIATION DOSIMETRY: MEASUREMENT STANDARDS AND APPLICATIONS

Table A.2. Electrons in graphite, $I = 81$ eV, $\rho = 2.265$ g cm⁻³.

T	S_{el}/ρ	S_{rad}/ρ	S_{tot}/ρ	ρr_0	Y	δ	$\partial(\log)/\partial(\log I)$		
MeV	MeV cm ² g ⁻¹			g cm ⁻²			S_{el}/ρ	ρr_0	Y
0.00100	1.048E+02	2.507E-03	1.048E+02	4.772E-06	1.197E-05	2.470E-04	-0.375	0.375	0.375
0.00150	8.057E+01	2.715E-03	8.057E+01	1.027E-05	1.760E-05	3.719E-04	-0.326	0.360	0.359
0.00200	6.617E+01	2.836E-03	6.617E+01	1.716E-05	2.278E-05	4.978E-04	-0.298	0.340	0.339
0.00300	4.956E+01	2.969E-03	4.957E+01	3.483E-05	3.235E-05	7.522E-04	-0.266	0.309	0.307
0.00400	4.012E+01	3.037E-03	4.012E+01	5.740E-05	4.123E-05	1.010E-03	-0.247	0.288	0.286
0.00500	3.395E+01	3.077E-03	3.395E+01	8.461E-05	4.963E-05	1.272E-03	-0.234	0.272	0.270
0.00600	2.958E+01	3.101E-03	2.958E+01	1.162E-04	5.766E-05	1.537E-03	-0.224	0.261	0.259
0.00800	2.374E+01	3.128E-03	2.374E+01	1.923E-04	7.286E-05	2.079E-03	-0.211	0.243	0.242
0.01000	1.999E+01	3.142E-03	1.999E+01	2.845E-04	8.720E-05	2.634E-03	-0.201	0.231	0.230
0.01500	1.460E+01	3.152E-03	1.461E+01	5.813E-04	1.204E-04	4.085E-03	-0.186	0.211	0.210
0.02000	1.169E+01	3.164E-03	1.169E+01	9.668E-04	1.512E-04	5.622E-03	-0.176	0.199	0.198
0.03000	8.572E+00	3.181E-03	8.575E+00	1.980E-03	2.080E-04	8.946E-03	-0.164	0.184	0.184
0.04000	6.908E+00	3.200E-03	6.911E+00	3.289E-03	2.604E-04	1.259E-02	-0.157	0.174	0.174
0.05000	5.866E+00	3.227E-03	5.869E+00	4.866E-03	3.097E-04	1.654E-02	-0.151	0.168	0.167
0.06000	5.150E+00	3.256E-03	5.153E+00	6.690E-03	3.566E-04	2.079E-02	-0.146	0.162	0.162
0.08000	4.226E+00	3.324E-03	4.229E+00	1.101E-02	4.449E-04	3.011E-02	-0.139	0.155	0.154
0.10000	3.654E+00	3.400E-03	3.657E+00	1.611E-02	5.277E-04	4.047E-02	-0.134	0.149	0.149
0.15000	2.871E+00	3.619E-03	2.875E+00	3.174E-02	7.175E-04	7.039E-02	-0.123	0.139	0.138
0.20000	2.473E+00	3.871E-03	2.476E+00	5.059E-02	8.913E-04	1.052E-01	-0.115	0.131	0.130
0.30000	2.076E+00	4.462E-03	2.081E+00	9.513E-02	1.213E-03	1.847E-01	-0.100	0.120	0.118
0.40000	1.886E+00	5.148E-03	1.891E+00	1.458E-01	1.518E-03	2.708E-01	-0.089	0.111	0.109
0.50000	1.778E+00	5.911E-03	1.784E+00	2.004E-01	1.818E-03	3.580E-01	-0.081	0.104	0.101
0.60000	1.712E+00	6.734E-03	1.718E+00	2.576E-01	2.117E-03	4.438E-01	-0.075	0.098	0.094
0.80000	1.640E+00	8.533E-03	1.648E+00	3.767E-01	2.724E-03	6.075E-01	-0.066	0.089	0.084
1.00000	1.606E+00	1.050E-02	1.617E+00	4.994E-01	3.345E-03	7.593E-01	-0.061	0.083	0.077
1.50000	1.582E+00	1.596E-02	1.598E+00	8.113E-01	4.970E-03	1.090E+00	-0.052	0.072	0.065
2.00000	1.586E+00	2.204E-02	1.608E+00	1.124E+00	6.686E-03	1.364E+00	-0.047	0.066	0.058
3.00000	1.609E+00	3.552E-02	1.645E+00	1.739E+00	1.033E-02	1.798E+00	-0.042	0.058	0.050
4.00000	1.634E+00	5.023E-02	1.684E+00	2.339E+00	1.417E-02	2.136E+00	-0.039	0.053	0.045
5.00000	1.656E+00	6.582E-02	1.722E+00	2.927E+00	1.814E-02	2.414E+00	-0.036	0.050	0.041
6.00000	1.675E+00	8.203E-02	1.757E+00	3.501E+00	2.219E-02	2.653E+00	-0.033	0.047	0.039
8.00000	1.706E+00	1.160E-01	1.822E+00	4.619E+00	3.043E-02	3.051E+00	-0.028	0.042	0.034
10.00000	1.729E+00	1.514E-01	1.881E+00	5.699E+00	3.876E-02	3.381E+00	-0.023	0.039	0.030
15.00000	1.770E+00	2.442E-01	2.014E+00	8.266E+00	5.953E-02	4.033E+00	-0.014	0.032	0.022
20.00000	1.797E+00	3.410E-01	2.138E+00	1.067E+01	7.980E-02	4.533E+00	-0.009	0.027	0.017
30.00000	1.832E+00	5.419E-01	2.374E+00	1.511E+01	1.181E-01	5.279E+00	-0.005	0.021	0.010
40.00000	1.856E+00	7.489E-01	2.605E+00	1.913E+01	1.533E-01	5.828E+00	-0.003	0.017	0.007
50.00000	1.874E+00	9.599E-01	2.834E+00	2.281E+01	1.854E-01	6.260E+00	-0.002	0.014	0.005
60.00000	1.888E+00	1.174E+00	3.062E+00	2.620E+01	2.147E-01	6.616E+00	-0.001	0.013	0.004
80.00000	1.911E+00	1.608E+00	3.519E+00	3.229E+01	2.665E-01	7.183E+00	-0.001	0.010	0.003
100.00000	1.928E+00	2.048E+00	3.977E+00	3.764E+01	3.106E-01	7.624E+00	0.000	0.009	0.002
150.00000	1.960E+00	3.166E+00	5.125E+00	4.868E+01	3.972E-01	8.429E+00	0.000	0.007	0.001
200.00000	1.982E+00	4.296E+00	6.278E+00	5.748E+01	4.612E-01	9.002E+00	0.000	0.006	0.001
300.00000	2.013E+00	6.574E+00	8.587E+00	7.105E+01	5.505E-01	9.811E+00	0.000	0.005	0.000
400.00000	2.035E+00	8.863E+00	1.090E+01	8.136E+01	6.107E-01	1.039E+01	0.000	0.004	0.000
500.00000	2.052E+00	1.116E+01	1.321E+01	8.968E+01	6.545E-01	1.083E+01	0.000	0.004	0.000
600.00000	2.066E+00	1.346E+01	1.553E+01	9.665E+01	6.882E-01	1.119E+01	0.000	0.003	0.000
800.00000	2.089E+00	1.809E+01	2.017E+01	1.079E+02	7.369E-01	1.177E+01	0.000	0.003	0.000
1000.00000	2.106E+00	2.272E+01	2.482E+01	1.168E+02	7.708E-01	1.222E+01	0.000	0.003	0.000

Appendix

Table A.3. Electrons in liquid water, $I = 78 \text{ eV}$, $\rho = 0.998 \text{ g cm}^{-3}$.

T	S_{el}/ρ	S_{rad}/ρ	S_{tot}/ρ	ρr_0	Y	δ	$\partial(\log)/\partial(\log I)$		
MeV	MeV cm ² g ⁻¹			g cm ⁻²			S_{el}/ρ	ρr_0	Y
0.0010	1.181E+02	2.830E-03	1.181E+02	4.235E-06	1.199E-05	0.000E+00	-0.370	0.370	0.370
0.0015	9.063E+01	3.122E-03	9.063E+01	9.118E-06	1.774E-05	0.000E+00	-0.322	0.355	0.354
0.0020	7.436E+01	3.307E-03	7.436E+01	1.524E-05	2.318E-05	0.000E+00	-0.295	0.336	0.334
0.0030	5.563E+01	3.530E-03	5.564E+01	3.098E-05	3.348E-05	0.000E+00	-0.263	0.306	0.303
0.0040	4.500E+01	3.657E-03	4.500E+01	5.110E-05	4.322E-05	0.000E+00	-0.245	0.285	0.282
0.0050	3.806E+01	3.737E-03	3.807E+01	7.536E-05	5.253E-05	0.000E+00	-0.232	0.270	0.267
0.0060	3.315E+01	3.791E-03	3.315E+01	1.036E-04	6.149E-05	0.000E+00	-0.223	0.258	0.255
0.0080	2.659E+01	3.855E-03	2.660E+01	1.714E-04	7.858E-05	0.000E+00	-0.210	0.241	0.239
0.0100	2.239E+01	3.890E-03	2.239E+01	2.537E-04	9.476E-05	0.000E+00	-0.200	0.229	0.227
0.0150	1.635E+01	3.928E-03	1.635E+01	5.189E-04	1.324E-04	0.000E+00	-0.185	0.210	0.209
0.0200	1.308E+01	3.939E-03	1.309E+01	8.632E-04	1.670E-04	0.000E+00	-0.176	0.198	0.197
0.0300	9.591E+00	3.965E-03	9.595E+00	1.769E-03	2.307E-04	0.000E+00	-0.165	0.184	0.183
0.0400	7.729E+00	3.987E-03	7.733E+00	2.939E-03	2.893E-04	0.000E+00	-0.158	0.175	0.174
0.0500	6.564E+00	4.011E-03	6.568E+00	4.348E-03	3.442E-04	0.000E+00	-0.152	0.168	0.168
0.0600	5.763E+00	4.044E-03	5.767E+00	5.978E-03	3.962E-04	0.000E+00	-0.149	0.163	0.163
0.0800	4.731E+00	4.122E-03	4.735E+00	9.833E-03	4.939E-04	0.000E+00	-0.143	0.156	0.156
0.1000	4.093E+00	4.211E-03	4.097E+00	1.439E-02	5.851E-04	0.000E+00	-0.139	0.151	0.151
0.1500	3.221E+00	4.471E-03	3.226E+00	2.833E-02	7.936E-04	0.000E+00	-0.132	0.143	0.142
0.2000	2.779E+00	4.771E-03	2.784E+00	4.512E-02	9.831E-04	0.000E+00	-0.127	0.138	0.137
0.3000	2.344E+00	5.482E-03	2.349E+00	8.464E-02	1.331E-03	0.000E+00	-0.121	0.131	0.130
0.4000	2.138E+00	6.309E-03	2.144E+00	1.294E-01	1.657E-03	0.000E+00	-0.116	0.126	0.125
0.5000	2.025E+00	7.228E-03	2.032E+00	1.774E-01	1.976E-03	0.000E+00	-0.113	0.123	0.122
0.6000	1.956E+00	8.221E-03	1.965E+00	2.275E-01	2.291E-03	1.501E-02	-0.093	0.119	0.116
0.8000	1.880E+00	1.039E-02	1.891E+00	3.315E-01	2.927E-03	1.005E-01	-0.074	0.107	0.102
1.0000	1.845E+00	1.276E-02	1.858E+00	4.384E-01	3.577E-03	2.086E-01	-0.061	0.097	0.090
1.5000	1.819E+00	1.934E-02	1.838E+00	7.096E-01	5.275E-03	4.982E-01	-0.044	0.080	0.068
2.0000	1.821E+00	2.666E-02	1.848E+00	9.811E-01	7.071E-03	7.703E-01	-0.036	0.068	0.055
3.0000	1.844E+00	4.286E-02	1.887E+00	1.517E+00	1.089E-02	1.231E+00	-0.028	0.055	0.041
4.0000	1.869E+00	6.056E-02	1.929E+00	2.041E+00	1.493E-02	1.601E+00	-0.024	0.047	0.034
5.0000	1.891E+00	7.922E-02	1.970E+00	2.554E+00	1.910E-02	1.906E+00	-0.022	0.042	0.029
6.0000	1.910E+00	9.865E-02	2.008E+00	3.057E+00	2.336E-02	2.165E+00	-0.021	0.039	0.026
8.0000	1.942E+00	1.392E-01	2.081E+00	4.035E+00	3.203E-02	2.589E+00	-0.019	0.034	0.023
10.0000	1.967E+00	1.816E-01	2.148E+00	4.980E+00	4.077E-02	2.928E+00	-0.018	0.031	0.021
15.0000	2.013E+00	2.924E-01	2.305E+00	7.226E+00	6.248E-02	3.567E+00	-0.015	0.026	0.017
20.0000	2.045E+00	4.079E-01	2.453E+00	9.327E+00	8.357E-02	4.039E+00	-0.013	0.022	0.015
30.0000	2.088E+00	6.472E-01	2.736E+00	1.318E+01	1.232E-01	4.734E+00	-0.009	0.018	0.011
40.0000	2.117E+00	8.936E-01	3.011E+00	1.667E+01	1.592E-01	5.252E+00	-0.006	0.016	0.009
50.0000	2.139E+00	1.145E+00	3.284E+00	1.985E+01	1.920E-01	5.665E+00	-0.005	0.014	0.007
60.0000	2.156E+00	1.399E+00	3.555E+00	2.277E+01	2.220E-01	6.010E+00	-0.003	0.012	0.006
80.0000	2.182E+00	1.914E+00	4.097E+00	2.801E+01	2.745E-01	6.563E+00	-0.002	0.010	0.004
100.0000	2.202E+00	2.437E+00	4.640E+00	3.259E+01	3.190E-01	6.998E+00	-0.001	0.009	0.003
150.0000	2.238E+00	3.763E+00	6.001E+00	4.204E+01	4.061E-01	7.796E+00	-0.001	0.007	0.002
200.0000	2.263E+00	5.103E+00	7.366E+00	4.955E+01	4.701E-01	8.367E+00	0.000	0.006	0.001
300.0000	2.298E+00	7.802E+00	1.010E+01	6.110E+01	5.590E-01	9.173E+00	0.000	0.005	0.001
400.0000	2.322E+00	1.051E+01	1.284E+01	6.986E+01	6.186E-01	9.747E+00	0.000	0.004	0.000
500.0000	2.341E+00	1.323E+01	1.558E+01	7.692E+01	6.620E-01	1.019E+01	0.000	0.004	0.000
600.0000	2.357E+00	1.596E+01	1.832E+01	8.284E+01	6.951E-01	1.056E+01	0.000	0.004	0.000
800.0000	2.382E+00	2.143E+01	2.381E+01	9.239E+01	7.431E-01	1.113E+01	0.000	0.003	0.000
1000.0000	2.401E+00	2.691E+01	2.931E+01	9.994E+01	7.764E-01	1.158E+01	0.000	0.003	0.000

KEY DATA FOR IONIZING-RADIATION DOSIMETRY: MEASUREMENT STANDARDS AND APPLICATIONS

Table A.4. Positrons in air, $I = 85.7$ eV, $\rho = 0.0012$ g cm⁻³.

T	S_{el}/ρ	S_{rad}/ρ	S_{tot}/ρ	ρr_0	Y	δ	$\partial(\log)/\partial(\log I)$		
MeV	MeV cm ² g ⁻¹			g cm ⁻²			S_{el}/ρ	ρr_0	Y
0.0010	1.235E+02	9.936E-04	1.235E+02	4.047E-06	4.021E-06	0.000E+00	-0.318	0.318	0.318
0.0015	9.305E+01	1.210E-03	9.305E+01	8.765E-06	6.180E-06	0.000E+00	-0.282	0.307	0.306
0.0020	7.551E+01	1.372E-03	7.551E+01	1.477E-05	8.527E-06	0.000E+00	-0.261	0.292	0.289
0.0030	5.576E+01	1.606E-03	5.577E+01	3.038E-05	1.350E-05	0.000E+00	-0.236	0.269	0.264
0.0040	4.475E+01	1.774E-03	4.475E+01	5.054E-05	1.868E-05	0.000E+00	-0.221	0.252	0.248
0.0050	3.764E+01	1.904E-03	3.764E+01	7.501E-05	2.397E-05	0.000E+00	-0.211	0.240	0.236
0.0060	3.264E+01	2.009E-03	3.264E+01	1.036E-04	2.932E-05	0.000E+00	-0.204	0.231	0.227
0.0080	2.601E+01	2.173E-03	2.602E+01	1.728E-04	4.012E-05	0.000E+00	-0.193	0.218	0.213
0.0100	2.179E+01	2.297E-03	2.180E+01	2.571E-04	5.099E-05	0.000E+00	-0.185	0.208	0.204
0.0150	1.578E+01	2.513E-03	1.579E+01	5.308E-04	7.812E-05	0.000E+00	-0.173	0.193	0.189
0.0200	1.256E+01	2.664E-03	1.256E+01	8.887E-04	1.050E-04	0.000E+00	-0.165	0.183	0.180
0.0300	9.125E+00	2.880E-03	9.128E+00	1.837E-03	1.580E-04	0.000E+00	-0.156	0.171	0.169
0.0400	7.306E+00	3.035E-03	7.309E+00	3.071E-03	2.100E-04	0.000E+00	-0.150	0.164	0.162
0.0500	6.171E+00	3.157E-03	6.174E+00	4.566E-03	2.607E-04	0.000E+00	-0.146	0.158	0.157
0.0600	5.393E+00	3.265E-03	5.396E+00	6.304E-03	3.103E-04	0.000E+00	-0.143	0.154	0.153
0.0800	4.393E+00	3.452E-03	4.396E+00	1.044E-02	4.067E-04	0.000E+00	-0.138	0.149	0.147
0.1000	3.776E+00	3.617E-03	3.780E+00	1.537E-02	4.997E-04	0.000E+00	-0.135	0.145	0.144
0.1500	2.935E+00	3.991E-03	2.939E+00	3.058E-02	7.200E-04	0.000E+00	-0.130	0.138	0.137
0.2000	2.510E+00	4.350E-03	2.514E+00	4.909E-02	9.265E-04	0.000E+00	-0.126	0.134	0.133
0.3000	2.091E+00	5.111E-03	2.096E+00	9.316E-02	1.313E-03	0.000E+00	-0.122	0.129	0.128
0.4000	1.893E+00	5.950E-03	1.899E+00	1.435E-01	1.681E-03	0.000E+00	-0.118	0.126	0.124
0.5000	1.784E+00	6.862E-03	1.791E+00	1.979E-01	2.041E-03	0.000E+00	-0.115	0.123	0.122
0.6000	1.719E+00	7.840E-03	1.727E+00	2.548E-01	2.399E-03	0.000E+00	-0.113	0.121	0.119
0.8000	1.652E+00	9.959E-03	1.662E+00	3.732E-01	3.115E-03	0.000E+00	-0.109	0.118	0.116
1.0000	1.626E+00	1.227E-02	1.638E+00	4.946E-01	3.839E-03	0.000E+00	-0.106	0.115	0.113
1.5000	1.621E+00	1.867E-02	1.639E+00	8.005E-01	5.700E-03	0.000E+00	-0.101	0.110	0.107
2.0000	1.642E+00	2.581E-02	1.668E+00	1.103E+00	7.630E-03	0.000E+00	-0.097	0.107	0.103
3.0000	1.695E+00	4.163E-02	1.737E+00	1.691E+00	1.165E-02	0.000E+00	-0.092	0.102	0.097
4.0000	1.744E+00	5.892E-02	1.803E+00	2.256E+00	1.582E-02	0.000E+00	-0.089	0.099	0.093
5.0000	1.786E+00	7.722E-02	1.863E+00	2.801E+00	2.007E-02	0.000E+00	-0.087	0.096	0.090
6.0000	1.823E+00	9.630E-02	1.919E+00	3.330E+00	2.436E-02	0.000E+00	-0.085	0.094	0.087
8.0000	1.883E+00	1.363E-01	2.019E+00	4.345E+00	3.299E-02	0.000E+00	-0.082	0.090	0.083
10.0000	1.931E+00	1.780E-01	2.109E+00	5.314E+00	4.159E-02	0.000E+00	-0.080	0.087	0.080
15.0000	2.020E+00	2.873E-01	2.308E+00	7.578E+00	6.263E-02	0.000E+00	-0.076	0.082	0.074
20.0000	2.085E+00	4.014E-01	2.486E+00	9.664E+00	8.278E-02	0.000E+00	-0.074	0.078	0.070
30.0000	2.176E+00	6.379E-01	2.814E+00	1.344E+01	1.202E-01	7.556E-03	-0.064	0.072	0.063
40.0000	2.232E+00	8.816E-01	3.114E+00	1.681E+01	1.540E-01	1.375E-01	-0.045	0.066	0.053
50.0000	2.269E+00	1.130E+00	3.399E+00	1.988E+01	1.849E-01	3.188E-01	-0.036	0.060	0.045
60.0000	2.297E+00	1.383E+00	3.680E+00	2.271E+01	2.132E-01	5.024E-01	-0.031	0.055	0.038
80.0000	2.337E+00	1.897E+00	4.234E+00	2.777E+01	2.632E-01	8.360E-01	-0.025	0.048	0.029
100.0000	2.367E+00	2.419E+00	4.786E+00	3.221E+01	3.061E-01	1.120E+00	-0.022	0.043	0.024
150.0000	2.418E+00	3.750E+00	6.168E+00	4.139E+01	3.910E-01	1.669E+00	-0.019	0.036	0.017
200.0000	2.452E+00	5.102E+00	7.555E+00	4.871E+01	4.542E-01	2.078E+00	-0.017	0.031	0.013
300.0000	2.500E+00	7.841E+00	1.034E+01	5.997E+01	5.431E-01	2.674E+00	-0.015	0.026	0.009
400.0000	2.532E+00	1.061E+01	1.314E+01	6.853E+01	6.035E-01	3.116E+00	-0.013	0.023	0.007
500.0000	2.556E+00	1.340E+01	1.595E+01	7.543E+01	6.477E-01	3.471E+00	-0.011	0.021	0.006
600.0000	2.575E+00	1.620E+01	1.877E+01	8.120E+01	6.817E-01	3.772E+00	-0.010	0.020	0.005
800.0000	2.603E+00	2.184E+01	2.444E+01	9.051E+01	7.312E-01	4.266E+00	-0.007	0.018	0.004
1000.0000	2.624E+00	2.752E+01	3.014E+01	9.787E+01	7.657E-01	4.667E+00	-0.005	0.017	0.003

Appendix

Table A.5. Positrons in graphite, $I = 81 \text{ eV}$, $\rho = 2.265 \text{ g cm}^{-3}$.

T	S_{el}/ρ	S_{rad}/ρ	S_{tot}/ρ	ρr_0	Y	δ	$\partial(\log)/\partial(\log I)$		
MeV	MeV cm ² g ⁻¹			g cm ⁻²			S_{el}/ρ	ρr_0	Y
0.00100	1.258E+02	9.929E-04	1.258E+02	3.973E-06	3.945E-06	2.470E-04	-0.312	0.312	0.312
0.00150	9.460E+01	1.183E-03	9.460E+01	8.610E-06	6.025E-06	3.719E-04	-0.277	0.301	0.301
0.00200	7.667E+01	1.322E-03	7.667E+01	1.452E-05	8.235E-06	4.978E-04	-0.257	0.287	0.285
0.00300	5.654E+01	1.517E-03	5.655E+01	2.991E-05	1.283E-05	7.522E-04	-0.233	0.264	0.261
0.00400	4.533E+01	1.655E-03	4.534E+01	4.980E-05	1.754E-05	1.010E-03	-0.218	0.249	0.245
0.00500	3.811E+01	1.760E-03	3.811E+01	7.396E-05	2.230E-05	1.272E-03	-0.208	0.237	0.233
0.00600	3.303E+01	1.844E-03	3.303E+01	1.022E-04	2.708E-05	1.537E-03	-0.201	0.228	0.224
0.00800	2.631E+01	1.974E-03	2.631E+01	1.706E-04	3.667E-05	2.079E-03	-0.190	0.215	0.211
0.01000	2.203E+01	2.071E-03	2.203E+01	2.540E-04	4.624E-05	2.634E-03	-0.183	0.205	0.202
0.01500	1.594E+01	2.241E-03	1.594E+01	5.248E-04	6.994E-05	4.085E-03	-0.170	0.190	0.187
0.02000	1.268E+01	2.364E-03	1.268E+01	8.792E-04	9.334E-05	5.622E-03	-0.163	0.180	0.178
0.03000	9.206E+00	2.533E-03	9.208E+00	1.819E-03	1.393E-04	8.946E-03	-0.153	0.168	0.166
0.04000	7.366E+00	2.655E-03	7.369E+00	3.042E-03	1.839E-04	1.259E-02	-0.147	0.161	0.159
0.05000	6.219E+00	2.754E-03	6.222E+00	4.526E-03	2.275E-04	1.654E-02	-0.142	0.155	0.154
0.06000	5.432E+00	2.840E-03	5.435E+00	6.251E-03	2.700E-04	2.079E-02	-0.139	0.151	0.150
0.08000	4.421E+00	2.989E-03	4.424E+00	1.036E-02	3.525E-04	3.011E-02	-0.133	0.145	0.144
0.10000	3.797E+00	3.119E-03	3.800E+00	1.526E-02	4.317E-04	4.047E-02	-0.129	0.141	0.139
0.15000	2.945E+00	3.420E-03	2.949E+00	3.040E-02	6.188E-04	7.039E-02	-0.120	0.132	0.131
0.20000	2.513E+00	3.713E-03	2.516E+00	4.888E-02	7.938E-04	1.052E-01	-0.113	0.126	0.125
0.30000	2.083E+00	4.342E-03	2.088E+00	9.301E-02	1.122E-03	1.847E-01	-0.100	0.117	0.115
0.40000	1.877E+00	5.041E-03	1.882E+00	1.437E-01	1.436E-03	2.708E-01	-0.090	0.109	0.106
0.50000	1.760E+00	5.804E-03	1.766E+00	1.987E-01	1.746E-03	3.580E-01	-0.082	0.102	0.099
0.60000	1.688E+00	6.623E-03	1.694E+00	2.566E-01	2.054E-03	4.438E-01	-0.076	0.097	0.093
0.80000	1.609E+00	8.405E-03	1.617E+00	3.778E-01	2.678E-03	6.075E-01	-0.068	0.089	0.084
1.00000	1.571E+00	1.034E-02	1.581E+00	5.030E-01	3.315E-03	7.593E-01	-0.062	0.083	0.077
1.50000	1.542E+00	1.574E-02	1.558E+00	8.225E-01	4.977E-03	1.090E+00	-0.053	0.073	0.066
2.00000	1.543E+00	2.175E-02	1.565E+00	1.143E+00	6.730E-03	1.364E+00	-0.049	0.066	0.059
3.00000	1.564E+00	3.507E-02	1.599E+00	1.776E+00	1.044E-02	1.798E+00	-0.043	0.059	0.051
4.00000	1.588E+00	4.964E-02	1.638E+00	2.393E+00	1.436E-02	2.136E+00	-0.040	0.054	0.046
5.00000	1.609E+00	6.512E-02	1.674E+00	2.997E+00	1.841E-02	2.414E+00	-0.037	0.051	0.042
6.00000	1.628E+00	8.124E-02	1.709E+00	3.588E+00	2.254E-02	2.653E+00	-0.034	0.048	0.040
8.00000	1.658E+00	1.150E-01	1.773E+00	4.737E+00	3.096E-02	3.051E+00	-0.028	0.043	0.035
10.00000	1.681E+00	1.503E-01	1.831E+00	5.847E+00	3.946E-02	3.381E+00	-0.024	0.040	0.031
15.00000	1.721E+00	2.430E-01	1.964E+00	8.481E+00	6.068E-02	4.033E+00	-0.015	0.033	0.023
20.00000	1.747E+00	3.396E-01	2.087E+00	1.095E+01	8.137E-02	4.533E+00	-0.009	0.028	0.017
30.00000	1.782E+00	5.406E-01	2.323E+00	1.549E+01	1.205E-01	5.279E+00	-0.005	0.021	0.011
40.00000	1.806E+00	7.483E-01	2.554E+00	1.959E+01	1.562E-01	5.828E+00	-0.003	0.017	0.007
50.00000	1.824E+00	9.605E-01	2.784E+00	2.334E+01	1.889E-01	6.260E+00	-0.002	0.015	0.005
60.00000	1.838E+00	1.176E+00	3.015E+00	2.679E+01	2.188E-01	6.616E+00	-0.001	0.013	0.004
80.00000	1.861E+00	1.616E+00	3.477E+00	3.296E+01	2.714E-01	7.183E+00	-0.001	0.011	0.003
100.00000	1.878E+00	2.063E+00	3.941E+00	3.836E+01	3.161E-01	7.624E+00	0.000	0.009	0.002
150.00000	1.910E+00	3.204E+00	5.113E+00	4.947E+01	4.038E-01	8.429E+00	0.000	0.007	0.001
200.00000	1.932E+00	4.365E+00	6.297E+00	5.827E+01	4.684E-01	9.002E+00	0.000	0.006	0.001
300.00000	1.963E+00	6.720E+00	8.683E+00	7.174E+01	5.581E-01	9.811E+00	0.000	0.005	0.000
400.00000	1.985E+00	9.104E+00	1.109E+01	8.191E+01	6.184E-01	1.039E+01	0.000	0.004	0.000
500.00000	2.002E+00	1.151E+01	1.351E+01	9.007E+01	6.622E-01	1.083E+01	0.000	0.004	0.000
600.00000	2.016E+00	1.392E+01	1.594E+01	9.687E+01	6.957E-01	1.119E+01	0.000	0.004	0.000
800.00000	2.038E+00	1.880E+01	2.083E+01	1.078E+02	7.441E-01	1.177E+01	0.000	0.003	0.000
1000.00000	2.055E+00	2.370E+01	2.576E+01	1.164E+02	7.776E-01	1.222E+01	0.000	0.003	0.000

KEY DATA FOR IONIZING-RADIATION DOSIMETRY: MEASUREMENT STANDARDS AND APPLICATIONS

Table A.6. Positrons in liquid water, $I = 78 \text{ eV}$, $\rho = 0.998 \text{ g cm}^{-3}$.

T	S_{el}/ρ	S_{rad}/ρ	S_{tot}/ρ	ρr_0	Y	δ	$\partial(\log)/\partial(\log I)$		
MeV	MeV cm ² g ⁻¹			g cm ⁻²			S_{el}/ρ	ρr_0	Y
0.0010	1.415E+02	1.012E-03	1.415E+02	3.534E-06	3.575E-06	0.000E+00	-0.309	0.309	0.309
0.0015	1.062E+02	1.219E-03	1.062E+02	7.661E-06	5.481E-06	0.000E+00	-0.275	0.298	0.297
0.0020	8.603E+01	1.377E-03	8.603E+01	1.293E-05	7.543E-06	0.000E+00	-0.255	0.284	0.282
0.0030	6.339E+01	1.608E-03	6.339E+01	2.665E-05	1.192E-05	0.000E+00	-0.231	0.262	0.258
0.0040	5.080E+01	1.775E-03	5.080E+01	4.440E-05	1.647E-05	0.000E+00	-0.217	0.246	0.242
0.0050	4.268E+01	1.904E-03	4.269E+01	6.597E-05	2.113E-05	0.000E+00	-0.207	0.235	0.231
0.0060	3.699E+01	2.009E-03	3.699E+01	9.121E-05	2.585E-05	0.000E+00	-0.200	0.226	0.222
0.0080	2.945E+01	2.172E-03	2.945E+01	1.523E-04	3.539E-05	0.000E+00	-0.189	0.213	0.209
0.0100	2.466E+01	2.297E-03	2.466E+01	2.268E-04	4.501E-05	0.000E+00	-0.182	0.204	0.200
0.0150	1.784E+01	2.516E-03	1.784E+01	4.689E-04	6.907E-05	0.000E+00	-0.170	0.189	0.186
0.0200	1.418E+01	2.665E-03	1.418E+01	7.857E-04	9.294E-05	0.000E+00	-0.163	0.180	0.177
0.0300	1.030E+01	2.881E-03	1.030E+01	1.626E-03	1.400E-04	0.000E+00	-0.154	0.168	0.166
0.0400	8.239E+00	3.036E-03	8.242E+00	2.720E-03	1.861E-04	0.000E+00	-0.148	0.161	0.159
0.0500	6.956E+00	3.158E-03	6.959E+00	4.046E-03	2.311E-04	0.000E+00	-0.144	0.156	0.154
0.0600	6.077E+00	3.266E-03	6.081E+00	5.588E-03	2.752E-04	0.000E+00	-0.141	0.152	0.151
0.0800	4.948E+00	3.455E-03	4.951E+00	9.261E-03	3.609E-04	0.000E+00	-0.136	0.147	0.145
0.1000	4.252E+00	3.621E-03	4.256E+00	1.364E-02	4.437E-04	0.000E+00	-0.133	0.143	0.142
0.1500	3.303E+00	4.000E-03	3.307E+00	2.715E-02	6.401E-04	0.000E+00	-0.128	0.137	0.136
0.2000	2.824E+00	4.363E-03	2.828E+00	4.360E-02	8.244E-04	0.000E+00	-0.125	0.133	0.132
0.3000	2.352E+00	5.132E-03	2.357E+00	8.278E-02	1.170E-03	0.000E+00	-0.120	0.128	0.127
0.4000	2.128E+00	5.978E-03	2.134E+00	1.276E-01	1.500E-03	0.000E+00	-0.117	0.124	0.123
0.5000	2.005E+00	6.896E-03	2.012E+00	1.760E-01	1.823E-03	0.000E+00	-0.114	0.122	0.120
0.6000	1.930E+00	7.879E-03	1.938E+00	2.267E-01	2.143E-03	1.501E-02	-0.094	0.118	0.116
0.8000	1.846E+00	1.002E-02	1.856E+00	3.324E-01	2.789E-03	1.005E-01	-0.075	0.107	0.102
1.0000	1.806E+00	1.234E-02	1.818E+00	4.414E-01	3.448E-03	2.086E-01	-0.063	0.097	0.090
1.5000	1.774E+00	1.881E-02	1.793E+00	7.191E-01	5.170E-03	4.982E-01	-0.045	0.080	0.069
2.0000	1.774E+00	2.601E-02	1.800E+00	9.976E-01	6.990E-03	7.703E-01	-0.037	0.069	0.056
3.0000	1.794E+00	4.198E-02	1.836E+00	1.548E+00	1.086E-02	1.231E+00	-0.028	0.056	0.042
4.0000	1.817E+00	5.947E-02	1.877E+00	2.087E+00	1.496E-02	1.601E+00	-0.025	0.048	0.034
5.0000	1.838E+00	7.795E-02	1.916E+00	2.614E+00	1.920E-02	1.906E+00	-0.023	0.043	0.030
6.0000	1.857E+00	9.722E-02	1.954E+00	3.131E+00	2.354E-02	2.165E+00	-0.021	0.039	0.027
8.0000	1.888E+00	1.376E-01	2.026E+00	4.136E+00	3.236E-02	2.589E+00	-0.019	0.034	0.023
10.0000	1.913E+00	1.798E-01	2.093E+00	5.107E+00	4.128E-02	2.928E+00	-0.018	0.031	0.021
15.0000	1.958E+00	2.905E-01	2.249E+00	7.410E+00	6.345E-02	3.567E+00	-0.015	0.026	0.018
20.0000	1.990E+00	4.061E-01	2.396E+00	9.563E+00	8.498E-02	4.039E+00	-0.013	0.023	0.015
30.0000	2.033E+00	6.462E-01	2.679E+00	1.351E+01	1.254E-01	4.734E+00	-0.009	0.019	0.012
40.0000	2.062E+00	8.937E-01	2.956E+00	1.706E+01	1.622E-01	5.252E+00	-0.006	0.016	0.009
50.0000	2.084E+00	1.146E+00	3.230E+00	2.029E+01	1.956E-01	5.665E+00	-0.005	0.014	0.007
60.0000	2.101E+00	1.403E+00	3.504E+00	2.327E+01	2.261E-01	6.010E+00	-0.003	0.013	0.006
80.0000	2.127E+00	1.925E+00	4.052E+00	2.857E+01	2.794E-01	6.563E+00	-0.002	0.011	0.004
100.0000	2.147E+00	2.456E+00	4.603E+00	3.320E+01	3.246E-01	6.998E+00	-0.001	0.009	0.003
150.0000	2.182E+00	3.810E+00	5.992E+00	4.269E+01	4.127E-01	7.796E+00	-0.001	0.007	0.002
200.0000	2.207E+00	5.185E+00	7.392E+00	5.019E+01	4.772E-01	8.367E+00	0.000	0.006	0.001
300.0000	2.242E+00	7.974E+00	1.022E+01	6.165E+01	5.665E-01	9.173E+00	0.000	0.005	0.001
400.0000	2.266E+00	1.079E+01	1.306E+01	7.029E+01	6.262E-01	9.747E+00	0.000	0.004	0.000
500.0000	2.286E+00	1.364E+01	1.592E+01	7.721E+01	6.694E-01	1.019E+01	0.000	0.004	0.000
600.0000	2.301E+00	1.649E+01	1.880E+01	8.299E+01	7.025E-01	1.056E+01	0.000	0.004	0.000
800.0000	2.326E+00	2.225E+01	2.457E+01	9.227E+01	7.500E-01	1.113E+01	0.000	0.003	0.000
1000.0000	2.345E+00	2.804E+01	3.039E+01	9.957E+01	7.830E-01	1.158E+01	0.000	0.003	0.000

Appendix

Table A.7. Protons in air, $I = 85.7$ eV, $\rho = 0.0012$ g cm⁻³.

T	S_{el}/ρ	S_{nuc}/ρ	S_{tot}/ρ	ρr_0	Detour factor	$\partial(\log)/\partial(\log I)$	
MeV		MeV cm ² g ⁻¹		g cm ⁻²		S_{el}/ρ	ρr_0
0.0010	1.197E+02	2.163E+01	1.413E+02	9.857E-06	0.3304	0.000	0.000
0.0015	1.467E+02	1.840E+01	1.651E+02	1.310E-05	0.3760	0.000	0.000
0.0020	1.693E+02	1.614E+01	1.854E+02	1.595E-05	0.4123	0.000	0.000
0.0030	2.074E+02	1.314E+01	2.205E+02	2.088E-05	0.4674	0.000	0.000
0.0040	2.395E+02	1.120E+01	2.507E+02	2.512E-05	0.5084	0.000	0.000
0.0050	2.678E+02	9.825E+00	2.776E+02	2.891E-05	0.5406	0.000	0.000
0.0060	2.933E+02	8.786E+00	3.021E+02	3.236E-05	0.5669	0.000	0.000
0.0080	3.387E+02	7.310E+00	3.460E+02	3.853E-05	0.6078	0.000	0.000
0.0100	3.787E+02	6.300E+00	3.850E+02	4.400E-05	0.6387	0.000	0.000
0.0150	4.504E+02	4.751E+00	4.552E+02	5.588E-05	0.6923	0.000	0.000
0.0200	5.067E+02	3.858E+00	5.106E+02	6.623E-05	0.7281	0.000	0.000
0.0300	5.905E+02	2.848E+00	5.933E+02	8.430E-05	0.7743	0.000	0.000
0.0400	6.483E+02	2.282E+00	6.506E+02	1.003E-04	0.8041	0.000	0.000
0.0500	6.877E+02	1.917E+00	6.896E+02	1.152E-04	0.8256	0.000	0.000
0.0600	7.132E+02	1.659E+00	7.149E+02	1.295E-04	0.8420	0.000	0.000
0.0800	7.341E+02	1.316E+00	7.354E+02	1.569E-04	0.8664	0.000	0.000
0.1000	7.290E+02	1.098E+00	7.301E+02	1.842E-04	0.8839	0.000	0.000
0.1500	6.672E+02	7.861E-01	6.680E+02	2.554E-04	0.9133	0.000	0.000
0.2000	5.922E+02	6.183E-01	5.928E+02	3.349E-04	0.9320	0.000	0.000
0.3000	4.765E+02	4.390E-01	4.770E+02	5.240E-04	0.9544	-0.204	0.031
0.4000	4.012E+02	3.435E-01	4.016E+02	7.536E-04	0.9668	-0.374	0.114
0.5000	3.498E+02	2.836E-01	3.501E+02	1.021E-03	0.9743	-0.412	0.190
0.6000	3.121E+02	2.423E-01	3.123E+02	1.324E-03	0.9793	-0.385	0.237
0.8000	2.587E+02	1.885E-01	2.589E+02	2.031E-03	0.9852	-0.348	0.281
1.0000	2.227E+02	1.548E-01	2.229E+02	2.867E-03	0.9886	-0.323	0.297
1.5000	1.682E+02	1.080E-01	1.683E+02	5.479E-03	0.9926	-0.286	0.299
2.0000	1.370E+02	8.340E-02	1.371E+02	8.792E-03	0.9943	-0.263	0.289
3.0000	1.017E+02	5.778E-02	1.018E+02	1.737E-02	0.9959	-0.237	0.269
4.0000	8.192E+01	4.444E-02	8.197E+01	2.839E-02	0.9966	-0.221	0.253
5.0000	6.905E+01	3.621E-02	6.909E+01	4.173E-02	0.9969	-0.210	0.241
6.0000	5.994E+01	3.061E-02	5.997E+01	5.731E-02	0.9972	-0.202	0.231
8.0000	4.781E+01	2.345E-02	4.783E+01	9.493E-02	0.9975	-0.190	0.217
10.0000	4.004E+01	1.905E-02	4.006E+01	1.408E-01	0.9976	-0.183	0.207
15.0000	2.892E+01	1.304E-02	2.894E+01	2.899E-01	0.9979	-0.170	0.191
20.0000	2.293E+01	9.953E-03	2.294E+01	4.855E-01	0.9980	-0.162	0.180
30.0000	1.652E+01	6.790E-03	1.653E+01	1.007E+00	0.9981	-0.152	0.168
40.0000	1.312E+01	5.173E-03	1.312E+01	1.691E+00	0.9982	-0.146	0.160
50.0000	1.098E+01	4.187E-03	1.099E+01	2.528E+00	0.9983	-0.142	0.155
60.0000	9.514E+00	3.523E-03	9.518E+00	3.509E+00	0.9983	-0.138	0.150
80.0000	7.618E+00	2.681E-03	7.620E+00	5.876E+00	0.9984	-0.133	0.144
100.0000	6.441E+00	2.169E-03	6.443E+00	8.744E+00	0.9985	-0.130	0.140
150.0000	4.815E+00	1.475E-03	4.816E+00	1.786E+01	0.9986	-0.124	0.133
200.0000	3.975E+00	1.122E-03	3.976E+00	2.937E+01	0.9986	-0.120	0.129
300.0000	3.117E+00	7.631E-04	3.118E+00	5.816E+01	0.9987	-0.115	0.123
400.0000	2.686E+00	5.806E-04	2.687E+00	9.293E+01	0.9988	-0.112	0.120
500.0000	2.431E+00	4.697E-04	2.431E+00	1.322E+02	0.9989	-0.110	0.117
600.0000	2.265E+00	3.951E-04	2.266E+00	1.749E+02	0.9990	-0.108	0.115
800.0000	2.069E+00	3.009E-04	2.069E+00	2.677E+02	0.9990	-0.105	0.112
1000.0000	1.962E+00	2.436E-04	1.963E+00	3.672E+02	0.9991	-0.102	0.110
1500.0000	1.850E+00	1.661E-04	1.850E+00	6.311E+02	0.9992	-0.097	0.105
2000.0000	1.820E+00	1.267E-04	1.820E+00	9.041E+02	0.9993	-0.094	0.102
3000.0000	1.828E+00	8.655E-05	1.828E+00	1.454E+03	0.9994	-0.089	0.098
4000.0000	1.861E+00	6.608E-05	1.861E+00	1.996E+03	0.9995	-0.085	0.095
5000.0000	1.898E+00	5.362E-05	1.898E+00	2.528E+03	0.9996	-0.083	0.093
6000.0000	1.934E+00	4.520E-05	1.934E+00	3.050E+03	0.9996	-0.081	0.091
8000.0000	1.998E+00	3.453E-05	1.998E+00	4.067E+03	0.9997	-0.078	0.088
10000.0000	2.052E+00	2.803E-05	2.053E+00	5.054E+03	0.9997	-0.075	0.086

KEY DATA FOR IONIZING-RADIATION DOSIMETRY: MEASUREMENT STANDARDS AND APPLICATIONS

Table A.8. Protons in graphite, $I = 81$ eV, $\rho = 2.265$ g cm⁻³.

T	S_{el}/ρ	S_{nuc}/ρ	S_{tot}/ρ	ρr_0	Detour factor	$\partial(\log)/\partial(\log I)$	
MeV		MeV cm ² g ⁻¹		g cm ⁻²		S_{el}/ρ	ρr_0
0.0010	1.185E+02	2.514E+01	1.436E+02	8.186E-06	0.3727	0.000	0.000
0.0015	1.451E+02	2.106E+01	1.662E+02	1.092E-05	0.4211	0.000	0.000
0.0020	1.676E+02	1.829E+01	1.859E+02	1.331E-05	0.4588	0.000	0.000
0.0030	2.053E+02	1.469E+01	2.200E+02	1.742E-05	0.5147	0.000	0.000
0.0040	2.370E+02	1.241E+01	2.494E+02	2.098E-05	0.5557	0.000	0.000
0.0050	2.650E+02	1.081E+01	2.758E+02	2.421E-05	0.5877	0.000	0.000
0.0060	2.903E+02	9.618E+00	2.999E+02	2.718E-05	0.6136	0.000	0.000
0.0080	3.352E+02	7.943E+00	3.431E+02	3.254E-05	0.6534	0.000	0.000
0.0100	3.748E+02	6.810E+00	3.816E+02	3.739E-05	0.6832	0.000	0.000
0.0150	4.353E+02	5.094E+00	4.404E+02	4.812E-05	0.7343	0.000	0.000
0.0200	4.811E+02	4.115E+00	4.852E+02	5.763E-05	0.7677	0.000	0.000
0.0300	5.503E+02	3.019E+00	5.533E+02	7.457E-05	0.8102	0.000	0.000
0.0400	6.110E+02	2.411E+00	6.134E+02	8.985E-05	0.8371	0.000	0.000
0.0500	6.630E+02	2.019E+00	6.650E+02	1.040E-04	0.8559	0.000	0.000
0.0600	6.976E+02	1.744E+00	6.993E+02	1.175E-04	0.8702	0.000	0.000
0.0800	7.267E+02	1.380E+00	7.281E+02	1.433E-04	0.8908	0.000	0.000
0.1000	7.266E+02	1.149E+00	7.277E+02	1.688E-04	0.9055	0.000	0.000
0.1500	6.712E+02	8.200E-01	6.720E+02	2.353E-04	0.9298	0.000	0.000
0.2000	5.986E+02	6.436E-01	5.992E+02	3.097E-04	0.9451	0.000	0.000
0.3000	4.833E+02	4.559E-01	4.837E+02	4.928E-04	0.9635	-0.080	0.005
0.4000	4.071E+02	3.561E-01	4.075E+02	7.190E-04	0.9736	-0.324	0.071
0.5000	3.546E+02	2.937E-01	3.549E+02	9.828E-04	0.9797	-0.406	0.154
0.6000	3.159E+02	2.505E-01	3.161E+02	1.282E-03	0.9836	-0.380	0.210
0.8000	2.624E+02	1.943E-01	2.626E+02	1.980E-03	0.9884	-0.343	0.263
1.0000	2.261E+02	1.594E-01	2.263E+02	2.803E-03	0.9910	-0.319	0.282
1.5000	1.711E+02	1.108E-01	1.712E+02	5.373E-03	0.9941	-0.281	0.289
2.0000	1.394E+02	8.541E-02	1.395E+02	8.628E-03	0.9955	-0.259	0.281
3.0000	1.035E+02	5.901E-02	1.036E+02	1.706E-02	0.9967	-0.233	0.263
4.0000	8.326E+01	4.529E-02	8.330E+01	2.789E-02	0.9972	-0.217	0.248
5.0000	7.010E+01	3.685E-02	7.014E+01	4.103E-02	0.9975	-0.207	0.236
6.0000	6.081E+01	3.112E-02	6.084E+01	5.638E-02	0.9977	-0.199	0.227
8.0000	4.845E+01	2.380E-02	4.847E+01	9.348E-02	0.9980	-0.188	0.213
10.0000	4.055E+01	1.932E-02	4.057E+01	1.388E-01	0.9981	-0.180	0.204
15.0000	2.925E+01	1.320E-02	2.926E+01	2.861E-01	0.9983	-0.168	0.188
20.0000	2.317E+01	1.006E-02	2.318E+01	4.796E-01	0.9984	-0.160	0.178
30.0000	1.668E+01	6.858E-03	1.669E+01	9.960E-01	0.9985	-0.150	0.166
40.0000	1.323E+01	5.221E-03	1.324E+01	1.674E+00	0.9986	-0.144	0.158
50.0000	1.107E+01	4.225E-03	1.108E+01	2.504E+00	0.9986	-0.139	0.152
60.0000	9.591E+00	3.553E-03	9.595E+00	3.477E+00	0.9987	-0.136	0.148
80.0000	7.675E+00	2.703E-03	7.678E+00	5.826E+00	0.9987	-0.130	0.142
100.0000	6.486E+00	2.186E-03	6.488E+00	8.673E+00	0.9988	-0.127	0.138
150.0000	4.843E+00	1.486E-03	4.845E+00	1.773E+01	0.9988	-0.120	0.130
200.0000	3.994E+00	1.130E-03	3.995E+00	2.917E+01	0.9989	-0.114	0.125
300.0000	3.125E+00	7.686E-04	3.126E+00	5.786E+01	0.9990	-0.106	0.117
400.0000	2.687E+00	5.847E-04	2.688E+00	9.257E+01	0.9990	-0.099	0.112
500.0000	2.426E+00	4.730E-04	2.427E+00	1.319E+02	0.9991	-0.092	0.107
600.0000	2.255E+00	3.979E-04	2.256E+00	1.747E+02	0.9991	-0.087	0.103
800.0000	2.050E+00	3.029E-04	2.050E+00	2.681E+02	0.9992	-0.077	0.095
1000.0000	1.936E+00	2.453E-04	1.936E+00	3.688E+02	0.9993	-0.071	0.089
1500.0000	1.806E+00	1.672E-04	1.806E+00	6.378E+02	0.9994	-0.060	0.079
2000.0000	1.760E+00	1.275E-04	1.761E+00	9.188E+02	0.9995	-0.053	0.072
3000.0000	1.744E+00	8.710E-05	1.744E+00	1.491E+03	0.9995	-0.046	0.063
4000.0000	1.756E+00	6.650E-05	1.756E+00	2.063E+03	0.9996	-0.042	0.058
5000.0000	1.775E+00	5.395E-05	1.775E+00	2.629E+03	0.9997	-0.039	0.054
6000.0000	1.796E+00	4.548E-05	1.796E+00	3.189E+03	0.9997	-0.037	0.051
8000.0000	1.834E+00	3.474E-05	1.834E+00	4.291E+03	0.9997	-0.034	0.047
10000.0000	1.868E+00	2.820E-05	1.868E+00	5.371E+03	0.9998	-0.031	0.044

Appendix

Table A.9. Protons in liquid water, $I = 78 \text{ eV}$, $\rho = 0.998 \text{ g cm}^{-3}$.

T	S_{el}/ρ	S_{nuc}/ρ	S_{tot}/ρ	ρr_0	Detour factor	$\partial(\log)/\partial(\log I)$	
MeV	MeV cm ² g ⁻¹			g cm ⁻²		S_{el}/ρ	ρr_0
0.0010	1.337E+02	4.315E+01	1.769E+02	6.319E-06	0.4555	0.000	0.000
0.0015	1.638E+02	3.460E+01	1.984E+02	8.969E-06	0.4906	0.000	0.000
0.0020	1.891E+02	2.927E+01	2.184E+02	1.137E-05	0.5197	0.000	0.000
0.0030	2.316E+02	2.281E+01	2.544E+02	1.560E-05	0.5647	0.000	0.000
0.0040	2.675E+02	1.894E+01	2.864E+02	1.930E-05	0.5986	0.000	0.000
0.0050	2.990E+02	1.631E+01	3.153E+02	2.262E-05	0.6254	0.000	0.000
0.0060	3.276E+02	1.439E+01	3.420E+02	2.567E-05	0.6473	0.000	0.000
0.0080	3.782E+02	1.175E+01	3.900E+02	3.113E-05	0.6813	0.000	0.000
0.0100	4.229E+02	1.000E+01	4.329E+02	3.599E-05	0.7070	0.000	0.000
0.0150	5.036E+02	7.400E+00	5.110E+02	4.657E-05	0.7514	0.000	0.000
0.0200	5.673E+02	5.939E+00	5.732E+02	5.578E-05	0.7808	0.000	0.000
0.0300	6.628E+02	4.325E+00	6.671E+02	7.187E-05	0.8187	0.000	0.000
0.0400	7.290E+02	3.437E+00	7.324E+02	8.613E-05	0.8429	0.000	0.000
0.0500	7.740E+02	2.870E+00	7.769E+02	9.935E-05	0.8602	0.000	0.000
0.0600	8.026E+02	2.473E+00	8.051E+02	1.120E-04	0.8735	0.000	0.000
0.0800	8.241E+02	1.951E+00	8.261E+02	1.364E-04	0.8931	0.000	0.000
0.1000	8.145E+02	1.620E+00	8.161E+02	1.607E-04	0.9073	0.000	0.000
0.1500	7.360E+02	1.152E+00	7.372E+02	2.249E-04	0.9310	0.000	0.000
0.2000	6.585E+02	9.016E-01	6.594E+02	2.967E-04	0.9460	-0.081	0.006
0.3000	5.435E+02	6.351E-01	5.441E+02	4.645E-04	0.9635	-0.295	0.077
0.4000	4.643E+02	4.928E-01	4.648E+02	6.640E-04	0.9731	-0.387	0.159
0.5000	4.065E+02	4.043E-01	4.069E+02	8.945E-04	0.9790	-0.394	0.220
0.6000	3.624E+02	3.438E-01	3.627E+02	1.155E-03	0.9829	-0.368	0.256
0.8000	2.997E+02	2.658E-01	3.000E+02	1.765E-03	0.9877	-0.334	0.288
1.0000	2.574E+02	2.173E-01	2.577E+02	2.487E-03	0.9905	-0.311	0.298
1.5000	1.934E+02	1.504E-01	1.936E+02	4.753E-03	0.9938	-0.276	0.294
2.0000	1.569E+02	1.157E-01	1.570E+02	7.639E-03	0.9952	-0.256	0.283
3.0000	1.160E+02	7.972E-02	1.161E+02	1.514E-02	0.9965	-0.231	0.262
4.0000	9.319E+01	6.113E-02	9.325E+01	2.482E-02	0.9971	-0.216	0.247
5.0000	7.842E+01	4.970E-02	7.847E+01	3.656E-02	0.9974	-0.206	0.235
6.0000	6.801E+01	4.195E-02	6.805E+01	5.028E-02	0.9976	-0.198	0.226
8.0000	5.417E+01	3.208E-02	5.420E+01	8.346E-02	0.9978	-0.187	0.212
10.0000	4.532E+01	2.603E-02	4.535E+01	1.240E-01	0.9980	-0.179	0.203
15.0000	3.269E+01	1.778E-02	3.271E+01	2.558E-01	0.9982	-0.167	0.187
20.0000	2.589E+01	1.356E-02	2.591E+01	4.289E-01	0.9983	-0.159	0.177
30.0000	1.864E+01	9.239E-03	1.865E+01	8.910E-01	0.9984	-0.150	0.165
40.0000	1.479E+01	7.034E-03	1.479E+01	1.498E+00	0.9985	-0.144	0.158
50.0000	1.238E+01	5.691E-03	1.238E+01	2.240E+00	0.9985	-0.140	0.152
60.0000	1.072E+01	4.786E-03	1.072E+01	3.111E+00	0.9986	-0.136	0.148
80.0000	8.578E+00	3.641E-03	8.581E+00	5.212E+00	0.9986	-0.132	0.142
100.0000	7.250E+00	2.944E-03	7.253E+00	7.759E+00	0.9987	-0.128	0.138
150.0000	5.417E+00	2.001E-03	5.419E+00	1.586E+01	0.9987	-0.123	0.131
200.0000	4.470E+00	1.522E-03	4.471E+00	2.609E+01	0.9988	-0.119	0.127
300.0000	3.504E+00	1.035E-03	3.505E+00	5.170E+01	0.9989	-0.114	0.122
400.0000	3.018E+00	7.870E-04	3.019E+00	8.263E+01	0.9990	-0.111	0.118
500.0000	2.731E+00	6.367E-04	2.732E+00	1.176E+02	0.9990	-0.109	0.116
600.0000	2.544E+00	5.355E-04	2.545E+00	1.556E+02	0.9991	-0.107	0.114
800.0000	2.323E+00	4.076E-04	2.324E+00	2.382E+02	0.9992	-0.104	0.111
1000.0000	2.203E+00	3.300E-04	2.204E+00	3.268E+02	0.9992	-0.096	0.108
1500.0000	2.065E+00	2.249E-04	2.065E+00	5.625E+02	0.9993	-0.066	0.095
2000.0000	2.017E+00	1.715E-04	2.017E+00	8.079E+02	0.9994	-0.052	0.084
3000.0000	1.998E+00	1.171E-04	1.998E+00	1.307E+03	0.9995	-0.038	0.069
4000.0000	2.010E+00	8.939E-05	2.010E+00	1.806E+03	0.9996	-0.031	0.059
5000.0000	2.029E+00	7.251E-05	2.030E+00	2.302E+03	0.9996	-0.027	0.052
6000.0000	2.050E+00	6.112E-05	2.050E+00	2.792E+03	0.9997	-0.024	0.048
8000.0000	2.090E+00	4.669E-05	2.090E+00	3.758E+03	0.9997	-0.021	0.041
10 000.0000	2.124E+00	3.788E-05	2.125E+00	4.707E+03	0.9998	-0.019	0.037

KEY DATA FOR IONIZING-RADIATION DOSIMETRY: MEASUREMENT STANDARDS AND APPLICATIONS

Table A.10. Alpha particles in air, $I = 85.7$ eV, $\rho = 0.0012$ g cm⁻³.

T	S_{el}/ρ	S_{nuc}/ρ	S_{tot}/ρ	ρr_0	Detour factor	$\partial(\log)/\partial(\log I)$	
MeV	MeV cm ² g ⁻¹			g cm ⁻²		S_{el}/ρ	ρr_0
0.0010	8.750E+01	1.340E+02	2.215E+02	5.377E-06	0.3767	0.000	0.000
0.0015	1.086E+02	1.256E+02	2.342E+02	7.562E-06	0.4012	0.000	0.000
0.0020	1.267E+02	1.178E+02	2.445E+02	9.651E-06	0.4220	0.000	0.000
0.0030	1.573E+02	1.049E+02	2.622E+02	1.360E-05	0.4558	0.000	0.000
0.0040	1.835E+02	9.494E+01	2.784E+02	1.730E-05	0.4830	0.000	0.000
0.0050	2.067E+02	8.698E+01	2.937E+02	2.079E-05	0.5058	0.000	0.000
0.0060	2.279E+02	8.047E+01	3.084E+02	2.411E-05	0.5255	0.000	0.000
0.0080	2.659E+02	7.036E+01	3.363E+02	3.032E-05	0.5580	0.000	0.000
0.0100	2.996E+02	6.283E+01	3.624E+02	3.605E-05	0.5841	0.000	0.000
0.0150	3.723E+02	5.018E+01	4.225E+02	4.879E-05	0.6327	0.000	0.000
0.0200	4.343E+02	4.220E+01	4.765E+02	5.992E-05	0.6670	0.000	0.000
0.0300	5.395E+02	3.250E+01	5.720E+02	7.900E-05	0.7136	0.000	0.000
0.0400	6.290E+02	2.672E+01	6.557E+02	9.530E-05	0.7447	0.000	0.000
0.0500	7.081E+02	2.284E+01	7.309E+02	1.097E-04	0.7674	0.000	0.000
0.0600	7.798E+02	2.002E+01	7.998E+02	1.228E-04	0.7850	0.000	0.000
0.0800	9.068E+02	1.619E+01	9.230E+02	1.460E-04	0.8107	0.000	0.000
0.1000	1.018E+03	1.368E+01	1.032E+03	1.665E-04	0.8289	0.000	0.000
0.1500	1.247E+03	9.991E+00	1.257E+03	2.101E-04	0.8585	0.000	0.000
0.2000	1.429E+03	7.956E+00	1.437E+03	2.472E-04	0.8768	0.000	0.000
0.3000	1.693E+03	5.736E+00	1.699E+03	3.108E-04	0.8993	0.000	0.000
0.4000	1.861E+03	4.531E+00	1.866E+03	3.667E-04	0.9134	0.000	0.000
0.5000	1.961E+03	3.767E+00	1.965E+03	4.188E-04	0.9234	0.000	0.000
0.6000	2.008E+03	3.235E+00	2.011E+03	4.690E-04	0.9310	0.000	0.000
0.8000	2.002E+03	2.540E+00	2.005E+03	5.682E-04	0.9424	0.000	0.000
1.0000	1.922E+03	2.103E+00	1.924E+03	6.698E-04	0.9507	0.000	0.000
1.5000	1.626E+03	1.487E+00	1.627E+03	9.518E-04	0.9647	-0.276	0.037
2.0000	1.381E+03	1.160E+00	1.383E+03	1.286E-03	0.9735	-0.414	0.123
3.0000	1.071E+03	8.157E-01	1.072E+03	2.115E-03	0.9834	-0.356	0.224
4.0000	8.858E+02	6.341E-01	8.864E+02	3.147E-03	0.9885	-0.323	0.261
5.0000	7.606E+02	5.211E-01	7.611E+02	4.368E-03	0.9915	-0.301	0.275
6.0000	6.696E+02	4.436E-01	6.700E+02	5.772E-03	0.9934	-0.285	0.279
8.0000	5.453E+02	3.437E-01	5.456E+02	9.101E-03	0.9955	-0.263	0.277
10.0000	4.634E+02	2.817E-01	4.637E+02	1.309E-02	0.9967	-0.248	0.270
15.0000	3.423E+02	1.959E-01	3.425E+02	2.581E-02	0.9980	-0.224	0.252
20.0000	2.747E+02	1.511E-01	2.748E+02	4.222E-02	0.9985	-0.210	0.238
30.0000	2.001E+02	1.045E-01	2.002E+02	8.546E-02	0.9990	-0.193	0.219
40.0000	1.593E+02	8.026E-02	1.593E+02	1.419E-01	0.9992	-0.182	0.206
50.0000	1.332E+02	6.533E-02	1.333E+02	2.108E-01	0.9993	-0.175	0.197
60.0000	1.150E+02	5.516E-02	1.151E+02	2.918E-01	0.9993	-0.170	0.190
80.0000	9.119E+01	4.215E-02	9.123E+01	4.885E-01	0.9994	-0.162	0.180
100.0000	7.614E+01	3.418E-02	7.618E+01	7.295E-01	0.9994	-0.156	0.173
150.0000	5.493E+01	2.324E-02	5.496E+01	1.514E+00	0.9995	-0.147	0.162
200.0000	4.369E+01	1.763E-02	4.370E+01	2.543E+00	0.9995	-0.141	0.154
300.0000	3.184E+01	1.191E-02	3.185E+01	5.265E+00	0.9996	-0.134	0.146
400.0000	2.563E+01	8.987E-03	2.564E+01	8.791E+00	0.9996	-0.130	0.140
500.0000	2.179E+01	7.219E-03	2.179E+01	1.304E+01	0.9996	-0.126	0.136
600.0000	1.917E+01	6.034E-03	1.917E+01	1.795E+01	0.9996	-0.124	0.133
800.0000	1.583E+01	4.533E-03	1.583E+01	2.951E+01	0.9996	-0.120	0.129
1000.0000	1.379E+01	3.635E-03	1.379E+01	4.310E+01	0.9997	-0.118	0.126

Appendix

Table A.11. Alpha particles in graphite, $I = 81$ eV, $\rho = 2.265$ g cm⁻³.

T	S_{el}/ρ	S_{nuc}/ρ	S_{tot}/ρ	ρr_0	Detour factor	$\partial(\log)/\partial(\log I)$	
MeV	MeV cm ² g ⁻¹			g cm ⁻²		S_{el}/ρ	ρr_0
0.0010	1.923E+02	1.557E+02	3.480E+02	3.630E-06	0.4644	0.000	0.000
0.0015	2.289E+02	1.441E+02	3.730E+02	5.010E-06	0.4941	0.000	0.000
0.0020	2.590E+02	1.340E+02	3.930E+02	6.315E-06	0.5185	0.000	0.000
0.0030	3.083E+02	1.179E+02	4.262E+02	8.755E-06	0.5565	0.000	0.000
0.0040	3.489E+02	1.058E+02	4.547E+02	1.102E-05	0.5856	0.000	0.000
0.0050	3.840E+02	9.627E+01	4.803E+02	1.316E-05	0.6091	0.000	0.000
0.0060	4.153E+02	8.860E+01	5.039E+02	1.520E-05	0.6288	0.000	0.000
0.0080	4.699E+02	7.685E+01	5.468E+02	1.900E-05	0.6602	0.000	0.000
0.0100	5.170E+02	6.822E+01	5.852E+02	2.253E-05	0.6846	0.000	0.000
0.0150	6.151E+02	5.395E+01	6.690E+02	3.050E-05	0.7280	0.000	0.000
0.0200	6.955E+02	4.509E+01	7.406E+02	3.760E-05	0.7575	0.000	0.000
0.0300	8.262E+02	3.446E+01	8.607E+02	5.008E-05	0.7963	0.000	0.000
0.0400	9.327E+02	2.820E+01	9.609E+02	6.105E-05	0.8213	0.000	0.000
0.0500	1.024E+03	2.402E+01	1.048E+03	7.100E-05	0.8393	0.000	0.000
0.0600	1.104E+03	2.101E+01	1.125E+03	8.021E-05	0.8530	0.000	0.000
0.0800	1.240E+03	1.693E+01	1.257E+03	9.699E-05	0.8727	0.000	0.000
0.1000	1.354E+03	1.427E+01	1.368E+03	1.122E-04	0.8866	0.000	0.000
0.1500	1.574E+03	1.038E+01	1.584E+03	1.460E-04	0.9087	0.000	0.000
0.2000	1.731E+03	8.249E+00	1.739E+03	1.760E-04	0.9222	0.000	0.000
0.3000	1.929E+03	5.931E+00	1.935E+03	2.302E-04	0.9385	0.000	0.000
0.4000	2.027E+03	4.677E+00	2.032E+03	2.805E-04	0.9485	0.000	0.000
0.5000	2.063E+03	3.884E+00	2.067E+03	3.292E-04	0.9555	0.000	0.000
0.6000	2.060E+03	3.333E+00	2.063E+03	3.776E-04	0.9608	0.000	0.000
0.8000	1.993E+03	2.613E+00	1.996E+03	4.759E-04	0.9683	0.000	0.000
1.0000	1.891E+03	2.161E+00	1.893E+03	5.788E-04	0.9736	0.000	0.000
1.5000	1.615E+03	1.526E+00	1.616E+03	8.644E-04	0.9818	-0.129	0.017
2.0000	1.387E+03	1.189E+00	1.388E+03	1.199E-03	0.9865	-0.270	0.070
3.0000	1.085E+03	8.350E-01	1.086E+03	2.021E-03	0.9916	-0.352	0.176
4.0000	8.982E+02	6.486E-01	8.988E+02	3.038E-03	0.9941	-0.319	0.229
5.0000	7.720E+02	5.327E-01	7.725E+02	4.242E-03	0.9956	-0.297	0.251
6.0000	6.804E+02	4.532E-01	6.808E+02	5.624E-03	0.9965	-0.281	0.260
8.0000	5.544E+02	3.509E-01	5.547E+02	8.899E-03	0.9976	-0.259	0.263
10.0000	4.712E+02	2.874E-01	4.715E+02	1.282E-02	0.9981	-0.244	0.259
15.0000	3.478E+02	1.996E-01	3.480E+02	2.533E-02	0.9988	-0.220	0.245
20.0000	2.788E+02	1.538E-01	2.789E+02	4.149E-02	0.9991	-0.207	0.232
30.0000	2.028E+02	1.062E-01	2.029E+02	8.414E-02	0.9993	-0.190	0.214
40.0000	1.612E+02	8.149E-02	1.613E+02	1.398E-01	0.9994	-0.180	0.202
50.0000	1.348E+02	6.626E-02	1.348E+02	2.079E-01	0.9995	-0.173	0.194
60.0000	1.163E+02	5.588E-02	1.164E+02	2.880E-01	0.9905	-0.167	0.187
80.0000	9.214E+01	4.266E-02	9.218E+01	4.826E-01	0.9995	-0.160	0.177
100.0000	7.689E+01	3.455E-02	7.693E+01	7.212E-01	0.9996	-0.154	0.171
150.0000	5.543E+01	2.344E-02	5.545E+01	1.499E+00	0.9996	-0.145	0.159
200.0000	4.405E+01	1.776E-02	4.407E+01	2.518E+00	0.9996	-0.139	0.152
300.0000	3.209E+01	1.198E-02	3.210E+01	5.219E+00	0.9996	-0.132	0.143
400.0000	2.581E+01	9.027E-03	2.582E+01	8.720E+00	0.9997	-0.126	0.137
500.0000	2.193E+01	7.251E-03	2.193E+01	1.294E+01	0.9997	-0.123	0.133
600.0000	1.928E+01	6.049E-03	1.929E+01	1.782E+01	0.9997	-0.119	0.130
800.0000	1.591E+01	4.548E-03	1.591E+01	2.932E+01	0.9997	-0.114	0.125
1000.0000	1.384E+01	3.643E-03	1.385E+01	4.285E+01	0.9997	-0.110	0.121

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Table A.12. Alpha particles in liquid water, $I = 78 \text{ eV}$, $\rho = 0.998 \text{ g cm}^{-3}$.

T	S_{el}/ρ	S_{nuc}/ρ	S_{tot}/ρ	ρr_0	Detour factor	$\partial(\log)/\partial(\log I)$	
MeV	MeV cm ² g ⁻¹			g cm ⁻²		S_{el}/ρ	ρr_0
0.0010	9.891E+01	2.282E+02	3.271E+02	3.273E-06	0.5212	0.000	0.000
0.0015	1.228E+02	2.078E+02	3.306E+02	4.789E-06	0.5364	0.000	0.000
0.0020	1.431E+02	1.912E+02	3.343E+02	6.294E-06	0.5501	0.000	0.000
0.0030	1.776E+02	1.660E+02	3.436E+02	9.247E-06	0.5737	0.000	0.000
0.0040	2.069E+02	1.477E+02	3.546E+02	1.211E-05	0.5934	0.000	0.000
0.0050	2.330E+02	1.336E+02	3.666E+02	1.489E-05	0.6102	0.000	0.000
0.0060	2.568E+02	1.225E+02	3.793E+02	1.757E-05	0.6250	0.000	0.000
0.0080	2.993E+02	1.056E+02	4.049E+02	2.267E-05	0.6497	0.000	0.000
0.0100	3.370E+02	9.341E+01	4.304E+02	2.746E-05	0.6700	0.000	0.000
0.0150	4.181E+02	7.347E+01	4.916E+02	3.831E-05	0.7080	0.000	0.000
0.0200	4.871E+02	6.123E+01	5.483E+02	4.793E-05	0.7352	0.000	0.000
0.0300	6.039E+02	4.665E+01	6.506E+02	6.462E-05	0.7724	0.000	0.000
0.0400	7.030E+02	3.811E+01	7.411E+02	7.900E-05	0.7973	0.000	0.000
0.0500	7.905E+02	3.242E+01	8.229E+02	9.179E-05	0.8155	0.000	0.000
0.0600	8.696E+02	2.834E+01	8.979E+02	1.034E-04	0.8296	0.000	0.000
0.0800	1.009E+03	2.281E+01	1.032E+03	1.241E-04	0.8503	0.000	0.000
0.1000	1.131E+03	1.921E+01	1.150E+03	1.425E-04	0.8649	0.000	0.000
0.1500	1.383E+03	1.397E+01	1.397E+03	1.817E-04	0.8886	0.000	0.000
0.2000	1.582E+03	1.109E+01	1.593E+03	2.151E-04	0.9032	0.000	0.000
0.3000	1.873E+03	7.969E+00	1.881E+03	2.725E-04	0.9212	0.000	0.000
0.4000	2.062E+03	6.282E+00	2.068E+03	3.230E-04	0.9323	0.000	0.000
0.5000	2.178E+03	5.214E+00	2.183E+03	3.699E-04	0.9402	0.000	0.000
0.6000	2.240E+03	4.473E+00	2.244E+03	4.150E-04	0.9462	0.000	0.000
0.8000	2.256E+03	3.506E+00	2.260E+03	5.034E-04	0.9551	0.000	0.000
1.0000	2.190E+03	2.898E+00	2.193E+03	5.931E-04	0.9615	0.000	0.000
1.5000	1.877E+03	2.045E+00	1.879E+03	8.385E-04	0.9722	-0.265	0.034
2.0000	1.599E+03	1.594E+00	1.600E+03	1.128E-03	0.9789	-0.398	0.117
3.0000	1.239E+03	1.119E+00	1.240E+03	1.844E-03	0.9867	-0.342	0.213
4.0000	1.021E+03	8.686E-01	1.022E+03	2.737E-03	0.9907	-0.312	0.250
5.0000	8.747E+02	7.132E-01	8.754E+02	3.798E-03	0.9931	-0.291	0.264
6.0000	7.687E+02	6.066E-01	7.693E+02	5.020E-03	0.9946	-0.276	0.269
8.0000	6.239E+02	4.694E-01	6.243E+02	7.924E-03	0.9963	-0.255	0.267
10.0000	5.290E+02	3.844E-01	5.294E+02	1.142E-02	0.9973	-0.241	0.261
15.0000	3.894E+02	2.667E-01	3.896E+02	2.258E-02	0.9983	-0.219	0.245
20.0000	3.118E+02	2.054E-01	3.121E+02	3.702E-02	0.9988	-0.205	0.232
30.0000	2.267E+02	1.417E-01	2.269E+02	7.514E-02	0.9992	-0.189	0.214
40.0000	1.802E+02	1.087E-01	1.803E+02	1.250E-01	0.9993	-0.179	0.202
50.0000	1.506E+02	8.833E-02	1.507E+02	1.859E-01	0.9991	-0.172	0.193
60.0000	1.300E+02	7.448E-02	1.301E+02	2.575E-01	0.9994	-0.167	0.186
80.0000	1.030E+02	5.679E-02	1.030E+02	4.317E-01	0.9995	-0.159	0.177
100.0000	8.593E+01	4.598E-02	8.597E+01	6.452E-01	0.9995	-0.154	0.170
150.0000	6.194E+01	3.121E-02	6.197E+01	1.341E+00	0.9996	-0.145	0.159
200.0000	4.923E+01	2.362E-02	4.925E+01	2.253E+00	0.9996	-0.140	0.152
300.0000	3.586E+01	1.593E-02	3.587E+01	4.670E+00	0.9996	-0.133	0.144
400.0000	2.885E+01	1.201E-02	2.886E+01	7.802E+00	0.9996	-0.128	0.138
500.0000	2.452E+01	9.637E-03	2.453E+01	1.158E+01	0.9997	-0.125	0.134
600.0000	2.156E+01	8.053E-03	2.157E+01	1.594E+01	0.9997	-0.123	0.131
800.0000	1.780E+01	6.052E-03	1.781E+01	2.622E+01	0.9997	-0.119	0.127
1000.0000	1.550E+01	4.848E-03	1.551E+01	3.830E+01	0.9997	-0.116	0.124

Appendix

Table A.13. Carbon ions in air, $I = 85.7$ eV, $\rho = 0.0012$ g cm⁻³.

T	S_{el}/ρ	S_{nuc}/ρ	S_{tot}/ρ	ρr_0	$\partial(\log)/\partial(\log I)$	
MeV		MeV cm ² g ⁻¹		g cm ⁻²	S_{el}/ρ	ρr_0
0.0010	1.188E+02	7.394E+02	8.582E+02	5.826E-07	0.000	0.000
0.0015	1.475E+02	7.705E+02	9.180E+02	1.141E-06	0.000	0.000
0.0020	1.720E+02	7.815E+02	9.535E+02	1.675E-06	0.000	0.000
0.0030	2.135E+02	7.794E+02	9.929E+02	2.700E-06	0.000	0.000
0.0040	2.504E+02	7.647E+02	1.015E+03	3.695E-06	0.000	0.000
0.0050	2.801E+02	7.460E+02	1.026E+03	4.675E-06	0.000	0.000
0.0060	3.091E+02	7.263E+02	1.035E+03	5.645E-06	0.000	0.000
0.0080	3.605E+02	6.881E+02	1.049E+03	7.563E-06	0.000	0.000
0.0100	4.061E+02	6.530E+02	1.059E+03	9.461E-06	0.000	0.000
0.0150	5.045E+02	5.801E+02	1.085E+03	1.413E-05	0.000	0.000
0.0200	5.885E+02	5.241E+02	1.113E+03	1.868E-05	0.000	0.000
0.0300	7.312E+02	4.431E+02	1.174E+03	2.743E-05	0.000	0.000
0.0400	8.530E+02	3.869E+02	1.240E+03	3.572E-05	0.000	0.000
0.0500	9.614E+02	3.451E+02	1.306E+03	4.357E-05	0.000	0.000
0.0600	1.060E+03	3.125E+02	1.373E+03	5.104E-05	0.000	0.000
0.0800	1.237E+03	2.646E+02	1.502E+03	6.496E-05	0.000	0.000
0.1000	1.394E+03	2.309E+02	1.625E+03	7.776E-05	0.000	0.000
0.1500	1.733E+03	1.776E+02	1.911E+03	1.061E-04	0.000	0.000
0.2000	2.023E+03	1.458E+02	2.169E+03	1.306E-04	0.000	0.000
0.3000	2.517E+03	1.091E+02	2.626E+03	1.723E-04	0.000	0.000
0.4000	2.939E+03	8.819E+01	3.027E+03	2.077E-04	0.000	0.000
0.5000	3.312E+03	7.446E+01	3.387E+03	2.389E-04	0.000	0.000
0.6000	3.648E+03	6.470E+01	3.713E+03	2.671E-04	0.000	0.000
0.8000	4.235E+03	5.165E+01	4.286E+03	3.170E-04	0.000	0.000
1.0000	4.727E+03	4.325E+01	4.771E+03	3.612E-04	0.000	0.000
1.5000	5.637E+03	3.117E+01	5.668E+03	4.566E-04	0.000	0.000
2.0000	6.191E+03	2.461E+01	6.215E+03	5.405E-04	0.000	0.000
3.0000	6.631E+03	1.757E+01	6.648E+03	6.947E-04	0.000	0.000
4.0000	6.573E+03	1.379E+01	6.586E+03	8.452E-04	0.000	0.000
5.0000	6.379E+03	1.141E+01	6.390E+03	9.993E-04	0.000	0.000
6.0000	6.178E+03	9.765E+00	6.188E+03	1.158E-03	0.000	0.000
8.0000	5.822E+03	7.630E+00	5.830E+03	1.491E-03	0.000	0.000
10.0000	5.525E+03	6.291E+00	5.531E+03	1.844E-03	0.000	0.000
15.0000	4.915E+03	4.423E+00	4.920E+03	2.804E-03	0.000	0.000
20.0000	4.425E+03	3.438E+00	4.428E+03	3.877E-03	0.000	0.000
30.0000	3.679E+03	2.404E+00	3.681E+03	6.363E-03	0.000	0.000
40.0000	3.155E+03	1.863E+00	3.157E+03	9.306E-03	-0.092	0.013
50.0000	2.754E+03	1.527E+00	2.755E+03	1.270E-02	-0.183	0.047
60.0000	2.425E+03	1.297E+00	2.426E+03	1.658E-02	-0.214	0.084
80.0000	1.948E+03	1.001E+00	1.949E+03	2.584E-02	-0.200	0.128
100.0000	1.639E+03	8.183E-01	1.639E+03	3.707E-02	-0.191	0.148
150.0000	1.191E+03	5.656E-01	1.191E+03	7.338E-02	-0.176	0.165
200.0000	9.467E+02	4.344E-01	9.471E+02	1.208E-01	-0.167	0.167
300.0000	6.839E+02	2.984E-01	6.842E+02	2.469E-01	-0.156	0.164
400.0000	5.430E+02	2.279E-01	5.433E+02	4.122E-01	-0.150	0.160
500.0000	4.545E+02	1.848E-01	4.547E+02	6.144E-01	-0.145	0.155
600.0000	3.935E+02	1.555E-01	3.936E+02	8.515E-01	-0.141	0.152
800.0000	3.143E+02	1.181E-01	3.145E+02	1.424E+00	-0.136	0.147
1000.0000	2.651E+02	9.524E-02	2.652E+02	2.120E+00	-0.132	0.142
1500.0000	1.967E+02	6.436E-02	1.968E+02	4.343E+00	-0.126	0.135
2000.0000	1.612E+02	4.853E-02	1.612E+02	7.171E+00	-0.122	0.131
3000.0000	1.247E+02	3.258E-02	1.247E+02	1.432E+01	-0.117	0.125
4000.0000	1.062E+02	2.448E-02	1.062E+02	2.307E+01	-0.114	0.121
5000.0000	9.506E+01	1.962E-02	9.508E+01	3.306E+01	-0.111	0.119
6000.0000	8.777E+01	1.637E-02	8.779E+01	4.403E+01	-0.109	0.116
8000.0000	7.896E+01	1.230E-02	7.897E+01	6.817E+01	-0.106	0.113
10 000.0000	7.402E+01	9.842E-03	7.403E+01	9.440E+01	-0.103	0.111

KEY DATA FOR IONIZING-RADIATION DOSIMETRY: MEASUREMENT STANDARDS AND APPLICATIONS

Table A.14. Carbon ions in graphite, $I = 81 \text{ eV}$, $\rho = 2.265 \text{ g cm}^{-3}$.

T	S_{el}/ρ	S_{nuc}/ρ	S_{tot}/ρ	ρr_0	$\partial(\log)/\partial(\log I)$	
MeV		MeV $\text{cm}^2 \text{g}^{-1}$		g cm^{-2}	S_{el}/ρ	ρr_0
0.0010	3.092E+02	8.506E+02	1.160E+03	4.311E-07	0.000	0.000
0.0015	3.701E+02	8.801E+02	1.250E+03	8.429E-07	0.000	0.000
0.0020	4.204E+02	8.877E+02	1.308E+03	1.233E-06	0.000	0.000
0.0030	5.032E+02	8.780E+02	1.381E+03	1.976E-06	0.000	0.000
0.0040	5.748E+02	8.563E+02	1.431E+03	2.686E-06	0.000	0.000
0.0050	6.302E+02	8.313E+02	1.461E+03	3.377E-06	0.000	0.000
0.0060	6.836E+02	8.063E+02	1.490E+03	4.054E-06	0.000	0.000
0.0080	7.755E+02	7.590E+02	1.534E+03	5.376E-06	0.000	0.000
0.0100	8.546E+02	7.167E+02	1.571E+03	6.664E-06	0.000	0.000
0.0150	1.019E+03	6.310E+02	1.650E+03	9.767E-06	0.000	0.000
0.0200	1.153E+03	5.666E+02	1.720E+03	1.273E-05	0.000	0.000
0.0300	1.370E+03	4.749E+02	1.845E+03	1.834E-05	0.000	0.000
0.0400	1.546E+03	4.124E+02	1.958E+03	2.360E-05	0.000	0.000
0.0500	1.696E+03	3.662E+02	2.062E+03	2.858E-05	0.000	0.000
0.0600	1.829E+03	3.305E+02	2.160E+03	3.331E-05	0.000	0.000
0.0800	2.062E+03	2.785E+02	2.340E+03	4.220E-05	0.000	0.000
0.1000	2.268E+03	2.422E+02	2.510E+03	5.045E-05	0.000	0.000
0.1500	2.719E+03	1.852E+02	2.904E+03	6.893E-05	0.000	0.000
0.2000	3.106E+03	1.516E+02	3.258E+03	8.516E-05	0.000	0.000
0.3000	3.747E+03	1.130E+02	3.860E+03	1.132E-04	0.000	0.000
0.4000	4.269E+03	9.108E+01	4.360E+03	1.376E-04	0.000	0.000
0.5000	4.709E+03	7.677E+01	4.786E+03	1.594E-04	0.000	0.000
0.6000	5.089E+03	6.663E+01	5.156E+03	1.795E-04	0.000	0.000
0.8000	5.711E+03	5.309E+01	5.764E+03	2.161E-04	0.000	0.000
1.0000	6.192E+03	4.440E+01	6.237E+03	2.494E-04	0.000	0.000
1.5000	6.973E+03	3.194E+01	7.005E+03	3.245E-04	0.000	0.000
2.0000	7.368E+03	2.519E+01	7.393E+03	3.938E-04	0.000	0.000
3.0000	7.591E+03	1.795E+01	7.609E+03	5.264E-04	0.000	0.000
4.0000	7.475E+03	1.408E+01	7.489E+03	6.585E-04	0.000	0.000
5.0000	7.227E+03	1.164E+01	7.238E+03	7.943E-04	0.000	0.000
6.0000	6.964E+03	9.958E+00	6.974E+03	9.350E-04	0.000	0.000
8.0000	6.484E+03	7.773E+00	6.492E+03	1.232E-03	0.000	0.000
10.0000	6.077E+03	6.406E+00	6.083E+03	1.551E-03	0.000	0.000
15.0000	5.278E+03	4.500E+00	5.282E+03	2.436E-03	0.000	0.000
20.0000	4.685E+03	3.495E+00	4.688E+03	3.442E-03	0.000	0.000
30.0000	3.827E+03	2.442E+00	3.830E+03	5.815E-03	0.000	0.000
40.0000	3.240E+03	1.891E+00	3.242E+03	8.662E-03	-0.089	0.013
50.0000	2.801E+03	1.549E+00	2.803E+03	1.199E-02	-0.180	0.048
60.0000	2.458E+03	1.315E+00	2.459E+03	1.580E-02	-0.211	0.085
80.0000	1.973E+03	1.015E+00	1.974E+03	2.494E-02	-0.198	0.128
100.0000	1.659E+03	8.291E-01	1.660E+03	3.604E-02	-0.188	0.148
150.0000	1.204E+03	5.724E-01	1.205E+03	7.192E-02	-0.174	0.164
200.0000	9.567E+02	4.393E-01	9.571E+02	1.188E-01	-0.165	0.166
300.0000	6.905E+02	3.012E-01	6.908E+02	2.437E-01	-0.154	0.162
400.0000	5.480E+02	2.301E-01	5.482E+02	4.075E-01	-0.148	0.158
500.0000	4.585E+02	1.864E-01	4.587E+02	6.079E-01	-0.143	0.153
600.0000	3.967E+02	1.566E-01	3.969E+02	8.430E-01	-0.139	0.150
800.0000	3.168E+02	1.188E-01	3.169E+02	1.411E+00	-0.133	0.144
1000.0000	2.670E+02	9.581E-02	2.671E+02	2.102E+00	-0.129	0.140
1500.0000	1.980E+02	6.463E-02	1.980E+02	4.309E+00	-0.122	0.132
2000.0000	1.621E+02	4.870E-02	1.621E+02	7.121E+00	-0.117	0.127
3000.0000	1.251E+02	3.265E-02	1.252E+02	1.424E+01	-0.109	0.120
4000.0000	1.064E+02	2.454E-02	1.064E+02	2.296E+01	-0.103	0.115
5000.0000	9.506E+01	1.966E-02	9.508E+01	3.294E+01	-0.097	0.110
6000.0000	8.759E+01	1.640E-02	8.761E+01	4.392E+01	-0.092	0.106
8000.0000	7.848E+01	1.232E-02	7.849E+01	6.817E+01	-0.082	0.099
10 000.0000	7.322E+01	9.855E-03	7.323E+01	9.466E+01	-0.084	0.098

Appendix

Table A.15. Carbon ions in liquid water, $I = 78$ eV, $\rho = 0.998$ g cm⁻³.

T	S_{el}/ρ	S_{nuc}/ρ	S_{tot}/ρ	ρr_0	$\partial(\log)/\partial(\log I)$	
MeV		MeV cm ² g ⁻¹		g cm ⁻²	S_{el}/ρ	ρr_0
0.0010	1.421E+02	1.037E+03	1.179E+03	4.241E-07	0.000	0.000
0.0015	1.773E+02	1.077E+03	1.254E+03	8.319E-07	0.000	0.000
0.0020	2.075E+02	1.090E+03	1.297E+03	1.224E-06	0.000	0.000
0.0030	2.589E+02	1.083E+03	1.342E+03	1.980E-06	0.000	0.000
0.0040	3.047E+02	1.059E+03	1.364E+03	2.718E-06	0.000	0.000
0.0050	3.416E+02	1.032E+03	1.374E+03	3.448E-06	0.000	0.000
0.0060	3.776E+02	1.003E+03	1.381E+03	4.175E-06	0.000	0.000
0.0080	4.412E+02	9.475E+02	1.389E+03	5.619E-06	0.000	0.000
0.0100	4.975E+02	8.973E+02	1.395E+03	7.056E-06	0.000	0.000
0.0150	6.183E+02	7.944E+02	1.413E+03	1.062E-05	0.000	0.000
0.0200	7.207E+02	7.160E+02	1.437E+03	1.413E-05	0.000	0.000
0.0300	8.930E+02	6.034E+02	1.496E+03	2.095E-05	0.000	0.000
0.0400	1.038E+03	5.258E+02	1.564E+03	2.749E-05	0.000	0.000
0.0500	1.166E+03	4.683E+02	1.634E+03	3.375E-05	0.000	0.000
0.0600	1.281E+03	4.236E+02	1.705E+03	3.974E-05	0.000	0.000
0.0800	1.488E+03	3.581E+02	1.847E+03	5.101E-05	0.000	0.000
0.1000	1.676E+03	3.122E+02	1.988E+03	6.144E-05	0.000	0.000
0.1500	2.099E+03	2.396E+02	2.339E+03	8.459E-05	0.000	0.000
0.2000	2.476E+03	1.966E+02	2.673E+03	1.046E-04	0.000	0.000
0.3000	3.131E+03	1.469E+02	3.278E+03	1.382E-04	0.000	0.000
0.4000	3.696E+03	1.186E+02	3.814E+03	1.664E-04	0.000	0.000
0.5000	4.198E+03	1.001E+02	4.298E+03	1.911E-04	0.000	0.000
0.6000	4.652E+03	8.697E+01	4.739E+03	2.132E-04	0.000	0.000
0.8000	5.446E+03	6.939E+01	5.515E+03	2.522E-04	0.000	0.000
1.0000	6.116E+03	5.808E+01	6.174E+03	2.864E-04	0.000	0.000
1.5000	7.364E+03	4.184E+01	7.406E+03	3.598E-04	0.000	0.000
2.0000	8.139E+03	3.302E+01	8.172E+03	4.238E-04	0.000	0.000
3.0000	8.791E+03	2.356E+01	8.815E+03	5.405E-04	0.000	0.000
4.0000	8.708E+03	1.848E+01	8.727E+03	6.540E-04	0.000	0.000
5.0000	8.372E+03	1.529E+01	8.387E+03	7.708E-04	0.000	0.000
6.0000	8.029E+03	1.309E+01	8.042E+03	8.926E-04	0.000	0.000
8.0000	7.422E+03	1.022E+01	7.433E+03	1.152E-03	0.000	0.000
10.0000	6.926E+03	8.428E+00	6.934E+03	1.430E-03	0.000	0.000
15.0000	5.980E+03	5.924E+00	5.986E+03	2.209E-03	0.000	0.000
20.0000	5.284E+03	4.603E+00	5.289E+03	3.100E-03	0.000	0.000
30.0000	4.297E+03	3.218E+00	4.300E+03	5.208E-03	0.000	0.000
40.0000	3.627E+03	2.493E+00	3.630E+03	7.749E-03	-0.089	0.013
50.0000	3.134E+03	2.043E+00	3.136E+03	1.072E-02	-0.179	0.048
60.0000	2.750E+03	1.735E+00	2.752E+03	1.413E-02	-0.210	0.084
80.0000	2.207E+03	1.339E+00	2.208E+03	2.230E-02	-0.196	0.127
100.0000	1.855E+03	1.094E+00	1.856E+03	3.222E-02	-0.188	0.147
150.0000	1.346E+03	7.563E-01	1.347E+03	6.432E-02	-0.173	0.163
200.0000	1.069E+03	5.806E-01	1.070E+03	1.063E-01	-0.165	0.165
300.0000	7.717E+02	3.986E-01	7.721E+02	2.180E-01	-0.154	0.162
400.0000	6.124E+02	3.045E-01	6.127E+02	3.646E-01	-0.148	0.157
500.0000	5.123E+02	2.468E-01	5.126E+02	5.438E-01	-0.143	0.153
600.0000	4.434E+02	2.076E-01	4.436E+02	7.542E-01	-0.139	0.150
800.0000	3.540E+02	1.576E-01	3.542E+02	1.263E+00	-0.134	0.145
1000.0000	2.984E+02	1.271E-01	2.985E+02	1.881E+00	-0.131	0.140
1500.0000	2.213E+02	8.582E-02	2.214E+02	3.855E+00	-0.125	0.134
2000.0000	1.813E+02	6.474E-02	1.814E+02	6.369E+00	-0.121	0.129
3000.0000	1.402E+02	4.342E-02	1.402E+02	1.273E+01	-0.116	0.124
4000.0000	1.193E+02	3.264E-02	1.193E+02	2.051E+01	-0.112	0.120
5000.0000	1.068E+02	2.615E-02	1.068E+02	2.940E+01	-0.110	0.117
6000.0000	9.860E+01	2.182E-02	9.862E+01	3.916E+01	-0.108	0.115
8000.0000	8.867E+01	1.639E-02	8.869E+01	6.066E+01	-0.105	0.112
10 000.0000	8.311E+01	1.312E-02	8.312E+01	8.401E+01	-0.102	0.110

References

- AAPM (1986). American Association of Physicists in Medicine. *Protocol for Heavy Charged—Particle Therapy Beam Dosimetry*, AAPM Report 16 (American Association of Physicists in Medicine, New York).
- Ahlen, S. P. (1980). “Theoretical and experimental aspects of the energy loss of relativistic heavily ionizing particles,” *Rev. Mod. Phys.* **52**, 121–173.
- Ajzenberg-Selov, F. (1990). “Energy levels of light nuclei $A = 11$ –12,” *Nucl. Phys. A* **506**, 1–158.
- Akkerman, A., and Akkerman, E. (1999). “Characteristics of electron inelastic interactions in organic compounds and water over the energy range 20–10 000 eV,” *J. Appl. Phys.* **86**, 5809–5816.
- Akkerman, A., Barak, J., Chadwick, M. B., Levinson, J., Murat, M., and Lifshitz, Y. (2001). “Updated NIEL calculations for estimating the damage induced by particles and γ -rays in Si and GaAs,” *Radiat. Phys. Chem.* **62**, 301–310.
- Ali, E. S. M., Spencer, B., McEwen, M. R., and Rogers, D. W. O. (2015). “Towards a quantitative, measurement-based estimate of the uncertainty in photon mass attenuation coefficients at radiation therapy energies,” *Phys. Med. Biol.* **60**, 1641–1654.
- Allen, S. J. M. (1926). “Absorption of x-rays from lambda 0.08 to 4.0 angstroms,” *Phys. Rev.* **28**, 907–922 (ibid 27, 266–276 (1926)).
- Almond, P. R., Biggs, P. J., Coursey, B. M., Hanson, W. F., Huq, M. S., Nath, R., and Rogers, D. W. O. (1999). “AAPM’s TG-51 protocol for clinical reference dosimetry of high-energy photon and electron beams,” *Med. Phys.* **26**, 1847–1870.
- Andreo, P. (2009). “On the clinical spatial resolution achievable with protons and heavier charged particle radiotherapy beams,” *Phys. Med. Biol.* **54**, N205–N215.
- Andreo, P., Medin, J., and Vynckier, S. (2000). “Reply to the comments of L J Verhey on ‘Comparison of dosimetry recommendations for clinical proton beams’,” *Phys. Med. Biol.* **45**, L52–L54.
- Andreo, P., Burns, D. T., and Salvat, F. (2012). “On the uncertainties of photon mass energy-absorption coefficients and their ratios for radiation dosimetry,” *Phys. Med. Biol.* **57**, 2117–2136.
- Andreo, P., Wulff, J., Burns, D. T., and Palmans, H. (2013). “Consistency in reference radiotherapy dosimetry: Resolution of an apparent conundrum when ^{60}Co is the reference quality for charged-particle and photon beams,” *Phys. Med. Biol.* **58**, 6593–6621.
- Andrews, C. L. (1938). “X-rays of wave-length $1.5 \leq \lambda \leq 8.3$ angstroms,” *Phys. Rev.* **54**, 994–999.
- Anton, M., Kapsch, R.-P., Krauss, A., von Voigts-Rhetz, P., Zink, K., and McEwen, M. (2013). “Difference in the relative response of the alanine dosimeter to megavoltage x-ray and electron beams,” *Phys. Med. Biol.* **58**, 3259–3282.
- Ashley, J. C. (1982). “Stopping power of liquid water for low-energy electrons,” *Radiat. Res.* **89**, 25–31.
- Attix, F. H. (1979a). “The partition of kerma to account for bremsstrahlung,” *Health Phys.* **36**, 347–354.
- Attix, F. H. (1979b). “Addendum to ‘The partition of kerma to account for bremsstrahlung’,” *Health Phys.* **36**, 536.
- Austerlitz, C., Mota, H., Almeida, C. E., Allison, R., and Sibata, C. (2007). “Quality assurance of HDR ^{192}Ir sources using a Fricke dosimeter,” *Med. Phys.* **34**, 1348–1353.
- Austerlitz, C., Mota, H. C., Sempau, J., Benhabib, S. M., Campos, D., Allison, R., deAlmeida, C. E., Zhu, D., and Sibata, C. H. (2008). “Determination of absorbed dose in water at the reference point $D(r_0, \theta_0)$ for an ^{192}Ir HDR brachytherapy source using a Fricke system,” *Med. Phys.* **35**, 5360–5365.
- Baek, W. Y., Both, G. H., Gassen, D., Neuwirth, W., and Zielinski, M. (1987). “Energy loss of ^{12}C projectiles in different carbon modifications,” *Phys. Rev. A* **35**, 51–59.
- Bakker, C. J., and Segrè, E. (1951). “Stopping power and energy loss for ion pair production for 340-MeV protons,” *Phys. Rev.* **81**, 489–492.
- Barkla, C. G., and Sadler, C. A. (1909). “The absorption of Röntgen rays,” *Philos. Mag.* **17**, 739–760.
- Barnard, G. P., Aston, G. H., and Marsh, A. R. S. (1960). “On the effect of variations in the ambient air on the calibration and use of ionization dosimeters,” *Br. J. Radiol.* **33**, 644–645.
- Bashandy, E. (1962). “Experimental determination of the absorption coefficients of gamma rays through different barriers,” *Int. J. Appl. Radiat. Isot.* **13**, 173–178.
- Batterman, B. W. (1958). “Use of polystyrene as an x-ray absorber—the mass absorption coefficient of carbon,” *Rev. Sci. Instrum.* **29**, 1132.
- Baumgart, H., Arnold, W., Berg, H., Huttel, E., and Clausnitzer, G. (1983). “Proton stopping powers in various gases,” *Nucl. Instrum. Meth.* **204**, 597–604.
- Bay, Z., Mann, W. B., Seliger, H. H., and Wyckoff, H. O. (1957). “Absolute measurement of W_{air} for sulfur-35 beta rays,” *Radiat. Res.* **7**, 558–569.
- Bearden, A. J. (1966). “X-ray photoeffect cross sections in low- and medium-Z absorbers for the energy range 852 eV to 40 keV,” *J. Appl. Phys.* **37**, 1681–1692.

- Bearden, J. A., and Burr, A. F. (1967). "Reevaluation of x-ray atomic energy levels," *Rev. Mod. Phys.* **39**, 125–142.
- Bell, R. J., Bish, D. R. B., and Gill, P. E. (1972). "Separate subshell contributions to the stopping power of rare gases," *J. Phys. B* **5**, 476–482.
- Berger, M. J. (1993). *ESTAR, PSTAR, ASTAR*, Report IAEA-NDS-144 (International Atomic Energy Agency, Vienna).
- Berger, M. J., and Hubbell, J. H. (1987). *XCOM: Photon cross sections on a personal computer*, Technical Report NBSIR 87-3597 (National Bureau of Standards, Washington).
- Berger, M. J., and Paul, H. (1995). "Stopping powers, ranges, and straggling," pp. 419–545 in *Atomic and Molecular Data for Radiotherapy and Radiation Research* (International Atomic Energy Agency, Vienna).
- Berger, M. J., Hubbell, J. H., Seltzer, S. M., Chang, J., Coursey, J. S., Sukumar, R., Zucker, D. S., and Olsen, K. (2010). *XCOM: Photon Cross Section Database (version 1.5)*. [Online] Available: <http://physics.nist.gov/xcom> (National Institute of Standards and Technology, Gaithersburg, MD).
- Bergstrom, P. M., Jr., and Pratt, R. H. (1997). "An overview of the theories used in Compton scattering calculations," *Radiat. Phys. Chem.* **50**, 3–29.
- Bergstrom, P. M., Jr., Surić, T., Pisk, K., and Pratt, R. H. (1993). "Compton scattering of photons from bound electrons: Full relativistic independent-particle-approximation calculations," *Phys. Rev. A* **48**, 1134–1162.
- Berry, A. A., and Lawrence, J. L. (1979). "X-ray attenuation coefficients of graphite in the range 0.40 to 1.54 angstroms," *Acta. Cryst. A* **35**, 316–318.
- Besemer, A., Paganetti, H., and Bednarz, B. (2013). "The clinical impact of uncertainties in the mean excitation energy of human tissues during proton therapy," *Phys. Med. Biol.* **58**, 887–902.
- Besenbacher, F., Andersen, H. H., Hvelplund, P., and Knudsen, H. (1979). "Stopping power of swift hydrogen and helium ions in gases," *Mat.-Fys. Medd.* **40**, 1–39.
- Bethe, H. (1930). "Zur Theorie des Durchgangs schneller Korpuskularstrahlen durch Materie," *Ann. Phys.* **397**, 325–400.
- Bethe, H. A. (1934). "The influence of screening on the creation and stopping of electrons," *Proc. Camb. Philos. Soc.* **30**, 524–539.
- Bethe, H. A., and Ashkin, J. (1953). "Passage of radiations through matter," in *Experimental Nuclear Physics*, Segré, E., Ed. (Wiley, New York).
- Bethe, H., and Heitler, W. (1934). "On the stopping of fast particles and on the creation of positive electrons," *Proc. R. Soc. A* **146**, 83–112.
- Bethe, H. A., and Maximon, L. C. (1954). "Theory of bremsstrahlung and pair production. I. Differential cross section," *Phys. Rev.* **93**, 768–784.
- Bewley, D. K. (1963). "The measurement of locally absorbed dose of megavoltage x rays by means of a carbon calorimeter," *Br. J. Radiol.* **36**, 865–878.
- Bhabha, H. J. (1936). "The scattering of positrons by electrons with exchange on Dirac's theory of the positron," *Proc. Roy. Soc. A* **154**, 195–206.
- Bichsel, H., and Hiraoka, T. (1992). "Energy loss of 70 MeV protons in elements," *Nucl. Instrum. Meth.* **B66**, 345–351.
- Bichsel, H., and Tschalär, C. (1965). "Range-difference measurements for protons in C, Al, sapphire, Si, quartz, Ge, and Ag," *Bull. Am. Phys. Soc.* **10**, 723.
- Bichsel, H., Hiraoka, T., and Omata, K. (2000). "Aspects of fast-ion dosimetry," *Radiat. Res.* **153**, 208–219.
- Biggs, F., Mendelsohn, L. B., and Mann, J. B. (1975). "Hartree–Fock Compton profiles for the elements," *At. Data Nucl. Data Tables* **16**, 201–309.
- Birge, R. T. (1932). "The calculation of errors by the method of least squares," *Phys. Rev.* **40**, 207–227.
- Bohr, N. (1913). "II. On the theory of the decrease of velocity of moving electrified particles on passing through matter," *Phil. Mag.* **25**, 10–31.
- Boutillon, M., and Perroche, A.-M. (1993). "Ionometric determination of absorbed dose to water for cobalt-60 gamma rays," *Phys. Med. Biol.* **38**, 439–454.
- Boutillon, M., and Perroche-Roux, A.-M. (1987). "Re-evaluation of the W value for electrons in dry air," *Phys. Med. Biol.* **32**, 213–219.
- Bradley, D. A., Chong, C. S., and Ghose, A. M. (1986). "Photon absorptiometric studies of elements, mixtures and substances of biomedical interest," *Phys. Med. Biol.* **31**, 267–273.
- Brolley, J. E., and Ribe, F. L. (1955). "Energy loss by 8.86-MeV deuterons and 4.43-MeV protons," *Phys. Rev.* **98**, 1112–1117.
- Brusa, D., Stutz, G., Riveros, J. A., Fernández-Varea, J. M., and Salvat, F. (1996). "Fast sampling algorithm for the simulation of photon Compton scattering," *Nucl. Instrum. Meth. A* **379**, 167–175.
- Brysk, H., and Zerby, C. D. (1968). "Photoelectric cross sections in the keV range," *Phys. Rev.* **171**, 292–298.
- Büermann, L., and Burns, D. T. (2009). "Air-kerma cavity standards," *Metrologia* **46**, S24–S28.
- Büermann, L., and Hilgers, G. (2007). "Significant discrepancies in air kerma rates measured with free-air and cavity ionization chambers," *Nucl. Instrum. Meth. A* **580**, 477–480.
- Büermann, L., Grosswendt, B., Kramer, H. M., Selbach, H. J., Gerlach, M., Hoffmann, M., and Krumrey, M. (2006). "Measurement of the x-ray mass energy-absorption coefficient of air using 3 keV to 10 keV synchrotron radiation," *Phys. Med. Biol.* **51**, 5125–5150.
- Büermann, L., Gargioni, E., Hilgers, G., and Krauss, A. (2007). "Comparison of ionometric and calorimetric determination of absorbed dose to water for cobalt-60 gamma rays," in *Workshop on Absorbed Dose and Air Kerma Primary Standards* (Laboratoire National de Métrologie et d'Essais, Paris).
- Buhr, H., Büermann, L., Gerlach, M., Krumrey, M., and Rabus, H. (2012). "Measurement of the mass energy-absorption coefficient of air for x-rays in the range from 3 to 60 keV," *Phys. Med. Biol.* **57**, 8231–8247.

References

- Burbidge, P. W. (1922). "The absorption of the K x-rays of silver in gases and gaseous mixtures," *Phil. Mag.* **43**, 381–392.
- Burns, D. T. (2009). "A re-evaluation of the I -value for graphite based on an analysis of recent work on $W_{s,c,a}$ and cavity perturbation corrections," *Metrologia* **46**, 585–590.
- Burns, D. T. (2012). "An analysis of existing data for W_{air} , I_c and the product $W_{air} s_{c,air}$," *Metrologia* **49**, 507–512.
- Burns, D. T., and Büermann, L. (2009). "Free-air ionization chambers," *Metrologia* **46**, S9–S23.
- Burns, D. T., Allisy, P. J., and Kessler, C. (2007). "Re-evaluation of the BIPM international standard for air kerma in ^{60}Co gamma radiation," *Metrologia* **44**, L53–L56.
- Burns, D. T., Picard, S., Kessler, C., and Roger, P. (2014). "Use of the BIPM calorimetric and ionometric standards in megavoltage photon beams to determine W_{air} and I_c ," *Phys. Med. Biol.* **59**, 1353–1365.
- Calvert, L. D., Killeen, R. C. G., and Mathieson, A. McL. (1975). "X-ray attenuation coefficient of carbon for Cu K-alpha radiation," *Acta Cryst. A* **31**, 855–856.
- Carlson, T. A. (1975). *Photoelectron and Auger Spectroscopy* (Plenum Press, New York).
- CCEMRI (1985). Consultative Committee for the Standards of Measurement of Ionizing Radiation. pp. R131–R132 in *Report of the 11th Session* (Bureau International des Poids et Mesures, Sèvres, France).
- Chantler, C. T. (1993). "Improvements in computation of form factors," *Radiat. Phys. Chem.* **41**, 759–766.
- Chantler, C. T. (1995). "Theoretical form factor, attenuation, and scattering tabulation for $Z=1-92$ from $E=1-10$ eV to $E=0.4-1.0$ MeV," *J. Phys. Chem. Ref. Data* **24**, 71–643.
- Chantler, C. T. (2000). "Detailed tabulation of atomic form factors, photoelectric absorption and scattering cross section, and mass attenuation coefficients in the vicinity of absorption edges in the soft x-ray ($Z=30-36$, $Z=60-89$, $E=0.1$ keV–10 keV), addressing convergence issues of earlier work," *J. Phys. Chem. Ref. Data* **29**, 597–1056.
- Chantler, C. T., Olsen, K., Dragoset, R. A., Chang, J., Kishore, A. R., Kotochigova, S. A., and Zucker, D. S. (2005). *X-Ray Form Factor, Attenuation and Scattering Tables* (version 2.1). [Online] Available: <http://physics.nist.gov/ffast> (National Institute of Standards and Technology, Gaithersburg, MD).
- Chatterjee, B. K., and Roy, S. C. (1998). "Tables of elastic scattering cross sections of photons in the energy range of 50–1500 keV for all elements in the range $13 \leq Z \leq 104$," *J. Phys. Chem. Data*, **27**, 1011–1215.
- Chipman, D. R. (1955). "Mass absorption coefficient of carbon for Cu K-alpha radiation," *J. Appl. Phys.* **26**, 1387.
- Cojocar, C. D., and Ross, C. K. (2012). "Extracting W_{air} from the 1976 electron beam measurements of Domen and Lamperti," *Med. Phys.* **39**, 4645–4645.
- Cojocar, C. D., Stucki, G., McEwen, M. R., and Ross, C. K. (2010). "Determination of absorbed dose to water in megavoltage electron beams using a calorimeter–Fricke hybrid system," pp. 99–109 in *Standards, Applications and Quality Assurance in Medical Radiation Dosimetry* (International Atomic Energy Agency, Vienna).
- Colgate, S. A. (1952). "Gamma-ray absorption measurements," *Phys. Rev.* **87**, 592–601.
- Combecher, D. (1980). "Measurement of W values of low-energy electrons in several gases," *Radiat. Res.* **84**, 189–218.
- Compton, A. H. (1923). "A quantum theory of the scattering of x-rays by light elements," *Phys. Rev.* **21**, 483–502.
- Conner, A. L., Atwater, H. F., Plassmann, E. H., and McCrary, J. H. (1970). "Gamma-ray attenuation coefficient measurements," *Phys. Rev. A* **1**, 539–544.
- Cottens, E., Janssens, A., Eggermont, G., and Jacobs, R. (1980). "Absorbed dose calorimetry with a graphite calorimeter and G-value determinations for the Fricke dosimeter in high energy electron beams," pp. 189–211 in *Proc. Int. Symp. on Biomedical Dosimetry: Physical Aspects, Instrumentation, Calibration, Paris 1980*, Report IAEA-SM-249/32 (International Atomic Energy Agency, Vienna).
- Cowan, C. L. (1948). "The absorption of gamma-radiation," *Phys. Rev.* **74**, 1841–1845.
- Cromer, D. T. (1965). "Anomalous dispersion corrections computed from self-consistent field relativistic Dirac–Slater wave functions," *Acta Crystallogr.* **18**, 17–23.
- Cromer, D., and Liberman, D. (1970). "Relativistic calculation of anomalous scattering factors for x rays," *J. Chem. Phys.* **53**, 1891–1898.
- Cromer, D. T., and Liberman, D. A. (1981). "Anomalous dispersion calculations near to and on the long-wavelength side of an absorption edge," *Acta Cryst. A* **37**, 267–268.
- Cromer, D. T., and Mann, J. B. (1968). "X-ray scattering factors computed from numerical Hartree–Fock wave functions," *Acta Cryst. A* **24**, 321–324.
- Crowther, J. A., and Orton, L. H. H. (1932). "On the absorption of x-rays in gases and vapours. I. Gases," *Phil. Mag.* **13**, 505–523 (see also *Phil. Mag.* **10**, 329–342 (1930)).
- Cullen, D. E. (1989). *Program SCATMAN: A Code Designed to Calculate Photon Coherent Scattering Anomalous Scattering Factors and Cross Sections*, Technical Report UCRL-ID-103422 (Lawrence Livermore National Laboratory, Livermore, CA).
- Cullen, D. E., Chen, M. H., Hubbell, J. H., Perkins, S. T., Plechaty, E. F., Rathkopf, J. A., and Scofield, J. H. (1989). *Tables and Graphs of Photon-Interaction Cross Sections from 10 eV to 100 GeV Derived from the LLNL Evaluated Photon Data Library (EPDL)*, Technical Report UCRL-50400, Vol. 6, Rev. 4, Pt. B (Lawrence Livermore National Laboratory, Livermore, CA).
- Cullen, D. E., Hubbell, J. H., and Kissel, L. (1997). *EPDL97: The Evaluated Photon Data Library, '97 Version*, Technical Report UCRL-50400, Vol. 6, Rev. 5 (Lawrence Livermore National Laboratory, Livermore, CA).
- Cuykendall, T. R. (1936). "The absorption of x-rays of wave-length $50 \leq \lambda \leq 150$ x.u. by elements of low atomic number," *Phys. Rev.* **50**, 105–109.

- Dalgarno, A., Degges, T., and Williams, D. A. (1967). "Dipole properties of molecular nitrogen," *Proc. Phys. Soc.* **92**, 291–295.
- Davidge, P. C., Tomlinson, M., and Wright, J. (1959). *The Oxidation of Graphite During Irradiation in Oxygen*, Atomic Energy Authority AERE C/R 1450 (United Kingdom Atomic Energy Authority, Harwell, UK).
- Davies, H., Bethe, H. A., and Maximon, L. C. (1954). "Theory of bremsstrahlung and pair production. II. Integral cross section for pair production," *Phys. Rev.* **93**, 788–795.
- Davis, S. D., Ross, C. K., Mobit, P. N., Van der Zwan, L., Chase, W. J., and Shortt, K. R. (2003). "The response of LiF thermoluminescence dosimeters to photon beams in the energy range from 30 kV x rays to ^{60}Co gamma rays," *Radiat. Prot. Dosim.* **106**, 33–43.
- Dehmer, J. L., Inokuti, M., and Saxon, R. P. (1975). "Systematics of moments of dipole oscillator-strength distributions for atoms of the first and second row," *Phys. Rev. A* **12**, 102–121.
- De Jong, M. D., Tran, C. Q., Chantler, C. T., Barnea, Z., Dhal, B. B., Cojckson, D. J., Lee, W.-K., and Mashayekhi, A. (2005). "Measurement of the x-ray mass attenuation coefficient and determination of the imaginary component of the atomic form factor of molybdenum," *Phys. Rev. A* **71**, 032702-1–16.
- Del Grande, N. K., Stinner, R. J., and Oliver, A. J. (1969). Personal communication to J. Hubbell (1/20/69) and listed in *Compilation of X-Ray Cross Sections, Section III*, McMaster, W. H., Kerr Del Grande, N., Mallett, J. H., and Hubbell, J. H., Report UCRL-50174 (Lawrence Radiation Laboratory, Livermore, CA).
- De Prez, L. A., and de Pooter, J. A. (2008). "The new NMi orthovolt x-rays absorbed dose to water primary standard based on water calorimetry," *Phys. Med. Biol.* **53**, 3531–3542.
- Ding, G. X., and Rogers, D. W. O. (1995). *Energy Spectra, Angular Spread, and Dose Distributions of Electron Beams from Various Accelerators Used in Radiotherapy*, NRC Report PIRS-0439 (National Research Council, Ottawa).
- Dingfelder, M., Hantke, D., Inokuti, M., and Paretzke, H. G. (1998). "Electron inelastic-scattering cross sections in liquid water," *Radiat. Phys. Chem.* **53**, 1–18.
- Domen, S. R. (1980). "Absorbed dose water calorimeter," *Med. Phys.* **7**, 157–159.
- Domen, S. R., and Lamperti, P. J. (1974). "A heat-loss-compensated calorimeter: Theory, design, and performance," *J. Res. Nat. Bur. Stand.* **78**, 595–610.
- Domen, S. R., and Lamperti, P. J. (1976). "Comparisons of calorimetric and ionometric measurements in graphite irradiated with electrons from 15 to 50 MeV," *Med. Phys.* **3**, 294–301.
- Eggarter, E. (1975). "Comprehensive optical and collision data for radiation action. II. Ar," *J. Chem. Phys.* **62**, 833–847.
- El Gamal, I., Cojocar, C., Mainegra-Hing, E., Ross, C., and McEwen, M. (2013). "Development of an absorbed dose to water primary standard for HDR Ir-192 brachytherapy based on the Fricke dosimetry system," *Med. Phys.* **40**, 432–432.
- Emfietzoglou, D. (2003). "Inelastic cross-sections for electron transport in liquid water: A comparison of dielectric models," *Radiat. Phys. Chem.* **66**, 373–385.
- Emfietzoglou, D., and Nikjoo, H. (2005). "The effect of model approximations on single-collision distributions of low-energy electrons in liquid water," *Radiat. Res.* **163**, 98–111.
- Emfietzoglou, D., Cucinotta, F. A., and Nikjoo, H. (2005). "A complete dielectric response model for liquid water: A solution of the Bethe ridge problem," *Radiat. Res.* **164**, 202–211.
- Emfietzoglou, D., Garcia-Molina, R., Kyriakou, I., Abril, I., and Nikjoo, H. (2009). "A dielectric response study of the electronic stopping power of liquid water for energetic protons and a new I -value for water," *Phys. Med. Biol.* **54**, 3451–3472.
- Engelke, B. A., and Hohlfeld, K. (1971). "A calorimeter as a standard device to measure absorbed doses and the determination of the mean energy absorbed to generate an ion-pair in air (translated from German)," *PTB Mitteilungen* **71**, 336–342.
- Ergun, S., and Tiensuu, V. (1958). "Determination of x-ray absorption coefficients of inhomogeneous materials," *J. Appl. Phys.* **29**, 946–949.
- Fano, U. (1963). "Penetration of protons, alpha particles, and mesons," *Ann. Rev. Nucl. Sci.*, **13**, 1–66.
- Fastrup, B., Borup, A., and Hvelplund, P. (1968). "Stopping cross section in atmospheric air of 0.2–0.5 MeV atoms with $6 \leq Z_1 \leq 24$," *Can. J. Phys.* **46**, 489–495.
- Feist, H. (1982). "Determination of the absorbed dose to water for high-energy photons and electrons by total absorption of electrons in ferrous sulphate solution," *Phys. Med. Biol.* **27**, 1435–1447.
- Feng, I. J., Pratt, R. H., and Tseng, H. K. (1981). "Positron bremsstrahlung," *Phys. Rev. A* **24**, 1358–1363.
- Fermi, E. (1940). "The ionization loss of energy in gases and in condensed materials," *Phys. Rev.* **57**, 485–493.
- Fletcher, J. W. (1982). *Radiation Chemistry of Water at Low Dose Rates with Emphasis on the Energy Balance*, Atomic Energy of Canada Ltd. Report AECL-7834 (Chalk River Nuclear Laboratories, Chalk River, ON).
- Fregene, A. O. (1967). "Calibration of the ferrous sulfate dosimeter by ionometric and calorimetric methods for radiations of a wide range of energy," *Radiat. Res.* **31**, 256–272.
- Fricke, H., and Hart, E. J. (1966). "Chemical dosimetry," pp. 167–239 in *Radiation Dosimetry, Vol. II. Instrumentation*, Attix, F. H., and Roesch, W. C., Eds. (Academic Press, New York).
- Garcia-Molina, R., Abril, I., Denton, C. D., Heredia-Avalos, S., Kyriakou, I., and Emfietzoglou, D. (2009). "Calculated depth-dose distributions for H^+ and He^+ beams in liquid water," *Nucl. Instrum. Meth. B* **267**, 2647–2652.
- Gerward, L. (1983). "X-ray attenuation coefficients of carbon in the energy range 5 to 20 keV," *Acta Cryst. A* **39**, 322–325.
- Gimm, H. A., and Hubbell, J. H. (1978). *Total Photon Absorption Cross Section Measurements, Theoretical Analysis and Evaluations for Energies above 10 MeV*, ICRU 2016 – All rights reserved.

References

- Technical Report NBS-TN-968 (National Bureau of Standards, Washington).
- Glover, J. L., Chantler, C. T., Barnea, Z., Rae, N. A., Tran, C. Q., Creagh, D. C., Paterson, D., and Dahl, B. B. (2008). "Measurements of the x-ray mass attenuation coefficient and the imaginary part of the form factor of copper," *Phys. Rev. A* **78**, 052901-1–13.
- Gopal, S., and Sanjeevaiah, B. (1973a). "Gamma-ray attenuation coefficient measurements," *Phys. Rev. A* **8**, 2814–2818.
- Gopal, S., and Sanjeevaiah, B. (1973b). "A method to determine the gamma-ray attenuation coefficients," *Nucl. Instrum. Methods* **107**, 221–225.
- Goswami, B., and Chaudhuri, N. (1971). "Measurements of gamma-ray mass attenuation coefficients of organic scintillators," *Nucl. Instrum. Methods* **92**, 433–434.
- Goswami, B., and Chaudhuri, N. (1973). "Measurements of gamma-ray attenuation coefficients," *Phys. Rev. A* **7**, 1912–1916.
- Gray, L. H. (1936). "An ionization method for the absolute measurement of γ -ray energy," *Proc. Roy. Soc. A* **156**, 578–596.
- Gross, W., Wingate, C., and Failla, G. (1957). "Average energy lost by sulfur-35 beta rays per ion pair produced in air," *Radiat. Res.* **7**, 570–580.
- Grosskurth, K. (1934). "Neubestimmung der Massenschwächungskoeffizienten monochromatischer Röntgenstrahlen für 16 Elemente und Paraffin zwischen 0.128 und 2.5 Angstrom," *Ann. Phys. (Leipzig)* **20**, 197–232.
- Guiho, J. P., and Simoen, J. P. (1975). "Détermination expérimentale de l'énergie moyenne nécessaire à la production d'une paire d'ions dans l'air," *J. Appl. Radiat. Isot.* **26**, 714–719.
- Hahn, T. M. (1934). "The absorption of monochromatic x-rays of short wave-length," *Phys. Rev.* **46**, 149–153.
- Hanke, C. C., and Bichsel, H. (1970). *Precision Energy Loss Measurements for Natural Alpha Particles in Argon*, Munksgaard, Denmark.
- Haug, E. (1975). "Bremsstrahlung and pair production in the field of free electrons," *Z. Naturforsch.* **30a**, 1099–1113.
- Haug, E. (2006). "Electron–electron and electron–positron bremsstrahlung," *Radiat. Phys. Chem.* **75**, 1330–1345.
- Haug, E., and Keppler, M. (1984). "Bremsstrahlung in the field of bound electrons," *J. Phys. B* **17**, 2075–2084.
- Haug, E., and Nakel, W. (2004). *The Elementary Process of Bremsstrahlung* (World Scientific, Singapore).
- Hayashi, H., Watanabe, N., Udagawa, Y., and Kao, C.-C. (2000). "The complete optical spectrum of liquid water measured by inelastic x-ray scattering," *Proc. Natl Acad. Sci. USA* **97**, 6264–6266.
- Heller, J. M., Hamm, R. N., Birkhoff, R. D., and Painter, L. R. (1974). "Collective oscillation in liquid water," *J. Chem. Phys.* **60**, 3483–3486.
- Henke, B. L., Elgin, R. L., Lent, R. E., and Ledingham, R. B. (1967). "X-ray absorption in the 2 to 200 angstrom region," *Norelco Reporter* **14**, 112–131.
- Henke, B. L., Gullikson, E. M., and Davis, J. C. (1993). "X-ray interactions: Photoabsorption, scattering, transmission, and reflection at $E = 50$ –30,000 eV, $Z = 1$ –92," *At. Data Nucl. Data Tables* **54**, 181–342.
- Henry, L. C., and Kennett, T. J. (1971). "Measurement of intermediate energy gamma-ray cross sections," *Can. J. Phys.* **49**, 1167–1178.
- Hewlett, C. W. (1921). "The mass absorption and mass scattering coefficients for homogeneous x rays of wavelength between 0.13 and 1.05 angstrom units in water, lithium, carbon, nitrogen, oxygen, aluminum and iron," *Phys. Rev.* **17**, 284–301.
- Hubbell, J. H. (1969). *Photon Cross Sections, Attenuation Coefficients, and Energy Absorption Coefficients from 10 keV to 100 GeV*, NBS Report NSRDS-NBS 29 (National Bureau of Standards, Washington).
- Hubbell, J. H. (1977). "Photon mass attenuation and mass energy-absorption coefficients for H, C, N, O, Ar, and 7 mixtures from 0.1 keV to 20 MeV," *Radiat. Res.* **70**, 58–81.
- Hubbell, J. H. (1982). "Photon mass attenuation and energy-absorption coefficients," *J. Appl. Radiat. Isot.* **33**, 1269–1290.
- Hubbell, J. H. (1999). "Review of photon interaction cross section data in the medical and biological context," *Phys. Med. Biol.* **44**, R1–R22.
- Hubbell, J. H. (2006). "Review and history of photon cross section calculations," *Phys. Med. Biol.* **51**, R245–R262.
- Hubbell, J. H., and Berger, M. J. (1966). *Attenuation Coefficients, Energy Absorption Coefficients and Related Quantities, and Photon Atomic Cross Sections*, NBS Report 8681 (National Bureau of Standards, Washington).
- Hubbell, J. H., and Bergstrom, P. M., Jr. (2004). *Delbrück Scattering: A Bibliography, and Some Numerical Comparisons with Other Photon Interaction Processes*, Report NISTIR 7115 (National Institute of Standards and Technology, Washington).
- Hubbell, J. H., and Øverbø, I. (1979). "Relativistic atomic form factors and photon coherent scattering cross sections," *J. Phys. Chem. Ref. Data* **8**, 69–106.
- Hubbell, J. H., Veigele, W. J., Briggs, E. A., Brown, R. T., Cromer, D. T., and Howerton, R. J. (1975). "Atomic form factors, incoherent scattering functions, and photon scattering cross sections," *J. Phys. Chem. Ref. Data* **4**, 471–538.
- Hubbell, J. H., Veigele, W. J., Briggs, E. A., Brown, R. T., Cromer, D. T., and Howerton, R. J. (1977). "Erratum: Atomic form factors, incoherent scattering functions, and photon scattering cross sections," *J. Phys. Chem. Ref. Data* **6**, 615–616.
- Hubbell, J. H., Gimm, H. A., and Øverbø, I. (1980). "Pair, triplet, and total atomic cross sections (and mass attenuation coefficients) for 1 MeV–100 GeV photons in elements $Z=1$ to 100," *J. Phys. Chem. Ref. Data* **9**, 1023–1148.
- Hvelplund, P. (1971). "Energy loss and straggling of 100–500 keV atoms with $2 \leq Z_1 \leq 12$ in various gases," *Dan. Vidensk. Selsk. Mat. Fys. Medd.* **38**, 3–25.
- IAEA (1995). International Atomic Energy Agency, *Atomic and Molecular Data for Radiotherapy and Radiation Research*, Technical Report TECDOC 799 (International Atomic Energy Agency, Vienna).
- IAEA (2000a). International Atomic Energy Agency, *Absorbed Dose Determination in External Beam Radiotherapy: An International Code of Practice for* ICRU 2016 – All rights reserved.

- Dosimetry Based on Standards of Absorbed Dose to Water*, Technical Report TRS-398 (International Atomic Energy Agency, Vienna).
- IAEA (2000b). International Atomic Energy Agency, *Handbook of Photonuclear Data for Applications: Cross Sections and Spectra*, Technical Report TECDOC 1178 (International Atomic Energy Agency, Vienna).
- ICRU (1970). International Commission on Radiation Units and Measurements. *Radiation Dosimetry: X Rays Generated at Potentials of 5 to 150 kV*, ICRU Report 17 (International Commission on Radiation Units and Measurements, Bethesda, MD).
- ICRU (1979). International Commission on Radiation Units and Measurements. *Average Energy Required to Produce an Ion Pair*, ICRU Report 31 (International Commission on Radiation Units and Measurements, Bethesda, MD).
- ICRU (1984a). International Commission on Radiation Units and Measurements. *Stopping Powers for Electrons and Positrons*, ICRU Report 37 (International Commission on Radiation Units and Measurements, Bethesda, MD).
- ICRU (1984b). International Commission on Radiation Units and Measurements. *Radiation Dosimetry: Electron Beams with Energies Between 1 and 50 MeV*, ICRU Report 35 (International Commission on Radiation Units and Measurements, Bethesda, MD).
- ICRU (1989). International Commission on Radiation Units and Measurements. *Tissue Substitutes in Radiation Dosimetry and Measurement*, ICRU Report 44 (International Commission on Radiation Units and Measurements, Bethesda, MD).
- ICRU (1992). International Commission on Radiation Units and Measurements. *Photon, Electron, Proton and Neutron Interaction Data for Body Tissues*, ICRU Report 46 (International Commission on Radiation Units and Measurements, Bethesda, MD).
- ICRU (1993). International Commission on Radiation Units and Measurements. *Stopping Power and Ranges for Protons and Alpha Particles*, ICRU Report 49 (International Commission on Radiation Units and Measurements, Bethesda, MD).
- ICRU (1998a). International Commission on Radiation Units and Measurements. *Clinical Proton Dosimetry—Part I: Beam Production, Beam Delivery and Measurement of Absorbed Dose*, ICRU Report 59 (International Commission on Radiation Units and Measurements, Bethesda, MD).
- ICRU (1998b). International Commission on Radiation Units and Measurements. *Fundamental Quantities and Units for Ionizing Radiation*, ICRU Report 60 (International Commission on Radiation Units and Measurements, Bethesda, MD).
- ICRU (2001). International Commission on Radiation Units and Measurements. *Dosimetry of High-Energy Photon Beams Based on Standards of Absorbed Dose to Water*, ICRU Report 64 *J. ICRU* **1** (Nuclear Technology Publishing, Ashford, UK).
- ICRU (2005). International Commission on Radiation Units and Measurements. *Stopping of Ions Heavier Than Helium*, ICRU Report 73, *J. ICRU* **5** (Oxford University Press, Oxford).
- ICRU (2007). International Commission on Radiation Units and Measurements. *Prescribing, Recording, and Reporting Proton-Beam Therapy*, ICRU Report 78, *J. ICRU* **7** (Oxford University Press, Oxford).
- ICRU (2011). International Commission on Radiation Units and Measurements. *Fundamental Quantities and Units for Ionizing Radiation*, ICRU Report 85a, *J. ICRU* **11** (Oxford University Press, Oxford).
- Inokuti, M. (1971). "Inelastic collisions of fast charged particles with atoms and molecules—the Bethe theory revisited," *Rev. Mod. Phys.* **43**, 297–347.
- Inokuti, M. (1975). "Ionization yields in gases under electron irradiation," *Radiat. Res.* **64**, 6–22.
- Inokuti, M., and Smith, D. Y. (1982). "Fermi density effect on the stopping power of metallic aluminum," *Phys. Rev. B* **25**, 61–66.
- Islam, M. T., Rae, N. A., Glover, J. L., Barnea, Z., de Jong, M. D., Tran, C. Q., Wang, J., and Chantler, C. T. (2010). "Measurement of the x-ray mass attenuation coefficients of gold in the 38–50 keV range," *Phys. Rev. A* **81**, 022903-1–9.
- ISO (1996). International Organization for Standardization. *X and gamma reference radiations for calibrating dose-meters and dose rate meters and for determining their response as a function of photon energy—Part 1: Radiation characteristics and production methods*, ISO Report 4037-1 (International Organization for Standardization, Geneva).
- IUPAC (1997). Compendium of Chemical Terminology, 2nd Ed. (the "Gold Book"), [Online] Available: <http://goldbook.iupac.org>, (Blackwell Scientific Publications, Oxford).
- Janni, J. F. (1982). "Proton range-energy tables, 1 keV–10 GeV, energy loss, range, path length, time-of-flight, straggling, multiple scattering, and nuclear interaction probability. In two parts. Part 1. For 63 compounds Part 2. For elements $1 \leq Z \leq 92$," *At. Data Nucl. Data Tables* **27**, 147–339.
- JCGM (2008). Joint Committee on Guides for Uncertainty. *Evaluation of Measurement Data—Guide to the Expression of Uncertainty in Measurement*, Publication JCGM 100 (Bureau International des Poids et Mesures, Sèvres, France).
- Jenkins, T. M., Nelson, W. R., Rindi, A., Nahum, A. E., and Rogers, D. W. O., Eds. (1989). *Monte Carlo Transport of Electrons and Photons* (Plenum Press, New York).
- Jesse, W. P. (1958). "Absolute energy to produce an ion pair in various gases by beta particles from S^{35} ," *Phys. Rev.* **109**, 2002–2004.
- Jhanwar, B. L., Meath, W. J., and MacDonald, J. C. F. (1986). "Radiation interaction parameters and their additivity for primary alcohols, 1-alkenes, and normal alkanes," *Radiat. Res.* **106**, 288–299.
- Jones, D. T. L. (2006). "The W-value in air for proton therapy beams," *Radiat. Phys. Chem.* **75**, 541–550.
- Joyet, G., Baudraz, A., and Joyet, M. L. (1974). "Determination of the electronic density and the average atomic number of tissues in man by gamma-ray attenuation," *Experientia* **30**, 1338–1341.
- Kamakura, S., Sakamoto, N., Ogawa, H., Tsuchida, H., and Inokuti, M. (2006). "Mean excitation energies for

References

- the stopping power of atoms and molecules evaluated from oscillator-strength spectra," *J. Appl. Phys.* **100**, 064905-1–12.
- Kane, P. P., Kissel, L., Pratt, R. H., and Roy, S. C. (1986). "Elastic scattering of γ -rays and x-rays by atoms," *Phys. Rep.* **140**, 75–159.
- Kato, M., Suzuki, I. H., Nohtomi, A., Morishita, Y., Kurosawa, T., and Saito, N. (2010). "Photon W-value of dry air determined using a cryogenic radiometer combined with a multi-electrode ion chamber for soft x-rays," *Radiat. Phys. Chem.* **79**, 397–404.
- KCDB (2013). *The BIPM Key Comparison Data Base*. [Online] Available: <http://kcdb.bipm.org>. (Bureau International de Poids et Mesures, Sèvres, France).
- Kellerer, A. M., Hahn, K., and Rossi, H. H. (1992). "Intermediate dosimetric quantities," *Radiat. Res.* **130**, 15–25.
- Kim, L., Pratt, R. H., Seltzer, S. M., and Berger, M. J. (1986). "Ratio of positron to electron bremsstrahlung energy loss: An approximate scaling law," *Phys. Rev. A* **33**, 3002–3009.
- Kissel, L., and MacCallum, C. (1977). *Benchmark Bremsstrahlung Energy Spectrum Data for 1 and 2 MeV*, Internal Report PITT-179 (University of Pittsburgh, Pittsburgh, PA).
- Kissel, L., Zhou, B., Roy, S. C., Sen Gupta, S. K., and Pratt, R. H. (1995). "The validity of form-factor, modified-form-factor and anomalous-scattering-factor approximations in elastic scattering calculations," *Acta Crystallogr. A* **51**, 271–288.
- Klassen, N. V., and Ross, C. K. (1988). "The radiation Chemistry of water at high LET," pp. 95–97 in *NRC Workshop on Water Calorimetry*, Ross, C. K., and Klassen, N. V., Eds. (National Research Council, Ottawa).
- Klassen, N. V., and Ross, C. K. (1991). "Absorbed dose calorimetry using various aqueous solutions," *Radiat. Phys. Chem.* **38**, 95–104.
- Klassen, N. V., and Ross, C. K. (1997). "Water calorimetry: The heat defect," *J. Res. Natl Inst. Stand. Technol.* **102**, 63–74.
- Klassen, N. V., and Ross, C. K. (2002). "Water calorimetry: A correction to the heat defect calculations," *J. Res. Natl Inst. Stand. Technol.* **107**, 171–178.
- Klassen, N. V., Shortt, K. R., Seuntjens, J., and Ross, C. K. (1999). "Fricke dosimetry: The difference between G (Fe^{3+}) for ^{60}Co gamma-rays and high-energy x-rays," *Phys. Med. Biol.* **44**, 1609–1624.
- Klein, O., and Nishina, Y. (1929). "Über die Streuung von Strahlung Durch Freie Elektronen nach der Neuen Relativistischen Quantendynamik von Dirac," *Z. Phys.* **52**, 853–868.
- Kodre, A., Padežnik Gomilšek, J., Hauko, R., Hall, M., and Arcon, I. (2013). "Absolute determination of the x-ray absorption coefficient of barium and the L region using a liquid absorption cell," *X-Ray Spectrom.* **42**, 63–67.
- Kohlrausch, K. W. F. (1917). "Die Absorption der Gamma-Strahlen von Radium, Parts I, II, III," *Sitzungsber. Akad. Wiss. Wien* **126**, 441–472 (Pt. 1), 683–704 (Pt. 2), 887–913 (Pt. 3).
- Kohn, W., and Sham, L. J. (1965). "Self-consistent equations including exchange and correlation effects," *Phys. Rev.* **140**, A1133–A1138.
- Krämer, M., and Scholz, M. (2000). "Treatment planning for heavy-ion radiotherapy: Calculation and optimization of biologically effective dose," *Phys. Med. Biol.* **45**, 3319–3330.
- Krauss, A. (2006). "The PTB water calorimeter for the absolute determination of absorbed dose to water in ^{60}Co radiation," *Metrologia* **43**, 259–272.
- Krauss, A., and Kramer, H. M. (2003). "The heat defect in the PTB water calorimeter: A discussion on uncertainty," pp. 15–16 in *Proc. Workshop on Recent Advances in Absorbed Dose Standards*, [Online] Available: <http://www.arpsa.gov.au/pubs/absdos/kramer.pdf>.
- Krauss, A., Büermann, L., Kramer, H.-M., and Selbach, H.-J. (2012). "Calorimetric determination of the absorbed dose to water for medium-energy x-rays with generating voltages from 70 to 280 kV," *Phys. Med. Biol.* **57**, 6245–6268.
- Kumar, A., and Meath, W. J. (1985). "Integrated dipole oscillator strengths and dipole properties for Ne, Ar, Kr, Xe, HF, HCl, and HBr," *Can. J. Chem.* **63**, 1616–1630.
- Kumar, A., Meath, W. J., Bündgen, P., and Thakkar, A. J. (1996). "Reliable anisotropic dipole properties, and dispersion energy coefficients, for O_2 evaluated using constrained dipole oscillator strength techniques," *J. Chem. Phys.* **105**, 4927–4937.
- Kumazaki, Y., Akagi, T., Yanou, T., Suga, D., Hishikawa, Y., and Teshima, T. (2007). "Determination of the mean excitation energy of water from proton beam ranges," *Radiat. Meas.* **42**, 1683–1691.
- La Russa, D. J., McEwen, M., and Rogers, D. W. O. (2007). "An experimental and computational investigation of the standard temperature-pressure correction factor for ion chambers in kilovoltage x rays," *Med. Phys.* **34**, 4690–4699.
- Lakshminarayana, V., and Jnanananda, S. (1961). "Interaction of gamma radiation with matter," *J. Sci. Industr. Res.* **B20**, 1–7 (see also *Proc. Phys. Soc. (London)* **77**, 593–598 (1961)).
- Langhoff, P. W., and Yates, A. C. (1972). "Moment theory bounds on the mean energies of stopping, straggling, and molecular excitation," *J. Phys. B* **5**, 1071–1082.
- LaVerne, J. A., and Mozumder, A. (1986). "Effect of phase on the stopping and range distribution of low-energy electrons in water," *J. Phys. Chem.* **90**, 3242–3247.
- LaVerne, J. A., and Schuler, R. H. (1996). "Radiolysis of the Fricke dosimeter with ^{58}Ni and ^{238}U ions: Response for particles of high linear energy transfer," *J. Phys. Chem.* **100**, 16034–16040.
- Lee, C. M., Kissel, L., Pratt, R. H., and Tseng, H. K. (1976). "Electron bremsstrahlung spectrum, 1-500 keV," *Phys. Rev. A* **13**, 1714–1727.
- Lee, P. A., Citrin, P. H., Eisenberger, P., and Kincaid, B. M. (1981). "Extended x-ray absorption fine structure—its strengths and limitations as a structural tool," *Rev. Mod. Phys.* **53**, 769–806.

- Liamsuwan, T., and Nikjoo, H. (2013). "Cross sections for bare and dressed carbon ions in water and neon," *Phys. Med. Biol.* **58**, 641–672.
- Liberman, D., Waber, J. T., and Cromer, D. T. (1965). "Self-consistent-field Dirac-Slater wave functions for atoms and ions. I. Comparison with previous calculations," *Phys. Rev.* **137**, A27–A34.
- Lindhard, J., and Sørensen, A. H. (1996). "Relativistic theory of stopping for heavy ions," *Phys. Rev. A* **53**, 2443–2456.
- Ma, C.-M., Rogers, D. W. O., Shortt, K. R., Ross, C. K., Nahum, A. E., and Bielajew, A. F. (1993). "Wall-correction and absorbed-dose conversion factors for Fricke dosimetry: Monte Carlo calculations and measurements," *Med. Phys.* **20**, 283–292.
- MacPherson, M. S. (1998). *Accurate Measurements of the Collision Stopping Powers for 5 to 30 MeV Electrons*, Ph.D. dissertation, Carlton University, Ottawa, and NRC Report PIRS-0626 (National Research Council, Ottawa).
- Mahmoud, K. A. (1957). "A new technique for measurement of absorption coefficients of gamma rays of different energies in different absorbers," *Proc. Third Arab Congress*, Cairo, 726–740.
- Mandel, J., and Paule, R. C. (1970). "Interlaboratory evaluation of a material with unequal numbers of replicates," *Anal. Chem.* **42**, 1194–1197.
- Martin, F. W., and Northcliffe, L. C. (1962). "Energy loss and effective charge of He, C, and Ar ions below 10 MeV/amu in gases," *Phys. Rev.* **128**, 1166–1174.
- Mattsson, O., Svensson, H., Wickman, G., Domen, S. R., Pruitt, J. S., and Loevinger, R. (1990). "Absorbed dose in water: Comparison of several methods using a liquid ionization chamber," *Acta Oncol.* **29**, 235–240.
- Maxon, M. S., and Corman, E. G. (1967). "Electron-electron bremsstrahlung from a quantum plasma ($Z=1$)," *Phys. Rev.* **163**, 156–162.
- Mayneord, W. V., and Cipriani, A. J. (1947). "The absorption of gamma-rays from Co^{60} ," *Can. J. Res.* **25A**, 303–314.
- Mayneord, M. V., and Roberts, J. E. (1935). "Absorption of short wave-length x-rays," *Nature* **136**, 793.
- McCrary, J. H., Plassmann, E. H., Puckett, J. M., Conner, A. L., and Zimmermann, G. W. (1967). "X-ray attenuation-coefficient measurements," *Phys. Rev.* **153**, 307–312.
- McCrary, J. H., Ziegler, L. H., and Looney, L. D. (1969). "Attenuation of 5.9 keV photons by air," *J. Appl. Phys.* **40**, 2690–2693. (1969).
- McCrary, J. H., Looney, L. D., Constanten, C. P., and Atwater, H. F. (1970). "Attenuation coefficients of gases for 4.5–145-keV photons," *Phys. Rev. A* **2**, 2489–2497.
- McDonald, J. C., and Domen, S. R. (1986). "A-150 plastic radiometric calorimeter for charged particles and other radiations," *Nucl. Instrum. Meth.* **252**, 35–40.
- McEwen, M. R. (2010). "Measurement of ionization chamber absorbed dose k_Q factors in megavoltage photon beams," *Med. Phys.* **37**, 2179–2193.
- McEwen, M. R., DuSautoy, A. R., and Williams, A. J. (1998). "The calibration of therapy level electron beam ionization chambers in terms of absorbed dose to water," *Phys. Med. Biol.* **43**, 2503–2519.
- McEwen, M., DeWerd, L., Ibbott, G., Followill, D., Rogers, D. W. O., Seltzer, S., and Seuntjens, J. (2014a). "Addendum to the AAPM's TG-51 protocol for clinical reference dosimetry of high-energy photon beams," *Med. Phys.* **41**, 04150-1–20.
- McEwen, M., El Gamal, I., Mainegra-Hing, E., and Cojocaru, C. (2014b). *Determination of the Radiation Chemical Yield (G) for the Fricke Chemical Dosimetry System in Photon and Electron Beams*, NRC Report PIRS-1980 (National Research Council, Ottawa).
- Medin, J., and Andreo, P. (1997). "Monte Carlo calculated stopping-power ratios, water/air, for clinical proton dosimetry (50–250 MeV)," *Phys. Med. Biol.* **42**, 89–105.
- Medin, J., Andreo, P., and Vynckier, S. (2000). "Comparison of dosimetry recommendations for clinical proton beams," *Phys. Med. Biol.*, **45**, 3195–3211.
- Millar, R. H., and Greening, J. R. (1974). "Experimental x-ray mass attenuation coefficients for materials of low atomic number in the energy range 4 to 25 keV," *J. Phys. B* **7**, 2332–2344.
- Mohan, R., Chui, C., and Lidofsky, L. (1985). "Energy and angular distributions of photons from medical linear accelerators," *Med. Phys.* **12**, 592–597.
- Mohr, P. J., Taylor, B. N., and Newell, D. B. (2012). "CODATA recommended values of the fundamental physical constants: 2010," *J. Phys. Chem. Ref. Data* **41**, 043109-1–84.
- Møller, C. (1932). "Zur Theorie des Durchgangs schneller Elektronen durch Materie," *Ann. Phys.* **406**, 531–585.
- Mora, G. M., Maio, A., and Rogers, D. W. O. (1999). "Monte Carlo simulation of a typical ^{60}Co therapy source," *Med. Phys.* **26**, 2494–2502.
- Mork, K. J. (1971). "Radiative corrections. II. Compton effect," *Phys. Rev. A* **4**, 917–927.
- Müller, J. W. (2000a). *Weighted medians*, Report BIPM-2000/6 (Bureau International des Poids et Mesures, Sèvres, France).
- Müller, J. W. (2000b). "Possible advantages of a robust evaluation of comparisons," *J. Res. Natl. Inst. Stand. Technol.* **105**, 551–556.
- Murty, R. C. (1964). "Gamma ray absorption coefficients," *Proc. Phys. Soc. (London)* **84**, 1032–1034.
- Murty, V. R. K., Rao, K. S., Arunaprasad, G., Parthasaradhi, K., Rao, J. R., and Lakshminarayana, V. (1977). "Empirical photoabsorption cross sections for C, N, O, F and Cl obtained from molecular total photon cross-sections in the energy region (5-1100) keV," *Nuovo Cimento A* **39**, 125–130.
- Myers, I. T., LeBlanc, W. H., Fleming, D. N., and Wyckoff, H. O. (1961). *An Adiabatic Calorimeter for High Precision Source Standardization and Determination of W (air)*, Hanford Laboratory Report HW-SA-2165 (Hanford Atomic Products Operation, Richland, WA).
- Nahum, A. E. (1978). "Water/air mass stopping power ratios for megavoltage photon and electron beams," *Phys. Med. Biol.* **23**, 24–38.
- Nathuram, R., Rao, I. S. S., and Mehta, M. K. (1984). "Experimental mass absorption coefficients for low energy photons in beryllium, carbon, aluminum, silicon, copper, and lead," *Indian J. Phys. A* **58**, 300–304.

References

- Nathuram, R., Rao, I. S. S., and Mehta, M. K. (1988). "Photoelectric cross sections for 6–20-keV photons in beryllium, carbon, magnesium, aluminum, silicon, copper, silver, and lead," *Phys. Rev. A* **37**, 4978–4981.
- Nečas, V., Käferböck, W., Rössler, W., and Bauer, P. (1993). "Electronic stopping of hydrogen ions in graphite and amorphous carbon," *Nucl. Instrum. Meth. B* **80–81**, 41–44.
- Neuwirth, W., Pietsch, W., Richter, K., and Hauser, U. (1975). "Electronic stopping cross sections of elements and compounds for swift Li ions," *Z. Phys.* **275**, 209–214.
- Niatel, M.-T. (1969). "Etude expérimentale de l'influence de la vapeur d'eau sur l'ionisation produite dans l'air," *C. R. Académie Sci. Paris B* **268**, 1650–1653.
- Niatel, M.-T. (1975). "Influence de la vapeur d'eau sur l'ionisation de l'air dans le cas d'une chambre à cavité," *C. R. Académie Sci. Paris B* **281**, 361–363.
- Niatel, M. T., Perroche-Roux, A. M., and Boutillon, M. (1985). "Two determinations of W for electrons in dry air," *Phys. Med. Biol.* **30**, 67–75.
- Nordin, J. A., and Henkelman, R. M. (1979). "Measurement of stopping power ratios for 60 MeV positive or negative pions," *Phys. Med. Biol.* **24**, 781–792.
- Northcliffe, L. C. (1960). "Energy loss and effective charge of heavy ions in aluminum," *Phys. Rev.* **120**, 1744–1757.
- Northcliffe, L. C. (1963). "Passage of heavy ions through matter," *Ann. Rev. Nucl. Sci.* **13**, 67–102.
- Ogier, W. T., Lucas, G. J., and Park, R. J. (1964). "Ultrafast x-ray absorption coefficients of Al, Be, C, O and F," *Appl. Phys. Lett.* **5**, 146–147.
- Olsen, H. (1955). "Outgoing and ingoing waves in final states and bremsstrahlung," *Phys. Rev.* **99**, 1335–1336.
- Olsen, H., Maximon, L. C., and Wergeland, H. (1957). "Theory of high-energy bremsstrahlung and pair production in a screened field," *Phys. Rev.* **106**, 27–46.
- Olson, A. R., Dershem, E., and Storch, H. H. (1923). "X-ray absorption coefficients of carbon, hydrogen and oxygen," *Phys. Rev.* **21**, 30–37.
- Olzanski, A., Klassen, N., Ross, C., and Shortt, K. (2002). "The IRS Fricke dosimetry system," NRC Report PIRS-0815 (National Research Council, Ottawa, Canada).
- Palmans, H. (2011). "Absolute and relative dosimetry for protons and ions—challenges and solutions," *Radiother. Oncol.* **99**, S141.
- Palmans, H., Verhaegen, F., Denis, J.-M., Vynckier, S., and Thierens, H. (2001). "Experimental p_{wall} and p_{cel} correction factors for ionization chambers in low-energy clinical proton beams," *Phys. Med. Biol.* **46**, 1187–1204.
- Panta, P. P., Zagorski, Z. P., and Gluszcwski, W. J. (1997). "Thermal defects of water, graphite and polystyrene affecting calorimetric response," pp. 499–509 in *Symposium on Radiation Technology for Conservation of the Environment* (International Atomic Energy Agency, Vienna).
- Paul, H. (2007). "The mean ionization potential of water, and its connection to the range of energetic carbon ions in water," *Nucl. Instrum. Meth. B* **255**, 435–437.
- Paul, H., and Schinner, A. (2001). "An empirical approach to the stopping power of solids and gases for ions from ^3Li to ^{18}Ar ," *Nucl. Instrum. Meth. B* **179**, 299–315.
- Paul, H., and Schinner, A. (2002). "An empirical approach to the stopping power of solids and gases for ions from ^3Li to ^{18}Ar —Part II," *Nucl. Instrum. Meth. B* **195**, 166–174.
- Paul, H., Geithner, O., and Jäkel, O. (2007). "The ratio of stopping powers of water and air for dosimetry applications in tumor therapy," *Nucl. Instrum. Meth. B* **256**, 561–564.
- Paule, R. C., and Mandel, J. (1982). "Consensus values and weighting factors," *J. Res. Natl. Bur. Stand.* **87**, 377–385.
- Peele, A. G., Chantler, C. T., Paterson, D., McMahon, P. J., Irving, T. H. K., Nugent, K. A., Brunton, A. N., and McNulty, I. (2002). "Measurement of mass attenuation coefficients in air by application of detector linearity tests," *Phys. Rev A* **66**, 042702-1–7.
- Perkins, S. T., Cullen, D. E., Chen, M. H., Hubbell, J. H., Rathkopf, J., and Scofield, J. (1991). *Tables and Graphs of Atomic Subshell and Relaxation Data Derived from the LLNL Evaluated Atomic Data Library (EADL), Z = 1–100*, Technical Report UCRL-50400, Vol. 30 (Lawrence Livermore National Laboratory, Livermore, CA).
- Petree, B., and Lamperti, P. (1967). "A comparison of absorbed dose determinations in graphite by cavity ionization measurements and by calorimetry," *J. Res. Natl. Bur. Stand.* **71**, 19–27.
- Picard, S., Burns, D. T., and Roger, P. (2009). *Construction of an absorbed-dose graphite calorimeter*, Report BIPM-2009/1 (Bureau International des Poids et Mesures, Sèvres, France).
- Picard, S., Burns, D. T., Roger, P., Allisy-Roberts, P. J., McEwen, M. R., Cojocar, C. D., and Ross, C. K. (2010). "Comparison of the standards for absorbed dose to water of the NRC and the BIPM for accelerator photon beams," *Metrologia* **47**, 06025.
- Plujko, V. A., Capote, R., and Gorbachenko, O. M. (2011). "Giant dipole resonance parameters with uncertainties from photonuclear cross sections," *At. Data Nucl. Data Tables* **97**, 567–585.
- Porter, L. E. (1994). "Additivity and effective charge studies with light projectiles and target gases of CH_4 , CO_2 , He, Ne, Ar, Kr, Xe, Cl_2 and Br_2 ," *Nucl. Instrum. Meth. B* **93**, 203–209.
- Pratt, R. H. (1960a). "Photoeffect from the L shell," *Phys. Rev.* **119**, 1619–1626.
- Pratt, R. H. (1960b). "Atomic photoelectric effect at high energies," *Phys. Rev.* **117**, 1017–1028.
- Pratt, R. H. (2014). "Photon absorption and photon scattering—What we do not know and why it matters," *Radiat. Phys. Chem.* **95**, 4–13.
- Pratt, R. H., and Tseng, H. K. (1972). "Behavior of electron wave functions near the atomic nucleus and normalization screening theory in the atomic photoeffect," *Phys. Rev. A* **5**, 1063–1072.

- Pratt, R. H., Ron, A., and Tseng, H. K. (1973). "Atomic photoelectric effect above 10 keV," *Rev. Mod. Phys.* **45**, 273–325.
- Pratt, R. H., Tseng, H. K., Lee, C. M., Kissel, L., MacCallum, C., and Riley, M. (1977). "Bremsstrahlung energy spectra from electrons of kinetic energy $1 \text{ keV} \leq T_1 \leq 2000 \text{ keV}$ incident on neutral atoms $2 \leq Z \leq 92$," *At. Data Nucl. Data Tables* **20**, 175–209.
- Pratt, R. H., LaJohn, L. A., Florecu, V., Surić, T., Chatterjee, B. K., and Roy, S. C. (2010). "Compton scattering revisited," *Radiat. Phys. Chem.* **79**, 124–131.
- Puttaswamy, K. S., Gowda, R., and Sanjeevaiah, B. (1979). "Photoelectric cross sections derived from the total absorption cross sections in the energy range 5–130 keV," *Can. J. Chem.* **57**, 92–98.
- Rae, N. A., Chantler, C. T., Barnea, Z., de Jong, M. D., Tran, C. Q., and Hester, J. R. (2010). "X-ray mass attenuation coefficient and the imaginary part of the atomic form factor of zinc over the energy range of 7.2–5.2 keV," *Phys. Rev. A* **81**, 022904-1–10.
- Rakavy, G., and Ron, A. (1967). "Atomic photoeffect in the range $E_\gamma = 1\text{--}2000 \text{ keV}$," *Phys. Rev.* **159**, 50–56.
- Ramanathan, G. (2008). *Possible Heat Defect in Graphite*, NPL Report AIR (RES) 017 (National Physical Laboratory, Teddington, UK).
- Rao, P. S., and Gregg, E. C. (1975). "Attenuation of monoenergetic gamma rays in tissues," *Am. J. Roentg.* **123**, 631–637.
- Rao, V. V., Shah Nawaz, and Rao, D. V. (1981). "Photoelectric cross sections for x-ray photons in low- and medium-Z elements," *Physica C* **111**, 107–110.
- Rao, A. S. N., Perumallu, A., and Rao, G. K. (1984). "Photon cross section measurements in compounds and elements in the energy range 30–660 keV," *Physica C* **124**, 96–104.
- Rao, D. V., Seltzer, S. M., and Bergstrom, P. M., Jr. (2004). "Compton scattering cross-sections for individual subshells for a few elements of biological interest in the energy region 5 keV–10 MeV," *Radiat. Phys. Chem.* **70**, 479–489.
- Rapp, B., Perichon, N., Denozziere, M., Daures, J., Ostrowski, A., and Bordy, J.-M. (2013). "The LNE-LNHB water calorimeter for primary measurement of absorbed dose at low depth in water: Application to medium-energy x-rays," *Phys. Med. Biol.* **58**, 2769–2786.
- Rehr, J. J., and Albers, R. C. (2000). "Theoretical approaches to x-ray absorption fine structure," *Rev. Mod. Phys.* **72**, 621–654.
- Reiter, G., Kniest, N., Pfaff, E., and Clausnitzer, G. (1990). "Proton and helium stopping cross sections in H_2 , He, N_2 , O_2 , Ne, Ar, Kr, Xe, CH_4 and CO_2 ," *Nucl. Instrum. Meth. B* **44**, 399–411.
- Ribberfors, R. (1975). "Relationship of the relativistic Compton cross section to the momentum distribution of bound electron states," *Phys. Rev. B* **12**, 2067–2074.
- Ritchie, R. H., Hamm, R. N., Turner, J. E., and Wright, H. A. (1978). "The interaction of swift electrons with liquid water," pp. 345–354 in *Symposium on Microdosimetry*, Booz, J., and Ebert, H. G., Eds. (Harwood Academic, Brussels).
- Rogers, D. W. O. (1993). *Uncertainties in the ^{60}Co Graphite-to-Air Stopping Power Ratio and a re-evaluation of $(W/e)_{\text{air}}$ values*, NRC Report PIRS-363 (National Research Council, Ottawa).
- Rogers, D. W. O., and Ross, C. K. (1988). "The role of humidity and other correction factors in the AAPM TG-21 dosimetry protocol," *Med. Phys.* **15**, 40–48.
- Rogers, D. W. O., Kawrakow, I., Seuntjens, J. P., Walters, B. R. B., and Mainegra-Hing, E. (2011). *NRC User Codes for EGSnrc*, NRC Report PIRS-702 (National Research Council, Ottawa).
- Roof, R. B., Jr. (1959). "X-ray absorption coefficients of the elements with Z 1 to 17 for Mo K-alpha radiation," *Phys. Rev.* **113**, 826–827.
- Ross, C. K., and Klassen, N. V. (1996). "Water calorimetry for radiation dosimetry," *Phys. Med. Biol.* **41**, 1–29.
- Ross, C. K., Klassen, N. V., and Smith, G. D. (1984). "The effect of various dissolved gases on the heat defect of water," *Med. Phys.* **11**, 653–658.
- Ross, C., Klassen, N., Shortt, K., and Smith, G. (1989). "A direct comparison of water calorimetry and Fricke dosimetry," *Phys. Med. Biol.* **34**, 23–42.
- Ross, C. K., McEwen, M. R., and Klassen, N. V. (2007). "Vessel designs and correction factors for water calorimetry," Absorbed Dose and Air Kerma Primary Standards Workshop, 9–11 May, Paris. http://www.nucleide.org/ADAKPS_WS/.
- Sabbatucci, L., and Salvat, F. (2016). "Theory and calculation of the atomic photoeffect," *Radiat. Phys. Chem.* **121**, 122–140.
- Säbel, M., Schmidt, T., and Pauly, H. (1974). "Heat defect of low energy X-rays in some materials of interest in absorbed dose calorimetry," *Radiat. Environ. Biophys.* **11**, 259–264.
- Sakamoto, N., Ogawa, H., and Shiomi-Tsuda, N. (1996). "Stopping powers of carbon for protons from 4 to 13 MeV," *Nucl. Instrum. Meth. B* **115**, 84–87.
- Salata, C., David, M., de Almeida, C., El Gamal, I., Cojocaru, C., Mainegra-Hing, E., and McEwen, M. (2014). "Comparison of absorbed dose to water standards for HDR Ir-192 brachytherapy between the LCR, Brazil, and NRC, Canada," *Med. Phys.* **41**, 388–388.
- Saloman, E. B., and Hubbell, J. H. (1986). *X-ray Attenuation Coefficients (Total Cross Sections): Comparison of the Experimental Data Base with the Recommended Values of Henke and the Theoretical Values of Scofield for Energies Between 0.1–100 keV*, Report NBSIR-86-3431 (National Bureau of Standards, Gaithersburg, MD).
- Saloman, E. B., Hubbell, J. H., and Scofield, J. H. (1988). "X-ray attenuation cross sections for energies 100 eV to 100 keV and elements $Z = 1$ to $Z = 92$," *At. Data Nucl. Data Tables* **38**, 1–196.
- Salvat, F. (2015a). "The PENELOPE code system. Specific features and recent improvements," *Ann. Nucl. Energy* **82**, 98–109.
- Salvat, F. (2015b). *PENELOPE-2014: A Code System for Monte Carlo Simulation of Electron and Photon Transport*, Publication NEA/NSC/DOC(2015)3, Nuclear Energy Agency Data Bank (Organization for Economic Co-operation and Development, Issy-les-Moulineaux, France).
- Salvat, F., and Fernández-Varea, J. M. (2009). "Overview of physical interaction models for photon and electron

References

- transport used in Monte Carlo codes," *Metrologia* **46**, S112–S138.
- Sauer, S. P. A., Sabin, J. R., and Oddershede, J. (1995). "Calculated molecular mean excitation energies for some small molecules," *Nucl. Instrum. Meth. B* **100**, 458–463.
- Schardt, D., Steidl, P., Krämer, M., Weber, U., Parodi, K., and Brons, S. (2008). "Precision Bragg-curve measurements for light ions in water," p. 373 in GSI Report 2008-1 (Gesellschaft für Schwerionenforschung, Darmstadt).
- Schmickley, R. D., and Pratt, R. H. (1967). "K-, L-, and M-shell atomic photoeffect for screened-potential models," *Phys. Rev.* **164**, 104–116.
- Schocken, K. (1929). "Über die Schwächungskoeffizienten einiger Gase für kurzwellige Röntgenstrahlen," *Z. Phys.* **58**, 39–47.
- Schoknecht, G. (1963). "Absorption measurements on x-ray contrast media with monochromatic radiation," *Biophysik* **1**, 114–122.
- Schulz, R. J., Venkataramanan, N., and Huq, M. S. (1990). "The thermal defect of A-150 plastic and graphite for low-energy protons," *Phys. Med. Biol.* **35**, 1563–1574.
- Scofield, J. H. (1973). *Theoretical Photoionization Cross Sections from 1 to 1500 keV*, Report UCRL-51326 (Lawrence Livermore Laboratory, Livermore, CA).
- Seltzer, S. M. (1993). "Calculation of photon mass energy-transfer and mass energy-absorption coefficients," *Radiat. Res.* **136**, 147–170.
- Seltzer, S. M., and Berger, M. J. (1985). "Bremsstrahlung spectra from electron interactions with screened atomic nuclei and orbital electrons," *Nucl. Instrum. Meth. B* **12**, 95–134.
- Seltzer, S. M., and Berger, M. J. (1986). "Bremsstrahlung energy spectra from electrons with kinetic energy 1 keV–10 GeV incident on screened nuclei and orbital electrons of neutral atoms with $Z = 1$ –100," *At. Data Nucl. Data Tables* **35**, 345–418.
- Seltzer, S. M., and Bergstrom, P. M., Jr. (2003). "Changes in the US primary standards for the air kerma from gamma-ray beams," *J. Res. Natl. Inst. Stand. Technol.* **108**, 359–382.
- Senemaud, G. (1969). "Mesure de l'affaiblissement du rayonnement x par le beryllium, le carbone et differents plastiques entre 4 et 17.5 angstrom," *J. Phys. (Paris)* **30**, 811–818.
- Seuntjens, J., and Duane, S. (2009). "Photon absorbed dose standards," *Metrologia* **46**, S39–S58.
- Seuntjens, J. P., Ross, C. K., Shortt, K. R., and Rogers, D. W. O. (2000). "Absorbed-dose beam quality conversion factors for cylindrical chambers in high energy photon beams," *Med. Phys.* **27**, 2763–2779.
- Shalek, R. J., and Smith, C. E. (1969). "Chemical dosimetry for the measurement of high-energy photons and electrons," *Ann. N. Y. Acad. Sci.* **161**, 44–62.
- Sharpe, P., Ed. (2009). "Special issue on radiation dosimetry," *Metrologia* **46**, S1–S138.
- Sheikh-Bagheri, D., and Rogers, D. W. O. (2002). "Monte Carlo calculation of nine megavoltage photon beam spectra using the BEAM code," *Med. Phys.* **29**, 391–402.
- Shimizu, S., Hanai, T., and Okamoto, S. (1952). "The absorption of gamma-rays from ^{60}Co ," *Phys. Rev.* **85**, 290–294.
- Shiomi-Tsuda, N., Sakamoto, N., Ogawa, H., Tanaka, M., Saitoh, M., and Kitoba, U. (1999). "Stopping powers of N_2 and O_2 for protons from 4.0 to 13.0 MeV," *Nucl. Instrum. Meth. B* **149**, 17–24.
- Shortt, K. R. (1989). "The temperature dependence of $G(\text{Fe}^{3+})$ for the Fricke dosimeter," *Phys. Med. Biol.* **34**, 1923–1926.
- Siegbahn, K. (1969). *ESCA Applied to Free Molecules* (North-Holland, Amsterdam).
- Sigmund, P., and Schinner, A. (2000). "Binary stopping theory for swift heavy ions," *Eur. Phys. J. D* **12**, 425–434.
- Sigmund, P., and Schinner, A. (2002). "Binary theory of electronic stopping," *Nucl. Instrum. Meth. B* **195**, 64–90.
- Sigmund, P., Schinner, A., and Paul, H. (2009). "Errata and Addenda: ICRU Report 73 (Stopping of Ions Heavier than Helium)," [Online] Available: http://icru.org/images/pdf/errata_and_addenda.pdf.
- Sihver, L., Schardt, D., and Kanai, T. (1998). "Depth-dose distributions of high-energy carbon, oxygen and neon beams in water," *Jpn. J. Med. Phys.* **18**, 1–21.
- Spencer, R. G. (1931). "Absorption of soft x-rays in gases," *Phys. Rev.* **38**, 1932–1937 (see also *Phys. Rev.* **39**, 178 (1932)).
- Spencer, L. V., and Attix, F. H. (1955). "A theory of cavity ionization," *Radiat. Res.* **3**, 239–254.
- Sternheimer, R. M. (1952). "The density effect for the ionization loss in various materials," *Phys. Rev.* **88**, 851–859.
- Sternheimer, R. M., Seltzer, S. M., and Berger, M. J. (1982). "Density effect for the ionization loss of charged particles in various substances," *Phys. Rev. B* **26**, 6067–6076.
- Stockmeyer, W. (1932). "Untersuchungen zur Anwendung der Ionisationsmessmethode bei Röntgen Strahlen," *Ann. Phys. (Leipzig)* **12**, 71–106.
- Storm, L., and Israel, H. I. (1970). "Photon cross sections from 1 keV to 100 MeV for elements $Z=1$ to $Z=100$," *At. Data Nucl. Data Tables* **7**, 565–681.
- Stucki, G., and Vörös, S. (2007). "Experimental k_{Q,Q_0} electron beam quality correction factors for the types NACP02 and PTW34001 plane-parallel chambers," in *Workshop on Absorbed Dose and Air Kerma Primary Standards* (Laboratoire national de métrologie et d'essais, Paris).
- Surić, T., Bergstrom, P. M., Jr., Pisk, K., and Pratt, R. H. (1991). "Compton scattering of photons by inner-shell electrons," *Phys. Rev. Lett.* **67**, 189–192.
- Svensson, H., and Brahme, A. (1986). "Recent Advances in Electron and Photon Dosimetry," pp. 87–170 in *Radiation Dosimetry*, Orton, C. G., Ed. (Springer, New York).
- Swint, J. B., Prior, R. M., and Ramirez, J. J. (1970). "Energy loss of protons in gases," *Nucl. Instrum. Meth.* **80**, 134–140.
- Takata, N., and Begum, A. (2008). "Corrections to air kerma and exposure measured with free air ionisation chambers for charge of photoelectrons, Compton electrons and Auger electrons," *Radiat. Prot. Dosim.* **130**, 410–418.

- Tan, Z., Xia, Y., Zhao, M., Liu, X., Li, F., Huang, B., and Ji, Y. (2004). "Electron stopping power and mean free path in organic compounds over the energy range of 20–10,000 eV," *Nucl. Instrum. Meth. B* **222**, 27–43.
- Taylor, E. G. (1922). "Absorption coefficients for homogeneous x-rays," *Phys. Rev.* **20**, 709–714.
- Tessier, F., and Kawrakow, I. (2008). "Calculation of the electron–electron bremsstrahlung cross-section in the field of atomic electrons," *Nucl. Instrum. Meth. B* **266**, 625–634.
- Thompson, T. J. (1952). *Effect of Chemical Structure on Stopping Powers for High-Energy Protons*, Report UCRL-1910 (Lawrence Radiation Laboratory, Berkeley, CA).
- Thomson, J. J. (1906). "On the number of corpuscles in an atom," *Phil. Mag.* **11**, 769–781.
- Thomson, R. M., and Rogers, D. W. O. (2010). "Re-evaluation of the product of W/e_{air} and the graphite to air stopping-power ratio for ^{60}Co air kerma standards," *Phys. Med. Biol.* **55**, 3577–3595.
- Tran, C. Q., Chantler, C. T., Barnea, Z., Paterson, D., and Cookson, D. J. (2003). "Measurement of the x-ray mass attenuation coefficient and the imaginary part of the form factor of silicon using synchrotron radiation," *Phys. Rev. A* **67**, 042716-1–12.
- Tran, C. Q., Chantler, C. T., Barnea, Z., de Jong, M. D., Dhal, B. B., Chung, C. T. Y., Paterson, D., and Wang, J. (2005). "Measurement of the x-ray mass attenuation coefficient of silver using the x-ray-extended range technique," *J. Phys. B* **38**, 89–107.
- Tseng, H. K., and Pratt, R. H. (1971). "Exact screened calculations of atomic-field bremsstrahlung," *Phys. Rev. A* **3**, 100–115.
- Umesh, T. K., Ranganathaiah, C., Gowda, R., Puttaswamy, K. S., and Sanjeevaiah, B. (1981). "Incoherent-scattering cross sections in low- and medium-Z elements derived from the measured total attenuation cross sections in compounds," *Phys. Rev. A* **23**, 2365–2373.
- Varlamov, A. V., Varlamov, V. V., Rudenko, D. S., and Stepanov, M. E. (1999). *Atlas of Giant Dipole Resonances: Parameters and Graphs of Photonuclear Reaction Cross Sections*, Report INDC(NDS)-394, IAEA Nuclear Data Section (International Atomic Energy Agency, Vienna).
- Vatnitskiy, S., Andreo, P., and Jones, D. T. L. (2011). "Recent advances in dosimetry in reference conditions for proton and light-ion beams," pp. 357–366 in *Standards, Applications and Quality Assurance in Medical Radiation Dosimetry* (International Atomic Energy Agency, Vienna).
- Verhaegen, F., and Palmans, H. (2001). "A systematic Monte Carlo study of secondary electron fluence perturbation in clinical proton beams (70–250 MeV) for cylindrical and spherical ion chambers," *Med. Phys.*, **28**, 2088–2095.
- Verhey, L. J. (2000). "Response to 'Comparison of dosimetry recommendations for clinical proton beams'," *Phys. Med. Biol.* **45**, L51–L52.
- Vynckier, S., Bonnett, D. E., and Jones, D. T. L. (1991). "Code of practice for clinical proton dosimetry," *Radiother. Oncol.* **20**, 53–63.
- Waibel, E., and Grosswendt, B. (1978). "Determination of W values and backscatter coefficients for slow electrons in air," *Radiat. Res.* **76**, 241–249.
- Waller, I., and Hartree, D. R. (1929). "On the intensity of total scattering of x-rays," *Proc. Roy. Soc. A* **124**, 119–142.
- Wambersie, A., DeLuca, P. M., Andreo, P., and Hendry, J. H. (2004). "'Light' or 'Heavy' ions: A debate of terminology," *Radiother. Oncol.* **73**, Supplement 2, iii.
- Weber, J., and Van den Berge, D. J. (1969). "The effective atomic number and the calculation of the composition of phantom materials," *Br. J. Radiol.* **42**, 378–383.
- Weissmantel, C., and Wünschmann, M. (1965). "Systematische Untersuchung der Schwächungskoeffizienten von Elementen der Ordnungszahlen 1 bis 30 bezüglich Gamma- bzw. Röntgenstrahlung im Energiebereich 6 bis 85 keV," *Z. Chem.* **5**, 191–193.
- Weisweiler, W. (1968). "Messung des Massenabsorptionskoeffizienten in Kohlenstoff im Wellenlängenbereich 10 bis 70 Angstromen," p.193 in *Proc. 5th Int. Congr. X-Ray Optics and Micro-analysis* (Tübingen, Germany).
- Wheeler, J. A., and Lamb, W. E. (1939). "Influence of atomic electrons on radiation and pair production," *Phys. Rev.* **55**, 858–862.
- Wiedenbeck, M. L. (1962). "Total x-ray attenuation coefficients from 40 keV to 412 keV," *Phys. Rev.* **126**, 1009–1010.
- Wingårdh, K. A. (1922). "Untersuchungen über die Absorption der Röntgenstrahlen. I," *Z. Phys.* **8**, 363–376.
- Woernle, B. (1930). "Die Absorption langwelliger Röntgenstrahlen von 2-10 Angstrom in leichten Elementen," *Ann. Phys. (Leipzig)* **5**, 475–506.
- Wrede, W. (1939). "Die Massenschwächungskoeffizienten monochromatischer Röntgenstrahlen von 24 Elemente zwischen C(6) und Ce(58) für Wellenlängen von 0.1279 bis 1.433 Angstromen," *Ann. Physik (Leipzig)* **36**, 681–695.
- Wynd, S. J. (1952). "Absorption coefficients of gamma-rays," *Phys. Rev.* **87**, 165 (see also *Proc. Phys. Soc. (London)* Ser. A **66**, 382–390 (1953)).
- Zeiss, G. D., Meath, W. J., MacDonald, J. C. F., and Dawson, D. J. (1977). "Dipole oscillator strength distributions, sums, and some related properties for Li, N, O, H₂, N₂, O₂, NH₃, H₂O, NO, and N₂O," *Can. J. Phys.* **55**, 2080–2100.
- Zeng, G. G., McEwen, M. R., Rogers, D. W. O., and Klassen, N. V. (2005). "An experimental and Monte Carlo investigation of the energy dependence of alanine/EPR dosimetry: II. Clinical electron beams," *Phys. Med. Biol.* **50**, 1119–1129.
- Ziegler, J. F. (1977). "The electronic and nuclear stopping of energetic ions," *Appl. Phys. Lett.* **31**, 544–546.
- Ziegler, J. F. (1978). *The Stopping and Ranges of Ions in Matter*, Vol. **4** (Pergamon, New York).
- Ziegler, J. F. (1999). "Stopping of energetic light ions in elemental matter," *J. Appl. Phys.* **85**, 1249–1272.
- Ziegler, J. F., Biersack, J. P., and Littmark, U. (1985). *The Stopping and Ranges of Ions in Matter*, Vol. **1** (Pergamon, New York).
- Ziegler, J. F., Ziegler, M. D., and Biersack, J. P. (2010). "SRIM—The stopping and range of ions in matter (2010)," *Nucl. Instrum. Meth. B* **268**, 1818–1823.

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Stopping Powers and Ranges for Protons and Alpha Particles, with Data Disk
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Stopping Powers and Ranges for Protons and Alpha Particles
Report no. 49, 1993

Phantoms and Computational Models in Therapy, Diagnosis and Protection
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Measurement of Dose Equivalents from External Photon and Electron Radiations
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Photon, Electron, Proton and Neutron Interaction Data for Body Tissues, with Data Disk
Report no. 46D, 1992

Photon, Electron, Proton and Neutron Interaction Data for Body Tissues
Report no. 46, 1992

Clinical Neutron Dosimetry—Part I: Determination of Absorbed Dose in a Patient Treated by External Beams of Fast Neutrons
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Tissue Substitutes in Radiation Dosimetry and Measurement
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Determination of Dose Equivalents from External Radiation Sources - Part 2
Report no. 43, 1988

Use of Computers in External Beam Radiotherapy Procedures with High-Energy Photons and Electrons
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Modulation Transfer Function of Screen-Film Systems
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Determination of Dose Equivalents Resulting from External Radiation Sources
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Stopping Powers for Electrons and Positrons
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Microdosimetry
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Radiation Dosimetry: Electron Beams with Energies Between 1 and 50 MeV
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The Dosimetry of Pulsed Radiation
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Radiation Quantities and Units
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Methods of Assessment of Absorbed Dose in Clinical Use of Radionuclides
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Average Energy Required to Produce an Ion Pair
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Quantitative Concepts and Dosimetry in Radiobiology
Report no. 30, 1979

Basic Aspects of High Energy Particle Interactions and Radiation Dosimetry
Report no. 28, 1978

Neutron Dosimetry for Biology and Medicine
Report no. 26, 1977

An International Neutron Dosimetry Intercomparison
Report no. 27, 1978

Conceptual Basis for the Determination of Dose Equivalent
Report no. 25, 1976

Determination of Absorbed Dose in a Patient Irradiated by Beams of X or Gamma Rays in Radiotherapy Procedures
Report no. 24, 1976

Measurement of Absorbed Dose in a Phantom Irradiated by a Single Beam of X or Gamma Rays
Report no. 23, 1973